

Solvable Quantum Many-Body Models in Condensed Matter

Mother Models, Equivalence Maps, and Unified Solution Methods

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Preface

The purpose of these lecture notes is not to present a long catalogue of isolated solvable models. Instead, the central aim is to construct a *mother-model and equivalence-map approach* to exactly solvable quantum many-body systems in condensed matter physics. The guiding strategy is to identify a small number of mathematically general and physically representative structures, solve them in detail, and then recover many familiar models as parameter limits, dual descriptions, transfer-matrix limits, Jordan–Wigner images, Bogoliubov–de Gennes representations, gauge-sector reductions, or special cases of Bethe-ansatz/Yang–Baxter integrability.

The defining principle of the notes is therefore

$\begin{array}{c} \textit{Solve a small number of mother models} \\ \implies \\ \textit{cover whole families by parameter choices or equivalence maps.} \end{array}$
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Thus the intended structure is not

$$\text{Ising} \rightarrow \text{XY} \rightarrow \text{Kitaev} \rightarrow \text{SSH} \rightarrow \cdots,$$

but rather

$\begin{array}{c} \text{free fermions/BdG} \rightarrow \text{transfer matrices} \rightarrow \text{Bethe ansatz/Yang–Baxter} \\ \rightarrow \text{commuting projectors} \rightarrow \text{Majorana–gauge solvability.} \end{array}$
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How to read these notes

A productive reading route is the following.

- (i) First master the unified tools in Part I: spin algebra, the Jordan–Wigner transformation, Fourier transformation on a lattice, BdG formalism, Bogoliubov diagonalization, transfer matrices, and duality.
- (ii) Then study the transverse-field XY chain in Part II. This is the first genuine mother model in the notes; it contains the TFIM, the XX chain, and the Kitaev chain after fermionization.
- (iii) Next read Part III to understand the transfer-matrix correspondence between one-dimensional quantum systems and two-dimensional classical statistical models.
- (iv) Then move to Part IV, which treats interacting but integrable systems such as the XXZ spin chain and the six-vertex model.
- (v) Finally, read Part V for two-dimensional quantum exact solvability, including the toric code, the Kitaev honeycomb model, Rokhsar–Kivelson points, and exact ground-state constructions.

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Chapter 1

Introduction: Solvability by Mother Models and Equivalence Maps

1.1 Motivation

Exactly solvable models in condensed matter physics have two complementary roles. First, they provide rare nonperturbative laboratories in which the spectrum, quasiparticles, correlation functions, order parameters, topological invariants, or partition functions can be computed with controlled mathematics. Second, they serve as conceptual fixed points around which more general many-body phenomena can be organized: quantum criticality, topological phases, fractionalization, duality, integrability, and entanglement structure.

The distinctive feature of these notes is that solvable models are organized by *mother models*. A mother model is a Hamiltonian, transfer matrix, algebraic structure, or exactly solvable construction general enough that several familiar models can be obtained from it through parameter choices, limiting procedures, duality transformations, fermionization, or gauge-sector reduction.

Principle 1.1 (Mother-model principle). If a family of models shares the same diagonalization mechanism, transfer matrix, quadratic fermion/BdG structure, Yang–Baxter algebra, stabilizer algebra, or Majorana-gauge reduction, one should first solve the mother structure rather than repeatedly solve each special case from scratch.

1.2 Four notions of exact solvability

The phrase “exactly solvable” is not unique. These notes distinguish four useful levels of solvability.

Table 1.1: Different strengths of exact solvability.

Level	Meaning	Representative models
A. Full spectral diagonalization	One can obtain the full spectrum, quasiparticles, partition function, or correlation-function structure.	XY chain, TFIM, Kitaev chain, 2D Ising model

Level	Meaning	Representative models
B. Free-fermion/ Pfaffian solvability	An interacting spin model or classical statistical model can be mapped by Jordan–Wigner, Majorana, Kasteleyn, or Pfaffian methods to a quadratic fermion problem.	TFIM, XY chain, Kitaev chain, dimer models, 2D Ising model, Kitaev honeycomb model in a fixed flux sector
C. Bethe/Yang–Baxter integrability	The model is not necessarily free, but it is solvable because of factorized scattering, commuting transfer matrices, or an underlying Yang–Baxter structure.	XXX, XXZ, XYZ, six-vertex, eight-vertex, 1D Hubbard model
D. Exact ground state or commuting-projector solvability	The ground state or the whole spectrum is controlled by projector, stabilizer, or frustration-free structure, although generic dynamics need not be integrable.	Toric code, cluster model, AKLT chain, RK dimer model

1.3 Core equivalence map

The central map of these notes is shown in [Fig. 1.1](#).

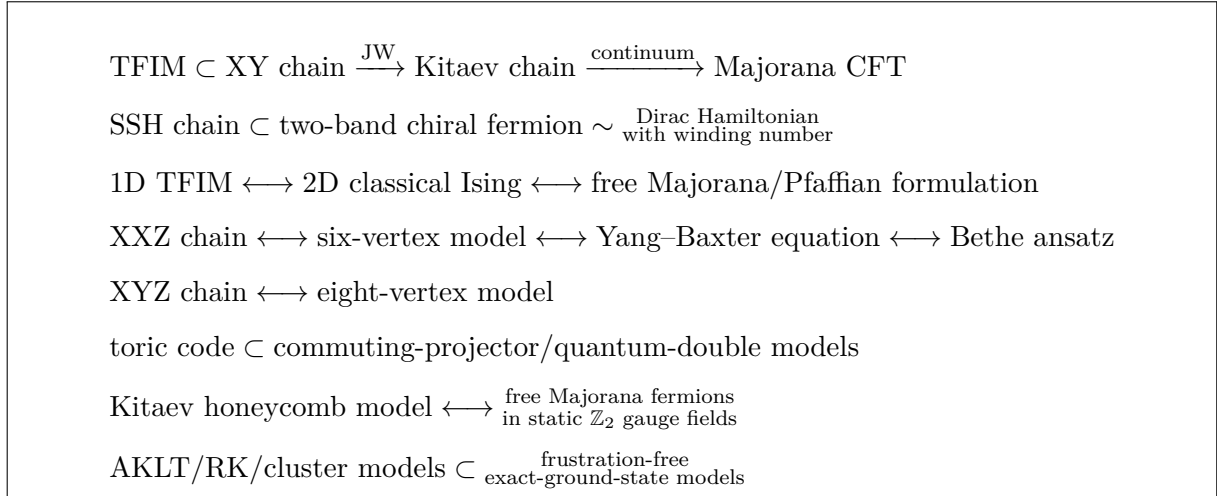


Figure 1.1: Core equivalence map among the main solvable structures.

1.4 Primary mother structures

The full notes are organized around five primary mother structures:

free-fermion/BdG solvability,
 transfer matrices and quantum–classical correspondence,
 Bethe ansatz and Yang–Baxter integrability,
 commuting-projector/stabilizer solvability,
 Majorana fermions coupled to static $\mathbb{Z}_2$ gauge fields.

(1.1)

Chapter 2

Model Taxonomy by Equivalence Class

2.1 One-dimensional quantum models

Table 2.1: Mother classes, representative models, and solution methods for one-dimensional quantum many-body systems.

Mother class	Representative models	Main solution method	Role in the notes
Quadratic fermion / BdG	TFIM, transverse-field XY chain, XX chain, Kitaev chain	Jordan–Wigner transformation, Fourier transform, Bogoliubov rotation	Essential main line
Two-band chiral free fermion	SSH chain, dimerized hopping chain, dimerized XX chain	Bloch Hamiltonian and winding number	Essential main line
Bethe-ansatz integrability	XXX chain, XXZ chain	Coordinate Bethe ansatz and algebraic Bethe ansatz	Essential main line
Yang–Baxter / vertex-model integrability	XXZ $\leftrightarrow$ six-vertex, XYZ $\leftrightarrow$ eight-vertex	Commuting transfer matrices, R -matrix, Yang–Baxter equation	Middle-to-advanced main line
Nested Bethe ansatz	1D Hubbard model	Lieb–Wu equations, nested Bethe ansatz	Advanced chapter
Long-range integrability	Haldane–Shastry model, Calogero–Sutherland model	Inverse-square interaction, fractional statistics, special conserved charges	Advanced chapter
Frustration-free / MPS solvability	AKLT chain, cluster model, Majumdar–Ghosh chain	Projector decomposition, exact MPS, stabilizer algebra	Supplementary but recommended
Continuum low-energy field theory	Luttinger liquid, Ising CFT, Majorana CFT, compact boson	Bosonization, CFT, scaling limit	Unifying low-energy chapter

2.2 Two-dimensional classical models relevant to quantum many-body physics

Table 2.2: Two-dimensional classical exactly solved models and their relations to quantum many-body systems.

Mother class	Representative models	Main solution method	Relation to quantum systems
Transfer matrix / free fermion	2D classical Ising model at zero field	Onsager solution, transfer matrix, fermionization	Anisotropic limit gives the 1D TFIM
Vertex integrability	Six-vertex model	Bethe ansatz, Yang–Baxter equation	Quantum Hamiltonian limit gives the XXZ chain
Vertex integrability	Eight-vertex model	Baxter solution, elliptic parametrization	Quantum Hamiltonian limit gives the XYZ chain
Pfaffian free-fermion structure	Classical dimer model	Kasteleyn orientation, Pfaffian determinant	Related to RK dimers and Majorana/Pfaffian methods
Loop/height/Coulomb-gas representation	$O(n)$ loop model, SOS/RSOS models	Height mapping, Coulomb gas, integrable points	Related to CFT, minimal models, and critical lattice models

2.3 Two-dimensional quantum models

Table 2.3: Classification of two-dimensional quantum exactly solvable models.

Mother class	Representative models	Main solution method	Remarks
Commuting projector	Toric code	Stabilizer diagonalization, local commuting projectors	Exactly solvable fixed point of topological order
Quantum double	$D(G)$ models, generalized toric codes	Group algebra, quantum double, anyon fusion	Non-Abelian generalization of the toric code structure
Majorana + static gauge field	Kitaev honeycomb model	Majorana representation, conserved $\mathbb{Z}_2$ gauge fields, free Majorana Hamiltonian in each flux sector	Central exactly solvable 2D spin-liquid model
Free Chern/BdG bands	Chern insulator, $p + ip$ superconductor	Band/BdG diagonalization, Chern number	Noninteracting many-body exact models
Rokhsar–Kivelson structure	Quantum dimer models at the RK point	Equal-weight ground state, stochastic/classical correspondence	Exact ground state; not necessarily full-spectrum solvable
Stabilizer SPT	2D cluster states and related graph states	Commuting stabilizers, symmetry-protected order	Useful examples for quantum information and SPT phases

Part I

Unified Mathematical Tools

Chapter 3

Spin- $\frac{1}{2}$ Algebra and Lattice Conventions

3.1 Purpose of this chapter

Before discussing Jordan–Wigner transformations, Bogoliubov rotations, transfer matrices, or Bethe ansatz, we must fix a small number of elementary conventions. For spin chains, many sign errors do not come from difficult physics; they come from inconsistent choices of

$$\sigma^\pm, \quad |\uparrow\rangle, |\downarrow\rangle, \quad \sigma^z \text{ versus } n, \quad \text{open versus periodic boundary conditions.} \quad (3.1)$$

This chapter fixes these conventions once and for all. The main convention used throughout the free-fermion part of the notes is

$$|0\rangle_j \equiv |\uparrow\rangle_j, \quad |1\rangle_j \equiv |\downarrow\rangle_j, \quad n_j = \frac{1 - \sigma_j^z}{2}. \quad (3.2)$$

Thus the fermionic occupied state corresponds to a down spin. With this choice, the Jordan–Wigner convention in the next chapter is

$$c_j = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z \right) \sigma_j^+, \quad c_j^\dagger = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z \right) \sigma_j^-. \quad (3.3)$$

The present chapter should therefore be read as the local algebraic dictionary on which all later fermionization formulae are based.

3.2 Local Hilbert space

A single spin-1/2 degree of freedom has local Hilbert space

$$\mathcal{H}_j \cong \mathbb{C}^2. \quad (3.4)$$

We use the ordered basis

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3.5)$$

The Pauli matrices are

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.6)$$

They act on the basis states as

$$\sigma^z |\uparrow\rangle = |\uparrow\rangle, \quad \sigma^z |\downarrow\rangle = -|\downarrow\rangle. \quad (3.7)$$

For a chain of length L , the many-body Hilbert space is the ordered tensor product

$$\mathcal{H}_L = \bigotimes_{j=1}^L \mathcal{H}_j \cong (\mathbb{C}^2)^{\otimes L}. \quad (3.8)$$

A basis vector is written as

$$|s_1, s_2, \dots, s_L\rangle, \quad s_j \in \{\uparrow, \downarrow\}. \quad (3.9)$$

Equivalently, with the occupation convention introduced above,

$$|s_1, s_2, \dots, s_L\rangle \equiv |n_1, n_2, \dots, n_L\rangle, \quad n_j = \begin{cases} 0, & s_j = \uparrow, \\ 1, & s_j = \downarrow. \end{cases} \quad (3.10)$$

The operator acting as σ^α on site j and as the identity elsewhere is

$$\sigma_j^\alpha = \mathbf{1}_1 \otimes \dots \otimes \mathbf{1}_{j-1} \otimes \sigma^\alpha \otimes \mathbf{1}_{j+1} \otimes \dots \otimes \mathbf{1}_L, \quad \alpha = x, y, z. \quad (3.11)$$

Operators supported on distinct sites commute:

$$[\sigma_i^\alpha, \sigma_j^\beta] = 0, \quad i \neq j. \quad (3.12)$$

3.3 Pauli algebra and spin operators

On one site, the Pauli matrices obey

$$\sigma^\alpha \sigma^\beta = \delta_{\alpha\beta} \mathbf{1} + i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \sigma^\gamma, \quad \alpha, \beta \in \{x, y, z\}. \quad (3.13)$$

Equivalently,

$$[\sigma^\alpha, \sigma^\beta] = 2i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \sigma^\gamma, \quad \{\sigma^\alpha, \sigma^\beta\} = 2\delta_{\alpha\beta} \mathbf{1}. \quad (3.14)$$

The physical spin operators are

$$S_j^\alpha = \frac{1}{2} \sigma_j^\alpha. \quad (3.15)$$

Therefore

$$[S_i^\alpha, S_j^\beta] = i\delta_{ij} \sum_{\gamma} \epsilon_{\alpha\beta\gamma} S_j^\gamma. \quad (3.16)$$

The total spin components are

$$S_{\text{tot}}^\alpha = \sum_{j=1}^L S_j^\alpha = \frac{1}{2} \sum_{j=1}^L \sigma_j^\alpha. \quad (3.17)$$

Remark 3.1 (Pauli normalization versus spin normalization). Many condensed-matter texts write Heisenberg-type Hamiltonians using S^α , whereas the free-fermion spin-chain literature often writes them using Pauli matrices. Since $S^\alpha = \sigma^\alpha/2$, one has

$$\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{4} (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \sigma_j^z \sigma_{j+1}^z). \quad (3.18)$$

Thus coupling constants differ by factors of 4 when translating between the two normalizations. Unless explicitly stated otherwise, the spin-chain Hamiltonians in these notes are written in Pauli normalization.

Useful trace identities are

$$\text{Tr } \sigma^\alpha = 0, \quad \text{Tr}(\sigma^\alpha \sigma^\beta) = 2\delta_{\alpha\beta}, \quad (3.19)$$

and

$$\text{Tr}(\sigma^\alpha \sigma^\beta \sigma^\gamma) = 2i\epsilon_{\alpha\beta\gamma}. \quad (3.20)$$

These identities are especially useful when deriving transfer matrices, density matrices, and two-band Bloch Hamiltonians.

3.4 Raising and lowering operators

We define

$$\sigma^+ = \frac{\sigma^x + i\sigma^y}{2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma^- = \frac{\sigma^x - i\sigma^y}{2} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (3.21)$$

Hence

$$\sigma^+ |\downarrow\rangle = |\uparrow\rangle, \quad \sigma^- |\uparrow\rangle = |\downarrow\rangle, \quad (3.22)$$

while

$$\sigma^+ |\uparrow\rangle = 0, \quad \sigma^- |\downarrow\rangle = 0. \quad (3.23)$$

The inverse relations are

$$\sigma^x = \sigma^+ + \sigma^-, \quad \sigma^y = i(\sigma^- - \sigma^+). \quad (3.24)$$

The local algebra is

$$(\sigma^+)^2 = (\sigma^-)^2 = 0, \quad \{\sigma^+, \sigma^-\} = \mathbf{1}, \quad [\sigma^+, \sigma^z] = \sigma^-, \quad (3.25)$$

and

$$[\sigma^z, \sigma^+] = 2\sigma^+, \quad [\sigma^z, \sigma^-] = -2\sigma^-. \quad (3.26)$$

The two local projectors are

$$\sigma^+ \sigma^- = \frac{1 + \sigma^z}{2} = |\uparrow\rangle\langle\uparrow|, \quad \sigma^- \sigma^+ = \frac{1 - \sigma^z}{2} = |\downarrow\rangle\langle\downarrow|. \quad (3.27)$$

3.5 Occupation convention and hard-core bosons

The Jordan–Wigner transformation used later identifies the spin-down state with fermion occupation. We therefore define

$$|0\rangle_j \equiv |\uparrow\rangle_j, \quad |1\rangle_j \equiv |\downarrow\rangle_j. \quad (3.28)$$

The local number operator is

$$n_j = |\downarrow\rangle_j \langle \downarrow| = \sigma_j^- \sigma_j^+ = \frac{1 - \sigma_j^z}{2}. \quad (3.29)$$

Thus

$$\sigma_j^z = 1 - 2n_j. \quad (3.30)$$

It is convenient to introduce local hard-core boson operators

$$b_j = \sigma_j^+, \quad b_j^\dagger = \sigma_j^-. \quad (3.31)$$

With the present convention, $b_j^\dagger$ creates a down spin, hence creates an occupied site:

$$b_j^\dagger |0\rangle_j = |1\rangle_j, \quad b_j |1\rangle_j = |0\rangle_j. \quad (3.32)$$

The number operator is

$$n_j = b_j^\dagger b_j. \quad (3.33)$$

On the same site, hard-core bosons obey

$$b_j^2 = (b_j^\dagger)^2 = 0, \quad \{b_j, b_j^\dagger\} = 1, \quad [b_j, b_j^\dagger] = 1 - 2n_j = \sigma_j^z. \quad (3.34)$$

On different sites, however, they commute rather than anticommute:

$$[b_i, b_j] = 0, \quad [b_i, b_j^\dagger] = 0, \quad [b_i^\dagger, b_j^\dagger] = 0, \quad i \neq j. \quad (3.35)$$

This mixed behavior is the reason a nonlocal Jordan–Wigner string is needed to convert hard-core bosons into canonical fermions.

Remark 3.2 (Why $b = \sigma^+$ but $b^\dagger = \sigma^-$). This convention may look reversed if one thinks of σ^+ as a creation operator. Here the created object is not spin-up, but an occupied fermionic site. Since occupation means spin down, the operator that creates occupation must send $|\uparrow\rangle$ to $|\downarrow\rangle$. Therefore $b^\dagger = \sigma^-$.

3.6 Useful two-site identities

The following identities are used repeatedly in the XY, XX, XXZ, and Kitaev-chain chapters. For nearest-neighbour sites j and $j+1$,

$$\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y = 2 (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+), \quad (3.36)$$

$$\sigma_j^x \sigma_{j+1}^x - \sigma_j^y \sigma_{j+1}^y = 2 (\sigma_j^+ \sigma_{j+1}^+ - \sigma_j^- \sigma_{j+1}^-). \quad (3.37)$$

Equivalently,

$$\sigma_j^x \sigma_{j+1}^x = \sigma_j^+ \sigma_{j+1}^+ + \sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+ + \sigma_j^- \sigma_{j+1}^-, \quad (3.38)$$

$$\sigma_j^y \sigma_{j+1}^y = -\sigma_j^+ \sigma_{j+1}^+ + \sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+ - \sigma_j^- \sigma_{j+1}^-. \quad (3.39)$$

In terms of the occupation convention,

$$\sigma_j^z \sigma_{j+1}^z = (1 - 2n_j)(1 - 2n_{j+1}) = 1 - 2n_j - 2n_{j+1} + 4n_j n_{j+1}. \quad (3.40)$$

This last identity explains why an Ising interaction in the z direction becomes a density–density interaction after fermionization, whereas the xx and yy terms of the XY chain become hopping and pairing terms.

3.7 Lattice conventions

Unless stated otherwise, a one-dimensional chain has sites

$$j = 1, 2, \dots, L, \quad (3.41)$$

with lattice spacing set to

$$a = 1. \quad (3.42)$$

A nearest-neighbour Hamiltonian is written using one of two bond sets:

$$\mathcal{B}_{\text{OBC}} = \{(j, j+1) \mid j = 1, 2, \dots, L-1\}, \quad (3.43)$$

$$\mathcal{B}_{\text{PBC}} = \{(j, j+1) \mid j = 1, 2, \dots, L\}, \quad L+1 \equiv 1. \quad (3.44)$$

Thus

$$\sum_{\langle j, j+1 \rangle \in \mathcal{B}_{\text{OBC}}} \text{ means } \sum_{j=1}^{L-1}, \quad (3.45)$$

and

$$\sum_{\langle j, j+1 \rangle \in \mathcal{B}_{\text{PBC}}} \text{ means } \sum_{j=1}^L \text{ with } L+1 \equiv 1. \quad (3.46)$$

When no boundary condition is explicitly stated, formulae for local bulk Hamiltonian densities should be read as open-chain formulae first. Periodic boundary conditions must be treated with additional care after the Jordan–Wigner transformation because the boundary spin term carries a fermion-parity-dependent sign.

For translation-invariant periodic chains, the one-site translation operator T is defined by

$$T\sigma_j^\alpha T^{-1} = \sigma_{j+1}^\alpha, \quad T^L = 1. \quad (3.47)$$

Momentum eigenstates satisfy

$$T|\psi_k\rangle = e^{ik}|\psi_k\rangle, \quad (3.48)$$

where the allowed values of k depend on the boundary condition and, after the Jordan–Wigner transformation, on the fermion-parity sector. The detailed Fourier convention is fixed in the Fourier-transform chapter.

Remark 3.3 (Ordering convention). The Jordan–Wigner string depends on a one-dimensional ordering of sites. In these notes the ordering is always

$$1 < 2 < \dots < L. \quad (3.49)$$

This is natural for one-dimensional chains. In higher-dimensional spin models, a linear ordering can still be imposed, but it usually destroys locality. This is one reason why two-dimensional exact solvability often requires other structures, such as commuting projectors, transfer matrices, or Majorana fermions coupled to static $\mathbb{Z}_2$ gauge fields.

3.8 Common spin Hamiltonians

We now list the spin-chain Hamiltonians that will appear repeatedly in these notes. The purpose here is not to solve them, but to fix signs and normalizations. The default Pauli-normalized nearest-neighbour XYZ Hamiltonian is

$$H_{\text{XYZ}} = - \sum_{\langle j, j+1 \rangle} (J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y + J_z \sigma_j^z \sigma_{j+1}^z) - h \sum_{j=1}^L \sigma_j^z. \quad (3.50)$$

With this convention, positive J_α favors ferromagnetic alignment in the α direction.

3.8.1 Ising and transverse-field Ising chains

The Ising interaction used in the free-fermion part of the notes is chosen along x , while the transverse field is along z :

$$H_{\text{TFIM}} = -J \sum_{\langle j, j+1 \rangle} \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^L \sigma_j^z. \quad (3.51)$$

The choice of x as the ordering direction and z as the transverse-field direction is adapted to the occupation convention $n_j = (1 - \sigma_j^z)/2$. It is unitarily equivalent to the alternative textbook convention $-J \sum_j \sigma_j^z \sigma_{j+1}^z - h \sum_j \sigma_j^x$ by a global spin rotation.

The purely classical Ising interaction is recovered by setting $h = 0$:

$$H_{\text{Ising}}^x = -J \sum_{\langle j, j+1 \rangle} \sigma_j^x \sigma_{j+1}^x. \quad (3.52)$$

A longitudinal field along the ordering direction,

$$-h_x \sum_j \sigma_j^x, \quad (3.53)$$

can be added, but it destroys the standard free-fermion solvability of the one-dimensional TFIM.

3.8.2 XY and XX chains

The transverse-field XY chain is the central free-fermion mother model of these notes:

$$H_{\text{XY}} = - \sum_{\langle j, j+1 \rangle} \left[\frac{J(1+\gamma)}{2} \sigma_j^x \sigma_{j+1}^x + \frac{J(1-\gamma)}{2} \sigma_j^y \sigma_{j+1}^y \right] - h \sum_{j=1}^L \sigma_j^z. \quad (3.54)$$

The parameter γ is the XY anisotropy. Important limits are

$$\gamma = 1 \implies H_{\text{XY}} = H_{\text{TFIM}}, \quad (3.55)$$

$$\gamma = 0 \implies H_{\text{XX}} = -\frac{J}{2} \sum_{\langle j, j+1 \rangle} (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y) - h \sum_j \sigma_j^z. \quad (3.56)$$

Using [Eq. \(3.36\)](#), the XX interaction can also be written as

$$H_{\text{XX}} = -J \sum_{\langle j, j+1 \rangle} (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) - h \sum_j \sigma_j^z. \quad (3.57)$$

This form makes the conservation of total z -magnetization manifest.

3.8.3 XXZ and Heisenberg chains

In the Pauli normalization compatible with the above XX convention, we write

$$H_{\text{XXZ}} = -\frac{J}{2} \sum_{\langle j, j+1 \rangle} (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta \sigma_j^z \sigma_{j+1}^z) - h \sum_j \sigma_j^z. \quad (3.58)$$

The isotropic point $\Delta = 1$ gives a Heisenberg chain in Pauli normalization:

$$H_{\text{Heis}} = -\frac{J}{2} \sum_{\langle j, j+1 \rangle} \boldsymbol{\sigma}_j \cdot \boldsymbol{\sigma}_{j+1} - h \sum_j \sigma_j^z. \quad (3.59)$$

In spin-operator normalization, the same interaction is written as

$$H_{\text{Heis}}^S = \mathcal{J} \sum_{\langle j, j+1 \rangle} \mathbf{S}_j \cdot \mathbf{S}_{j+1} - h_S \sum_j S_j^z. \quad (3.60)$$

The conversion is

$$\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{4} \boldsymbol{\sigma}_j \cdot \boldsymbol{\sigma}_{j+1}, \quad S_j^z = \frac{1}{2} \sigma_j^z. \quad (3.61)$$

The Bethe-ansatz chapters may use the conventional antiferromagnetic sign $\mathcal{J} > 0$; when this happens, the sign and normalization will be stated explicitly.

3.8.4 Bond-dependent Kitaev-type spin Hamiltonians

In two dimensions, the Kitaev honeycomb model uses bond-dependent Ising interactions:

$$H_{\text{Kitaev}} = -J_x \sum_{\langle ij \rangle_x} \sigma_i^x \sigma_j^x - J_y \sum_{\langle ij \rangle_y} \sigma_i^y \sigma_j^y - J_z \sum_{\langle ij \rangle_z} \sigma_i^z \sigma_j^z. \quad (3.62)$$

Here $\langle ij \rangle_\alpha$ denotes a nearest-neighbour bond of type α . This model is not solved by the one-dimensional Jordan–Wigner transformation. Instead, it becomes exactly solvable through a Majorana representation and static $\mathbb{Z}_2$ gauge fields.

3.9 Symmetries and conserved quantities

Symmetry is one of the main tools for block diagonalization. We collect the most important symmetries here.

3.9.1 Translation symmetry

For periodic boundary conditions, a translation-invariant Hamiltonian satisfies

$$[H, T] = 0. \quad (3.63)$$

The Hilbert space can then be decomposed into momentum sectors labelled by the eigenvalues e^{ik} of T . Translation symmetry is essential for the Fourier transform and the reduction of quadratic Hamiltonians into independent momentum blocks.

3.9.2 Fermion parity and Ising spin flip

Define the global z -parity operator

$$P_z = \prod_{j=1}^L \sigma_j^z. \quad (3.64)$$

It satisfies

$$P_z^2 = 1, \quad P_z \sigma_j^x P_z = -\sigma_j^x, \quad P_z \sigma_j^y P_z = -\sigma_j^y, \quad P_z \sigma_j^z P_z = \sigma_j^z. \quad (3.65)$$

Therefore the TFIM and the XY chain commute with P_z :

$$[H_{\text{TFIM}}, P_z] = 0, \quad [H_{\text{XY}}, P_z] = 0. \quad (3.66)$$

Under the occupation convention [Eq. \(3.29\)](#),

$$P_z = \prod_{j=1}^L (1 - 2n_j) = (-1)^N, \quad N = \sum_{j=1}^L n_j. \quad (3.67)$$

Thus the global Ising spin-flip symmetry of the spin chain becomes fermion parity after the Jordan–Wigner transformation.

3.9.3 $U(1)$ spin-rotation symmetry

The total z -magnetization is

$$M_z = \sum_{j=1}^L \sigma_j^z = L - 2N. \quad (3.68)$$

A Hamiltonian has $U(1)$ spin-rotation symmetry about the z axis if

$$[H, M_z] = 0. \quad (3.69)$$

The XX, XXZ, and Heisenberg chains have this symmetry. Equivalently, they are invariant under

$$U_z(\theta) = \exp\left(-\frac{i\theta}{2} \sum_j \sigma_j^z\right), \quad (3.70)$$

which acts as

$$U_z(\theta) \sigma_j^\pm U_z(\theta)^{-1} = e^{\mp i\theta} \sigma_j^\pm. \quad (3.71)$$

The anisotropic XY chain with $\gamma \neq 0$ does not conserve M_z because [Eq. \(3.37\)](#) contains pair creation and pair annihilation terms. Nevertheless, it still conserves $P_z = (-1)^N$.

3.9.4 Full spin-rotation symmetry

The isotropic Heisenberg chain at zero field has full $SU(2)$ symmetry:

$$[H_{\text{Heis}}, S_{\text{tot}}^\alpha] = 0, \quad \alpha = x, y, z. \quad (3.72)$$

An external field along z breaks this to the $U(1)$ subgroup generated by S_{tot}^z .

3.9.5 Boundary-sector subtlety after Jordan–Wigner

For an open chain, the Jordan–Wigner transformation maps local nearest-neighbour bulk terms to local fermion bilinears without any boundary ambiguity. For a periodic spin chain, the boundary term connecting L and 1 depends on the parity sector

$$P_z = (-1)^N = p, \quad p = \pm 1. \quad (3.73)$$

With the Jordan–Wigner convention used in these notes, the effective fermionic boundary condition in a fixed parity sector is

$$c_{L+1} = -p c_1. \quad (3.74)$$

Thus

$$p = +1 \implies c_{L+1} = -c_1, \quad p = -1 \implies c_{L+1} = c_1. \quad (3.75)$$

That is, even fermion parity corresponds to anti-periodic fermions, while odd fermion parity corresponds to periodic fermions. The derivation is given in the Jordan–Wigner chapter.

3.10 Summary of conventions

The conventions fixed in this chapter are summarized as follows:

$$|0\rangle_j = |\uparrow\rangle_j, \quad |1\rangle_j = |\downarrow\rangle_j, \quad n_j = \frac{1 - \sigma_j^z}{2}. \quad (3.76)$$

The raising and lowering operators are

$$\sigma^+ = \frac{\sigma^x + i\sigma^y}{2}, \quad \sigma^- = \frac{\sigma^x - i\sigma^y}{2}, \quad (3.77)$$

with

$$\sigma^+ |\downarrow\rangle = |\uparrow\rangle, \quad \sigma^- |\uparrow\rangle = |\downarrow\rangle. \quad (3.78)$$

The hard-core boson convention is

$$b_j = \sigma_j^+, \quad b_j^\dagger = \sigma_j^-, \quad n_j = b_j^\dagger b_j. \quad (3.79)$$

The principal free-fermion mother model is the transverse-field XY chain

$$H_{XY} = - \sum_{\langle j, j+1 \rangle} \left[\frac{J(1+\gamma)}{2} \sigma_j^x \sigma_{j+1}^x + \frac{J(1-\gamma)}{2} \sigma_j^y \sigma_{j+1}^y \right] - h \sum_j \sigma_j^z, \quad (3.80)$$

which contains the TFIM at $\gamma = 1$ and the XX chain at $\gamma = 0$. The global operator

$$P_z = \prod_j \sigma_j^z = (-1)^N \quad (3.81)$$

will become fermion parity in the Jordan–Wigner representation. These conventions are the reference point for all later equivalence maps.

Chapter 4

Jordan–Wigner Transformation

4.1 Definition

The Jordan–Wigner transformation maps a one-dimensional spin-1/2 chain to a chain of spinless fermions. The essential difficulty is that spin operators on different sites commute, whereas fermionic operators on different sites must anticommute. The nonlocal Jordan–Wigner string supplies precisely the required fermionic signs.

Throughout this chapter we use the convention

$$|0\rangle_j \equiv |\uparrow\rangle_j, \quad |1\rangle_j \equiv |\downarrow\rangle_j. \quad (4.1)$$

Thus the occupied fermionic state corresponds to spin down. The Pauli matrices are

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (4.2)$$

The spin raising and lowering operators are

$$\sigma^+ = \frac{\sigma^x + i\sigma^y}{2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma^- = \frac{\sigma^x - i\sigma^y}{2} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (4.3)$$

They act as

$$\sigma^+ |\downarrow\rangle = |\uparrow\rangle, \quad \sigma^- |\uparrow\rangle = |\downarrow\rangle. \quad (4.4)$$

With the convention that spin down is occupied, it is natural to identify the local hard-core boson operators as

$$b_j = \sigma_j^+, \quad b_j^\dagger = \sigma_j^-. \quad (4.5)$$

The local number operator is then

$$n_j = b_j^\dagger b_j = \sigma_j^- \sigma_j^+ = \frac{1 - \sigma_j^z}{2}. \quad (4.6)$$

Equivalently,

$$\sigma_j^z = 1 - 2n_j. \quad (4.7)$$

We define the Jordan–Wigner string by

$$K_j = \prod_{\ell=1}^{j-1} \sigma_\ell^z = \prod_{\ell=1}^{j-1} (1 - 2n_\ell). \quad (4.8)$$

Since n_ℓ has eigenvalues 0 and 1, we may also write

$$K_j = \exp\left(i\pi \sum_{\ell=1}^{j-1} n_\ell\right). \quad (4.9)$$

This exponential form is often useful because

$$e^{i\pi n_\ell} = 1 - 2n_\ell = \sigma_\ell^z. \quad (4.10)$$

The Jordan–Wigner transformation used in these notes is

$$c_j = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z\right) \sigma_j^+, \quad c_j^\dagger = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z\right) \sigma_j^-. \quad (4.11)$$

Equivalently,

$$c_j = K_j \sigma_j^+ = K_j b_j, \quad c_j^\dagger = K_j \sigma_j^- = K_j b_j^\dagger. \quad (4.12)$$

The inverse transformation is

$$\sigma_j^+ = K_j c_j, \quad \sigma_j^- = K_j c_j^\dagger, \quad (4.13)$$

or, equivalently,

$$b_j = K_j c_j, \quad b_j^\dagger = K_j c_j^\dagger. \quad (4.14)$$

The string satisfies

$$K_j^\dagger = K_j, \quad K_j^2 = 1, \quad K_{j+1} = K_j \sigma_j^z = K_j (1 - 2n_j). \quad (4.15)$$

Because K_j only acts on sites strictly to the left of j , it commutes with all operators on site j :

$$[K_j, \sigma_j^\pm] = 0, \quad [K_j, \sigma_j^z] = 0, \quad [K_j, c_j] = 0, \quad [K_j, c_j^\dagger] = 0. \quad (4.16)$$

Using the inverse transformation, the spin operators become

$$\sigma_j^x = K_j (c_j^\dagger + c_j), \quad \sigma_j^y = K_j i(c_j^\dagger - c_j), \quad \sigma_j^z = 1 - 2c_j^\dagger c_j. \quad (4.17)$$

Remark 4.1. A common source of sign confusion is the alternative string $\prod_{\ell < j} (-\sigma_\ell^z)$. This differs from the convention used here by a site-dependent sign:

$$\prod_{\ell=1}^{j-1} (-\sigma_\ell^z) = (-1)^{j-1} \prod_{\ell=1}^{j-1} \sigma_\ell^z. \quad (4.18)$$

It therefore corresponds to a staggered fermion redefinition

$$c_j^{\text{alt}} = (-1)^{j-1} c_j. \quad (4.19)$$

In this chapter and in the rest of these notes, we consistently use $K_j = \prod_{\ell < j} \sigma_\ell^z$.

4.2 Canonical anticommutation relations

We now prove that the operators defined by the Jordan–Wigner transformation obey the canonical fermionic anticommutation relations

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = 0, \quad \{c_i^\dagger, c_j^\dagger\} = 0. \quad (4.20)$$

First consider the same site. Since K_j commutes with $\sigma_j^\pm$ and $K_j^2 = 1$,

$$c_j c_j^\dagger = K_j \sigma_j^+ K_j \sigma_j^- = \sigma_j^+ \sigma_j^-, \quad (4.21)$$

$$c_j^\dagger c_j = K_j \sigma_j^- K_j \sigma_j^+ = \sigma_j^- \sigma_j^+. \quad (4.22)$$

Therefore

$$\{c_j, c_j^\dagger\} = \sigma_j^+ \sigma_j^- + \sigma_j^- \sigma_j^+ = 1. \quad (4.23)$$

Also,

$$c_j^2 = (\sigma_j^+)^2 = 0, \quad (c_j^\dagger)^2 = (\sigma_j^-)^2 = 0. \quad (4.24)$$

Now take $i < j$. Define

$$R_{ij} = \prod_{\ell=i+1}^{j-1} \sigma_\ell^z, \quad (4.25)$$

so that

$$K_j = K_i \sigma_i^z R_{ij}. \quad (4.26)$$

All operators at different spin sites commute. The only nontrivial sign comes from the factor σ_i^z , which is present in the string of the operator at site j .

For two annihilation operators,

$$\begin{aligned} c_i c_j &= K_i \sigma_i^+ K_j \sigma_j^+ \\ &= K_i \sigma_i^+ K_i \sigma_i^z R_{ij} \sigma_j^+ \\ &= \sigma_i^+ \sigma_i^z R_{ij} \sigma_j^+ \\ &= -\sigma_i^+ R_{ij} \sigma_j^+, \end{aligned} \quad (4.27)$$

where we used

$$\sigma_i^+ \sigma_i^z = -\sigma_i^+. \quad (4.28)$$

On the other hand,

$$\begin{aligned} c_j c_i &= K_j \sigma_j^+ K_i \sigma_i^+ \\ &= K_i \sigma_i^z R_{ij} \sigma_j^+ K_i \sigma_i^+ \\ &= \sigma_i^z \sigma_i^+ R_{ij} \sigma_j^+ \\ &= +\sigma_i^+ R_{ij} \sigma_j^+, \end{aligned} \quad (4.29)$$

where

$$\sigma_i^z \sigma_i^+ = +\sigma_i^+. \quad (4.30)$$

Hence

$$c_i c_j = -c_j c_i, \quad i \neq j. \quad (4.31)$$

Thus

$$\{c_i, c_j\} = 0. \quad (4.32)$$

Similarly,

$$\begin{aligned}
c_i c_j^\dagger &= K_i \sigma_i^+ K_j \sigma_j^- \\
&= \sigma_i^+ \sigma_i^z R_{ij} \sigma_j^- \\
&= -\sigma_i^+ R_{ij} \sigma_j^-,
\end{aligned} \tag{4.33}$$

while

$$\begin{aligned}
c_j^\dagger c_i &= K_j \sigma_j^- K_i \sigma_i^+ \\
&= \sigma_i^z \sigma_i^+ R_{ij} \sigma_j^- \\
&= +\sigma_i^+ R_{ij} \sigma_j^-.
\end{aligned} \tag{4.34}$$

Therefore

$$c_i c_j^\dagger = -c_j^\dagger c_i, \quad i \neq j. \tag{4.35}$$

Together with the same-site result, this proves

$$\{c_i, c_j^\dagger\} = \delta_{ij}. \tag{4.36}$$

Finally, for two creation operators,

$$\begin{aligned}
c_i^\dagger c_j^\dagger &= K_i \sigma_i^- K_j \sigma_j^- \\
&= \sigma_i^- \sigma_i^z R_{ij} \sigma_j^- \\
&= +\sigma_i^- R_{ij} \sigma_j^-,
\end{aligned} \tag{4.37}$$

where

$$\sigma_i^- \sigma_i^z = +\sigma_i^-, \tag{4.38}$$

whereas

$$\begin{aligned}
c_j^\dagger c_i^\dagger &= K_j \sigma_j^- K_i \sigma_i^- \\
&= \sigma_i^z \sigma_i^- R_{ij} \sigma_j^- \\
&= -\sigma_i^- R_{ij} \sigma_j^-.
\end{aligned} \tag{4.39}$$

Hence

$$c_i^\dagger c_j^\dagger = -c_j^\dagger c_i^\dagger, \quad i \neq j. \tag{4.40}$$

This completes the proof of the canonical fermionic anticommutation relations.

4.3 Boundary conditions and fermion parity

For an open chain, the Jordan–Wigner transformation is unambiguous: the string always extends from the left boundary to the site immediately before j . No operator wraps around the chain.

For a periodic spin chain, however, a boundary term connecting site L to site 1 carries a fermion-parity-dependent sign. The total fermion number and fermion parity are

$$N = \sum_{j=1}^L n_j, \quad P_F = (-1)^N. \tag{4.41}$$

Since $1 - 2n_j = \sigma_j^z$, we have

$$P_F = \prod_{j=1}^L (1 - 2n_j) = \prod_{j=1}^L \sigma_j^z. \quad (4.42)$$

The parity operator satisfies

$$P_F^2 = 1, \quad P_F c_j P_F = -c_j, \quad P_F c_j^\dagger P_F = -c_j^\dagger. \quad (4.43)$$

The formal Jordan–Wigner string at site $L + 1$ is

$$K_{L+1} = \prod_{j=1}^L \sigma_j^z = P_F. \quad (4.44)$$

Thus, if the spin variables are periodic,

$$\sigma_{L+1}^\pm \equiv \sigma_1^\pm, \quad (4.45)$$

the formal extension of the Jordan–Wigner definition gives

$$c_{L+1}^{\text{form}} = K_{L+1} \sigma_{L+1}^+ = P_F \sigma_1^+ = P_F c_1. \quad (4.46)$$

This equation is a formal extension of the operator definition. It should not yet be identified with the effective boundary condition of a quadratic fermion Hamiltonian. The effective boundary condition is obtained only after the boundary spin bilinears are rewritten in the same algebraic form as the bulk nearest-neighbour bilinears.

For example,

$$\begin{aligned} b_L^\dagger b_1 &= K_L c_L^\dagger c_1 \\ &= \left(\prod_{\ell=1}^{L-1} \sigma_\ell^z \right) c_L^\dagger c_1 \\ &= P_F \sigma_L^z c_L^\dagger c_1 \\ &= -P_F c_L^\dagger c_1. \end{aligned} \quad (4.47)$$

Similarly,

$$b_L^\dagger b_1^\dagger = -P_F c_L^\dagger c_1^\dagger. \quad (4.48)$$

Therefore, within a fixed fermion-parity sector $P_F = p = \pm 1$, rewriting the boundary term in the same form as the bulk terms is equivalent to imposing the effective fermionic boundary condition

$$c_{L+1} = -p c_1. \quad (4.49)$$

Consequently,

$$p = +1 \quad \implies \quad c_{L+1} = -c_1, \quad (4.50)$$

so the even-parity sector corresponds to anti-periodic fermions, whereas

$$p = -1 \quad \implies \quad c_{L+1} = c_1, \quad (4.51)$$

so the odd-parity sector corresponds to periodic fermions.

4.4 Useful Jordan–Wigner identities

This section collects the identities most frequently used when translating one-dimensional spin-chain Hamiltonians into fermionic language. All formulae below use the convention

$$K_j = \prod_{\ell=1}^{j-1} \sigma_{\ell}^z = \prod_{\ell=1}^{j-1} (1 - 2n_{\ell}), \quad c_j = K_j \sigma_j^+, \quad c_j^{\dagger} = K_j \sigma_j^-. \quad (4.52)$$

Local spin, hard-core boson, and number operators

The hard-core boson convention is

$$\sigma_j^+ = b_j, \quad \sigma_j^- = b_j^{\dagger}, \quad n_j = b_j^{\dagger} b_j = c_j^{\dagger} c_j. \quad (4.53)$$

Therefore

$$\sigma_j^x = b_j^{\dagger} + b_j, \quad \sigma_j^y = i(b_j^{\dagger} - b_j), \quad \sigma_j^z = 1 - 2b_j^{\dagger} b_j = 1 - 2c_j^{\dagger} c_j. \quad (4.54)$$

The local hard-core algebra is

$$\{b_j, b_j^{\dagger}\} = 1, \quad \{b_j, b_j\} = 0, \quad \{b_j^{\dagger}, b_j^{\dagger}\} = 0, \quad (4.55)$$

while for $i \neq j$,

$$[b_i, b_j] = 0, \quad [b_i, b_j^{\dagger}] = 0, \quad [b_i^{\dagger}, b_j^{\dagger}] = 0. \quad (4.56)$$

The fermions satisfy

$$\{c_i, c_j^{\dagger}\} = \delta_{ij}, \quad \{c_i, c_j\} = 0, \quad \{c_i^{\dagger}, c_j^{\dagger}\} = 0. \quad (4.57)$$

String identities

The string identities most often used are

$$K_j^2 = 1, \quad K_j^{\dagger} = K_j, \quad K_{j+1} = K_j(1 - 2n_j). \quad (4.58)$$

Moreover,

$$c_j^{\dagger}(1 - 2n_j) = c_j^{\dagger}, \quad c_j(1 - 2n_j) = -c_j, \quad (4.59)$$

and

$$(1 - 2n_j)c_j^{\dagger} = -c_j^{\dagger}, \quad (1 - 2n_j)c_j = c_j. \quad (4.60)$$

These four identities are the main source of the signs in nearest-neighbour JW formulae.

Hard-core boson bilinears

Using $b_j = K_j c_j$, $b_j^{\dagger} = K_j c_j^{\dagger}$, and $K_{j+1} = K_j(1 - 2n_j)$, one obtains

$$b_j^{\dagger} b_j = c_j^{\dagger} c_j = n_j, \quad (4.61)$$

$$b_j^{\dagger} b_{j+1}^{\dagger} = c_j^{\dagger} (1 - 2n_j) c_{j+1}^{\dagger} = c_j^{\dagger} c_{j+1}^{\dagger}, \quad (4.62)$$

$$b_j^{\dagger} b_{j+1} = c_j^{\dagger} (1 - 2n_j) c_{j+1} = c_j^{\dagger} c_{j+1}, \quad (4.63)$$

$$b_j b_{j+1} = c_j (1 - 2n_j) c_{j+1} = -c_j c_{j+1}, \quad (4.64)$$

$$b_j b_{j+1}^{\dagger} = c_j (1 - 2n_j) c_{j+1}^{\dagger} = -c_j c_{j+1}^{\dagger}. \quad (4.65)$$

Equivalently, after reordering fermions on different sites,

$$-c_j c_{j+1}^{\dagger} = c_{j+1}^{\dagger} c_j, \quad -c_j c_{j+1} = c_{j+1} c_j. \quad (4.66)$$

Spin operators in fermions

The single-site spin operators are

$$\sigma_j^x = K_j(c_j^\dagger + c_j), \quad \sigma_j^y = K_j i(c_j^\dagger - c_j), \quad \sigma_j^z = 1 - 2n_j. \quad (4.67)$$

The nearest-neighbour xx bilinear is

$$\begin{aligned} \sigma_j^x \sigma_{j+1}^x &= (b_j^\dagger + b_j)(b_{j+1}^\dagger + b_{j+1}) \\ &= c_j^\dagger c_{j+1}^\dagger + c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j + c_{j+1} c_j \\ &= c_j^\dagger c_{j+1}^\dagger + c_j^\dagger c_{j+1} + \text{H.c.} \end{aligned} \quad (4.68)$$

The nearest-neighbour yy bilinear is

$$\begin{aligned} \sigma_j^y \sigma_{j+1}^y &= -(b_j^\dagger - b_j)(b_{j+1}^\dagger - b_{j+1}) \\ &= -c_j^\dagger c_{j+1}^\dagger + c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j - c_{j+1} c_j \\ &= -\left(c_j^\dagger c_{j+1}^\dagger - c_j^\dagger c_{j+1} + \text{H.c.}\right). \end{aligned} \quad (4.69)$$

The nearest-neighbour zz bilinear is

$$\begin{aligned} \sigma_j^z \sigma_{j+1}^z &= (1 - 2n_j)(1 - 2n_{j+1}) \\ &= 1 - 2n_j - 2n_{j+1} + 4n_j n_{j+1}. \end{aligned} \quad (4.70)$$

Useful linear combinations are

$$\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y = 2 \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right), \quad (4.71)$$

$$\sigma_j^x \sigma_{j+1}^x - \sigma_j^y \sigma_{j+1}^y = 2 \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right). \quad (4.72)$$

Tensor-product representation

For direct matrix construction on a finite chain of length L , the JW operators are

$$c_j = \left(\bigotimes_{\ell=1}^{j-1} \sigma_\ell^z \right) \otimes \sigma_j^+ \otimes \left(\bigotimes_{\ell=j+1}^L \mathbf{1}_\ell \right), \quad (4.73)$$

and

$$c_j^\dagger = \left(\bigotimes_{\ell=1}^{j-1} \sigma_\ell^z \right) \otimes \sigma_j^- \otimes \left(\bigotimes_{\ell=j+1}^L \mathbf{1}_\ell \right). \quad (4.74)$$

For example, for $L = 4$,

$$c_1 = \sigma_1^+ \otimes \mathbf{1}_2 \otimes \mathbf{1}_3 \otimes \mathbf{1}_4, \quad (4.75)$$

$$c_2 = \sigma_1^z \otimes \sigma_2^+ \otimes \mathbf{1}_3 \otimes \mathbf{1}_4, \quad (4.76)$$

$$c_3 = \sigma_1^z \otimes \sigma_2^z \otimes \sigma_3^+ \otimes \mathbf{1}_4, \quad (4.77)$$

$$c_4 = \sigma_1^z \otimes \sigma_2^z \otimes \sigma_3^z \otimes \sigma_4^+. \quad (4.78)$$

The creation operators are obtained by replacing σ_j^+ with σ_j^- .

Compact summary

The essential dictionary is

$$n_j = c_j^\dagger c_j = \frac{1 - \sigma_j^z}{2}, \quad \sigma_j^z = 1 - 2n_j, \quad (4.79)$$

$$K_j = \prod_{\ell=1}^{j-1} \sigma_\ell^z = \prod_{\ell=1}^{j-1} (1 - 2n_\ell), \quad (4.80)$$

$$c_j = K_j \sigma_j^+, \quad c_j^\dagger = K_j \sigma_j^-, \quad (4.81)$$

$$\sigma_j^x = K_j (c_j^\dagger + c_j), \quad \sigma_j^y = K_j i(c_j^\dagger - c_j), \quad \sigma_j^z = 1 - 2c_j^\dagger c_j, \quad (4.82)$$

and

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = 0, \quad \{c_i^\dagger, c_j^\dagger\} = 0. \quad (4.83)$$

These identities are the basic toolkit for translating one-dimensional spin chains into fermionic language.

Chapter 5

Fourier Transform and Momentum-Space Blocks

5.1 Purpose of this chapter

After the Jordan–Wigner transformation, the spin chains of interest become fermionic lattice Hamiltonians. For open chains, the resulting quadratic Hamiltonians can be diagonalized directly in real space. For periodic and translation-invariant chains, however, the natural variables are momentum modes. The purpose of this chapter is to fix the Fourier-transform convention used in these notes and to explain why quadratic Hamiltonians decompose into independent momentum-space blocks.

There are three convention-dependent points that must be fixed carefully:

- (i) the sign convention in the Fourier transform;
- (ii) the allowed momenta, which depend on the fermionic boundary condition;
- (iii) the harmless but useful global phase that may be included in the Fourier transform.

The third point is especially convenient in superconducting chains, where the pairing gap may be complex. A uniform phase of the pairing gap can often be moved between the Hamiltonian and the definition of the momentum-space fermions by a constant gauge rotation.

5.2 Discrete Fourier transform with a boundary twist

Consider a chain of length L with fermionic operators $c_j, c_j^\dagger$, $j = 1, \dots, L$, satisfying

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \quad (5.1)$$

We impose the general boundary condition

$$c_{j+L} = e^{i\vartheta} c_j, \quad \vartheta \in [0, 2\pi). \quad (5.2)$$

The phase ϑ is a boundary twist. Periodic boundary conditions correspond to $\vartheta = 0$, while anti-periodic boundary conditions correspond to $\vartheta = \pi$.

The allowed momenta are the solutions of

$$e^{ikL} = e^{i\vartheta}. \quad (5.3)$$

Thus

$$k_m = \frac{2\pi m + \vartheta}{L}, \quad m = 0, 1, \dots, L-1, \quad (5.4)$$

where momenta are understood modulo 2π . We denote this finite set by

$$\mathcal{K}_\vartheta = \left\{ \frac{2\pi m + \vartheta}{L} \mid m = 0, 1, \dots, L-1 \right\}. \quad (5.5)$$

Definition 5.1 (Fourier transform used in these notes). For $k \in \mathcal{K}_\vartheta$, we define

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_{j=1}^L e^{-ikj} c_j, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_{k \in \mathcal{K}_\vartheta} e^{ikj} c_k. \quad (5.6)$$

Here ϕ is a constant Fourier-gauge phase. Unless otherwise stated, the default convention is $\phi = 0$.

The phase ϕ in Eq. (5.6) is the phase shown in the convention

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_{j=1}^L e^{-ikj} c_j, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_k e^{ikj} c_k. \quad (5.7)$$

It is not a boundary twist. The boundary twist is ϑ , which changes the allowed values of k . The constant phase ϕ only redefines every momentum operator by the same $U(1)$ phase.

The inverse transform follows from the finite orthogonality relations

$$\frac{1}{L} \sum_{j=1}^L e^{i(k-k')j} = \delta_{k,k'}, \quad \frac{1}{L} \sum_{k \in \mathcal{K}_\vartheta} e^{ik(j-j')} = \delta_{j,j'}. \quad (5.8)$$

The second identity is valid for $j, j' = 1, \dots, L$. The common phase ϕ drops out of the orthogonality relations.

Proposition 5.1 (Canonical algebra in momentum space). *The operators defined in Eq. (5.6) obey*

$$\{c_k, c_{k'}^\dagger\} = \delta_{k,k'}, \quad \{c_k, c_{k'}\} = \{c_k^\dagger, c_{k'}^\dagger\} = 0. \quad (5.9)$$

Proof. Using the real-space canonical anticommutation relations,

$$\begin{aligned} \{c_k, c_{k'}^\dagger\} &= \frac{1}{L} \sum_{j,j'=1}^L e^{-ikj} e^{ik'j'} \{c_j, c_{j'}^\dagger\} \\ &= \frac{1}{L} \sum_{j=1}^L e^{i(k'-k)j} = \delta_{k,k'}. \end{aligned} \quad (5.10)$$

The phases $e^{-i\phi}$ and $e^{i\phi}$ cancel. The other anticommutators vanish in the same way. $\square$

Remark 5.1 (Choice of lattice origin). The transform above uses e^{ikj} with $j = 1, \dots, L$. Some books use $e^{ik(j-1)}$ instead. This is only a k -dependent redefinition $c_k \mapsto e^{ik} c_k$. It can shift phase factors between the Fourier transform and the momentum-space Hamiltonian, but it cannot change spectra, correlation functions, or topological invariants. In these notes we keep the $j = 1, \dots, L$ convention to match the Jordan–Wigner chapters.

5.3 Allowed momenta in Jordan–Wigner parity sectors

For a periodic spin chain, the Jordan–Wigner transformation does not produce one single fermionic boundary condition. Instead, the effective boundary condition depends on the fermion-parity sector. With the convention used in these notes, [Eq. \(4.49\)](#) gives

$$c_{L+1} = -p c_1, \quad p = (-1)^N = \pm 1. \quad (5.11)$$

Therefore

fermion parity	p	boundary condition	allowed momenta
even	+1	$c_{L+1} = -c_1$	$k_m = \frac{(2m+1)\pi}{L}$
odd	-1	$c_{L+1} = c_1$	$k_m = \frac{2\pi m}{L}$

(5.12)

with $m = 0, 1, \dots, L-1$.

This distinction is negligible for many bulk thermodynamic quantities in the limit $L \rightarrow \infty$, but it is essential for finite-size spectra, exact diagonal forms, and parity-sensitive ground-state questions. In particular, periodic fermions may contain self-conjugate momenta satisfying

$$k \equiv -k \pmod{2\pi}. \quad (5.13)$$

For even L and periodic boundary conditions, these are $k = 0$ and $k = \pi$. Such modes must be treated separately in a BdG decomposition because they are not paired with a distinct momentum $-k$.

In the thermodynamic limit, for a smooth function $f(k)$,

$$\frac{1}{L} \sum_{k \in \mathcal{K}_\vartheta} f(k) \longrightarrow \int_{-\pi}^{\pi} \frac{dk}{2\pi} f(k). \quad (5.14)$$

The boundary angle ϑ only shifts the mesh by an amount of order $1/L$.

5.4 Translation-invariant hopping terms

The main reason for using Fourier modes is that translation-invariant hopping terms are diagonal in momentum. Consider

$$H_t = \sum_{j=1}^L \sum_r t_r c_j^\dagger c_{j+r}, \quad (5.15)$$

where the boundary condition is understood and the hopping amplitudes satisfy

$$t_{-r} = t_r^* \quad (5.16)$$

so that H_t is Hermitian. Substituting the Fourier transform gives

$$\sum_{j=1}^L c_j^\dagger c_{j+r} = \sum_{k \in \mathcal{K}_\vartheta} e^{ikr} c_k^\dagger c_k. \quad (5.17)$$

Therefore

$$H_t = \sum_{k \in \mathcal{K}_\vartheta} \xi(k) c_k^\dagger c_k, \quad \xi(k) = \sum_r t_r e^{ikr}. \quad (5.18)$$

The Fourier-gauge phase ϕ cancels from all number-conserving bilinears.

Example 5.1 (Nearest-neighbour tight-binding chain). For

$$H_t = -t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - \mu \sum_j c_j^\dagger c_j, \quad (5.19)$$

we obtain

$$H_t = \sum_k \xi(k) c_k^\dagger c_k, \quad \xi(k) = -\mu - 2t \cos k. \quad (5.20)$$

This is the normal-state dispersion that appears in the Kitaev chain and in the fermionic image of the transverse-field XY chain.

5.5 Translation-invariant pairing terms

Superconducting or Jordan–Wigner-transformed anisotropic spin chains contain terms that do not conserve fermion number but do conserve fermion parity. A convenient translation-invariant pairing Hamiltonian is

$$H_\Delta = \frac{1}{2} \sum_{j=1}^L \sum_r \left(\Delta_r c_j^\dagger c_{j+r}^\dagger + \Delta_r^* c_{j+r} c_j \right). \quad (5.21)$$

For spinless fermions, only the antisymmetric part of the pairing is physical, so one may take

$$\Delta_{-r} = -\Delta_r. \quad (5.22)$$

The Fourier transform gives

$$H_\Delta = \frac{1}{2} \sum_{k \in \mathcal{K}_\vartheta} \left[\Delta_\phi(k) c_k^\dagger c_{-k}^\dagger + \Delta_\phi(k)^* c_{-k} c_k \right], \quad (5.23)$$

where

$$\Delta_\phi(k) = e^{-2i\phi} \sum_r \Delta_r e^{ikr}. \quad (5.24)$$

Thus the additional Fourier phase does not affect hopping terms, but it rotates the superconducting gap by twice the phase. This factor of two is simply the charge-two nature of a Cooper-pair operator: a pair creation operator contains two fermion creation operators.

If the pairing is written instead in the annihilation form

$$\sum_{j,r} \left(\tilde{\Delta}_r c_j c_{j+r} + \tilde{\Delta}_r^* c_{j+r}^\dagger c_j^\dagger \right), \quad (5.25)$$

then the coefficient of the annihilation term transforms with $e^{+2i\phi}$. The two statements are the same convention expressed from opposite sides of the Hermitian conjugate pair.

5.6 BdG blocks for spinless superconducting chains

Combining the hopping and pairing terms, define the Nambu spinor

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}. \quad (5.26)$$

Then a general spinless quadratic Hamiltonian can be written as

$$H = \frac{1}{2} \sum_{k \in \mathcal{K}_\vartheta} \Psi_k^\dagger \mathcal{H}_{\text{BdG}}(k) \Psi_k + E_0, \quad (5.27)$$

with

$$\mathcal{H}_{\text{BdG}}(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(-k) \end{pmatrix}. \quad (5.28)$$

For most nearest-neighbour chains considered in these notes, $\xi(-k) = \xi(k)$ and $\Delta_\phi(-k) = -\Delta_\phi(k)$.

Equivalently, after dropping any term proportional to the identity matrix, the BdG block can be written as

$$\mathcal{H}_{\text{BdG}}(k) = d_x(k)\tau_x + d_y(k)\tau_y + d_z(k)\tau_z, \quad (5.29)$$

with

$$d_x(k) = \text{Re } \Delta_\phi(k), \quad d_y(k) = -\text{Im } \Delta_\phi(k), \quad d_z(k) = \xi(k). \quad (5.30)$$

The quasiparticle energies are

$$E_\pm(k) = \pm \sqrt{\xi(k)^2 + |\Delta_\phi(k)|^2}. \quad (5.31)$$

Hence a constant Fourier-gauge phase changes the orientation of the vector (d_x, d_y) in the pairing plane but does not change the spectrum.

5.7 Independent momentum-space blocks

A translation-invariant number-conserving quadratic Hamiltonian has the form

$$H = \sum_k \xi(k) c_k^\dagger c_k. \quad (5.32)$$

It is already decomposed into independent one-mode blocks labelled by k .

A pairing Hamiltonian is different: it couples k and $-k$. Therefore its basic blocks are not single momenta but pairs $(k, -k)$. Let $\mathcal{K}_+$ be a choice of one representative from each pair $\{k, -k\}$, excluding self-conjugate momenta. Then

$$H = \sum_{k \in \mathcal{K}_+} H_{k, -k} + H_{\text{self}} + E_0. \quad (5.33)$$

Here H_{self} contains possible self-conjugate modes such as $k = 0$ and $k = \pi$ under periodic boundary conditions. For odd-parity spinless pairing, $\Delta_\phi(k) = 0$ at self-conjugate momenta, so these modes remain unpaired.

For a non-self-conjugate pair, the even-parity local Hilbert space is spanned by

$$|0\rangle_{k, -k}, \quad c_k^\dagger c_{-k}^\dagger |0\rangle_{k, -k}. \quad (5.34)$$

In this subspace, the BdG block describes a two-level system. This is why the transverse-field Ising chain, the transverse-field XY chain, and the Kitaev chain all reduce to products of independent two-level problems in momentum space.

Principle 5.1 (Momentum-space block principle). A translation-invariant quadratic fermion Hamiltonian decomposes into independent momentum blocks. Number-conserving Hamiltonians decompose into k blocks. Spinless superconducting Hamiltonians decompose into $(k, -k)$ BdG blocks, plus possible self-conjugate modes.

5.8 Physical meaning of the extra Fourier phase

The constant phase ϕ in Eq. (5.6) implements the operator redefinition

$$c_k^{(\phi)} = e^{-i\phi} c_k^{(0)}. \quad (5.35)$$

In Nambu space,

$$\Psi_k^{(\phi)} = \begin{pmatrix} e^{-i\phi} & 0 \\ 0 & e^{i\phi} \end{pmatrix} \Psi_k^{(0)} = e^{-i\phi\tau_z} \Psi_k^{(0)}. \quad (5.36)$$

Therefore BdG matrices in the two conventions are related by the unitary gauge transformation

$$\mathcal{H}_{\text{BdG}}^{(\phi)}(k) = e^{-i\phi\tau_z} \mathcal{H}_{\text{BdG}}^{(0)}(k) e^{i\phi\tau_z}. \quad (5.37)$$

This rotates the τ_x - τ_y components of the BdG vector by a constant angle and leaves $d_z(k)$ invariant.

The distinction between ϑ and ϕ is important:

- ϑ changes the boundary condition and therefore changes the allowed momenta at finite L ;
- ϕ is a uniform gauge choice for momentum-space fermions and does not change the allowed momenta.

A boundary twist can represent a physical flux through a ring. A constant Fourier-gauge phase by itself cannot.

5.9 Example: complex pairing phase in the Kitaev chain

Consider the Kitaev chain written in the creation-pair convention

$$H_K = -\mu \sum_j \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \sum_j \left(\Delta c_j^\dagger c_{j+1}^\dagger + \Delta^* c_{j+1} c_j \right), \quad (5.38)$$

where

$$\Delta = |\Delta| e^{i\theta_\Delta}. \quad (5.39)$$

Using Eq. (5.24), the off-diagonal BdG gap is

$$\Delta_\phi(k) = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (5.40)$$

The normal dispersion is

$$\xi(k) = -\mu - 2t \cos k. \quad (5.41)$$

Thus

$$\mathcal{H}_K(k) = \begin{pmatrix} \xi(k) & 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k \\ -2i |\Delta| e^{-i(\theta_\Delta - 2\phi)} \sin k & -\xi(k) \end{pmatrix}. \quad (5.42)$$

The spectrum is

$$E_\pm(k) = \pm \sqrt{(\mu + 2t \cos k)^2 + 4 |\Delta|^2 \sin^2 k}, \quad (5.43)$$

which is independent of θ_Δ and ϕ .

Equation Eq. (5.40) shows the role of the extra Fourier phase. Choosing

$$\phi = \frac{\theta_\Delta}{2} \quad (5.44)$$

removes the explicit superconducting phase from the gap and gives the standard purely imaginary p -wave form

$$\Delta_\phi(k) = 2i|\Delta|\sin k. \quad (5.45)$$

Alternatively, one may choose a different ϕ to make the off-diagonal gap point along another fixed direction in the τ_x - τ_y plane.

For an isolated uniform Kitaev chain, the global phase θ_Δ is therefore not an independent bulk observable. It is a superconducting gauge phase. The bulk spectrum and the topological phase boundary

$$|\mu| = 2|t| \quad (5.46)$$

are unchanged by it. What is physical is a gauge-invariant relative phase: for example, the phase difference between two superconductors in a Josephson junction, a spatial gradient of the superconducting phase, or a boundary twist that cannot be removed globally. In such situations the phase is tied to current, flux, or junction physics and cannot be discarded as a mere convention.

From the BdG-vector viewpoint, a uniform complex phase of Δ rotates the closed curve

$$k \mapsto (d_x(k), d_y(k), d_z(k)) \quad (5.47)$$

around the d_z axis. This does not change whether the curve winds around the origin in the pairing plane. Hence the topological classification of the uniform Kitaev chain is unaffected by the constant gap phase, even though the explicit matrix representation of the BdG block changes.

5.10 Summary of conventions

The Fourier convention used in the rest of these notes is

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_{j=1}^L e^{-ikj} c_j, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_{k \in \mathcal{K}_\vartheta} e^{ikj} c_k, \quad (5.48)$$

with default $\phi = 0$. The allowed momenta are

$$k_m = \frac{2\pi m + \vartheta}{L}, \quad m = 0, 1, \dots, L-1. \quad (5.49)$$

For Jordan–Wigner fermions obtained from a periodic spin chain,

sector	boundary condition	momenta
$(-1)^N = +1$	anti-periodic	$k_m = \frac{(2m+1)\pi}{L}$
$(-1)^N = -1$	periodic	$k_m = \frac{2\pi m}{L}$

(5.50)

Number-conserving hopping terms give one-mode blocks labelled by k . Pairing terms give BdG blocks labelled by $(k, -k)$. The extra phase ϕ is a useful Fourier-gauge convention: it cancels from hopping terms but shifts the phase of a pairing gap by twice its value.

Chapter 6

Bogoliubov–de Gennes Formalism

The Jordan–Wigner transformation maps several spin-1/2 chains to quadratic fermion Hamiltonians. If the fermion number is conserved, Fourier transformation already diagonalizes the problem into one-mode momentum blocks. If pairing terms are present, however, the Hamiltonian conserves only fermion parity, and the elementary block is no longer a single mode k but a paired sector $(k, -k)$. The Bogoliubov–de Gennes (BdG) formalism is the systematic language for this situation.

The purpose of this chapter is to fix the algebraic conventions used later in the XY chain, the transverse-field Ising chain, and the Kitaev chain. We do not yet perform the full Bogoliubov rotation; that will be done in the next chapter. Here we explain how a quadratic pairing Hamiltonian is encoded as a matrix in Nambu space, why the resulting matrix has a built-in particle–hole redundancy, and how this Nambu-space structure differs from an ordinary two-band Bloch Hamiltonian such as the SSH model.

6.1 Real-space quadratic fermion Hamiltonians

Let $c_j, c_j^\dagger$ be canonical spinless fermions on a chain of length L , obeying

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \quad (6.1)$$

The most general quadratic Hamiltonian preserving fermion parity is

$$H = \sum_{i,j=1}^L h_{ij} c_i^\dagger c_j + \frac{1}{2} \sum_{i,j=1}^L \left(\Delta_{ij} c_i^\dagger c_j^\dagger + \Delta_{ij}^* c_j c_i \right) + E_{\text{cl}}. \quad (6.2)$$

Hermiticity requires

$$h^\dagger = h. \quad (6.3)$$

Fermi statistics require the pairing matrix to be antisymmetric,

$$\Delta^T = -\Delta, \quad (6.4)$$

because only the antisymmetric part of $\Delta_{ij} c_i^\dagger c_j^\dagger$ survives.

It is useful to introduce the real-space Nambu spinor

$$\Upsilon = \begin{pmatrix} c_1 \\ \vdots \\ c_L \\ c_1^\dagger \\ \vdots \\ c_L^\dagger \end{pmatrix}, \quad \Upsilon^\dagger = \begin{pmatrix} c_1^\dagger & \cdots & c_L^\dagger & c_1 & \cdots & c_L \end{pmatrix}. \quad (6.5)$$

Then Eq. (6.2) can be written as

$$H = \frac{1}{2} \Upsilon^\dagger \mathbb{H}_{\text{BdG}} \Upsilon + E_{\text{const}}, \quad (6.6)$$

where

$$\mathbb{H}_{\text{BdG}} = \begin{pmatrix} h & \Delta \\ -\Delta^* & -h^T \end{pmatrix}. \quad (6.7)$$

The constant differs from the normal-ordering constant in Eq. (6.2); explicitly,

$$E_{\text{const}} = E_{\text{cl}} + \frac{1}{2} \text{Tr } h. \quad (6.8)$$

This shift has no effect on eigenvectors or quasiparticle energies, but it must be retained when computing the absolute ground-state energy.

Remark 6.1 (Why the factor of 1/2 appears). The Nambu spinor contains both particles and holes. Therefore the degrees of freedom are doubled. The prefactor 1/2 in Eq. (6.6) compensates this doubling. Without this factor the normal hopping term and the pairing term would be counted twice.

6.2 Intrinsic particle–hole redundancy

The BdG matrix is not an arbitrary Hermitian $2L \times 2L$ matrix. It satisfies an intrinsic particle–hole constraint. Let

$$\tau_x = \begin{pmatrix} 0 & \mathbf{1}_L \\ \mathbf{1}_L & 0 \end{pmatrix}, \quad \mathcal{C} = \tau_x K, \quad (6.9)$$

where K denotes complex conjugation. Then

$$\tau_x \mathbb{H}_{\text{BdG}}^* \tau_x = -\mathbb{H}_{\text{BdG}}, \quad \text{equivalently} \quad \mathcal{C} \mathbb{H}_{\text{BdG}} \mathcal{C}^{-1} = -\mathbb{H}_{\text{BdG}}. \quad (6.10)$$

Thus if

$$\mathbb{H}_{\text{BdG}} \Phi = E \Phi, \quad (6.11)$$

then

$$\mathbb{H}_{\text{BdG}} (\mathcal{C} \Phi) = -E (\mathcal{C} \Phi). \quad (6.12)$$

The spectrum is symmetric about zero. This symmetry is not an optional physical symmetry imposed on the model; it is a redundancy of the Nambu description.

Principle 6.1 (BdG redundancy principle). A BdG Hamiltonian doubles the Hilbert-space variables by treating c and $c^\dagger$ as independent components of a Nambu spinor. The particle–hole constraint removes this artificial doubling: the negative-energy BdG states are particle–hole partners of the positive-energy states, not independent physical single-particle bands.

6.3 Translation-invariant BdG blocks

For a translation-invariant chain, the Fourier convention of the previous chapter is

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_{j=1}^L e^{-ikj} c_j, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_{k \in \mathcal{K}_\vartheta} e^{ikj} c_k, \quad (6.13)$$

with

$$\mathcal{K}_\vartheta = \left\{ k_m = \frac{2\pi m + \vartheta}{L} \mid m = 0, 1, \dots, L-1 \right\}. \quad (6.14)$$

The boundary twist ϑ fixes the allowed momenta, while the constant phase ϕ is a Fourier-gauge convention.

Consider the momentum-space Hamiltonian

$$H = \sum_{k \in \mathcal{K}_\vartheta} \xi(k) c_k^\dagger c_k + \frac{1}{2} \sum_{k \in \mathcal{K}_\vartheta} \left[\Delta_\phi(k) c_k^\dagger c_{-k}^\dagger + \Delta_\phi(k)^* c_{-k} c_k \right] + E_{\text{cl}}. \quad (6.15)$$

For spinless fermions,

$$\Delta_\phi(-k) = -\Delta_\phi(k). \quad (6.16)$$

The natural Nambu spinor for the pair $(k, -k)$ is

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \quad \Psi_k^\dagger = \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix}. \quad (6.17)$$

Then

$$H = \frac{1}{2} \sum_{k \in \mathcal{K}_\vartheta} \Psi_k^\dagger \mathcal{H}_{\text{BdG}}(k) \Psi_k + E_0, \quad (6.18)$$

where

$$\mathcal{H}_{\text{BdG}}(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(-k) \end{pmatrix}. \quad (6.19)$$

The corresponding particle-hole constraint is

$$\tau_x \mathcal{H}_{\text{BdG}}(-k)^* \tau_x = -\mathcal{H}_{\text{BdG}}(k). \quad (6.20)$$

For the nearest-neighbour chains studied later, one usually has

$$\xi(-k) = \xi(k), \quad \Delta_\phi(-k) = -\Delta_\phi(k), \quad (6.21)$$

and the matrix simplifies to

$$\mathcal{H}_{\text{BdG}}(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(k) \end{pmatrix}. \quad (6.22)$$

6.4 Two-level representation

Let τ_x, τ_y, τ_z denote Pauli matrices acting in Nambu space, not in the original spin space. For the common even-dispersion case [Eq. \(6.22\)](#), the BdG block can be written as

$$\mathcal{H}_{\text{BdG}}(k) = d_x(k) \tau_x + d_y(k) \tau_y + d_z(k) \tau_z, \quad (6.23)$$

where

$$d_x(k) = \text{Re } \Delta_\phi(k), \quad d_y(k) = -\text{Im } \Delta_\phi(k), \quad d_z(k) = \xi(k). \quad (6.24)$$

The BdG spectrum is therefore

$$E_\pm(k) = \pm E(k), \quad E(k) = \sqrt{\xi(k)^2 + |\Delta_\phi(k)|^2} = \sqrt{d_x(k)^2 + d_y(k)^2 + d_z(k)^2}. \quad (6.25)$$

A bulk gap closes only if

$$d_x(k) = d_y(k) = d_z(k) = 0 \quad (6.26)$$

for some allowed momentum k in the thermodynamic limit.

Remark 6.2 (Meaning of the negative branch). The negative branch $-E(k)$ is not a second independent physical species. It is the particle-hole partner of the positive branch $E(k)$. The physical quasiparticle spectrum after Bogoliubov diagonalization consists of the positive energies $E(k)$, together with a ground-state energy shift.

6.5 Independent $(k, -k)$ sectors

For $k \not\equiv -k$ modulo the reciprocal lattice, the four-dimensional Fock space generated by $c_k^\dagger$ and $c_{-k}^\dagger$ splits into even and odd fermion-parity sectors. The even sector is

$$\mathcal{H}_{k,-k}^{\text{even}} = \text{span} \left\{ |0\rangle_{k,-k}, c_k^\dagger c_{-k}^\dagger |0\rangle_{k,-k} \right\}, \quad (6.27)$$

whereas the odd sector is

$$\mathcal{H}_{k,-k}^{\text{odd}} = \text{span} \left\{ c_k^\dagger |0\rangle_{k,-k}, c_{-k}^\dagger |0\rangle_{k,-k} \right\}. \quad (6.28)$$

Pairing terms connect the two states in the even sector but do not connect states of opposite fermion parity. Thus the BdG problem is equivalent to a two-level problem in each even-parity $(k, -k)$ sector.

If the allowed momentum set contains self-conjugate momenta satisfying

$$k \equiv -k \pmod{2\pi}, \quad (6.29)$$

then $k = 0$ or $k = \pi$. For spinless odd-parity pairing,

$$\Delta_\phi(0) = 0, \quad \Delta_\phi(\pi) = 0, \quad (6.30)$$

so these modes are not paired by the superconducting term. They contribute ordinary number-operator pieces of the form

$$\xi(k) c_k^\dagger c_k. \quad (6.31)$$

Such self-conjugate modes are often irrelevant in the thermodynamic bulk spectrum, but they matter in finite-size calculations and in the parity-sector bookkeeping of Jordan-Wigner-transformed spin chains.

6.6 Gauge phase in Nambu space

The previous chapter allowed a constant phase ϕ in the Fourier transform. This phase acts on momentum-space fermions as

$$c_k^{(\phi)} = e^{-i\phi} c_k^{(0)}. \quad (6.32)$$

Consequently,

$$\Psi_k^{(\phi)} = e^{-i\phi\tau_z} \Psi_k^{(0)}. \quad (6.33)$$

The BdG matrix transforms as

$$\mathcal{H}_{\text{BdG}}^{(\phi)}(k) = e^{-i\phi\tau_z} \mathcal{H}_{\text{BdG}}^{(0)}(k) e^{i\phi\tau_z}. \quad (6.34)$$

Therefore ϕ rotates the τ_x - τ_y components of the BdG vector while leaving $d_z(k) = \xi(k)$ and the spectrum invariant. Equivalently, the pairing gap transforms as

$$\Delta_\phi(k) = e^{-2i\phi} \Delta_0(k). \quad (6.35)$$

The factor of two reflects the fact that the pairing operator carries charge 2 under the fermion $U(1)$ phase rotation.

This distinction is important:

$$\vartheta \text{ changes the allowed momenta, whereas } \phi \text{ only changes the Nambu gauge.} \quad (6.36)$$

A boundary twist ϑ can represent a physical flux through a ring. A constant ϕ is a basis convention unless it is tied to a relative phase between distinct superconducting regions or to a spatially varying order parameter.

6.7 Example: Kitaev-chain BdG matrix

Consider the spinless Kitaev chain

$$H_K = -\mu \sum_j \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \sum_j \left(\Delta c_j^\dagger c_{j+1}^\dagger + \Delta^* c_{j+1} c_j \right), \quad (6.37)$$

with

$$\Delta = |\Delta| e^{i\theta_\Delta}. \quad (6.38)$$

Using the Fourier convention [Eq. \(6.13\)](#), one finds

$$\xi(k) = -\mu - 2t \cos k, \quad \Delta_\phi(k) = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (6.39)$$

The BdG block is

$$\mathcal{H}_K(k) = \begin{pmatrix} -\mu - 2t \cos k & 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k \\ -2i |\Delta| e^{-i(\theta_\Delta - 2\phi)} \sin k & \mu + 2t \cos k \end{pmatrix}. \quad (6.40)$$

The spectrum is

$$E_\pm(k) = \pm \sqrt{(\mu + 2t \cos k)^2 + 4 |\Delta|^2 \sin^2 k}. \quad (6.41)$$

It is independent of the uniform phase θ_Δ and of the Fourier-gauge phase ϕ .

Choosing

$$\phi = \frac{\theta_\Delta}{2} \quad (6.42)$$

removes the explicit superconducting phase from the off-diagonal matrix element:

$$\Delta_\phi(k) = 2i |\Delta| \sin k. \quad (6.43)$$

For a single uniform Kitaev chain this is only a gauge choice. In contrast, relative superconducting phases between different segments, Josephson junctions, or spatial phase gradients are physical because they cannot be removed by one constant global choice of ϕ .

For $\phi = \theta_\Delta/2$, the BdG vector is

$$\mathbf{d}(k) = (0, -2 |\Delta| \sin k, -\mu - 2t \cos k), \quad (6.44)$$

using the convention $d_y = -\text{Im } \Delta_\phi$. The gap closes at

$$\sin k = 0, \quad \mu + 2t \cos k = 0, \quad (6.45)$$

so the thermodynamic phase boundaries are

$$\mu = -2t \quad (k = 0), \quad \mu = 2t \quad (k = \pi). \quad (6.46)$$

Equivalently, for real nonzero t ,

$$|\mu| = 2|t| \quad (6.47)$$

marks the bulk gap closing between the trivial and topological phases.

6.8 Example: BdG form of the transverse-field XY chain

The transverse-field XY chain later studied as a mother model has the Pauli-form Hamiltonian

$$H_{\text{XY}} = - \sum_{j=1}^L \left[\frac{1+\gamma}{2} \sigma_j^x \sigma_{j+1}^x + \frac{1-\gamma}{2} \sigma_j^y \sigma_{j+1}^y \right] - h \sum_{j=1}^L \sigma_j^z, \quad (6.48)$$

up to the boundary-sector subtleties already fixed in the Jordan–Wigner chapter. With the convention

$$n_j = c_j^\dagger c_j = \frac{1 - \sigma_j^z}{2}, \quad (6.49)$$

the bulk Jordan–Wigner image is a quadratic pairing Hamiltonian. In the common normalization used in these notes, its BdG vector may be chosen as

$$\mathbf{d}_{\text{XY}}(k) = (0, -2\gamma \sin k, 2(h - \cos k)), \quad (6.50)$$

so that

$$E_{\text{XY}}(k) = 2\sqrt{(h - \cos k)^2 + \gamma^2 \sin^2 k}. \quad (6.51)$$

This form makes the common free-fermion structure transparent:

$$\text{XY chain} \longleftrightarrow \text{spinless BdG chain} \longleftrightarrow \text{independent } (k, -k) \text{ two-level systems.} \quad (6.52)$$

The transverse-field Ising chain is the special case $\gamma = 1$. The XX chain is the special case $\gamma = 0$, where the pairing component vanishes and the problem reduces to a number-conserving free-fermion hopping model.

Remark 6.3 (Normalization warning). Different references place factors of 2 in different locations in Eq. (6.48). The invariant statement is the shape of the spectrum and the gap-closing conditions. With the above convention, the critical lines are obtained from

$$h - \cos k = 0, \quad \gamma \sin k = 0. \quad (6.53)$$

For $\gamma \neq 0$, this gives $h = 1$ at $k = 0$ and $h = -1$ at $k = \pi$.

6.9 Nambu space versus sublattice space

A two-component Hamiltonian of the form

$$\mathcal{H}(k) = d_x(k)\tau_x + d_y(k)\tau_y + d_z(k)\tau_z \quad (6.54)$$

may describe very different physics depending on what the two components mean. For the Kitaev and XY chains, the two components are Nambu components,

$$\Psi_k = (c_k, c_{-k}^\dagger)^T. \quad (6.55)$$

For the SSH chain, the two components are sublattice components,

$$\psi_k = (a_k, b_k)^T, \quad (6.56)$$

where a_k and b_k annihilate fermions on two distinct sites within the unit cell. The formal 2×2 matrix notation is similar, but the physical meaning is different.

	BdG/Nambu two-level system	SSH/sublattice two-band system
Two-component spinor	$(c_k, c_{-k}^\dagger)^T$	$(a_k, b_k)^T$
Physical components	particle and hole of related momenta	two independent orbitals/sublattices in one unit cell
Reason for two bands	artificial Nambu doubling	genuine Bloch-band structure
Built-in symmetry	particle-hole redundancy $\tau_x \mathcal{H}(-k)^* \tau_x = -\mathcal{H}(k)$	no BdG redundancy; SSH has chiral symmetry only for special hopping structure
Prefactor in Hamiltonian	$\frac{1}{2} \sum_k \Psi_k^\dagger \mathcal{H}_{\text{BdG}}(k) \Psi_k$	$\sum_k \psi_k^\dagger h_{\text{Bloch}}(k) \psi_k$
Meaning of negative energy	hole partner of positive-energy quasiparticle	physical valence band
Ground state	Bogoliubov/BCS quasiparticle vacuum	filled-band Slater determinant at fixed particle number

This distinction is central to the organization of these notes. The Kitaev, TFIM, and XY chains are solved through BdG pairing blocks. The SSH model is a number-conserving free-fermion Bloch problem. Both can be represented by maps from momentum space to a vector $\mathbf{d}(k)$, and both can carry topological winding numbers in one dimension, but the Hilbert-space interpretation of the same-looking Pauli matrices is not the same.

6.10 Summary

The BdG formalism rewrites a quadratic parity-preserving fermion Hamiltonian in Nambu space. In real space,

$$\mathbb{H}_{\text{BdG}} = \begin{pmatrix} h & \Delta \\ -\Delta^* & -h^T \end{pmatrix}, \quad h^\dagger = h, \quad \Delta^T = -\Delta. \quad (6.57)$$

In a translation-invariant spinless chain, the basic momentum block is

$$\mathcal{H}_{\text{BdG}}(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(-k) \end{pmatrix}. \quad (6.58)$$

For $\xi(-k) = \xi(k)$ this becomes a two-level Hamiltonian

$$\mathcal{H}_{\text{BdG}}(k) = \mathbf{d}(k) \cdot \boldsymbol{\tau}, \quad \mathbf{d}(k) = (\text{Re } \Delta_\phi(k), -\text{Im } \Delta_\phi(k), \xi(k)), \quad (6.59)$$

with eigenvalues

$$E_\pm(k) = \pm \sqrt{\xi(k)^2 + |\Delta_\phi(k)|^2}. \quad (6.60)$$

The particle-hole relation

$$\tau_x \mathcal{H}_{\text{BdG}}(-k)^* \tau_x = -\mathcal{H}_{\text{BdG}}(k) \quad (6.61)$$

is an intrinsic Nambu redundancy. The next chapter will use this structure to construct the explicit Bogoliubov transformation, the quasiparticle operators, and the BCS product ground state.

Chapter 7

Bogoliubov Transformation and Quasiparticles

The previous chapter formulated quadratic pairing Hamiltonians as BdG matrices. A BdG matrix makes the particle-hole structure transparent, but it is still not the final solution of the many-body problem. The final step is the *Bogoliubov transformation*: a canonical change of fermionic operators that turns each BdG block into a collection of independent quasiparticle number operators.

This chapter gives the explicit diagonalization used throughout the later chapters on the transverse-field XY chain, the transverse-field Ising chain, and the Kitaev chain. The main idea is simple. For each independent momentum pair $(k, -k)$, pairing mixes the empty state and the doubly occupied state. The Bogoliubov transformation is the fermionic rotation that chooses the correct linear combinations of particles and holes so that the Hamiltonian becomes a collection of free quasiparticles.

7.1 Independent positive-momentum sectors

For the spinless BdG chains considered in these notes, the momentum-space Hamiltonian has the form

$$H = \sum_{k \in \mathcal{K}_\vartheta} \xi(k) c_k^\dagger c_k + \frac{1}{2} \sum_{k \in \mathcal{K}_\vartheta} \left[\Delta_\phi(k) c_k^\dagger c_{-k}^\dagger + \Delta_\phi(k)^* c_{-k} c_k \right] + E_{\text{cl}}, \quad (7.1)$$

where

$$\xi(-k) = \xi(k), \quad \Delta_\phi(-k) = -\Delta_\phi(k). \quad (7.2)$$

The allowed set $\mathcal{K}_\vartheta$ is fixed by the boundary twist introduced in the Fourier-transform chapter. Let $\mathcal{K}_+$ denote a set containing exactly one representative from each non-self-conjugate pair $\{k, -k\}$. Self-conjugate momenta, if present, satisfy

$$k \equiv -k \pmod{2\pi}, \quad (7.3)$$

and are therefore $k = 0$ or $k = \pi$.

For $k \in \mathcal{K}_+$, the contribution of the pair $(k, -k)$ is

$$H_k = \xi_k \left(c_k^\dagger c_k + c_{-k}^\dagger c_{-k} - 1 \right) + \Delta_k c_k^\dagger c_{-k}^\dagger + \Delta_k^* c_{-k} c_k, \quad (7.4)$$

where, for compactness,

$$\xi_k \equiv \xi(k), \quad \Delta_k \equiv \Delta_\phi(k). \quad (7.5)$$

Equivalently, with the Nambu spinor

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \quad \Psi_k^\dagger = \begin{pmatrix} c_k^\dagger & c_{-k} \end{pmatrix}, \quad (7.6)$$

one has

$$H_k = \Psi_k^\dagger \mathcal{H}_{\text{BdG}}(k) \Psi_k, \quad \mathcal{H}_{\text{BdG}}(k) = \begin{pmatrix} \xi_k & \Delta_k \\ \Delta_k^* & -\xi_k \end{pmatrix}. \quad (7.7)$$

Notice that no additional factor of $1/2$ appears in Eq. (7.7), because $\mathcal{K}_+$ already contains only one momentum from each pair. This is equivalent to the all-momentum BdG formula with a factor of $1/2$.

The positive BdG energy is

$$E_k = \sqrt{\xi_k^2 + |\Delta_k|^2}. \quad (7.8)$$

The goal is to find new canonical fermions γ_k, γ_{-k} such that

$$H_k = E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right). \quad (7.9)$$

7.2 Bogoliubov angle and coherence factors

Write the complex pairing amplitude as

$$\Delta_k = |\Delta_k| e^{i\alpha_k}, \quad \alpha_k = \text{Arg } \Delta_k, \quad (7.10)$$

whenever $\Delta_k \neq 0$. Introduce the polar angle $\theta_k \in [0, \pi]$ by

$$\cos \theta_k = \frac{\xi_k}{E_k}, \quad \sin \theta_k = \frac{|\Delta_k|}{E_k}. \quad (7.11)$$

The standard coherence factors are

$$u_k = \cos \frac{\theta_k}{2} = \sqrt{\frac{E_k + \xi_k}{2E_k}}, \quad v_k = -e^{i\alpha_k} \sin \frac{\theta_k}{2} = -e^{i\alpha_k} \sqrt{\frac{E_k - \xi_k}{2E_k}}. \quad (7.12)$$

They obey

$$|u_k|^2 + |v_k|^2 = 1, \quad (7.13)$$

and, with the above phase convention,

$$\frac{v_k}{u_k} = -\frac{\Delta_k}{E_k + \xi_k} \quad (7.14)$$

whenever $E_k + \xi_k \neq 0$.

Remark 7.1 (Gauge convention at singular points). The expression Eq. (7.12) is a convenient local gauge. At points where $E_k + \xi_k = 0$, the ratio in Eq. (7.14) is not a good coordinate even though the physical state is well defined. One may instead use an equivalent gauge based on $E_k - \xi_k$. This is the same type of gauge singularity encountered when representing a two-level eigenvector on a Bloch sphere.

7.3 Canonical Bogoliubov transformation

Define the quasiparticle operators by

$$\gamma_k = u_k c_k - v_k c_{-k}^\dagger, \quad \gamma_{-k} = u_k c_{-k} + v_k c_k^\dagger. \quad (7.15)$$

Equivalently,

$$\begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix} = U_k \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}, \quad U_k = \begin{pmatrix} u_k & -v_k \\ v_k^* & u_k \end{pmatrix}. \quad (7.16)$$

The matrix U_k is unitary because of Eq. (7.13). Therefore the transformation preserves the canonical anticommutation relations:

$$\{\gamma_q, \gamma_{q'}^\dagger\} = \delta_{q,q'}, \quad \{\gamma_q, \gamma_{q'}\} = 0. \quad (7.17)$$

The inverse transformation is

$$c_k = u_k \gamma_k + v_k \gamma_{-k}^\dagger, \quad c_{-k} = u_k \gamma_{-k} - v_k \gamma_k^\dagger. \quad (7.18)$$

Proposition 7.1 (Diagonalization of one BdG block). *With u_k, v_k given by Eq. (7.12), the Bogoliubov transformation Eq. (7.15) diagonalizes the pair Hamiltonian:*

$$H_k = E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right). \quad (7.19)$$

Proof. Using the Nambu notation of Eq. (7.6), the transformation is

$$\Gamma_k = U_k \Psi_k, \quad \Gamma_k = \begin{pmatrix} \gamma_k \\ \gamma_{-k}^\dagger \end{pmatrix}. \quad (7.20)$$

A direct multiplication gives

$$U_k \mathcal{H}_{\text{BdG}}(k) U_k^\dagger = \begin{pmatrix} E_k & 0 \\ 0 & -E_k \end{pmatrix} = E_k \tau_z. \quad (7.21)$$

Hence

$$\begin{aligned} H_k &= \Psi_k^\dagger \mathcal{H}_{\text{BdG}}(k) \Psi_k = \Gamma_k^\dagger E_k \tau_z \Gamma_k \\ &= E_k \gamma_k^\dagger \gamma_k - E_k \gamma_{-k} \gamma_{-k}^\dagger = E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right), \end{aligned} \quad (7.22)$$

where the last equality uses $\gamma_{-k} \gamma_{-k}^\dagger = 1 - \gamma_{-k}^\dagger \gamma_{-k}$. $\square$

Thus the BdG eigenvalues $\pm E_k$ are not two independent particle species. They are the particle and hole branches of one canonical pair of quasiparticle modes. The physical excitation energies are nonnegative and are given by E_k .

7.4 Even-sector two-level problem

The same diagonalization can be seen directly in the many-body Hilbert space. For $k \neq -k$, the even-parity subspace is spanned by

$$|0\rangle_{k,-k}, \quad |11\rangle_{k,-k} = c_k^\dagger c_{-k}^\dagger |0\rangle_{k,-k}. \quad (7.23)$$

In this basis the pair Hamiltonian is represented by

$$H_k^{\text{even}} = \begin{pmatrix} -\xi_k & \Delta_k^* \\ \Delta_k & \xi_k \end{pmatrix}. \quad (7.24)$$

Its eigenvalues are $-E_k$ and $+E_k$. The lower eigenstate is

$$|\text{GS}\rangle_k = u_k |0\rangle_{k,-k} + v_k c_k^\dagger c_{-k}^\dagger |0\rangle_{k,-k}, \quad (7.25)$$

with the same u_k, v_k as in Eq. (7.12). Indeed,

$$\frac{v_k}{u_k} = -\frac{\Delta_k}{E_k + \xi_k} \quad (7.26)$$

is exactly the lower-eigenvector equation for Eq. (7.24).

The odd-parity states

$$c_k^\dagger |0\rangle_{k,-k}, \quad c_{-k}^\dagger |0\rangle_{k,-k} \quad (7.27)$$

are not mixed by the pairing term. Relative to the ground state of the pair, they are one-quasiparticle excitations with energy cost E_k . The doubly excited state $\gamma_k^\dagger \gamma_{-k}^\dagger |\text{GS}\rangle_k$ has energy cost $2E_k$.

7.5 BCS product ground state

The full quasiparticle vacuum is defined by

$$\gamma_q |\text{GS}\rangle = 0 \quad \text{for all quasiparticle modes } q. \quad (7.28)$$

For the non-self-conjugate momentum pairs, it is the product of the pair ground states:

$$|\text{GS}\rangle = \prod_{k \in \mathcal{K}_+} \left(u_k + v_k c_k^\dagger c_{-k}^\dagger \right) |0\rangle, \quad (7.29)$$

up to the separate treatment of self-conjugate modes. The ordering of the product over k must be fixed once and for all, since fermion creation operators anticommute. A different ordering changes only an overall sign convention for the many-body state.

The diagonal Hamiltonian is

$$H = E_{\text{vac}} + \sum_{k \in \mathcal{K}_+} E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} \right) + H_{\text{self}}, \quad (7.30)$$

where

$$E_{\text{vac}} = E_{\text{cl}} - \sum_{k \in \mathcal{K}_+} E_k \quad (7.31)$$

modulo the constant convention used in the original Hamiltonian. The term H_{self} denotes possible $k = 0$ or $k = \pi$ contributions, discussed below.

Principle 7.1 (Meaning of the BCS product). The BCS ground state is not a fixed-particle-number state. It is a coherent superposition of states with different fermion numbers but the same fermion parity. This is natural for BdG Hamiltonians, which conserve fermion parity rather than fermion number.

For each pair $(k, -k)$, the occupation and anomalous expectation values are

$$\langle c_k^\dagger c_k \rangle = |v_k|^2, \quad \langle c_{-k}^\dagger c_{-k} \rangle = |v_k|^2, \quad \langle c_{-k} c_k \rangle = u_k v_k. \quad (7.32)$$

Using [Eq. \(7.12\)](#), these become

$$|v_k|^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right), \quad u_k v_k = -\frac{\Delta_k}{2E_k}. \quad (7.33)$$

These contractions are the input for the covariance-matrix and Pfaffian methods developed in the next chapter.

7.6 Self-conjugate momenta and parity bookkeeping

If the allowed momentum set contains $q = 0$ or $q = \pi$, then $q = -q$ modulo 2π . For spinless odd-parity pairing,

$$\Delta_q = 0, \quad (7.34)$$

so such a mode cannot be paired with itself. Its BdG contribution is instead an ordinary one-mode problem. In a symmetric BdG convention one writes

$$H_q = \xi_q \left(c_q^\dagger c_q - \frac{1}{2} \right). \quad (7.35)$$

Equivalently,

$$H_q = |\xi_q| \left(\eta_q^\dagger \eta_q - \frac{1}{2} \right), \quad (7.36)$$

where

$$\eta_q = \begin{cases} c_q, & \xi_q > 0, \\ c_q^\dagger, & \xi_q < 0. \end{cases} \quad (7.37)$$

Thus the ground-state occupation of the self-conjugate mode is

$$n_q^{\text{GS}} = \begin{cases} 0, & \xi_q > 0, \\ 1, & \xi_q < 0. \end{cases} \quad (7.38)$$

These modes are negligible for most thermodynamic bulk quantities but essential for finite-size spectra and fermion-parity selection.

In Jordan–Wigner images of spin chains, the physical Hilbert space splits into fermion-parity sectors. The parity sector fixes the boundary condition, and the boundary condition fixes whether self-conjugate momenta are present. Therefore a careful finite-size solution requires the following sequence:

$$\begin{aligned} \text{spin parity sector} &\longrightarrow \text{fermion boundary condition} \longrightarrow \text{allowed momentum set} \\ &\longrightarrow \text{Bogoliubov diagonalization.} \end{aligned} \quad (7.39)$$

Only after this sequence can one compare the vacuum energies of different parity sectors to determine the true finite-size ground state.

7.7 Role of the Fourier-gauge phase

The Fourier convention used in these notes allows

$$c_k^{(\phi)} = e^{-i\phi} c_k^{(0)}, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_k e^{ikj} c_k^{(\phi)}. \quad (7.40)$$

As explained in the previous chapter, this is a Nambu gauge choice. It changes

$$\Delta_k^{(\phi)} = e^{-2i\phi} \Delta_k^{(0)}, \quad (7.41)$$

but leaves the spectrum invariant:

$$E_k = \sqrt{\xi_k^2 + |\Delta_k^{(\phi)}|^2}. \quad (7.42)$$

Since the coherence factor v_k carries the phase of Δ_k , it transforms as

$$v_k^{(\phi)} = e^{-2i\phi} v_k^{(0)}, \quad u_k^{(\phi)} = u_k^{(0)}, \quad (7.43)$$

within the convention [Eq. \(7.12\)](#). The BCS factor transforms covariantly:

$$u_k^{(\phi)} + v_k^{(\phi)} c_k^{(\phi)\dagger} c_{-k}^{(\phi)\dagger} = u_k^{(0)} + v_k^{(0)} c_k^{(0)\dagger} c_{-k}^{(0)\dagger}. \quad (7.44)$$

Thus the many-body state is unchanged; only its coordinate representation in Nambu space has been rotated.

This observation is especially useful for a Kitaev chain with complex pairing,

$$\Delta = |\Delta| e^{i\theta_\Delta}. \quad (7.45)$$

The gap function derived in the previous chapter is

$$\Delta_\phi(k) = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (7.46)$$

Choosing

$$\phi = \frac{\theta_\Delta}{2} \quad (7.47)$$

sets

$$\Delta_\phi(k) = 2i |\Delta| \sin k. \quad (7.48)$$

For a single uniform Kitaev chain, this removes the uniform superconducting phase from the BdG block. The phase becomes physical only when it is relative to another phase, spatially dependent, or tied to a boundary twist or flux that cannot be removed by a single global gauge choice.

7.8 Application: quasiparticles of the Kitaev chain

For the Kitaev chain in the convention of the previous chapter,

$$\xi_k = -\mu - 2t \cos k, \quad \Delta_k = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (7.49)$$

The quasiparticle dispersion is

$$E_k = \sqrt{(\mu + 2t \cos k)^2 + 4 |\Delta|^2 \sin^2 k}. \quad (7.50)$$

The Bogoliubov angle is determined by

$$\cos \theta_k = \frac{-\mu - 2t \cos k}{E_k}, \quad \sin \theta_k = \frac{2|\Delta| |\sin k|}{E_k}. \quad (7.51)$$

The phase entering v_k is

$$\alpha_k = \text{Arg} [2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k]. \quad (7.52)$$

In the gauge $\phi = \theta_\Delta/2$, this reduces to the phase of $i \sin k$. Hence the only remaining phase discontinuity is the sign change of $\sin k$, which reflects the odd-parity nature of spinless p -wave pairing.

The gap closes when both terms inside Eq. (7.50) vanish:

$$\sin k = 0, \quad \mu + 2t \cos k = 0. \quad (7.53)$$

Thus, for real nonzero t , the bulk phase boundaries are

$$\mu = -2t \quad (k = 0), \quad \mu = 2t \quad (k = \pi). \quad (7.54)$$

The Bogoliubov quasiparticles become gapless at these transition points.

7.9 Application: quasiparticles of the XY and TFIM chains

For the transverse-field XY chain in the normalization fixed earlier, the BdG vector can be chosen as

$$\mathbf{d}_{XY}(k) = (0, -2\gamma \sin k, 2(h - \cos k)). \quad (7.55)$$

Therefore

$$\xi_k = 2(h - \cos k), \quad \Delta_k = 2i\gamma \sin k, \quad (7.56)$$

up to the same Nambu-gauge conventions discussed above. The quasiparticle energy is

$$E_k = 2\sqrt{(h - \cos k)^2 + \gamma^2 \sin^2 k}. \quad (7.57)$$

The transverse-field Ising chain is obtained by setting $\gamma = 1$:

$$E_k^{\text{TFIM}} = 2\sqrt{(h - \cos k)^2 + \sin^2 k} = 2\sqrt{1 + h^2 - 2h \cos k}. \quad (7.58)$$

The gap closes at $h = 1$ and $k = 0$ in the thermodynamic limit. With the normalization used here, the excitation gap behaves as

$$\Delta_{\text{gap}} = 2|1 - h| \quad (7.59)$$

near the critical point.

The XX chain corresponds to $\gamma = 0$. In that case the pairing vanishes,

$$\Delta_k = 0, \quad (7.60)$$

and the Bogoliubov transformation reduces to an ordinary choice of occupied and empty fermion modes. Thus the XX chain is a number-conserving free-fermion problem, while the TFIM and generic XY chain are genuine BdG pairing problems.

7.10 Summary

Each non-self-conjugate momentum pair $(k, -k)$ of a spinless BdG chain is solved by a canonical Bogoliubov transformation. Starting from

$$H_k = \xi_k (n_k + n_{-k} - 1) + \Delta_k c_k^\dagger c_{-k}^\dagger + \Delta_k^* c_{-k} c_k, \quad (7.61)$$

one defines

$$E_k = \sqrt{\xi_k^2 + |\Delta_k|^2}, \quad u_k = \sqrt{\frac{E_k + \xi_k}{2E_k}}, \quad v_k = -e^{i \text{Arg } \Delta_k} \sqrt{\frac{E_k - \xi_k}{2E_k}}. \quad (7.62)$$

The quasiparticles are

$$\gamma_k = u_k c_k - v_k c_{-k}^\dagger, \quad \gamma_{-k} = u_k c_{-k} + v_k c_k^\dagger, \quad (7.63)$$

and the Hamiltonian becomes

$$H_k = E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right). \quad (7.64)$$

The quasiparticle vacuum is the BCS product state

$$|\text{GS}\rangle = \prod_{k \in \mathcal{K}_+} \left(u_k + v_k c_k^\dagger c_{-k}^\dagger \right) |0\rangle, \quad (7.65)$$

with separate finite-size bookkeeping for possible $k = 0$ and $k = \pi$ modes. The phase of the pairing gap appears in v_k and in the anomalous correlator, but a uniform Fourier-gauge phase changes only the representation, not the physical spectrum or state. This diagonal form is the starting point for the Gaussian correlation-function machinery developed in the next chapter.

Chapter 8

Correlation Functions, Covariance Matrices, and Pfaffians

The preceding chapters reduced the transverse-field Ising chain, the XY chain, and the Kitaev chain to quadratic fermion or BdG Hamiltonians. Diagonalizing the Hamiltonian gives the spectrum and the quasiparticle vacuum. For many-body physics, however, the spectrum is only half of the solution. One also wants order parameters, spin-spin correlators, string correlators, entanglement entropies, and finite-temperature response functions.

The essential reason quadratic fermion models are exactly solvable at the level of correlation functions is that their ground states and thermal states are *fermionic Gaussian states*. In such states all higher-order correlation functions are fixed by two-point functions. For local fermionic observables this statement is Wick's theorem. For spin observables obtained after the Jordan–Wigner transformation, the nonlocal strings turn spin correlators into products of many Majorana operators, and Wick's theorem organizes those products as Pfaffians. In translation-invariant one-dimensional problems, these Pfaffians often further reduce to Toeplitz determinants.

This chapter gives the correlation-function machinery used throughout the rest of the notes. The conventions are those of the previous chapters:

$$n_j = c_j^\dagger c_j = \frac{1 - \sigma_j^z}{2}, \quad \sigma_j^z = 1 - 2n_j, \quad (8.1)$$

and the Jordan–Wigner convention is

$$c_j = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^+, \quad c_j^\dagger = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^-. \quad (8.2)$$

Thus the occupied fermion state corresponds to spin down.

8.1 Gaussian fermionic states and Wick theorem

A fermionic state is called Gaussian if its density matrix can be written, up to normalization, as the exponential of a quadratic form in fermion operators. In a complex-fermion basis this means schematically

$$\rho = \frac{1}{Z} \exp \left[- \sum_{i,j} c_i^\dagger K_{ij} c_j - \frac{1}{2} \sum_{i,j} \left(L_{ij} c_i^\dagger c_j^\dagger + L_{ij}^* c_j c_i \right) \right], \quad (8.3)$$

where $K^\dagger = K$ and $L^T = -L$. Equivalently, in a Majorana basis to be introduced below,

$$\rho = \frac{1}{Z} \exp \left(\frac{i}{4} \sum_{m,n} W_{mn} a_m a_n \right), \quad W^T = -W, \quad (8.4)$$

with W real for a physical Hermitian quadratic exponent.

The ground state of any gapped quadratic Hamiltonian is Gaussian. A thermal state of a quadratic Hamiltonian is also Gaussian. This is the algebraic reason why free fermion and BdG models are not merely spectrally solvable, but also correlation-function solvable.

Theorem 8.1 (Wick theorem for parity-even Gaussian states). *Let ρ be a parity-even fermionic Gaussian state, and let $O_1, \dots, O_{2N}$ be fermionic operators linear in c_j and $c_j^\dagger$. Then*

$$\langle O_1 O_2 \cdots O_{2N} \rangle = \sum_{\text{pairings } P} \text{sgn}(P) \prod_{(r,s) \in P} \langle O_r O_s \rangle. \quad (8.5)$$

All odd products have vanishing expectation value:

$$\langle O_1 O_2 \cdots O_{2N+1} \rangle = 0. \quad (8.6)$$

The sign $\text{sgn}(P)$ is the fermionic sign obtained by bringing the paired operators next to one another.

For example,

$$\langle O_1 O_2 O_3 O_4 \rangle = \langle O_1 O_2 \rangle \langle O_3 O_4 \rangle - \langle O_1 O_3 \rangle \langle O_2 O_4 \rangle + \langle O_1 O_4 \rangle \langle O_2 O_3 \rangle. \quad (8.7)$$

The alternating signs are not optional. They are the main bookkeeping issue in fermionic correlation functions.

Sketch of proof. For a Gaussian density matrix, introduce Grassmann sources η_m coupled to a basis of Majorana operators a_m . The generating function has the form

$$Z[\eta] = \left\langle \exp \left(\sum_m \eta_m a_m \right) \right\rangle = \exp \left(\frac{1}{2} \sum_{m,n} \eta_m G_{mn} \eta_n \right), \quad (8.8)$$

where G_{mn} is the two-point contraction matrix. Differentiating $Z[\eta]$ with respect to the Grassmann sources gives all correlation functions. Since $Z[\eta]$ is the exponential of a quadratic form, every nonzero derivative pairs sources into two-point contractions. This gives Eq. (8.5). Odd derivatives vanish because the generating function is even in the Grassmann sources. $\square$

For practical calculations it is usually inefficient to explicitly enumerate all pairings. The compact object that automatically includes all pairings and their fermionic signs is the Pfaffian.

8.2 Pfaffians

Let M be a $2N \times 2N$ antisymmetric matrix. Its Pfaffian is defined by

$$\text{Pf}(M) = \frac{1}{2^N N!} \sum_{\pi \in S_{2N}} \text{sgn}(\pi) \prod_{r=1}^N M_{\pi(2r-1), \pi(2r)}. \quad (8.9)$$

Equivalently, it is the signed sum over all pairings of the $2N$ indices. It obeys the fundamental identity

$$\text{Pf}(M)^2 = \det M. \quad (8.10)$$

For the smallest cases,

$$\text{Pf} \begin{pmatrix} 0 & m_{12} \\ -m_{12} & 0 \end{pmatrix} = m_{12}, \quad (8.11)$$

and

$$\text{Pf} \begin{pmatrix} 0 & m_{12} & m_{13} & m_{14} \\ -m_{12} & 0 & m_{23} & m_{24} \\ -m_{13} & -m_{23} & 0 & m_{34} \\ -m_{14} & -m_{24} & -m_{34} & 0 \end{pmatrix} = m_{12}m_{34} - m_{13}m_{24} + m_{14}m_{23}. \quad (8.12)$$

Thus Eq. (8.12) is precisely the four-point Wick expansion Eq. (8.7).

If a_m are Majorana operators and the indices $m_1, \dots, m_{2N}$ are all distinct, Wick theorem can be written as

$$\langle a_{m_1} a_{m_2} \cdots a_{m_{2N}} \rangle = \text{Pf}(G_{\mathcal{I}}), \quad \mathcal{I} = \{m_1, \dots, m_{2N}\}, \quad (8.13)$$

where the antisymmetric contraction matrix is

$$(G_{\mathcal{I}})_{rs} = \begin{cases} \langle a_{m_r} a_{m_s} \rangle, & r < s, \\ -\langle a_{m_s} a_{m_r} \rangle, & r > s, \\ 0, & r = s. \end{cases} \quad (8.14)$$

The next section rewrites this formula in terms of the covariance matrix Γ .

8.3 Majorana covariance matrix

For each lattice site define two Hermitian Majorana operators

$$a_{2j-1} = c_j + c_j^\dagger, \quad a_{2j} = i(c_j - c_j^\dagger). \quad (8.15)$$

They satisfy

$$a_m^\dagger = a_m, \quad \{a_m, a_n\} = 2\delta_{mn}. \quad (8.16)$$

The inverse relations are

$$c_j = \frac{1}{2}(a_{2j-1} - ia_{2j}), \quad c_j^\dagger = \frac{1}{2}(a_{2j-1} + ia_{2j}). \quad (8.17)$$

Definition 8.1 (Majorana covariance matrix). For a state ρ , the Majorana covariance matrix is the real antisymmetric matrix

$$\Gamma_{mn} = \frac{i}{2} \langle [a_m, a_n] \rangle. \quad (8.18)$$

Equivalently,

$$\langle a_m a_n \rangle = \delta_{mn} - i\Gamma_{mn}. \quad (8.19)$$

The covariance matrix is the natural object for Gaussian-state calculations. For a parity-even Gaussian state, all correlation functions are determined by Γ . In particular, for distinct Majorana indices $m_1, \dots, m_{2N}$,

$$\langle a_{m_1} a_{m_2} \cdots a_{m_{2N}} \rangle = \text{Pf}(-i\Gamma_{\mathcal{I}}), \quad \mathcal{I} = \{m_1, \dots, m_{2N}\}. \quad (8.20)$$

Here $\Gamma_{\mathcal{I}}$ denotes the submatrix of Γ restricted to the ordered list $\mathcal{I} = (m_1, \dots, m_{2N})$. The order matters: permuting the indices changes the Pfaffian by the sign of the permutation, exactly as required for fermionic operators.

Remark 8.1 (Pure Gaussian states). For a pure Gaussian state, the covariance matrix obeys

$$\Gamma^2 = -\mathbf{1}. \quad (8.21)$$

For a mixed Gaussian state, its eigenvalues come in pairs $\pm i\nu_\alpha$ with $0 \leq \nu_\alpha \leq 1$. The pure-state limit corresponds to $\nu_\alpha = 1$ for every mode.

Relation to normal and anomalous correlators

It is often useful to express Γ in terms of the complex-fermion correlation matrices

$$C_{ij} = \langle c_i^\dagger c_j \rangle, \quad F_{ij} = \langle c_i c_j \rangle, \quad F_{ij} = -F_{ji}. \quad (8.22)$$

Using [Eq. \(8.15\)](#), one obtains

$$\Gamma_{2i-1, 2j-1} = i(C_{ij} - C_{ji} + F_{ij} - F_{ij}^*), \quad (8.23)$$

$$\Gamma_{2i, 2j} = i(C_{ij} - C_{ji} - F_{ij} + F_{ij}^*), \quad (8.24)$$

$$\Gamma_{2i-1, 2j} = \delta_{ij} - C_{ij} - C_{ji} - F_{ij} + F_{ij}^*. \quad (8.25)$$

The remaining block follows from antisymmetry:

$$\Gamma_{2i, 2j-1} = -\Gamma_{2j-1, 2i}. \quad (8.26)$$

For a translation-invariant BCS ground state written as

$$|\text{GS}\rangle = \prod_{k \in \mathcal{K}_+} (u_k + v_k c_k^\dagger c_{-k}^\dagger) |0\rangle, \quad (8.27)$$

with the Fourier convention of the previous chapter,

$$c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_k e^{ikj} c_k, \quad (8.28)$$

the normal and anomalous kernels may be written as

$$C_r = \langle c_j^\dagger c_{j+r} \rangle = \frac{1}{L} \sum_k e^{ikr} |v_k|^2, \quad (8.29)$$

$$F_r = \langle c_j c_{j+r} \rangle = \frac{e^{2i\phi}}{L} \sum_k e^{-ikr} \kappa_k, \quad \kappa_k = \langle c_k c_{-k} \rangle = -u_k v_k. \quad (8.30)$$

The phase ϕ is the Fourier-gauge phase discussed earlier. It rotates the representation of anomalous correlators but leaves gauge-invariant spin and fermion-parity observables unchanged.

Covariance matrix from a quadratic Majorana Hamiltonian

A general quadratic Majorana Hamiltonian can be written as

$$H = \frac{i}{4} \sum_{m,n=1}^{2L} A_{mn} a_m a_n, \quad A^T = -A, \quad (8.31)$$

where A is real antisymmetric. For its thermal state at inverse temperature β , the covariance matrix is

$$\Gamma_\beta = i \tanh \left(\frac{\beta i A}{2} \right). \quad (8.32)$$

In the zero-temperature limit for a gapped Hamiltonian,

$$\Gamma_0 = i \operatorname{sgn}(iA). \quad (8.33)$$

This formula is useful for numerical checks because it bypasses the explicit construction of many-body wavefunctions.

Example 8.1 (One fermionic mode). Let

$$H = \varepsilon \left(c^\dagger c - \frac{1}{2} \right). \quad (8.34)$$

Then

$$H = -\frac{i\varepsilon}{2} a_1 a_2, \quad A = \begin{pmatrix} 0 & -\varepsilon \\ \varepsilon & 0 \end{pmatrix}. \quad (8.35)$$

For $\varepsilon > 0$, the zero-temperature state is the empty state, and

$$\Gamma_{12} = 1, \quad \langle i a_1 a_2 \rangle = 1. \quad (8.36)$$

This agrees with $\sigma^z = 1 - 2n$ under the spin-down-as-occupied convention.

8.4 Local observables from the covariance matrix

The simplest observables are local density and magnetization. From [Eq. \(8.17\)](#),

$$n_j = c_j^\dagger c_j = \frac{1}{2} (1 - i a_{2j-1} a_{2j}), \quad (8.37)$$

and therefore

$$\sigma_j^z = 1 - 2n_j = i a_{2j-1} a_{2j}. \quad (8.38)$$

Taking the expectation value gives

$$\langle \sigma_j^z \rangle = \Gamma_{2j-1, 2j}, \quad \langle n_j \rangle = \frac{1}{2} (1 - \Gamma_{2j-1, 2j}). \quad (8.39)$$

For $i \neq j$, the longitudinal spin correlator is a four-Majorana expectation:

$$\langle \sigma_i^z \sigma_j^z \rangle = -\operatorname{Pf} \left(-i \Gamma_{\{2i-1, 2i, 2j-1, 2j\}} \right). \quad (8.40)$$

Equivalently, expanding the four-point Pfaffian gives

$$\langle \sigma_i^z \sigma_j^z \rangle = \Gamma_{2i-1, 2i} \Gamma_{2j-1, 2j} - \Gamma_{2i-1, 2j-1} \Gamma_{2i, 2j} + \Gamma_{2i-1, 2j} \Gamma_{2i, 2j-1}. \quad (8.41)$$

Thus even density-density and longitudinal spin correlations are already encoded in the same covariance matrix.

8.5 Spin correlators as Pfaffians

The nontrivial feature of spin correlators is the Jordan–Wigner string. Define

$$S_j = \prod_{\ell < j} \sigma_\ell^z = \prod_{\ell < j} i a_{2\ell-1} a_{2\ell}. \quad (8.42)$$

With the convention used in these notes,

$$\sigma_j^x = S_j a_{2j-1}, \quad \sigma_j^y = -S_j a_{2j}, \quad \sigma_j^z = i a_{2j-1} a_{2j}. \quad (8.43)$$

For $i < j$, multiplying the strings gives finite products of Majorana operators:

$$\sigma_i^x \sigma_j^x = i^{j-i} a_{2i} \left(\prod_{m=i+1}^{j-1} a_{2m-1} a_{2m} \right) a_{2j-1}, \quad (8.44)$$

$$\sigma_i^y \sigma_j^y = -i^{j-i} a_{2i-1} \left(\prod_{m=i+1}^{j-1} a_{2m-1} a_{2m} \right) a_{2j}, \quad (8.45)$$

$$\sigma_i^x \sigma_j^y = -i^{j-i} a_{2i} \left(\prod_{m=i+1}^{j-1} a_{2m-1} a_{2m} \right) a_{2j}. \quad (8.46)$$

The products are ordered from left to right as written. The string length grows linearly with the spin separation, but the expectation value is still exactly computable: it is a Pfaffian of a finite submatrix of Γ .

For example, define the ordered index list

$$\mathcal{I}_{xx}(i, j) = (2i, 2i+1, 2i+2, \dots, 2j-2, 2j-1), \quad i < j. \quad (8.47)$$

Then

$$\langle \sigma_i^x \sigma_j^x \rangle = i^{j-i} \text{Pf} \left(-i \Gamma_{\mathcal{I}_{xx}(i, j)} \right). \quad (8.48)$$

Similarly, with

$$\mathcal{I}_{yy}(i, j) = (2i-1, 2i+1, 2i+2, \dots, 2j-2, 2j), \quad (8.49)$$

one has

$$\langle \sigma_i^y \sigma_j^y \rangle = -i^{j-i} \text{Pf} \left(-i \Gamma_{\mathcal{I}_{yy}(i, j)} \right). \quad (8.50)$$

These equations are one of the most important practical outputs of the Jordan–Wigner solution: a highly nonlocal spin correlator has been reduced to a finite-dimensional linear-algebra problem.

Remark 8.2 (Why Pfaffians rather than determinants?). A determinant appears when the Wick contractions have a bipartite structure, for example when only contractions of the form $\langle B_m A_n \rangle$ enter. A Pfaffian is the more general object. It remains valid when the state has complex pairing, broken time-reversal symmetry, finite-size boundary subtleties, or mixed AA , BB , and AB contractions.

8.6 Toeplitz determinant form in translation-invariant chains

For many real XY-chain conventions, symmetries imply a simpler determinant formula. Introduce

$$A_j = c_j + c_j^\dagger = a_{2j-1}, \quad B_j = c_j^\dagger - c_j = i a_{2j}. \quad (8.51)$$

These satisfy

$$A_j^2 = 1, \quad B_j^2 = -1, \quad \{A_i, B_j\} = 0. \quad (8.52)$$

The local parity is

$$\sigma_j^z = A_j B_j, \quad (8.53)$$

and the transverse correlator for $r > 0$ becomes

$$\sigma_j^x \sigma_{j+r}^x = B_j A_{j+1} B_{j+1} A_{j+2} \cdots B_{j+r-1} A_{j+r}. \quad (8.54)$$

If the Gaussian state is translation invariant and its only nontrivial contractions in this basis are

$$G_r = \langle B_j A_{j+r} \rangle, \quad (8.55)$$

then Wick theorem reduces Eq. (8.54) to the Toeplitz determinant

$$\langle \sigma_j^x \sigma_{j+r}^x \rangle = \det \begin{pmatrix} G_1 & G_2 & \cdots & G_r \\ G_0 & G_1 & \cdots & G_{r-1} \\ \vdots & \vdots & \ddots & \vdots \\ G_{2-r} & G_{3-r} & \cdots & G_1 \end{pmatrix}. \quad (8.56)$$

This is a Toeplitz determinant because the matrix entries depend only on the index difference. Analogous determinant formulas exist for $\langle \sigma_j^y \sigma_{j+r}^y \rangle$ and mixed transverse correlators, with shifted kernels. In conventions or models where the bipartite contraction structure is absent, one should return to the Pfaffian formulas Eqs. (8.48) and (8.50).

Remark 8.3 (Thermodynamic limit). In the thermodynamic limit, G_r is usually represented by a Fourier integral

$$G_r = \int_{-\pi}^{\pi} \frac{dk}{2\pi} e^{-ikr} g(k), \quad (8.57)$$

where $g(k)$ is the symbol determined by the BdG ground state. Long-distance spin correlations then reduce to the asymptotic analysis of Toeplitz determinants. This is the bridge between free-fermion diagonalization and exact critical exponents in the Ising and XY chains.

8.7 Reduced density matrices and entanglement from covariance matrices

Although the present notes mainly use covariance matrices for correlation functions, the same object also determines entanglement in Gaussian states. Let A be a subsystem consisting of ℓ lattice sites, hence 2ℓ Majorana operators. Restrict the full covariance matrix to this subsystem:

$$\Gamma_A = \Gamma|_A. \quad (8.58)$$

The eigenvalues of Γ_A come in pairs $\pm i\nu_\alpha$, with $0 \leq \nu_\alpha \leq 1$, $\alpha = 1, \dots, \ell$. The reduced density matrix ρ_A is Gaussian, and its von Neumann entropy is

$$S_A = -\text{Tr} \rho_A \log \rho_A = \sum_{\alpha=1}^{\ell} \left[-\frac{1+\nu_\alpha}{2} \log \frac{1+\nu_\alpha}{2} - \frac{1-\nu_\alpha}{2} \log \frac{1-\nu_\alpha}{2} \right]. \quad (8.59)$$

Thus the same matrix Γ simultaneously encodes spin correlators, fermionic correlators, and entanglement spectra.

8.8 Practical recipe

For a quadratic spin chain solved by Jordan–Wigner, Fourier transform, and Bogoliubov transformation, the correlation-function workflow is:

- (i) Diagonalize the BdG Hamiltonian and obtain the coherence factors u_k, v_k or, equivalently, the occupied negative-energy BdG projector.
- (ii) Construct the normal and anomalous correlators C_{ij} and F_{ij} , either in momentum space using Eqs. (8.29) and (8.30) or directly in real space.
- (iii) Convert C and F into the Majorana covariance matrix Γ using Eqs. (8.23) to (8.25).
- (iv) Compute local fermionic observables directly from C and F , or from the relevant submatrix of Γ .
- (v) Compute spin correlators by rewriting the Jordan–Wigner strings as ordered Majorana products and evaluating the Pfaffian $\text{Pf}(-i\Gamma_{\mathcal{I}})$.
- (vi) In translation-invariant cases with the appropriate symmetry, simplify the Pfaffian to a Toeplitz determinant and use known determinant asymptotics for long-distance behavior.

The main conceptual point is that the nonlocality introduced by the Jordan–Wigner transformation does not destroy exact solvability. It only changes the final algebraic object from a two-point function to a Pfaffian of a finite covariance submatrix. This Pfaffian structure is the common language behind the exact correlation functions of the TFIM, the XY chain, Kitaev-chain spin representations, the two-dimensional Ising model, and more general free-Majorana solvable systems.

Chapter 9

Transfer Matrix and Quantum–Classical Correspondence

The previous chapters developed the free-fermion machinery used to diagonalize one-dimensional quantum spin chains: Jordan–Wigner transformation, Fourier transform, BdG blocks, Bogoliubov quasiparticles, and Gaussian correlation functions. Transfer matrices provide a complementary route to exact solvability. They are indispensable for two related reasons:

- (i) a classical statistical model in one higher dimension can often be interpreted as an imaginary-time path integral of a quantum model; and
- (ii) a family of commuting transfer matrices can generate a quantum Hamiltonian by a logarithmic derivative or Hamiltonian limit.

The first point is the quantum–classical correspondence. The second point is the bridge to vertex-model and Bethe-ansatz integrability in later chapters.

9.1 Classical transfer matrices

Consider a classical statistical model on a strip with M layers. Let $\mathbf{s}_m$ denote the entire configuration on the m th layer. If the Boltzmann weight couples only neighbouring layers, the partition function can be written as

$$Z = \sum_{\mathbf{s}_1, \dots, \mathbf{s}_M} T_{\mathbf{s}_1 \mathbf{s}_2} T_{\mathbf{s}_2 \mathbf{s}_3} \cdots T_{\mathbf{s}_M \mathbf{s}_1} = \text{Tr } T^M. \quad (9.1)$$

The matrix T is the *transfer matrix*. Its matrix element $T_{\mathbf{s}\mathbf{s}'}$ is the Boltzmann weight for two adjacent layers $\mathbf{s}$ and $\mathbf{s}'$, including a chosen assignment of intra-layer weights. The trace implements periodic boundary conditions in the transfer direction.

If T is finite-dimensional and diagonalizable, with eigenvalues

$$\lambda_0, \lambda_1, \lambda_2, \dots, \quad |\lambda_0| \geq |\lambda_1| \geq \dots, \quad (9.2)$$

then

$$Z = \sum_n \lambda_n^M. \quad (9.3)$$

In the thermodynamic limit $M \rightarrow \infty$, the largest eigenvalue controls the free energy per transfer step. If each layer contains L sites, then

$$f = -\frac{1}{\beta_{\text{cl}} L} \lim_{M \rightarrow \infty} \frac{1}{M} \log Z = -\frac{1}{\beta_{\text{cl}} L} \log \lambda_0, \quad (9.4)$$

up to the convention-dependent normalization of the classical inverse temperature β_{cl} .

Transfer matrices also encode correlation lengths. For a layer observable O ,

$$\langle O_m O_0 \rangle = \frac{1}{Z} \text{Tr} (OT^m OT^{M-m}). \quad (9.5)$$

After inserting the spectral resolution of T , the connected correlation function has the large-distance form

$$\langle O_m O_0 \rangle_c \sim \left(\frac{\lambda_1}{\lambda_0} \right)^m, \quad \xi^{-1} = \log \frac{\lambda_0}{|\lambda_1|}, \quad (9.6)$$

provided O has nonzero matrix element between the leading eigenstate and the first relevant subleading eigenstate. Thus the spectrum of T plays the same role for a classical model that the spectrum of a Hamiltonian plays for a quantum model.

9.2 Warm-up: the one-dimensional classical Ising chain

For the one-dimensional classical Ising model with spins $s_j = \pm 1$ and periodic boundary conditions,

$$Z_L(K) = \sum_{s_1, \dots, s_L = \pm 1} \exp \left(K \sum_{j=1}^L s_j s_{j+1} \right), \quad s_{L+1} = s_1, \quad (9.7)$$

the transfer matrix is the 2×2 matrix

$$T_{ss'} = e^{Kss'}, \quad T = \begin{pmatrix} e^K & e^{-K} \\ e^{-K} & e^K \end{pmatrix}. \quad (9.8)$$

Its eigenvalues are

$$\lambda_+ = e^K + e^{-K} = 2 \cosh K, \quad \lambda_- = e^K - e^{-K} = 2 \sinh K. \quad (9.9)$$

Therefore

$$Z_L(K) = \lambda_+^L + \lambda_-^L. \quad (9.10)$$

For finite K , the correlation length is finite:

$$\xi^{-1} = \log \frac{\lambda_+}{\lambda_-} = \log \coth K. \quad (9.11)$$

This elementary example already contains the main lesson: thermodynamics and correlations are spectral properties of the transfer matrix.

9.3 Transfer matrix as imaginary-time evolution

A quantum thermal partition function is

$$Z_q(\beta) = \text{Tr} e^{-\beta H}. \quad (9.12)$$

If imaginary time is discretized into M steps,

$$\beta = M\delta\tau, \quad (9.13)$$

then

$$Z_q(\beta) = \text{Tr} \left(e^{-\delta\tau H} \right)^M. \quad (9.14)$$

Thus the short imaginary-time evolution operator

$$T_q(\delta\tau) = e^{-\delta\tau H} \quad (9.15)$$

is itself a transfer matrix. Conversely, whenever a positive transfer matrix T can be written as

$$T = C e^{-a_\tau H}, \quad (9.16)$$

with $C > 0$ and a_τ interpreted as an imaginary-time lattice spacing, it defines the quantum Hamiltonian

$$H = -\frac{1}{a_\tau} \log T + \frac{\log C}{a_\tau} \mathbf{1}. \quad (9.17)$$

The additive constant proportional to $\mathbf{1}$ shifts the zero of energy and does not affect normalized expectation values or energy gaps.

If $T|n\rangle = \lambda_n|n\rangle$ and $T = C e^{-a_\tau H}$, then

$$\lambda_n = C e^{-a_\tau E_n}, \quad E_n - E_0 = -\frac{1}{a_\tau} \log \frac{\lambda_n}{\lambda_0}. \quad (9.18)$$

Therefore transfer-matrix gaps become quantum energy gaps. In particular,

$$\xi_\tau^{-1} = \log \frac{\lambda_0}{|\lambda_1|} = a_\tau (E_1 - E_0) \quad (9.19)$$

when the first subleading eigenvalue controls the decay.

Remark 9.1 (Hermiticity and positivity). A symmetric positive transfer matrix defines a Hermitian Hamiltonian by $H = -a_\tau^{-1} \log T$, up to a constant. More general transfer matrices may be non-Hermitian. Such matrices are still useful, for example in stochastic and non-equilibrium problems, but the direct interpretation as a Hermitian quantum Hamiltonian then requires additional structure.

9.4 Trotter decomposition and the extra dimension

The relation between a d -dimensional quantum model and a $(d+1)$ -dimensional classical model follows from the Trotter product formula. Suppose

$$H = A + B, \quad (9.20)$$

where A and B do not commute but have simple matrix elements in appropriate bases. Then

$$e^{-\beta H} = \lim_{M \rightarrow \infty} \left(e^{-\delta\tau A} e^{-\delta\tau B} \right)^M, \quad \delta\tau = \frac{\beta}{M}. \quad (9.21)$$

A symmetric version is

$$e^{-\delta\tau(A+B)} = e^{-\frac{\delta\tau}{2}A} e^{-\delta\tau B} e^{-\frac{\delta\tau}{2}A} + O(\delta\tau^3). \quad (9.22)$$

Inserting complete sets of states between the M imaginary-time factors produces a classical sum over configurations on M time slices. The additional discrete index is imaginary time. Schematically,

$$Z_{d\text{-dimensional quantum}} = \text{Tr} e^{-\beta H} \longleftrightarrow Z_{(d+1)\text{-dimensional classical}}. \quad (9.23)$$

This correspondence is exact in the Trotter limit $M \rightarrow \infty$ and becomes a controlled approximation at finite M .

The correspondence is not a statement that the quantum model and the classical model are identical microscopic objects. Rather, their partition functions and correlation functions are two representations of the same Euclidean statistical sum. Quantum noncommutativity is converted into classical coupling along the imaginary-time direction.

9.5 From the anisotropic two-dimensional Ising model to the TFIM

The most important example for these notes is the correspondence between the anisotropic two-dimensional classical Ising model and the one-dimensional TFIM. Let

$$Z_{2D} = \sum_{s_{j,m}=\pm 1} \exp \left[K_x \sum_{j,m} s_{j,m} s_{j+1,m} + K_\tau \sum_{j,m} s_{j,m} s_{j,m+1} \right], \quad (9.24)$$

where $j = 1, \dots, L$ is the spatial coordinate and $m = 1, \dots, M$ is the transfer direction. The couplings K_x and K_τ are classical dimensionless couplings.

To match the Pauli convention used in these notes, take the classical variable $s_j = \pm 1$ to label the eigenvalue of σ_j^x :

$$\sigma_j^x |s_j\rangle_x = s_j |s_j\rangle_x. \quad (9.25)$$

The horizontal classical coupling is diagonal in this basis and gives the operator

$$\exp \left(K_x \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right). \quad (9.26)$$

The vertical one-site Boltzmann weight is

$$W_{ss'} = e^{K_\tau ss'} = \begin{pmatrix} e^{K_\tau} & e^{-K_\tau} \\ e^{-K_\tau} & e^{K_\tau} \end{pmatrix}_{s,s'}. \quad (9.27)$$

In the σ^x eigenbasis, the operator σ^z flips s . Hence

$$W = A e^{\kappa \sigma^z}, \quad \tanh \kappa = e^{-2K_\tau}, \quad A = \sqrt{2 \sinh(2K_\tau)}. \quad (9.28)$$

Indeed,

$$A \cosh \kappa = e^{K_\tau}, \quad A \sinh \kappa = e^{-K_\tau}. \quad (9.29)$$

A convenient symmetric row-to-row transfer matrix is therefore

$$T = A^L \exp \left(\frac{K_x}{2} \sum_j \sigma_j^x \sigma_{j+1}^x \right) \exp \left(\kappa \sum_j \sigma_j^z \right) \exp \left(\frac{K_x}{2} \sum_j \sigma_j^x \sigma_{j+1}^x \right). \quad (9.30)$$

Using the symmetric Trotter formula, the strongly anisotropic limit is obtained by setting

$$K_x = \delta\tau J, \quad \kappa = \delta\tau h, \quad \tanh(\delta\tau h) = e^{-2K_\tau}. \quad (9.31)$$

Then

$$T = A^L \exp \left[-\delta\tau H_{\text{TFIM}} + O(\delta\tau^3) \right], \quad (9.32)$$

with

$$H_{\text{TFIM}} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z. \quad (9.33)$$

This is precisely the TFIM convention fixed earlier in these notes.

For small $\delta\tau$,

$$e^{-2K_\tau} = \tanh(\delta\tau h) = \delta\tau h + O(\delta\tau^3), \quad (9.34)$$

so the imaginary-time coupling K_τ is large. The quantum limit is therefore a strongly anisotropic limit of the classical two-dimensional model:

$$K_x \rightarrow 0, \quad K_\tau \rightarrow \infty, \quad K_x e^{2K_\tau} \text{ fixed}. \quad (9.35)$$

The classical critical condition of the anisotropic two-dimensional Ising model,

$$\sinh(2K_x) \sinh(2K_\tau) = 1, \quad (9.36)$$

then becomes

$$\frac{J}{h} = 1, \quad \text{or equivalently} \quad h = J. \quad (9.37)$$

In the common normalization $J = 1$, this gives the TFIM critical point $h = 1$.

Remark 9.2 (Why the basis matters). Many textbooks write the TFIM as $-J \sum_j \sigma_j^z \sigma_{j+1}^z - \Gamma \sum_j \sigma_j^x$. These notes use the rotated convention $-J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z$ because it is adapted to the Jordan–Wigner occupation convention $n_j = (1 - \sigma_j^z)/2$. The two forms are related by a global spin rotation. In the transfer-matrix derivation above, this rotation is implemented by labeling the classical Ising variable as the eigenvalue of σ^x rather than σ^z .

9.6 Operator insertions and Euclidean correlators

The transfer-matrix correspondence also maps correlation functions. A diagonal classical row observable $O(\mathbf{s})$ becomes an operator O acting on the row Hilbert space. Its correlation along the transfer direction is

$$\langle O_m O_0 \rangle_{\text{cl}} = \frac{1}{Z} \text{Tr} (O T^m O T^{M-m}). \quad (9.38)$$

If $T = C e^{-a_\tau H}$, then

$$\langle O_m O_0 \rangle_{\text{cl}} = \frac{1}{Z_q} \text{Tr} (O(\tau) O(0) e^{-\beta H}), \quad \tau = m a_\tau, \quad (9.39)$$

where

$$O(\tau) = e^{\tau H} O e^{-\tau H}. \quad (9.40)$$

At zero temperature, this becomes the ground-state Euclidean correlator. Its large- τ decay is controlled by the lowest quantum excitation carrying the quantum numbers of O :

$$\langle 0 | O(\tau) O(0) | 0 \rangle_c \sim e^{-\tau(E_1 - E_0)}. \quad (9.41)$$

Thus the transfer-matrix spectrum, the classical correlation length, and the quantum excitation gap are the same spectral information in different languages.

9.7 Quantum-to-classical dictionary

For the TFIM/Ising example in the conventions of these notes, the essential parameter dictionary is

$$\boxed{K_x = \delta\tau J, \quad e^{-2K_\tau} = \tanh(\delta\tau h)} \quad (9.42)$$

with row transfer matrix

$$T \propto \exp\left(\frac{\delta\tau J}{2} \sum_j \sigma_j^x \sigma_{j+1}^x\right) \exp\left(\delta\tau h \sum_j \sigma_j^z\right) \exp\left(\frac{\delta\tau J}{2} \sum_j \sigma_j^x \sigma_{j+1}^x\right). \quad (9.43)$$

The proportionality constant shifts the quantum free energy by an additive constant. The operator in the exponential is $-\delta\tau H_{\text{TFIM}}$ to leading order.

The following table summarizes the correspondence.

Concept	Classical transfer-matrix language	Quantum language
Transfer direction	Layer index m	Imaginary time $\tau = ma_\tau$
Row configuration	Classical spin row $\mathbf{s}_m$	Basis vector of the quantum Hilbert space
Transfer matrix	$T_{\mathbf{s}\mathbf{s}'}$	$e^{-a_\tau H}$ up to normalization
Largest eigenvalue	Dominant Boltzmann weight	Ground-state energy density
Eigenvalue ratio	Correlation length $\xi^{-1} = \log(\lambda_0/ \lambda_1)$	Energy gap $a_\tau(E_1 - E_0)$
Classical anisotropy	Different K_x, K_τ	Spatial velocity and time-lattice spacing
Criticality	Divergent correlation length	Closing quantum gap

9.8 Hamiltonian limits of commuting transfer matrices

In integrable lattice models one often has a one-parameter family of commuting transfer matrices,

$$[T(u), T(v)] = 0, \quad (9.44)$$

where u is a spectral parameter. If $T(u)$ is regular at a point $u = u_0$, the associated quantum Hamiltonian is commonly obtained from the logarithmic derivative

$$H \propto \frac{d}{du} \log T(u) \Big|_{u=u_0} = T(u)^{-1} \frac{dT(u)}{du} \Big|_{u=u_0}, \quad (9.45)$$

plus an additive and multiplicative normalization. Because all $T(u)$ commute, the Hamiltonian obtained this way commutes with the entire transfer-matrix family:

$$[H, T(v)] = 0. \quad (9.46)$$

This is the transfer-matrix origin of infinitely many conserved quantities in Bethe-ansatz integrable chains. Later, the six-vertex transfer matrix will yield the XXZ Hamiltonian in precisely this sense.

9.9 Practical summary

The transfer-matrix method should be read as a spectral equivalence principle:

$$\boxed{\text{classical layer-to-layer transfer} \iff \text{quantum imaginary-time evolution}}. \quad (9.47)$$

For the purposes of these notes, the essential workflow is:

- (i) Write the classical partition function as $Z = \text{Tr } T^M$.
- (ii) Interpret the layer configuration as a quantum basis state.
- (iii) If possible, write T as $Ce^{-a_\tau H}$ or take a controlled Hamiltonian limit.
- (iv) Extract quantum energies from transfer-matrix eigenvalue ratios.
- (v) Translate classical correlation lengths into quantum gaps.
- (vi) Use anisotropic limits to connect two-dimensional classical models to one-dimensional quantum Hamiltonians.

The 2D classical Ising model and the 1D TFIM are the canonical example. The same logic also underlies the relation between the six-vertex model and the XXZ chain, the eight-vertex model and the XYZ chain, and more general tensor-network interpretations of Euclidean path integrals.

Chapter 10

Duality

Duality is one of the most efficient organizing principles in exactly solvable models. A duality rewrites the same partition function, Hamiltonian, operator algebra, or low-energy field theory in variables adapted to a different physical regime. In the Ising family, the essential effect is the exchange

$$\begin{aligned} \text{ordered variables} &\longleftrightarrow \text{domain-wall or disorder variables,} \\ \text{strong coupling} &\longleftrightarrow \text{weak coupling.} \end{aligned} \quad (10.1)$$

Thus duality is not merely a computational trick. It explains why an apparently nonlocal description can become local after changing variables, why the transverse-field Ising critical point is located at the self-dual coupling, and why the two phases of the Ising chain are naturally described as order and disorder phases.

This chapter develops duality at three levels:

- (i) Kramers–Wannier duality of the two-dimensional classical Ising model;
- (ii) the operator Kramers–Wannier duality of the one-dimensional TFIM; and
- (iii) the precise logical status of self-duality as a criterion for criticality.

The conventions are matched to the TFIM Hamiltonian used in these notes,

$$H_{\text{TFIM}} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z, \quad J, h \geq 0. \quad (10.2)$$

The ordered phase is therefore the phase with long-range order in σ^x , while the transverse-field or disordered phase is polarized by σ^z .

10.1 Classical Kramers–Wannier duality

Consider the anisotropic two-dimensional classical Ising model on a square lattice,

$$Z(K_x, K_y) = \sum_{s_r = \pm 1} \exp \left[K_x \sum_{\langle rr' \rangle_x} s_r s_{r'} + K_y \sum_{\langle rr' \rangle_y} s_r s_{r'} \right], \quad (10.3)$$

where K_x and K_y are dimensionless couplings along the two lattice directions. The high-temperature expansion uses

$$e^{K s_r s_{r'}} = \cosh K (1 + s_r s_{r'} \tanh K). \quad (10.4)$$

After expanding all bonds and summing over the Ising spins, only configurations in which an even number of occupied bonds touches each site survive. These are closed loops on the original lattice. Schematically,

$$Z(K_x, K_y) = 2^N (\cosh K_x)^{E_x} (\cosh K_y)^{E_y} \sum_{\mathcal{C} \text{ closed}} (\tanh K_x)^{n_x(\mathcal{C})} (\tanh K_y)^{n_y(\mathcal{C})}. \quad (10.5)$$

Here N is the number of spins, E_x, E_y are the numbers of bonds in the two directions, and $n_x(\mathcal{C}), n_y(\mathcal{C})$ count occupied loop edges.

The low-temperature expansion instead starts from the two ferromagnetic ground states. A configuration is specified by domain walls on the dual lattice. A domain wall segment crossing a bond of coupling K costs a Boltzmann factor e^{-2K} . Matching the loop weights in the high-temperature expansion to the domain-wall weights of a dual Ising model gives the dual couplings

$$e^{-2K_x^*} = \tanh K_y, \quad e^{-2K_y^*} = \tanh K_x. \quad (10.6)$$

Equivalently,

$$\sinh(2K_x^*) \sinh(2K_y) = 1, \quad \sinh(2K_y^*) \sinh(2K_x) = 1. \quad (10.7)$$

The exchange of directions in Eq. (10.6) comes from the fact that a horizontal bond of the dual lattice crosses a vertical bond of the original lattice, and conversely.

Up to a non-singular prefactor depending on the couplings and the boundary conditions, the partition function of the original model equals the partition function of the dual model,

$$Z(K_x, K_y) \propto Z(K_x^*, K_y^*). \quad (10.8)$$

The proportionality factor affects the analytic part of the free energy but not the location of a nonanalytic singularity in the thermodynamic limit.

For the isotropic model $K_x = K_y = K$, the self-dual condition is

$$e^{-2K_c} = \tanh K_c, \quad \sinh(2K_c) = 1, \quad K_c = \frac{1}{2} \log(1 + \sqrt{2}). \quad (10.9)$$

For the anisotropic model, the self-dual curve is

$$\sinh(2K_x) \sinh(2K_y) = 1. \quad (10.10)$$

This is the same condition that appeared in the transfer-matrix derivation of the quantum–classical correspondence in the previous chapter. In the strongly anisotropic limit, it reduces to the TFIM condition $h = J$.

10.2 Operator duality in the transverse-field Ising chain

The quantum version of Kramers–Wannier duality is an exact change of operator variables. We first work on an infinite chain or in the bulk of a long open chain, where boundary subtleties do not obscure the local algebra. Define dual Pauli operators on the bonds of the original lattice by

$$\mu_{j+\frac{1}{2}}^z = \sigma_j^x \sigma_{j+1}^x, \quad \mu_{j+\frac{1}{2}}^x = \prod_{\ell \leq j} \sigma_\ell^z. \quad (10.11)$$

The first operator measures whether neighbouring σ^x spins are aligned; the second is a semi-infinite string of transverse-field operators and is called a *disorder operator*.

The operators in Eq. (10.11) obey the Pauli algebra. Indeed,

$$(\mu_{j+\frac{1}{2}}^x)^2 = (\mu_{j+\frac{1}{2}}^z)^2 = \mathbf{1}, \quad \{\mu_{j+\frac{1}{2}}^x, \mu_{j+\frac{1}{2}}^z\} = 0, \quad (10.12)$$

and dual operators on distinct dual sites commute. The anticommutation at the same dual site occurs because $\mu_{j+1/2}^x$ overlaps with $\sigma_j^x \sigma_{j+1}^x$ on exactly one site. For different dual sites, the number of overlaps is either zero or two, so the operators commute.

The inverse local identities are

$$\sigma_j^x \sigma_{j+1}^x = \mu_{j+\frac{1}{2}}^z, \quad \sigma_j^z = \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x. \quad (10.13)$$

Substituting these into Eq. (10.2) gives

$$H_{\text{TFIM}} = -J \sum_j \mu_{j+\frac{1}{2}}^z - h \sum_j \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x. \quad (10.14)$$

If we rotate the dual spin axes by defining

$$\tilde{\sigma}_{j+\frac{1}{2}}^x = \mu_{j+\frac{1}{2}}^z, \quad \tilde{\sigma}_{j+\frac{1}{2}}^z = \mu_{j+\frac{1}{2}}^x, \quad (10.15)$$

then the dual Hamiltonian becomes

$$H_{\text{TFIM}}(J, h) \mapsto -h \sum_j \tilde{\sigma}_{j-\frac{1}{2}}^z \tilde{\sigma}_{j+\frac{1}{2}}^z - J \sum_j \tilde{\sigma}_{j+\frac{1}{2}}^x. \quad (10.16)$$

This is the same TFIM form after relabeling the axes, with the two couplings exchanged:

$$(J, h) \longleftrightarrow (h, J). \quad (10.17)$$

Equivalently, in terms of the dimensionless ratio $g = h/J$,

$$g \mapsto g^* = \frac{1}{g}, \quad (10.18)$$

up to an overall energy scale.

Remark 10.1 (Why the dual lattice is shifted). The dual spins live on bonds $j + 1/2$, not on the original sites j . This shift is not cosmetic. It encodes the fact that the elementary dual variable is a domain-wall variable: it naturally lives between two neighbouring order variables. In the continuum limit, this shift becomes invisible, but on the lattice it is essential for getting the operator algebra and boundary conditions right.

10.3 Order and disorder operators

The duality map makes the distinction between order and disorder completely explicit. In the original spin language, the ferromagnetic order parameter is σ_j^x . In the dual language, a two-point order correlator becomes a string of dual domain-wall variables:

$$\sigma_i^x \sigma_j^x = \prod_{m=i}^{j-1} \mu_{m+\frac{1}{2}}^z, \quad i < j. \quad (10.19)$$

Conversely, the dual order operator μ^x is a disorder string in the original variables. Its two-point function is

$$\mu_{i+\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x = \prod_{\ell=i+1}^j \sigma_{\ell}^z, \quad i < j. \quad (10.20)$$

Thus Kramers–Wannier duality exchanges a local order operator with a nonlocal string operator.

In the ordered phase $J > h$, the original spins have long-range σ^x order, while the dual variables are disordered. In the paramagnetic phase $h > J$, the original order is absent, but the disorder operator has long-range order:

$$\begin{aligned} J > h & : \quad \langle \sigma^x \rangle \neq 0, & \langle \mu^x \rangle &= 0, \\ h > J & : \quad \langle \sigma^x \rangle = 0, & \langle \mu^x \rangle &\neq 0. \end{aligned} \quad (10.21)$$

This is the quantum-chain version of the classical statement that the low-temperature expansion in domain walls is dual to the high-temperature expansion in closed loops.

10.4 Finite chains, constraints, and boundary conditions

The local formulas above are exact in the bulk. On a finite chain, duality also acts on global sectors and boundary conditions. This point is important because many solvable-chain calculations in these notes use periodic boundary conditions, parity sectors, and momentum quantization.

For an open chain with L sites,

$$H_{\text{OBC}} = -J \sum_{j=1}^{L-1} \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^L \sigma_j^z, \quad (10.22)$$

define

$$\mu_{j+\frac{1}{2}}^z = \sigma_j^x \sigma_{j+1}^x, \quad \mu_{j+\frac{1}{2}}^x = \prod_{\ell=1}^j \sigma_{\ell}^z, \quad j = 1, \dots, L-1. \quad (10.23)$$

Then

$$\sigma_j^z = \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x, \quad \mu_{\frac{1}{2}}^x \equiv \mathbf{1}, \quad \mu_{L+\frac{1}{2}}^x \equiv P_z, \quad (10.24)$$

where

$$P_z = \prod_{j=1}^L \sigma_j^z \quad (10.25)$$

is the global Ising $\mathbb{Z}_2$ symmetry generator. Therefore the dual open-chain Hamiltonian contains boundary terms involving the fixed endpoint variables $\mu_{1/2}^x$ and $\mu_{L+1/2}^x$. In a fixed P_z sector these are ordinary boundary conditions for the dual chain.

For a periodic chain, the same issue appears as a relation between symmetry sectors and boundary twists. The definitions

$$\mu_{j+\frac{1}{2}}^z = \sigma_j^x \sigma_{j+1}^x, \quad \mu_{j+\frac{1}{2}}^x = \prod_{\ell=1}^j \sigma_{\ell}^z \quad (10.26)$$

lead to

$$\prod_{j=1}^L \mu_{j+\frac{1}{2}}^z = \prod_{j=1}^L \sigma_j^x \sigma_{j+1}^x = \mathbf{1}, \quad (10.27)$$

and

$$\mu_{L+\frac{1}{2}}^x = P_z \mu_{\frac{1}{2}}^x. \quad (10.28)$$

Hence a periodic original chain maps to a dual chain with a constraint and a boundary condition determined by the original global $\mathbb{Z}_2$ charge. Conversely, the global symmetry sector of the dual model determines the boundary condition of the original model. This is the spin-chain analogue of the parity-dependent fermion boundary conditions encountered in the Jordan–Wigner transformation.

Remark 10.2 (Duality is not always an on-site unitary). On an infinite chain, Kramers–Wannier duality is a faithful isomorphism of local operator algebras. On a finite periodic chain, it is not simply an on-site unitary map from one unconstrained 2^L -dimensional spin Hilbert space to another unconstrained 2^L -dimensional spin Hilbert space. One must keep track of constraints, global sectors, and boundary twists. Ignoring this point is a common source of erroneous factors of two in partition functions and spectra.

10.5 Self-duality and the TFIM critical point

The TFIM is self-dual when $J = h$. Since duality exchanges J and h , the point $J = h$ is invariant under the transformation. The exact free-fermion spectrum of Eq. (10.2) is

$$E(k) = 2\sqrt{(h - J \cos k)^2 + (J \sin k)^2}. \quad (10.29)$$

For $J, h > 0$, the excitation gap is obtained at $k = 0$:

$$\Delta_{\text{gap}} = 2|h - J|. \quad (10.30)$$

Thus the self-dual point is indeed the quantum critical point,

$$h = J. \quad (10.31)$$

At this point the order and disorder descriptions are equally natural, the Majorana mass in the continuum limit changes sign, and the scaling limit is the Ising conformal field theory.

This conclusion can also be obtained from the two-dimensional classical Ising model. The transfer-matrix chapter showed that the anisotropic classical critical condition

$$\sinh(2K_x) \sinh(2K_\tau) = 1 \quad (10.32)$$

reduces, in the strongly anisotropic quantum limit, to $h = J$. Kramers–Wannier duality and the quantum–classical correspondence therefore locate the same critical point.

10.6 When does self-duality determine a critical point?

Self-duality is powerful, but it is not by itself a proof of criticality. The precise statement is conditional.

Principle 10.1 (Self-duality criterion). Suppose a model depends on a coupling ratio g , and an exact duality maps $g \mapsto 1/g$. If the model has exactly one phase transition separating two phases exchanged by the duality, and if the transition is not hidden by an extended critical region or by additional symmetry-breaking/topological structure, then the transition must occur at the self-dual point $g = 1$.

The assumptions in this principle matter. A self-dual point can fail to be a unique continuous critical point for several reasons:

- (i) there may be more than one transition, appearing in dual pairs away from the self-dual point;
- (ii) there may be a first-order transition at the self-dual point rather than a continuous critical point;
- (iii) there may be an extended critical phase rather than an isolated critical point; or
- (iv) the duality may exchange sectors or boundary conditions rather than mapping a single finite-size Hamiltonian strictly to itself.

For the ferromagnetic TFIM with $J, h > 0$, the extra spectral information in [Eq. \(10.29\)](#) verifies that there is a single gap closing at $J = h$. Thus in this model self-duality is not merely suggestive; it correctly identifies the exact critical point.

Remark 10.3 (Sign conventions). The sign of h in [Eq. \(10.2\)](#) is removable by the global unitary $\prod_j \sigma_j^x$, which flips $\sigma_j^z \mapsto -\sigma_j^z$ and leaves $\sigma_j^x \sigma_{j+1}^x$ invariant. On a bipartite chain, the sign of J can also be changed by a staggered rotation, up to boundary subtleties for periodic chains. Therefore the essential ferromagnetic TFIM phase diagram is usually drawn in terms of $|h|/|J|$, with the critical point at $|h| = |J|$.

10.7 Relation to fermions and Majorana mass

Under the Jordan–Wigner transformation, the TFIM becomes a quadratic Majorana or BdG Hamiltonian. In the convention of these notes, the single-particle gap is controlled by the difference $h - J$. Near $k = 0$ and near $h = J$, the BdG spectrum has the relativistic form

$$E(k) \simeq 2\sqrt{(h - J)^2 + Jh k^2} \quad (10.33)$$

up to higher-order lattice corrections. The continuum theory is therefore a massive Majorana theory with mass proportional to

$$m \propto h - J. \quad (10.34)$$

Kramers–Wannier duality exchanges the two signs of this mass. The ordered Ising phase and the disordered Ising phase are therefore two massive Majorana phases separated by the massless Majorana point $m = 0$.

This observation is also the bridge to the Kitaev-chain interpretation. The TFIM ordered/disordered transition corresponds, after fermionization, to the transition between the topological and trivial phases of a one-dimensional spinless p -wave superconductor. In the spin language this distinction is expressed as order versus disorder; in the fermion language it is expressed as a change in the Majorana mass sign and, for open chains, the presence or absence of unpaired boundary Majorana modes. The next chapter will formulate this same transition in terms of winding numbers and BdG topology.

10.8 Summary

The essential identities of this chapter are

$$\mu_{j+\frac{1}{2}}^z = \sigma_j^x \sigma_{j+1}^x, \quad \mu_{j+\frac{1}{2}}^x = \prod_{\ell \leq j} \sigma_\ell^z, \quad \sigma_j^z = \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x. \quad (10.35)$$

They imply the bulk Hamiltonian mapping

$$-J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z \quad \longleftrightarrow \quad -h \sum_j \tilde{\sigma}_j^x \tilde{\sigma}_{j+1}^x - J \sum_j \tilde{\sigma}_j^z, \quad (10.36)$$

with the two couplings exchanged. Consequently the TFIM is self-dual at $h = J$, and the exact spectrum confirms that this point is the unique continuous critical point of the ferromagnetic chain.

The conceptual lesson is that duality turns nonlocal degrees of freedom into local ones. The same physical theory can be described either by the original order variables σ^x , by the disorder variables μ^x , by Jordan–Wigner fermions, or by Majorana fields. This is precisely the equivalence-map viewpoint that underlies the organization of these lecture notes.

Chapter 11

Topology of Two-Band Hamiltonians

The previous chapters developed the algebraic machinery needed to reduce many one-dimensional spin models to quadratic fermions and BdG momentum-space blocks. Once the Hamiltonian has been written as a family of finite-dimensional matrices depending on momentum, exact solvability is not only a question of finding the spectrum. It also becomes meaningful to ask whether the family of Bloch or BdG Hamiltonians can be continuously deformed into a trivial one while keeping the bulk gap open and preserving the required symmetries.

This chapter gives the minimal topological toolkit needed for the later chapters on the Kitaev chain, the SSH chain, and the XY/TFIM mother model. The central object is a two-band Hamiltonian

$$H(k) = d_0(k)\mathbf{1} + \mathbf{d}(k) \cdot \boldsymbol{\sigma}, \quad \mathbf{d}(k) = (d_x(k), d_y(k), d_z(k)), \quad (11.1)$$

where k belongs to the Brillouin zone. For BdG Hamiltonians the Pauli matrices act in Nambu space, while for ordinary band Hamiltonians they may act in sublattice, orbital, or pseudospin space. The same formulas therefore apply to two physically different but mathematically parallel settings:

$$\begin{aligned} \text{SSH chain} & : \text{ sublattice two-band Hamiltonian,} \\ \text{Kitaev chain} & : \text{ BdG two-band Hamiltonian in Nambu space.} \end{aligned} \quad (11.2)$$

The distinction between these two realizations is important: the Pauli matrices have the same algebra, but the symmetries and the meaning of the two components of the spinor are different.

11.1 Two-band Hamiltonians and maps to the Bloch sphere

The spectrum of [Eq. \(11.1\)](#) is

$$E_{\pm}(k) = d_0(k) \pm |\mathbf{d}(k)|. \quad (11.3)$$

Thus a two-band model is gapped at momentum k precisely when

$$|\mathbf{d}(k)| \neq 0. \quad (11.4)$$

For topology the scalar term $d_0(k)\mathbf{1}$ is usually irrelevant: it shifts the two energies by the same amount but does not change the eigenvectors. If the model is gapped over the entire Brillouin zone, one may define the normalized vector

$$\hat{\mathbf{d}}(k) = \frac{\mathbf{d}(k)}{|\mathbf{d}(k)|} \in S^2. \quad (11.5)$$

Therefore a gapped two-band Hamiltonian defines a map

$$\hat{\mathbf{d}}: \text{BZ} \longrightarrow S^2. \quad (11.6)$$

A topological phase is a homotopy class of such maps, possibly restricted by symmetry.

The projectors onto the upper and lower bands of $\mathbf{d} \cdot \boldsymbol{\sigma}$ are

$$P_+(k) = \frac{1 + \hat{\mathbf{d}}(k) \cdot \boldsymbol{\sigma}}{2}, \quad P_-(k) = \frac{1 - \hat{\mathbf{d}}(k) \cdot \boldsymbol{\sigma}}{2}. \quad (11.7)$$

If the lower band is occupied, the occupied-band geometry is completely encoded in $P_-(k)$.

Principle 11.1 (Gap closing is the only way to change topology). For a two-band Hamiltonian, the topological invariant can change only if there exists a momentum k_0 such that

$$\mathbf{d}(k_0) = 0. \quad (11.8)$$

At this point the two bands touch, the normalized vector $\hat{\mathbf{d}}(k)$ is undefined, and the homotopy class of the map can change.

This principle is the topological version of the gap-closing conditions already encountered in the BdG chapter. For example, the Kitaev-chain spectrum

$$E(k) = \sqrt{\xi(k)^2 + |\Delta(k)|^2} \quad (11.9)$$

can close only when both $\xi(k)$ and $\Delta(k)$ vanish. Since the nearest-neighbor p -wave gap is proportional to $\sin k$, the possible gap closings occur at $k = 0, \pi$.

11.2 Berry connection, Berry curvature, and Chern number

Although much of this chapter focuses on one-dimensional models, the general two-band language is most transparent if we first state the two-dimensional Berry-curvature formula. Let $|u_-(\mathbf{k})\rangle$ be a normalized eigenvector of the lower band. The Berry connection and Berry curvature are

$$A_a(\mathbf{k}) = i \langle u_-(\mathbf{k}) | \partial_{k_a} u_-(\mathbf{k}) \rangle, \quad F_{xy} = \partial_{k_x} A_y - \partial_{k_y} A_x. \quad (11.10)$$

Equivalently, in a manifestly gauge-invariant form,

$$F_{xy}(\mathbf{k}) = i \text{Tr} \left(P_-(\mathbf{k}) [\partial_{k_x} P_-(\mathbf{k}), \partial_{k_y} P_-(\mathbf{k})] \right). \quad (11.11)$$

Substituting [Eq. \(11.7\)](#) gives the standard two-band formula

$$F_{xy}(\mathbf{k}) = \frac{1}{2} \hat{\mathbf{d}} \cdot \left(\partial_{k_x} \hat{\mathbf{d}} \times \partial_{k_y} \hat{\mathbf{d}} \right). \quad (11.12)$$

The first Chern number of the occupied band is

$$C = \frac{1}{2\pi} \int_{\text{BZ}} F_{xy}(\mathbf{k}) d^2k = \frac{1}{4\pi} \int_{\text{BZ}} \hat{\mathbf{d}} \cdot (\partial_{k_x} \hat{\mathbf{d}} \times \partial_{k_y} \hat{\mathbf{d}}) d^2k. \quad (11.13)$$

Geometrically, C counts how many times the Brillouin-zone torus wraps the unit sphere under the map $\hat{\mathbf{d}}$.

Example 11.1 (A standard lattice Chern-insulator block). Consider

$$\mathbf{d}(\mathbf{k}) = (\sin k_x, \sin k_y, m + \cos k_x + \cos k_y). \quad (11.14)$$

The gap can close only at the four high-symmetry points $(0,0)$, $(\pi,0)$, $(0,\pi)$, and (π,π) , giving critical values $m = -2, 0, 2$. With the orientation convention in Eq. (11.13), the occupied-band Chern number is

$$C = \begin{cases} 0, & m > 2, \\ -1, & 0 < m < 2, \\ +1, & -2 < m < 0, \\ 0, & m < -2. \end{cases} \quad (11.15)$$

This example is not one of the main solvable chains of these notes, but it is a useful reference point: in two dimensions the full sphere S^2 can be wrapped, whereas in one-dimensional chiral models topology usually comes from restricting $\hat{\mathbf{d}}$ to an equator.

11.3 Why one-dimensional two-band topology needs symmetry

For a generic one-dimensional two-band Hamiltonian, the map is

$$S_{\text{BZ}}^1 \longrightarrow S^2. \quad (11.16)$$

Since $\pi_1(S^2) = 0$, every loop on the sphere can be contracted to a point. Therefore a generic gapped one-dimensional two-band Hamiltonian has no stable integer winding number.

A nontrivial one-dimensional invariant appears when a symmetry forces $\hat{\mathbf{d}}(k)$ to lie on a great circle of the sphere. The most important case for these notes is chiral symmetry.

Definition 11.1 (Chiral symmetry). A two-band Hamiltonian has chiral symmetry if there exists a unitary matrix S with $S^2 = \mathbf{1}$ such that

$$SH(k)S^{-1} = -H(k). \quad (11.17)$$

For the conventional choice $S = \sigma_z$, chiral symmetry forbids the identity and σ_z terms, so that

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y. \quad (11.18)$$

The vector $\mathbf{d}(k)$ now lies in the d_x – d_y plane. If the system is gapped, the complex number

$$q(k) = d_x(k) + i d_y(k) \quad (11.19)$$

never vanishes. Thus $q(k)$ defines a map

$$S_{\text{BZ}}^1 \longrightarrow \mathbb{C} \setminus \{0\} \simeq S^1. \quad (11.20)$$

This map is classified by an integer winding number.

11.4 Winding number of chiral two-band Hamiltonians

For the chiral Hamiltonian

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y = \begin{pmatrix} 0 & q(k)^* \\ q(k) & 0 \end{pmatrix}, \quad q(k) = d_x(k) + i d_y(k), \quad (11.21)$$

where the second equality follows from the standard Pauli matrices. The winding number is

$$\nu = \frac{1}{2\pi i} \int_{\text{BZ}} dk \partial_k \log q(k). \quad (11.22)$$

Equivalently, writing

$$q(k) = |q(k)| e^{i\varphi(k)}, \quad (11.23)$$

one obtains

$$\nu = \frac{1}{2\pi} \int_{\text{BZ}} dk \partial_k \varphi(k) = \frac{\varphi(2\pi) - \varphi(0)}{2\pi} \in \mathbb{Z}. \quad (11.24)$$

Thus ν counts how many times the curve

$$k \mapsto (d_x(k), d_y(k)) \quad (11.25)$$

winds around the origin.

Proposition 11.1 (Topological stability of the winding number). *If $q_s(k)$ is a smooth one-parameter family with $q_s(k) \neq 0$ for all k and all $s \in [0, 1]$, then the winding number Eq. (11.22) is independent of s .*

Proof. The s -derivative of the winding number is

$$\partial_s \nu = \frac{1}{2\pi i} \int_{\text{BZ}} dk \partial_k (q^{-1} \partial_s q). \quad (11.26)$$

Since the Brillouin zone is a circle and $q^{-1} \partial_s q$ is single-valued when q never vanishes, the integral of the total derivative is zero. $\square$

The winding number can therefore change only when $q(k_0) = 0$ for some momentum. This is the one-dimensional chiral version of the general gap-closing principle.

11.5 Zak phase and its relation to winding

For a one-dimensional occupied band, the Berry phase over the Brillouin zone is called the Zak phase:

$$\gamma_{\text{Zak}} = \oint_{\text{BZ}} dk i \langle u_-(k) | \partial_k u_-(k) \rangle \quad \text{mod } 2\pi. \quad (11.27)$$

For a chiral two-band Hamiltonian, one may choose the lower-band eigenvector as

$$|u_-(k)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -e^{-i\varphi(k)} \\ 1 \end{pmatrix}, \quad q(k) = |q(k)| e^{i\varphi(k)}. \quad (11.28)$$

This gives

$$\gamma_{\text{Zak}} = +\frac{1}{2} \int_{\text{BZ}} dk \partial_k \varphi(k) = +\pi \nu \quad \text{mod } 2\pi. \quad (11.29)$$

The sign depends on the eigenvector and Pauli-matrix convention, but the quantized state-ment is invariant:

$$\gamma_{\text{Zak}} = 0 \text{ or } \pi \pmod{2\pi} \quad (11.30)$$

for a chiral or inversion-symmetric two-band chain. Thus the parity of the winding number is visible as a Berry phase.

11.6 SSH chain as a chiral two-band model

The Su–Schrieffer–Heeger chain is the canonical ordinary-band example. It has two sublattices A and B in each unit cell, with intracell hopping t_1 and intercell hopping t_2 :

$$H_{\text{SSH}} = \sum_j \left(t_1 a_j^\dagger b_j + t_2 a_{j+1}^\dagger b_j + \text{h.c.} \right). \quad (11.31)$$

Using the Bloch spinor

$$\Psi_k = \begin{pmatrix} a_k \\ b_k \end{pmatrix}, \quad (11.32)$$

the Hamiltonian becomes

$$H_{\text{SSH}} = \sum_k \Psi_k^\dagger \begin{pmatrix} 0 & q_{\text{SSH}}(k)^* \\ q_{\text{SSH}}(k) & 0 \end{pmatrix} \Psi_k, \quad q_{\text{SSH}}(k) = t_1 + t_2 e^{ik}. \quad (11.33)$$

Equivalently,

$$d_x(k) = t_1 + t_2 \cos k, \quad d_y(k) = t_2 \sin k, \quad d_z(k) = 0. \quad (11.34)$$

The curve $q_{\text{SSH}}(k)$ is a circle of radius $|t_2|$ centered at t_1 on the real axis. It winds around the origin if and only if the origin lies inside this circle. For $t_1, t_2 > 0$ and the convention in [Eq. \(11.33\)](#),

$$\nu_{\text{SSH}} = \begin{cases} 0, & |t_1| > |t_2|, \\ 1, & |t_2| > |t_1|. \end{cases} \quad (11.35)$$

The transition occurs at $|t_1| = |t_2|$, where the gap closes at $k = \pi$ for $t_1 = t_2$.

Remark 11.1 (Unit-cell convention and the sign of winding). Changing the unit-cell convention multiplies $q(k)$ by a phase e^{ink} . This shifts the integer assigned to a particular Bloch basis. The robust physical statement for the SSH chain with a fixed boundary termination is whether the dimerization leaves unpaired edge sites. In the convention above, $|t_2| > |t_1|$ is the topological phase for the usual termination with weak intracell bonds at the boundary.

11.6.1 Edge zero modes

For an open chain in the topological regime $|t_2| > |t_1|$, the left-edge zero mode has support only on one sublattice. A zero-energy ansatz of the form

$$|\psi_L\rangle = \sum_{j \geq 1} \alpha_j a_j^\dagger |0\rangle \quad (11.36)$$

obeys the recursion

$$t_1 \alpha_j + t_2 \alpha_{j+1} = 0, \quad \alpha_{j+1} = -\frac{t_1}{t_2} \alpha_j. \quad (11.37)$$

Therefore

$$\alpha_j = \left(-\frac{t_1}{t_2}\right)^{j-1} \alpha_1, \quad (11.38)$$

which is normalizable precisely when $|t_1/t_2| < 1$. The bulk winding and the boundary zero mode therefore encode the same phase distinction.

11.7 BdG topology: Kitaev chain and Majorana zero modes

The Kitaev chain is also a two-band Hamiltonian, but the two components of the spinor are not sublattices. They are particle and hole components of a Nambu spinor. In the convention of the BdG chapter,

$$\mathcal{H}_K(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(k) \end{pmatrix}, \quad \xi(k) = -\mu - 2t \cos k, \quad (11.39)$$

with

$$\Delta_\phi(k) = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (11.40)$$

The uniform phase θ_Δ and the Fourier/Nambu gauge phase ϕ enter only through the combination $\theta_\Delta - 2\phi$. For a single uniform chain, one may choose

$$\phi = \frac{\theta_\Delta}{2} \quad (11.41)$$

so that

$$\Delta_\phi(k) = 2i |\Delta| \sin k. \quad (11.42)$$

In this real pairing gauge,

$$\mathcal{H}_K(k) = d_y(k)\tau_y + d_z(k)\tau_z, \quad d_y(k) = -2|\Delta| \sin k, \quad d_z(k) = \xi(k). \quad (11.43)$$

Here the Pauli matrices τ_a act in Nambu space. Since the vector lies in a plane, the real Kitaev chain has a winding-number description analogous to a chiral two-band model.

A convenient complex function is

$$q_K(k) = \xi(k) + 2i |\Delta| \sin k = -\mu - 2t \cos k + 2i |\Delta| \sin k. \quad (11.44)$$

Its winding number is

$$\nu_{\text{BDI}} = \frac{1}{2\pi i} \int_{\text{BZ}} dk \partial_k \log q_K(k). \quad (11.45)$$

Geometrically, $q_K(k)$ traces an ellipse centered at $-\mu$ on the real axis. The origin lies inside this ellipse if and only if

$$|\mu| < 2|t|. \quad (11.46)$$

Thus the Kitaev chain is topological for $|\mu| < 2|t|$ and trivial for $|\mu| > 2|t|$, assuming $\Delta \neq 0$.

11.7.1 Class D Pfaffian invariant

The real gauge above makes an additional chiral symmetry manifest and places the nearest-neighbor Kitaev chain in class BDI. More generally, a superconducting BdG Hamiltonian always has particle-hole symmetry. In the Nambu convention used earlier,

$$\tau_x \mathcal{H}_{\text{BdG}}(k)^* \tau_x = -\mathcal{H}_{\text{BdG}}(-k). \quad (11.47)$$

Without a real gauge or time-reversal-like symmetry, the one-dimensional superconductor is in class D and has a $\mathbb{Z}_2$ invariant rather than an integer winding number.

For a two-band Kitaev chain the class-D invariant reduces to the sign of the normal-state term at the particle-hole-invariant momenta $k = 0, \pi$, because the odd-parity pairing vanishes there:

$$\mathcal{M} = \text{sgn}[\xi(0)\xi(\pi)] = \text{sgn}[(-\mu - 2t)(-\mu + 2t)] = \text{sgn}(\mu^2 - 4t^2). \quad (11.48)$$

The nontrivial phase is

$$\mathcal{M} = -1 \iff |\mu| < 2|t|. \quad (11.49)$$

This is the $\mathbb{Z}_2$ remnant of the integer BDI winding. Uniform complex phases of Δ do not change Eq. (11.48); they can be removed by a global Nambu gauge choice. Relative superconducting phases, spatially varying phases, or fluxes that cannot be removed globally can reduce the manifest symmetry to class D, but the $\mathbb{Z}_2$ invariant remains well-defined as long as particle-hole symmetry and the bulk gap persist.

11.7.2 Exactly solvable Majorana limit

The boundary meaning of Eq. (11.46) is clearest at the exactly solvable point

$$\mu = 0, \quad t = |\Delta| > 0, \quad (11.50)$$

with open boundary conditions. Using Majorana operators

$$a_{2j-1} = c_j + c_j^\dagger, \quad a_{2j} = i(c_j - c_j^\dagger), \quad (11.51)$$

the Hamiltonian becomes, up to an overall sign convention fixed by the phase of Δ ,

$$H_K = -it \sum_{j=1}^{L-1} a_{2j} a_{2j+1}. \quad (11.52)$$

All bulk Majoranas are paired on bonds, while a_1 and a_{2L} are absent from the Hamiltonian. These two unpaired Majorana operators form a nonlocal boundary fermion

$$f_{\text{edge}} = \frac{a_1 + ia_{2L}}{2}. \quad (11.53)$$

The two occupations of f_{edge} are degenerate in the thermodynamic limit. Away from the exactly solvable point but within the same gapped topological phase, the Majorana zero modes remain exponentially localized at the edges.

Principle 11.2 (Bulk-boundary correspondence in one-dimensional two-band models). A nontrivial bulk winding or class-D $\mathbb{Z}_2$ invariant predicts protected boundary zero modes, provided the boundary termination respects the protecting symmetry and the bulk gap remains open.

For the SSH chain the boundary zero modes are ordinary complex fermion states localized on sublattices. For the Kitaev chain they are Majorana zero modes in Nambu space. The algebraic similarity of the two-band Hamiltonians should not obscure this physical difference.

11.8 Relation to the TFIM and the Majorana mass sign

The duality chapter showed that the transverse-field Ising chain separates two phases exchanged by Kramers–Wannier duality. The Jordan–Wigner and BdG chapters showed that the same model is equivalent to a quadratic Majorana chain. Topology supplies a third language for the same transition.

Near a gap-closing momentum, a one-dimensional BdG Hamiltonian takes the continuum Dirac/Majorana form

$$\mathcal{H}(q) = vq \tau_y + m \tau_z + \text{higher-order terms}, \quad (11.54)$$

where q measures momentum relative to the gap-closing point. A topological transition occurs when the mass m changes sign. For the Kitaev chain,

$$m_0 = \xi(0) = -\mu - 2t, \quad m_\pi = \xi(\pi) = -\mu + 2t. \quad (11.55)$$

The class-D invariant is simply the relative sign of these two masses:

$$\mathcal{M} = \text{sgn}(m_0 m_\pi). \quad (11.56)$$

The nontrivial phase has opposite signs at $k = 0$ and $k = \pi$.

In the TFIM image, the topological Majorana phase corresponds to the ordered Ising phase for open chains. The edge Majoranas encode the same twofold structure that appears as symmetry breaking in the spin variables. Thus the following descriptions are equivalent ways of describing the same physics:

$$\begin{aligned} \text{spin language} & : \text{ordered versus disordered Ising phase,} \\ \text{duality language} & : \text{order variable versus disorder variable,} \\ \text{fermion language} & : \text{trivial versus topological Majorana chain,} \\ \text{continuum language} & : \text{sign of a Majorana mass.} \end{aligned} \quad (11.57)$$

11.9 Symmetry classes relevant to the mother models

The full tenfold classification is beyond the scope of these notes, but the following restricted table is sufficient for the solvable models treated here.

Class	Symmetry content	1D invariant	Representative model in these notes
AIII	Chiral symmetry only	$\mathbb{Z}$ winding	SSH chain with chiral sublattice symmetry
BDI	Particle-hole plus real time-reversal-like symmetry, hence chiral symmetry	$\mathbb{Z}$ winding	Real Kitaev chain / JW image of XY-type chains in a real gauge
D	Particle-hole symmetry only	$\mathbb{Z}_2$ Pfaffian invariant	Kitaev chain with only BdG particle-hole symmetry manifest
A	No protecting symmetry	No stable 1D two-band invariant; $\mathbb{Z}$ Chern number in 2D	Two-dimensional Chern-insulator block

The table should be read operationally. The same matrix expression may belong to different classes depending on what physical structure the Pauli matrices represent and which perturbations are allowed. For example, adding a sublattice staggering term $m\sigma_z$ to the SSH chain breaks chiral symmetry and destroys the integer winding classification. Adding a perturbation to a BdG Hamiltonian must still respect particle-hole redundancy, so the allowed deformations are different.

11.10 Gauge choices, phases, and what is physically invariant

The Fourier and BdG chapters emphasized that a constant phase in the Fourier transform,

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_j e^{-ikj} c_j, \quad (11.58)$$

is a gauge convention. In BdG language it rotates the Nambu spinor and changes the phase of the gap function as

$$\Delta_\phi(k) = e^{-2i\phi} \Delta_0(k). \quad (11.59)$$

Consequently the components (d_x, d_y) of a BdG vector may rotate in the transverse plane under a gauge change. This rotation does not change the energy spectrum, the existence of a gap, or the class-D invariant [Eq. \(11.48\)](#).

For an ordinary chiral band Hamiltonian such as the SSH chain, a change of Bloch basis can also multiply $q(k)$ by a phase. If the phase is periodic and smooth, it does not change the physically meaningful distinction between phases with and without protected boundary states for a fixed termination. If the phase has the form e^{ink} , it reflects a shift of unit-cell convention and changes the integer assigned to the winding by a convention-dependent offset. Boundary physics is recovered only after the unit cell and termination are specified.

The correct hierarchy is therefore

$$\begin{aligned} \text{basis-dependent} & : d_x(k), d_y(k), d_z(k) \text{ as components,} \\ \text{gauge-covariant} & : q(k) \text{ and its phase,} \\ \text{physically invariant} & : \begin{array}{l} \text{gap closings, Berry phases modulo } 2\pi, \\ \text{winding differences, } \mathbb{Z}_2 \text{ parity, and edge modes} \end{array} \end{aligned} \quad (11.60)$$

This distinction is especially important for the Kitaev chain with a complex superconducting gap: a uniform complex phase is removable by a global gauge choice, while a relative phase across a junction or a spatial phase twist can represent physical superconducting phase bias or flux.

11.11 Practical diagnostic formulas

For later use, we collect the most useful diagnostic formulas.

Chiral two-band chain

For

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y, \quad q(k) = d_x(k) + id_y(k), \quad (11.61)$$

the integer invariant is

$$\nu = \frac{1}{2\pi i} \oint_{\text{BZ}} dk \partial_k \log q(k). \quad (11.62)$$

A gap closing occurs when $q(k) = 0$.

SSH chain

For

$$q_{\text{SSH}}(k) = t_1 + t_2 e^{ik}, \quad (11.63)$$

the usual topological regime for the standard open-chain termination is

$$|t_2| > |t_1|. \quad (11.64)$$

Kitaev chain

For

$$\xi(k) = -\mu - 2t \cos k, \quad \Delta(k) = 2i |\Delta| \sin k \quad (11.65)$$

in a real pairing gauge, the BDI winding may be computed from

$$q_K(k) = \xi(k) + 2i |\Delta| \sin k. \quad (11.66)$$

The class-D invariant is

$$\mathcal{M} = \text{sgn}[\xi(0)\xi(\pi)] = \text{sgn}(\mu^2 - 4t^2). \quad (11.67)$$

The topological Majorana phase is

$$\mathcal{M} = -1 \quad \Longleftrightarrow \quad |\mu| < 2|t|. \quad (11.68)$$

Two-dimensional two-band block

For a two-dimensional occupied lower band,

$$C = \frac{1}{4\pi} \int_{\text{BZ}} \hat{\mathbf{d}} \cdot \left(\partial_{k_x} \hat{\mathbf{d}} \times \partial_{k_y} \hat{\mathbf{d}} \right) d^2k. \quad (11.69)$$

11.12 Summary

The essential lesson of this chapter is that a solvable two-band Hamiltonian contains more information than its dispersion. The eigenvectors define a map from momentum space to a target space, and the topology of this map constrains possible phases and boundary states.

The main results are:

- (i) A gapped two-band Hamiltonian defines a normalized vector $\hat{\mathbf{d}}(k)$ and can change topology only when $\mathbf{d}(k) = 0$.
- (ii) In one dimension, a generic two-band Hamiltonian is topologically trivial unless a symmetry restricts the target space.
- (iii) Chiral two-band chains have an integer winding number $\nu = (2\pi i)^{-1} \int \partial_k \log q(k)$.

- (iv) The SSH chain realizes this winding in sublattice space.
- (v) The Kitaev chain realizes an analogous structure in Nambu space; in class D the robust invariant is the Pfaffian sign $\mathcal{M} = \text{sgn}[\xi(0)\xi(\pi)]$.
- (vi) Majorana edge modes are the boundary manifestation of the nontrivial Kitaev-chain invariant.
- (vii) The TFIM transition, the Kitaev-chain topological transition, and the sign change of a continuum Majorana mass are different languages for the same solvable critical structure.

Chapter 12

Preview of Bethe Ansatz and Yang–Baxter Integrability

The preceding chapters developed two main solution mechanisms. First, transfer matrices convert a classical statistical model into a spectral problem. Second, Jordan–Wigner and Bogoliubov transformations solve a large class of spin chains because their fermionic images are quadratic. The next integrability mechanism is different. Bethe-ansatz integrability solves genuinely interacting models, not by reducing them to independent one-particle modes, but by exploiting the factorization of many-body scattering and the existence of an infinite commuting family of transfer matrices.

This chapter is a conceptual and notational preview. Its purpose is to explain what the Bethe ansatz solves, why the Yang–Baxter equation appears, and how the row-to-row transfer matrices of vertex models become quantum Hamiltonians such as the XXZ chain. The complete coordinate and algebraic Bethe-ansatz derivations are postponed to later chapters.

12.1 From free particles to factorized scattering

A free-fermion Hamiltonian is solvable because it can be diagonalized into independent quasiparticles,

$$H = E_0 + \sum_k E_k \gamma_k^\dagger \gamma_k. \quad (12.1)$$

The many-body spectrum is then obtained by filling one-particle levels. Bethe ansatz uses a weaker but still powerful property: particles interact, but the scattering is elastic and factorized.

Definition 12.1 (Factorized scattering). Consider a one-dimensional many-body system with stable asymptotic quasiparticles labelled by momenta or rapidities. The scattering is called *factorized* if an M -body collision can be decomposed into an ordered product of two-body scattering processes, with no particle production and no diffraction.

In one spatial dimension the ordering of particles is meaningful. For coordinates

$$x_1 < x_2 < \cdots < x_M, \quad (12.2)$$

a coordinate Bethe wave function takes the form

$$\Psi(x_1, \dots, x_M) = \sum_{P \in S_M} A(P) \exp \left(i \sum_{a=1}^M k_{P_a} x_a \right), \quad (12.3)$$

where P permutes the momenta among the ordered particles. Interactions do not change the set of momenta. They only relate amplitudes associated with different permutations. For neighboring exchanges one writes schematically

$$A(P \circ s_a) = S(k_{P_a}, k_{P_{a+1}})A(P), \quad (12.4)$$

where s_a exchanges the a th and $(a+1)$ st momenta. For a scalar scattering phase, factorization is almost automatic because complex numbers commute. For particles with internal spin, color, or auxiliary indices, S becomes a matrix, and consistency is highly nontrivial.

The basic physical meaning is the following. In a generic interacting model, a many-particle collision can redistribute momenta in ways that cannot be described as a sequence of independent two-body events. In an integrable one-dimensional model, the infinite family of conservation laws is strong enough to forbid such diffraction. The only allowed effect of scattering is to permute the rapidities and multiply the wave function by two-body scattering matrices.

Principle 12.1 (Bethe-ansatz solvability). A one-dimensional interacting model is Bethe-ansatz solvable when its eigenstates can be assembled from plane waves whose amplitudes are related by consistent factorized two-body scattering.

12.2 A two-magnon preview

The simplest way to see the Bethe idea is to consider a spin chain above a ferromagnetic reference state. Let

$$|0\rangle = |\uparrow\uparrow\cdots\uparrow\rangle \quad (12.5)$$

and let a down spin be interpreted as a magnon. A one-magnon state is

$$|k\rangle = \sum_{x=1}^L e^{ikx} \sigma_x^- |0\rangle. \quad (12.6)$$

For a translation-invariant chain, it is an eigenstate in the one-magnon sector. The nontrivial part begins in the two-magnon sector, where the wave function has one expression for $x_1 < x_2$:

$$\Psi(x_1, x_2) = A_{12} e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)}. \quad (12.7)$$

The interaction only matters when the two magnons approach each other. The Schrodinger equation at contact fixes the ratio

$$A_{21} = S(k_1, k_2) A_{12}. \quad (12.8)$$

For a periodic chain, moving one magnon around the ring makes it scatter once with every other magnon. This gives the schematic Bethe equations

$$e^{ik_j L} = \prod_{\ell \neq j} S(k_j, k_\ell), \quad j = 1, \dots, M. \quad (12.9)$$

Thus periodic boundary conditions quantize the momenta in an interaction-dependent way. Free particles would have $S = \pm 1$ and hence ordinary integer or half-integer momenta. Interacting Bethe-ansatz particles acquire phase shifts from all other particles.

Example 12.1 (The XXZ two-body scattering phase). For one common Pauli-matrix convention, the XXZ chain may be written as

$$H_{\text{XXZ}} = J \sum_{j=1}^L (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1)). \quad (12.10)$$

In the two-magnon coordinate Bethe ansatz, the contact equation gives a scattering phase of the form

$$S(k_1, k_2) = -\frac{1 + e^{i(k_1+k_2)} - 2\Delta e^{ik_1}}{1 + e^{i(k_1+k_2)} - 2\Delta e^{ik_2}}. \quad (12.11)$$

The precise overall sign and energy convention depend on the normalization of Eq. (12.10), but the structural message is universal: the interaction is encoded in a two-body phase shift, and the many-body momenta are fixed by Eq. (12.9).

It is often more convenient to replace momenta by rapidities. In the XXX limit, the Bethe equations are commonly written as

$$\left(\frac{\lambda_j + i/2}{\lambda_j - i/2} \right)^L = \prod_{\ell \neq j} \frac{\lambda_j - \lambda_\ell + i}{\lambda_j - \lambda_\ell - i}. \quad (12.12)$$

For the gapless XXZ chain with $\Delta = \cos \gamma$, a standard rapidity form is

$$\left[\frac{\sinh(\lambda_j + i\gamma/2)}{\sinh(\lambda_j - i\gamma/2)} \right]^L = \prod_{\ell \neq j} \frac{\sinh(\lambda_j - \lambda_\ell + i\gamma)}{\sinh(\lambda_j - \lambda_\ell - i\gamma)}. \quad (12.13)$$

The left-hand side is free propagation around the ring. The right-hand side is the accumulated product of two-body scattering phases.

12.3 Why a three-body consistency condition is needed

For three particles there are multiple ways to exchange the same initial and final orderings. For instance, the permutation reversing $(1, 2, 3)$ to $(3, 2, 1)$ can be decomposed as

$$s_1 s_2 s_1 = s_2 s_1 s_2. \quad (12.14)$$

If the scattering amplitudes are scalar phases, this identity causes no serious constraint. If the two-body scatterings are matrices acting on internal degrees of freedom, the two possible sequences must give the same operator. This is the origin of the Yang–Baxter equation.

Let $R_{ab}(u)$ denote a two-body scattering or vertex operator acting nontrivially on spaces a and b , with spectral parameter difference u . The Yang–Baxter equation is

$$R_{12}(u-v) R_{13}(u-w) R_{23}(v-w) = R_{23}(v-w) R_{13}(u-w) R_{12}(u-v). \quad (12.15)$$

It is an equality of operators on a triple tensor product. Diagrammatically it states that two different orders of resolving a three-line crossing are equivalent. Algebraically it is precisely the consistency condition required for factorized scattering with internal degrees of freedom.

Remark 12.1 (Yang–Baxter versus ordinary commutativity). The Yang–Baxter equation is not the statement that two matrices commute. It is a cubic relation among three two-body operators acting on three tensor factors. Its consequence, after constructing a monodromy matrix and tracing over an auxiliary space, is the commutativity of transfer matrices.

12.4 R-matrix as vertex weight and as scattering matrix

The same R -matrix has two complementary meanings.

First, in a two-dimensional classical vertex model, R assigns a local Boltzmann weight to a vertex. The two tensor indices may be viewed as spin or arrow states living on the lines entering and leaving the vertex. The partition function is built by multiplying local R -matrix weights and summing over internal indices.

Second, in a one-dimensional quantum chain, R is the algebraic object that controls two-body scattering and the commuting transfer matrices. The spectral parameter u plays the role of an auxiliary rapidity. Taking a logarithmic derivative of the transfer matrix produces a quantum Hamiltonian.

For the six-vertex model, a standard trigonometric or hyperbolic R -matrix is

$$R(u) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u) & c(u) & 0 \\ 0 & c(u) & b(u) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix}, \quad (12.16)$$

where, for the hyperbolic parametrization,

$$a(u) = \sinh(u + \eta), \quad b(u) = \sinh u, \quad c(u) = \sinh \eta. \quad (12.17)$$

The nonzero entries encode the ice rule: the number of incoming arrows equals the number of outgoing arrows at each vertex. In the associated XXZ chain, the anisotropy parameter is controlled by

$$\Delta = \cosh \eta \quad (12.18)$$

for this hyperbolic convention. In the gapless convention one uses sines instead of hyperbolic sines and obtains $\Delta = \cos \gamma$.

12.5 Monodromy matrix and commuting transfer matrices

To build a length- L quantum chain, introduce an auxiliary space a and quantum spaces $1, \dots, L$. The monodromy matrix is the ordered product

$$M_a(u) = R_{aL}(u - \xi_L) R_{a,L-1}(u - \xi_{L-1}) \cdots R_{a1}(u - \xi_1), \quad (12.19)$$

where ξ_j are inhomogeneity parameters. The homogeneous chain is obtained by setting all $\xi_j = 0$. The transfer matrix is the auxiliary trace

$$\tau(u) = \text{Tr}_a M_a(u). \quad (12.20)$$

The Yang–Baxter equation implies the RTT relation

$$R_{ab}(u - v) M_a(u) M_b(v) = M_b(v) M_a(u) R_{ab}(u - v). \quad (12.21)$$

Taking the trace over the two auxiliary spaces a and b gives

$$[\tau(u), \tau(v)] = 0, \quad \text{for all } u, v. \quad (12.22)$$

This is the central algebraic output of Yang–Baxter integrability.

Theorem 12.1 (Commuting transfer matrices). *If the R -matrix satisfies the Yang–Baxter equation and the monodromy matrix is constructed as in Eq. (12.19), then the transfer matrices Eq. (12.20) commute for all values of the spectral parameter:*

$$[\tau(u), \tau(v)] = 0. \quad (12.23)$$

Proof. The Yang–Baxter equation implies the RTT relation Eq. (12.21). Taking $\text{Tr}_a \text{Tr}_b$ of both sides and using cyclicity of the trace in the auxiliary spaces gives

$$\text{Tr}_a \text{Tr}_b [R_{ab}(u-v) M_a(u) M_b(v)] = \text{Tr}_a \text{Tr}_b [M_b(v) M_a(u) R_{ab}(u-v)]. \quad (12.24)$$

Cyclicity moves $R_{ab}(u-v)$ through the auxiliary trace, leaving

$$\text{Tr}_a M_a(u) \text{Tr}_b M_b(v) = \text{Tr}_b M_b(v) \text{Tr}_a M_a(u). \quad (12.25)$$

This is exactly $\tau(u)\tau(v) = \tau(v)\tau(u)$. □

The physical meaning of Eq. (12.22) is that $\tau(u)$ is a generating function for conserved quantities. Expanding the logarithm around a regular point gives

$$\log \tau(u) = \sum_{n=0}^{\infty} (u - u_0)^n Q_n, \quad [Q_m, Q_n] = 0. \quad (12.26)$$

One of these charges is the Hamiltonian; the rest are higher conserved charges. This is the transfer-matrix form of quantum integrability.

12.6 Hamiltonian limit

A local quantum Hamiltonian is obtained from a regular transfer matrix. The key regularity property is

$$R(0) = \rho P, \quad (12.27)$$

where P is the permutation operator on two neighboring spin spaces and ρ is a scalar normalization. Then the logarithmic derivative of the transfer matrix at $u = 0$ produces a sum of local nearest-neighbor terms:

$$H = C \left. \frac{d}{du} \log \tau(u) \right|_{u=0} + C_0 = \sum_{j=1}^L h_{j,j+1} + C_0. \quad (12.28)$$

For the six-vertex R -matrix, this Hamiltonian is the spin-1/2 XXZ chain, up to normalization, additive constants, and boundary conventions:

$$H_{\text{XXZ}} = J \sum_{j=1}^L (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z) + \text{constant}. \quad (12.29)$$

This is the basic bridge between the six-vertex model and the XXZ quantum chain. It is the interacting analogue of the quantum–classical correspondence discussed in the transfer-matrix chapter.

12.7 Why commuting transfer matrices imply many conserved quantities

Suppose $\tau(u)$ is analytic near u_0 and $[\tau(u), \tau(v)] = 0$ for all u, v . Then every coefficient in the expansion of $\log \tau(u)$ commutes with every other coefficient. More explicitly, define

$$Q_n = \left. \frac{d^n}{du^n} \log \tau(u) \right|_{u=u_0}. \quad (12.30)$$

Since the whole family $\tau(u)$ commutes, one obtains

$$[Q_m, Q_n] = 0, \quad [Q_n, H] = 0. \quad (12.31)$$

The first nontrivial local charge is usually the Hamiltonian. Higher charges have increasing spatial range. In a local integrable chain one may schematically write

$$Q_2 = H, \quad Q_3 = \sum_j q_{j,j+1,j+2}, \quad Q_4 = \sum_j q_{j,j+1,j+2,j+3}, \quad \dots \quad (12.32)$$

The existence of these charges is the algebraic reason why scattering is elastic and factorized.

Remark 12.2 (Integrability is stronger than one conservation law). Translation invariance gives momentum conservation, and time-translation invariance gives energy conservation. These two conservation laws are not enough to solve an interacting many-body problem. Bethe-ansatz integrability requires an infinite tower of mutually commuting charges, generated by a commuting family of transfer matrices.

12.8 Coordinate Bethe ansatz versus algebraic Bethe ansatz

There are two complementary ways to use this structure.

12.8.1 Coordinate Bethe ansatz

The coordinate Bethe ansatz works directly with wave functions such as [Eq. \(12.3\)](#). Its main steps are:

- (i) choose a simple reference state;
- (ii) construct M -particle wave functions in ordered coordinate sectors;
- (iii) impose contact conditions when particles become adjacent;
- (iv) extract two-body scattering phases;
- (v) impose periodic boundary conditions to obtain Bethe equations.

This method is concrete and physically transparent. It is especially useful for the XXX and XXZ chains and will be developed in the coordinate Bethe-ansatz chapter.

12.8.2 Algebraic Bethe ansatz

The algebraic Bethe ansatz starts from the monodromy matrix. For a spin-1/2 chain the monodromy matrix is a 2×2 matrix in the auxiliary space,

$$M_a(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix}_a, \quad (12.33)$$

where A, B, C, D are operators acting on the quantum spin chain. The transfer matrix is

$$\tau(u) = A(u) + D(u). \quad (12.34)$$

One chooses a reference state $|0\rangle$ such that

$$C(u)|0\rangle = 0, \quad A(u)|0\rangle = a(u)|0\rangle, \quad D(u)|0\rangle = d(u)|0\rangle. \quad (12.35)$$

The operator $B(u)$ creates Bethe excitations:

$$|\lambda_1, \dots, \lambda_M\rangle = B(\lambda_1) \cdots B(\lambda_M) |0\rangle. \quad (12.36)$$

The RTT relation determines how $A(u)$ and $D(u)$ commute through the B 's. The unwanted terms vanish exactly when the rapidities satisfy the Bethe equations. Then Eq. (12.36) is an eigenstate of $\tau(u)$ and therefore also of the Hamiltonian extracted from $\log \tau(u)$.

Approach	Main object	Best use
Coordinate Bethe ansatz	Piecewise plane-wave many-body wave function	Direct derivation of scattering phases and Bethe equations
Algebraic Bethe ansatz	Monodromy matrix and RTT algebra	Systematic construction of eigenstates from $B(u)$ operators
Transfer-matrix method	Commuting family $\tau(u)$	Conserved charges and Hamiltonian limits of vertex models

12.9 Relation to free-fermion solvability

It is useful to compare the free-fermion and Bethe-ansatz mechanisms.

Feature	Free-fermion/BdG solvability	Bethe/Yang–Baxter integrability
Elementary objects	Independent fermionic modes or quasiparticles	Interacting Bethe quasiparticles with rapidities
Many-body construction	Fill independent modes	Solve coupled Bethe equations
Scattering	Trivial exchange signs or BdG mixing	Nontrivial two-body phase or matrix scattering
Conserved quantities	Quadratic mode occupations	Infinite tower from $\log \tau(u)$
Representative models	XY, TFIM, Kitaev, SSH	XXX, XXZ, XYZ, six-vertex, Hubbard
Core algebra	Canonical anticommutation and Bogoliubov rotation	Yang–Baxter equation and RTT relation

The XX chain sits at the overlap between these viewpoints. It is the $\Delta = 0$ limit of the XXZ chain and is also free after the Jordan–Wigner transformation. From the Bethe-ansatz perspective, its scattering phase reduces to the fermionic exchange sign. From the free-fermion perspective, it is simply a number-conserving tight-binding chain. This overlap is a useful consistency check between the two solution languages.

12.10 Physical interpretation of rapidities

In relativistic field theory a rapidity directly parametrizes energy and momentum. In lattice Bethe ansatz, the word is used more broadly: a rapidity is a spectral parameter that uniformizes the dispersion and scattering. The physical momentum is a function of rapidity,

$$p = p(\lambda), \quad (12.37)$$

and the two-body scattering depends only on rapidity differences in many homogeneous integrable models,

$$S(\lambda_i, \lambda_j) = S(\lambda_i - \lambda_j). \quad (12.38)$$

This difference property is one reason why the Yang–Baxter equation becomes a tractable functional equation for $R(u)$.

For the XXX chain one often uses

$$e^{ip(\lambda)} = \frac{\lambda + i/2}{\lambda - i/2}. \quad (12.39)$$

Then Eq. (12.12) has the direct interpretation

$$e^{ip(\lambda_j)L} = \prod_{\ell \neq j} S(\lambda_j - \lambda_\ell), \quad S(\lambda) = \frac{\lambda + i}{\lambda - i}. \quad (12.40)$$

12.11 What Yang–Baxter integrability does not mean

The term “integrable” is used in several related but distinct senses. In these notes, Yang–Baxter integrability will mean the existence of an R -matrix satisfying the Yang–Baxter equation, a commuting transfer-matrix family, and a Hamiltonian obtained from a logarithmic derivative. This is stronger than merely knowing many exact eigenstates or finding one special solvable point.

There are several useful cautions.

- (i) A model may be exactly solvable by free fermions without requiring a nontrivial Yang–Baxter structure.
- (ii) A model may have a transfer matrix but not a commuting family $\tau(u)$ for all spectral parameters.
- (iii) Self-duality may identify a candidate critical point, but it is not the same as Bethe-ansatz integrability.
- (iv) Topological two-band classification concerns eigenvector winding or Chern structure; Yang–Baxter integrability concerns factorized scattering and commuting transfer matrices.

The value of the mother-model perspective is that these solution mechanisms can be compared without conflating them.

12.12 Roadmap to the later integrable chapters

The material in this preview will be used later in three directions.

- (i) The six-vertex model will provide the canonical example of an R -matrix satisfying the Yang–Baxter equation.
- (ii) The Hamiltonian limit of the six-vertex transfer matrix will give the XXZ spin chain.
- (iii) The coordinate Bethe ansatz will solve the XXZ chain by deriving the many-body Bethe equations from factorized scattering.
- (iv) The algebraic Bethe ansatz will solve the same chain from the RTT relation and the operator algebra of $A(u), B(u), C(u), D(u)$.
- (v) The Hubbard chain will require a nested Bethe ansatz because its quasiparticles carry more than one kind of internal degree of freedom.

12.13 Summary

The essential ideas are:

- (i) Bethe ansatz replaces independent free modes by interacting rapidities whose many-body scattering factorizes into two-body scattering.
- (ii) The coordinate Bethe wave function is a sum of plane waves over momentum permutations, with amplitudes related by two-body scattering matrices or phases.
- (iii) Periodic boundary conditions give Bethe equations of the form $e^{ik_j L} = \prod_{\ell \neq j} S(k_j, k_\ell)$.
- (iv) For matrix-valued scattering, three-body consistency is the Yang–Baxter equation.
- (v) The R -matrix is both a local vertex weight in a two-dimensional classical model and the algebraic scattering/intertwining object of a quantum chain.
- (vi) The Yang–Baxter equation implies the RTT relation, which implies $[\tau(u), \tau(v)] = 0$.
- (vii) Expanding $\log \tau(u)$ produces mutually commuting conserved charges; the quantum Hamiltonian appears as one of these charges.
- (viii) The six-vertex model and the XXZ chain are the basic interacting examples connecting transfer matrices, Bethe ansatz, and Yang–Baxter integrability.

Part II

Free-Fermion Mother Models

Chapter 13

The Transverse-Field XY Chain as a Mother Model

The transverse-field XY chain is the first genuine mother model in these notes. It is not merely one more spin chain in a list of examples. It contains, as parameter limits or equivalent fermionic descriptions, the transverse-field Ising model, the XX chain, and the spinless Kitaev chain. It is also the simplest setting in which all technical tools developed in Part I meet in one calculation: Jordan–Wigner fermionization, parity-dependent boundary sectors, Fourier transformation, BdG blocks, Bogoliubov quasiparticles, Gaussian correlators, duality, and two-band topology.

The purpose of this chapter is therefore twofold. First, we solve the XY chain completely as a quadratic fermion problem. Second, we organize its special limits so that the later chapters can be read as controlled reductions of one solved mother model.

13.1 Hamiltonian and conventions

The transverse-field XY chain is defined by

$$H_{\text{XY}} = - \sum_{j=1}^L \left[\frac{J(1+\gamma)}{2} \sigma_j^x \sigma_{j+1}^x + \frac{J(1-\gamma)}{2} \sigma_j^y \sigma_{j+1}^y \right] - h \sum_{j=1}^L \sigma_j^z. \quad (13.1)$$

Here J sets the exchange scale, h is the transverse field, and γ is the XY anisotropy. The isotropic XX chain is obtained at $\gamma = 0$, while the transverse-field Ising chain is obtained at $\gamma = 1$.

Unless otherwise stated, the bulk formulas below are independent of whether one starts from open or periodic spin boundary conditions. Boundary conditions only enter when the Jordan–Wigner string wraps around the chain. This distinction is central enough that we keep it explicit throughout the derivation.

The spin and fermion conventions are those fixed in Part I:

$$c_j = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z \right) \sigma_j^+, \quad c_j^\dagger = \left(\prod_{\ell=1}^{j-1} \sigma_\ell^z \right) \sigma_j^-, \quad (13.2)$$

with

$$n_j = c_j^\dagger c_j = \frac{1 - \sigma_j^z}{2}, \quad \sigma_j^z = 1 - 2n_j. \quad (13.3)$$

Thus a spin-down state is interpreted as an occupied fermion state. This is the convention responsible for the sign of the chemical potential in the Kitaev-chain dictionary below.

It is useful to rewrite the exchange part of Eq. (13.1) as

$$\frac{J(1+\gamma)}{2}\sigma_j^x\sigma_{j+1}^x + \frac{J(1-\gamma)}{2}\sigma_j^y\sigma_{j+1}^y = \frac{J}{2}(\sigma_j^x\sigma_{j+1}^x + \sigma_j^y\sigma_{j+1}^y) + \frac{J\gamma}{2}(\sigma_j^x\sigma_{j+1}^x - \sigma_j^y\sigma_{j+1}^y). \quad (13.4)$$

The first bracket becomes fermion hopping; the second becomes p -wave pairing. This is the algebraic reason why the XY chain is a BdG mother model.

13.2 Special limits and the mother-model viewpoint

The most important reductions of Eq. (13.1) are summarized in Table 13.1.

Table 13.1: Special limits and equivalent descriptions of the transverse-field XY chain.

Parameter choice or map	Model obtained	Interpretation
$\gamma = 1$	Transverse-field Ising model	Maximal anisotropy; Ising criticality at $h = \pm J$ in the convention of this chapter
$\gamma = 0$	XX chain in a transverse field	Number-conserving free fermions; gapless for $ h < J $
$h = 0$	Zero-field anisotropic XY chain	Pure competition between hopping and pairing after Jordan–Wigner transformation
General h, γ	Spinless BdG chain	Independent $(k, -k)$ two-level systems after Fourier transformation
Jordan–Wigner image	Kitaev p -wave chain	$t = J$, $\mu = -2h$, and pairing amplitude fixed by $J\gamma$ up to pairing convention
Near $h = \pm J$, $\gamma \neq 0$	Massive Majorana theory	Ising universality class with a sign-changing Majorana mass
Near $\gamma = 0$, $ h < J $	Two Fermi points	XX/Luttinger-liquid universality class rather than Ising criticality

The role of the XY chain as a mother model can be summarized as

$$\begin{aligned} \text{XY spin chain} &\xrightarrow{\text{JW}} \text{spinless BdG chain} \\ &\xrightarrow{\text{Fourier}} \prod_{k>0} \text{two-level BdG blocks} \xrightarrow{\text{Bogoliubov}} \text{free quasiparticles}. \end{aligned} \quad (13.5)$$

All exact results in the later TFIM, XX, and Kitaev-chain chapters are obtained by specializing this chain of maps.

13.3 Jordan–Wigner image

We now derive the fermionic Hamiltonian. The nearest-neighbor Jordan–Wigner identities established earlier give

$$\sigma_j^x\sigma_{j+1}^x + \sigma_j^y\sigma_{j+1}^y = 2\left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j\right), \quad (13.6)$$

$$\sigma_j^x\sigma_{j+1}^x - \sigma_j^y\sigma_{j+1}^y = 2\left(c_j^\dagger c_{j+1}^\dagger + c_{j+1}c_j\right), \quad (13.7)$$

$$\sigma_j^z = 1 - 2c_j^\dagger c_j. \quad (13.8)$$

Substituting these identities into [Eq. \(13.1\)](#) gives the bulk Jordan–Wigner image

$$H_{XY}^{\text{bulk}} = -J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - J\gamma \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right) - h \sum_j (1 - 2n_j). \quad (13.9)$$

Equivalently,

$$H_{XY}^{\text{bulk}} = -hL + 2h \sum_j n_j - J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - J\gamma \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right). \quad (13.10)$$

The field term therefore plays the role of a chemical-potential term for the Jordan–Wigner fermions.

Remark 13.1 (Why the anisotropy becomes pairing). The isotropic part $\sigma^x \sigma^x + \sigma^y \sigma^y$ conserves the number of spin flips and becomes fermion hopping. The anisotropic part $\sigma^x \sigma^x - \sigma^y \sigma^y$ creates or annihilates spin flips in pairs and therefore becomes a superconducting pairing term. Fermion number is not conserved for $\gamma \neq 0$, but fermion parity is conserved.

For an open chain, [Eq. \(13.9\)](#) is the full fermionic Hamiltonian, with the sums running over $j = 1, \dots, L-1$.

For a periodic spin chain, the boundary term carries the Jordan–Wigner parity sign. Let

$$P_F = (-1)^N, \quad N = \sum_{j=1}^L n_j, \quad (13.11)$$

and fix a parity sector

$$P_F = p, \quad p = \pm 1. \quad (13.12)$$

As shown in the Jordan–Wigner chapter, the periodic spin chain is represented in that sector by the effective fermionic boundary condition

$$c_{L+1} = -p c_1. \quad (13.13)$$

Thus

$$p = +1 \iff c_{L+1} = -c_1 \quad (\text{anti-periodic fermions}), \quad (13.14)$$

whereas

$$p = -1 \iff c_{L+1} = c_1 \quad (\text{periodic fermions}). \quad (13.15)$$

In a fixed sector, one may use [Eq. \(13.9\)](#) with $j = 1, \dots, L$ provided the boundary condition [Eq. \(13.13\)](#) is imposed. The exact finite-size spin partition function or spectrum is obtained only after combining the two parity sectors with the correct parity projection. In the thermodynamic bulk spectrum, this distinction affects only finite-size momentum quantization.

13.4 Momentum quantization and Fourier transform

In a fixed parity sector p , the allowed momenta are

$$k_m = \frac{2\pi m + \vartheta_p}{L}, \quad m = 0, 1, \dots, L-1, \quad (13.16)$$

where

$$\vartheta_p = \begin{cases} \pi, & p = +1 \quad (\text{even sector, APBC}), \\ 0, & p = -1 \quad (\text{odd sector, PBC}). \end{cases} \quad (13.17)$$

We write the corresponding momentum set as $\mathcal{K}_{\vartheta_p}$.

We use the Fourier convention developed earlier,

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_{j=1}^L e^{-ikj} c_j, \quad c_j = \frac{e^{i\phi}}{\sqrt{L}} \sum_{k \in \mathcal{K}_{\vartheta_p}} e^{ikj} c_k. \quad (13.18)$$

The twist ϑ_p is physical boundary data. The constant phase ϕ is a Fourier-gauge convention; it rotates the phase of the pairing gap but not the spectrum.

The hopping and field terms give

$$-J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + 2h \sum_j n_j = \sum_{k \in \mathcal{K}_{\vartheta_p}} \xi(k) c_k^\dagger c_k, \quad (13.19)$$

with

$$\xi(k) = 2(h - J \cos k). \quad (13.20)$$

The pairing term gives an odd gap function. With the real-space sign in [Eq. \(13.9\)](#), the direct Fourier transform gives

$$\Delta_\phi^{\text{raw}}(k) = -2iJ\gamma e^{-2i\phi} \sin k. \quad (13.21)$$

This sign is a convention for the orientation of the Nambu pairing plane. Many BdG summaries, including some compact formulas in these notes, use the equivalent orientation

$$\Delta_\phi(k) = +2iJ\gamma e^{-2i\phi} \sin k. \quad (13.22)$$

The two choices are related by a harmless reversal of the pairing-plane orientation, for example by $\gamma \mapsto -\gamma$ or by an equivalent Nambu-gauge relabelling. They have the same $|\Delta(k)|$, the same quasiparticle spectrum, the same critical lines, and the same absolute winding number. Only orientation-sensitive signs of winding numbers change. In the rest of this chapter we display formulas with the convention [Eq. \(13.22\)](#), because it matches the BdG-vector convention used in the preceding chapters.

With this convention, the momentum-space Hamiltonian is

$$H_{XY}^{(p)} = -hL + \sum_{k \in \mathcal{K}_{\vartheta_p}} \xi(k) c_k^\dagger c_k + \frac{1}{2} \sum_{k \in \mathcal{K}_{\vartheta_p}} \left[\Delta_\phi(k) c_k^\dagger c_{-k}^\dagger + \Delta_\phi(k)^* c_{-k} c_k \right], \quad (13.23)$$

up to the finite-size parity projection already discussed.

13.5 BdG block and quasiparticle spectrum

Introduce the Nambu spinor

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}. \quad (13.24)$$

For $k \not\equiv -k$ modulo 2π , the pair $(k, -k)$ is governed by the BdG block

$$\mathcal{H}_{XY}(k) = \begin{pmatrix} \xi(k) & \Delta_\phi(k) \\ \Delta_\phi(k)^* & -\xi(k) \end{pmatrix}. \quad (13.25)$$

In the gauge $\phi = 0$ and with Eq. (13.22), this becomes

$$\mathcal{H}_{XY}(k) = \begin{pmatrix} 2(h - J \cos k) & 2iJ\gamma \sin k \\ -2iJ\gamma \sin k & -2(h - J \cos k) \end{pmatrix}. \quad (13.26)$$

Equivalently,

$$\mathcal{H}_{XY}(k) = \mathbf{d}_{XY}(k) \cdot \boldsymbol{\tau}, \quad \mathbf{d}_{XY}(k) = (0, -2J\gamma \sin k, 2(h - J \cos k)), \quad (13.27)$$

where τ_x, τ_y, τ_z act in Nambu space. The sign of the d_y component follows from the convention $d_y = -\text{Im} \Delta(k)$.

The quasiparticle energies are

$$E_{\pm}(k) = \pm E(k), \quad E(k) = 2\sqrt{(h - J \cos k)^2 + (J\gamma \sin k)^2}. \quad (13.28)$$

This is the central exact result of the chapter. All phase boundaries and low-energy limits follow from the zeros of this dispersion.

Remark 13.2 (BdG eigenvalues versus many-body excitations). The negative branch $-E(k)$ is the particle-hole partner of the positive branch. It is not an independent physical particle. After the Bogoliubov transformation, the many-body Hamiltonian is written in terms of positive-energy quasiparticles, with a ground-state energy shift.

13.6 Bogoliubov angle and diagonal form

For each independent non-self-conjugate pair $(k, -k)$, define

$$\xi_k = 2(h - J \cos k), \quad \Delta_k = 2iJ\gamma e^{-2i\phi} \sin k, \quad E_k = \sqrt{\xi_k^2 + |\Delta_k|^2}. \quad (13.29)$$

The Bogoliubov polar angle is

$$\cos \theta_k = \frac{\xi_k}{E_k}, \quad \sin \theta_k = \frac{|\Delta_k|}{E_k}. \quad (13.30)$$

Writing

$$\Delta_k = |\Delta_k| e^{i\alpha_k}, \quad (13.31)$$

the coherence factors may be chosen as

$$u_k = \sqrt{\frac{E_k + \xi_k}{2E_k}}, \quad v_k = -e^{i\alpha_k} \sqrt{\frac{E_k - \xi_k}{2E_k}}. \quad (13.32)$$

The quasiparticle operators are

$$\gamma_k = u_k c_k - v_k c_{-k}^\dagger, \quad \gamma_{-k} = u_k c_{-k} + v_k c_k^\dagger. \quad (13.33)$$

They satisfy canonical anticommutation relations and diagonalize the pair block:

$$H_k = E_k \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right). \quad (13.34)$$

The normalized even-parity ground state of the pair block is

$$|\Omega_k\rangle = \left(u_k + v_k c_k^\dagger c_{-k}^\dagger \right) |0\rangle_{k,-k}. \quad (13.35)$$

Thus, ignoring possible self-conjugate modes for the moment, the BCS ground state in a fixed boundary sector is

$$|\Omega\rangle = \prod_{k \in \mathcal{K}_+} \left(u_k + v_k c_k^\dagger c_{-k}^\dagger \right) |0\rangle, \quad (13.36)$$

where $\mathcal{K}_+$ contains one representative from each pair $\{k, -k\}$.

The corresponding ground-state energy in the thermodynamic limit is

$$e_0 = \lim_{L \rightarrow \infty} \frac{E_0}{L} = -\frac{1}{2\pi} \int_0^\pi E(k) dk, \quad (13.37)$$

with $E(k)$ given in Eq. (13.28). This formula includes the constant $-hL$ in the original spin Hamiltonian. For example, in the large-field limit $h \gg J$, it gives $e_0 \simeq -h$, as expected for the fully $+z$ polarized spin state.

13.7 Self-conjugate momenta and finite-size parity bookkeeping

The momenta $k = 0$ and $k = \pi$ satisfy $k \equiv -k$ modulo 2π . They occur only in the periodic fermion sector and only for compatible system sizes. Since the spinless pairing gap is odd in momentum,

$$\Delta(0) = 0, \quad \Delta(\pi) = 0, \quad (13.38)$$

these modes are not paired. They must be treated as ordinary fermionic modes. Their normal energies are

$$\xi(0) = 2(h - J), \quad \xi(\pi) = 2(h + J). \quad (13.39)$$

Occupying such a mode changes fermion parity and therefore affects the exact finite-size matching between the spin Hilbert space and the fermionic sector.

For bulk thermodynamics, the contribution of these isolated modes is negligible as $L \rightarrow \infty$. For exact diagonalization, finite-size spectra, and parity selection, they are essential. A safe finite-size procedure is:

- (i) choose a spin boundary condition;
- (ii) split the fermionic problem into $P_F = +1$ and $P_F = -1$ sectors;
- (iii) impose APBC in the even sector and PBC in the odd sector;
- (iv) diagonalize each quadratic Hamiltonian;
- (v) keep only states whose quasiparticle occupations have the required fermion parity.

13.8 Excitation gap and phase diagram

The bulk gap closes when

$$h - J \cos k = 0, \quad J\gamma \sin k = 0. \quad (13.40)$$

Assume $J \neq 0$. For $\gamma \neq 0$, the second equation forces

$$k = 0 \quad \text{or} \quad k = \pi. \quad (13.41)$$

The first equation then gives the two Ising critical lines

$$h = J \quad (k = 0), \quad h = -J \quad (k = \pi). \quad (13.42)$$

Near $h = J$, set $k \ll 1$. Then

$$E(k) \simeq 2 \sqrt{\left(h - J + \frac{Jk^2}{2}\right)^2 + (J\gamma k)^2}. \quad (13.43)$$

At the critical point $h = J$ with $\gamma \neq 0$, the leading dispersion is linear,

$$E(k) \simeq 2 |J\gamma| |k|, \quad (13.44)$$

which is the lattice signature of an Ising/Majorana critical point with dynamical exponent $z = 1$.

Similarly, near $h = -J$ one expands around $k = \pi$. Writing $q = k - \pi$, one finds another Ising critical point with the same low-energy structure.

The situation changes qualitatively when $\gamma = 0$. The pairing term vanishes and the Hamiltonian is number conserving. The dispersion reduces to

$$\xi(k) = 2(h - J \cos k). \quad (13.45)$$

The system is gapless whenever there are Fermi points satisfying

$$\cos k_F = \frac{h}{J}. \quad (13.46)$$

Thus, for $J > 0$,

$$\gamma = 0, \quad |h| < J \quad (13.47)$$

defines the gapless XX region. This is not an Ising critical line. It is a number-conserving free-fermion critical phase with two Fermi points and central charge $c = 1$ in the continuum limit. By contrast, the Ising lines at $\gamma \neq 0$ have a single Majorana critical mode and central charge $c = 1/2$.

The phase structure is therefore:

$$\begin{aligned} \gamma \neq 0, \quad h = J & : \text{ Ising critical line at } k = 0, \\ \gamma \neq 0, \quad h = -J & : \text{ Ising critical line at } k = \pi, \\ \gamma = 0, \quad |h| < J & : \text{ gapless XX/free-fermion critical phase,} \\ |h| > J & : \text{ gapped polarized phase,} \\ |h| < J, \quad \gamma \neq 0 & : \text{ gapped paired/ordered phase.} \end{aligned} \quad (13.48)$$

At the multicritical points $(h, \gamma) = (\pm J, 0)$, the linear pairing velocity vanishes and the dispersion along the $\gamma = 0$ direction becomes quadratic. This is why the multicritical point is not described by the same scaling limit as a generic Ising point.

13.9 Relation to the transverse-field Ising model

Setting $\gamma = 1$ in [Eq. \(13.1\)](#) gives

$$H_{\text{TFIM}} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z. \quad (13.49)$$

The quasiparticle spectrum becomes

$$E_{\text{TFIM}}(k) = 2\sqrt{(h - J \cos k)^2 + J^2 \sin^2 k} = 2\sqrt{h^2 + J^2 - 2hJ \cos k}. \quad (13.50)$$

For $J > 0$ and $h \geq 0$, the physically standard transition is at

$$h = J. \quad (13.51)$$

The excitation gap is

$$\Delta_{\text{gap}} = 2|h - J|. \quad (13.52)$$

The sign of $h - J$ becomes the sign of the continuum Majorana mass. This is the fermionic counterpart of the order-disorder transition of the Ising chain.

The second line $h = -J$ is present in the full XY parameter plane. For the usual TFIM convention one often restricts to $J > 0$ and $h \geq 0$, in which case this second line is outside the standard positive-field discussion. Keeping it in the mother-model phase diagram is useful because it is required by the full lattice BdG structure.

13.10 Relation to the XX chain

Setting $\gamma = 0$ removes the pairing term in [Eq. \(13.9\)](#). The Hamiltonian becomes

$$H_{\text{XX}} = -J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - hL + 2h \sum_j n_j. \quad (13.53)$$

This is a tight-binding model with dispersion

$$\xi(k) = 2(h - J \cos k). \quad (13.54)$$

The ground state is a Fermi sea, not a BCS paired state. For $J > 0$, modes with

$$\xi(k) < 0 \quad \Longleftrightarrow \quad \cos k > \frac{h}{J} \quad (13.55)$$

are occupied. When $|h| < J$, the two Fermi points are

$$k = \pm k_F, \quad k_F = \arccos\left(\frac{h}{J}\right). \quad (13.56)$$

Linearizing near the Fermi points gives two chiral fermion modes, or equivalently a compact boson/Luttinger-liquid continuum theory. This is why the XX critical region has central charge $c = 1$, not the Ising value $c = 1/2$.

13.11 Relation to the Kitaev chain

The bulk fermionic image of the XY chain is a spinless p -wave superconducting chain. To compare conventions cleanly, write the Kitaev chain as

$$H_{\text{K}} = -t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - \mu \sum_j \left(n_j - \frac{1}{2} \right) + \Delta \sum_j \left(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger \right), \quad (13.57)$$

with real Δ for the moment. Since

$$c_{j+1}^\dagger c_j^\dagger = -c_j^\dagger c_{j+1}^\dagger, \quad c_j c_{j+1} = -c_{j+1} c_j, \quad (13.58)$$

comparison with Eq. (13.10) gives the bulk dictionary

$$t = J, \quad \Delta = J\gamma, \quad \mu = -2h. \quad (13.59)$$

The constant term also matches:

$$-\mu \sum_j \left(n_j - \frac{1}{2} \right) = 2h \sum_j n_j - hL \quad \text{when } \mu = -2h. \quad (13.60)$$

Some references instead define the pairing term as

$$\Delta_{\text{cr}} \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right). \quad (13.61)$$

In that convention the same XY Hamiltonian corresponds to

$$\Delta_{\text{cr}} = -J\gamma. \quad (13.62)$$

This is only a convention for how the Hermitian conjugate pair is written. The physical spectrum depends on $|\Delta|$ and on the relative orientation of the BdG vector, not on this bookkeeping sign alone.

Using Eq. (13.59), the Kitaev-chain topological condition

$$|\mu| < 2|t| \quad (13.63)$$

becomes

$$|h| < |J|. \quad (13.64)$$

This is precisely the region between the two Ising critical lines of the XY mother model when $\gamma \neq 0$. In the spin language it is the ordered region for the Ising-anisotropic chain; in the fermion language it is the topological superconducting phase with Majorana edge modes under open boundary conditions.

13.12 Topological interpretation of the BdG vector

For real J , h , and γ , and in a real pairing gauge, the BdG vector lies in a two-dimensional plane:

$$\mathbf{d}(k) = (0, -2J\gamma \sin k, 2(h - J \cos k)). \quad (13.65)$$

Equivalently, one may encode the loop by the complex function

$$q(k) = d_z(k) + i d_y(k) = 2(h - J \cos k) - 2iJ\gamma \sin k. \quad (13.66)$$

As k runs around the Brillouin zone, $q(k)$ traces an ellipse centered at $2h$ on the real axis. The origin is enclosed if and only if

$$|h| < |J| \quad \text{and} \quad \gamma \neq 0. \quad (13.67)$$

The corresponding winding number may be written as

$$\nu = \frac{1}{2\pi i} \int_{-\pi}^{\pi} dk \partial_k \log q(k). \quad (13.68)$$

With the orientation convention in Eq. (13.66),

$$\nu = \begin{cases} -\text{sgn}(J\gamma), & |h| < |J|, \\ 0, & |h| > |J|. \end{cases} \quad (13.69)$$

Changing the sign convention for the pairing reverses the sign of ν but does not change whether the phase is topological. The robust class-D statement is the corresponding $\mathbb{Z}_2$ distinction:

$$|h| < |J| \iff \text{nontrivial Kitaev phase.} \quad (13.70)$$

13.13 Continuum limits

The same exact lattice dispersion produces different continuum theories in different regions of parameter space.

Ising/Majorana scaling near $h = J$

Take $J > 0$, $\gamma \neq 0$, and expand near $k = 0$ and $h = J$. Keeping the leading terms gives

$$\mathcal{H}_{XY}(k) \simeq 2(h - J)\tau_z - 2J\gamma k \tau_y, \quad (13.71)$$

up to the sign convention of the pairing plane. This is the momentum-space form of a massive Majorana Hamiltonian. The mass is proportional to

$$m_+ = 2(h - J), \quad (13.72)$$

and the velocity is proportional to

$$v = 2J\gamma. \quad (13.73)$$

The critical theory at $m_+ = 0$ has central charge $c = 1/2$.

Near $h = -J$, one expands around $k = \pi$. Setting $k = \pi + q$ gives another Majorana theory with mass proportional to

$$m_- = 2(h + J). \quad (13.74)$$

XX scaling for $\gamma = 0$

For $\gamma = 0$ and $|h| < J$, expand around the two Fermi points $\pm k_F$. The Fermi velocity is

$$v_F = \left| \frac{d\xi(k)}{dk} \right|_{k=k_F} = 2J \sin k_F = 2J \sqrt{1 - \left(\frac{h}{J}\right)^2}. \quad (13.75)$$

The resulting low-energy theory is a massless Dirac fermion, or equivalently a compact boson. Adding a small nonzero γ pairs the two Fermi points and gaps the XX critical phase, except at the Ising critical endpoints.

13.14 Summary

The transverse-field XY chain is solved by the following sequence:

- (i) Jordan–Wigner transformation maps the spin chain to a quadratic spinless fermion Hamiltonian with hopping, chemical potential, and p -wave pairing.
- (ii) Periodic spin boundary conditions split the fermionic problem into parity sectors: even parity gives APBC, odd parity gives PBC.
- (iii) Fourier transformation decomposes the Hamiltonian into independent $(k, -k)$ BdG blocks.
- (iv) Each block has

$$\xi(k) = 2(h - J \cos k), \quad |\Delta(k)| = 2 |J\gamma \sin k|. \quad (13.76)$$

- (v) The exact quasiparticle dispersion is

$$E(k) = 2\sqrt{(h - J \cos k)^2 + (J\gamma \sin k)^2}. \quad (13.77)$$

- (vi) For $\gamma \neq 0$, the bulk gap closes at $h = \pm J$, giving Ising critical lines.
- (vii) For $\gamma = 0$, the chain is number conserving and is gapless for $|h| < |J|$, giving the XX/free-fermion critical phase.
- (viii) Under the Kitaev-chain dictionary $t = J$, $\mu = -2h$, and $\Delta = J\gamma$ up to pairing convention, the region $|h| < |J|$ is the topological superconducting phase.

This single solution therefore supplies the exact solution of the TFIM, the XX chain, and the Kitaev chain as special cases or equivalent descriptions.

Chapter 14

The Transverse-Field Ising Model

The transverse-field Ising model is the most important one-parameter reduction of the transverse-field XY mother model. It is simple enough to display the full solution explicitly, but rich enough to exhibit the main structures of the lecture notes: Jordan–Wigner fermionization, BdG quasiparticles, parity-sector bookkeeping, Kramers–Wannier duality, order–disorder variables, and the Majorana continuum limit.

In the present convention the Hamiltonian is

$$H_{\text{TFIM}} = -J \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^L \sigma_j^z. \quad (14.1)$$

It is obtained from the XY chain by setting $\gamma = 1$. In most physical applications one takes $J > 0$ and $h \geq 0$. The full lattice solution, however, also makes the second critical line at $h = -J$ visible. We therefore keep signs explicit when doing so is useful.

14.1 Reduction from the XY mother model

Setting $\gamma = 1$ in the XY-chain Hamiltonian gives

$$H_{\text{XY}}|_{\gamma=1} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z, \quad (14.2)$$

which is exactly [Eq. \(14.1\)](#). The exact solution is therefore obtained by specializing the XY-chain BdG data:

$$\xi(k) = 2(h - J \cos k), \quad \Delta(k) = 2iJ \sin k \quad (14.3)$$

in the standard BdG orientation used in the preceding chapter. The quasiparticle dispersion is

$$E_{\text{TFIM}}(k) = 2\sqrt{(h - J \cos k)^2 + J^2 \sin^2 k} = 2\sqrt{h^2 + J^2 - 2hJ \cos k}. \quad (14.4)$$

For $J > 0$ and $h \geq 0$, the minimum occurs at $k = 0$ and the excitation gap is

$$\Delta_{\text{gap}} = 2|h - J|. \quad (14.5)$$

Thus the bulk gap closes at

$$h = J. \quad (14.6)$$

If negative fields are also retained, a second gap closing occurs at $h = -J$ and $k = \pi$.

Remark 14.1 (Why the TFIM is not merely a classical Ising chain). The exchange term in Eq. (14.1) is diagonal in the σ^x basis, whereas the field term is diagonal in the σ^z basis. These two parts do not commute. The transition at $h = J$ is therefore a genuine quantum phase transition controlled by the competition between ordering in the x direction and quantum fluctuations generated by the transverse field.

14.2 Jordan–Wigner form and boundary sectors

Using the Jordan–Wigner convention fixed earlier,

$$c_j = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^+, \quad c_j^\dagger = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^-, \quad (14.7)$$

with $n_j = (1 - \sigma_j^z)/2$, the open-chain or bulk fermionic Hamiltonian is

$$H_{\text{TFIM}}^{\text{bulk}} = -J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - J \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right) - h \sum_j (1 - 2n_j). \quad (14.8)$$

This is a spinless p -wave superconducting chain with hopping and pairing of equal magnitude. Fermion number is not conserved, but fermion parity

$$P_F = (-1)^N, \quad N = \sum_j n_j, \quad (14.9)$$

is conserved.

For a periodic spin chain, the Jordan–Wigner string turns the boundary term into a parity-dependent fermionic boundary condition. In a fixed parity sector $P_F = p$, one imposes

$$c_{L+1} = -p c_1. \quad (14.10)$$

Thus the even sector has anti-periodic fermions, while the odd sector has periodic fermions:

$$p = +1 : \quad k = \frac{2\pi m + \pi}{L}, \quad p = -1 : \quad k = \frac{2\pi m}{L}. \quad (14.11)$$

The thermodynamic spectrum is insensitive to this distinction, but the exact finite-size spectrum and the spin partition function require the correct parity projection.

14.3 BdG block and Bogoliubov diagonalization

For each non-self-conjugate momentum pair $(k, -k)$, define the Nambu spinor

$$\Psi_k = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}. \quad (14.12)$$

The BdG block is

$$\mathcal{H}_{\text{TFIM}}(k) = \begin{pmatrix} 2(h - J \cos k) & 2iJ \sin k \\ -2iJ \sin k & -2(h - J \cos k) \end{pmatrix}. \quad (14.13)$$

Equivalently,

$$\mathcal{H}_{\text{TFIM}}(k) = \mathbf{d}_{\text{TFIM}}(k) \cdot \boldsymbol{\tau}, \quad \mathbf{d}_{\text{TFIM}}(k) = (0, -2J \sin k, 2(h - J \cos k)). \quad (14.14)$$

The Bogoliubov angle may be chosen as

$$\cos \theta_k = \frac{h - J \cos k}{\sqrt{(h - J \cos k)^2 + J^2 \sin^2 k}}, \quad \sin \theta_k = \frac{J |\sin k|}{\sqrt{(h - J \cos k)^2 + J^2 \sin^2 k}}. \quad (14.15)$$

The diagonal many-body form is

$$H_{\text{TFIM}}^{(p)} = E_0^{(p)} + \sum_{k \in \mathcal{K}_{\vartheta_p}} E_{\text{TFIM}}(k) \gamma_k^\dagger \gamma_k, \quad (14.16)$$

where the allowed momentum set depends on the parity sector as in Eq. (14.11). Equivalently, after choosing one representative from each pair $\{k, -k\}$,

$$H_k = E_{\text{TFIM}}(k) \left(\gamma_k^\dagger \gamma_k + \gamma_{-k}^\dagger \gamma_{-k} - 1 \right). \quad (14.17)$$

The BCS ground state in a fixed sector is the $\gamma = 1$ specialization of the XY-chain ground state.

14.4 Phases and order parameters

For $J > 0$ and $h \geq 0$, the two phases are:

$$\begin{aligned} h > J & : \text{paramagnetic phase polarized along } z, \\ 0 \leq h < J & : \text{ferromagnetic Ising phase ordered along } x. \end{aligned} \quad (14.18)$$

In the paramagnetic phase the field dominates and the ground state is smoothly connected to

$$|\uparrow_z \uparrow_z \cdots \uparrow_z\rangle. \quad (14.19)$$

In the ferromagnetic phase, the thermodynamic ground space breaks the global Ising symmetry

$$U_Z = \prod_{j=1}^L \sigma_j^z, \quad U_Z \sigma_j^x U_Z^{-1} = -\sigma_j^x. \quad (14.20)$$

The two symmetry-broken states are approximately the two x -polarized states at $h = 0$.

The exact spontaneous magnetization for $J > 0$ and $0 \leq h < J$ is

$$m_x = \lim_{r \rightarrow \infty} \sqrt{\langle \sigma_j^x \sigma_{j+r}^x \rangle} = \left[1 - \left(\frac{h}{J} \right)^2 \right]^{1/8}. \quad (14.21)$$

It vanishes for $h \geq J$. The exponent $1/8$ is the two-dimensional Ising order-parameter exponent, reflecting the quantum–classical correspondence between the one-dimensional TFIM and the two-dimensional classical Ising model.

14.5 Kramers–Wannier duality

The operator duality is defined on the dual lattice sites $j + 1/2$ by

$$\mu_{j+\frac{1}{2}}^z = \sigma_j^x \sigma_{j+1}^x, \quad \mu_{j+\frac{1}{2}}^x = \prod_{\ell \leq j} \sigma_\ell^z. \quad (14.22)$$

These operators satisfy the same Pauli algebra on the dual lattice. In particular,

$$\sigma_j^z = \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x. \quad (14.23)$$

Ignoring boundary subtleties for the moment, the Hamiltonian becomes

$$H_{\text{TFIM}}(J, h) = -J \sum_j \mu_{j+\frac{1}{2}}^z - h \sum_j \mu_{j-\frac{1}{2}}^x \mu_{j+\frac{1}{2}}^x. \quad (14.24)$$

After a relabelling of the dual lattice and a spin-axis relabelling, this has the same form as the original Hamiltonian with the couplings exchanged:

$$H_{\text{TFIM}}(J, h) \longleftrightarrow H_{\text{TFIM}}(h, J). \quad (14.25)$$

The self-dual point is therefore

$$h = J. \quad (14.26)$$

Together with the exact solution, this confirms that the self-dual point is the actual critical point for $J, h > 0$.

The dual order parameter is a disorder operator in the original spin variables:

$$\langle \mu_{j+\frac{1}{2}}^x \mu_{j+r+\frac{1}{2}}^x \rangle = \left\langle \prod_{\ell=j+1}^{j+r} \sigma_\ell^z \right\rangle. \quad (14.27)$$

Thus the ferromagnetic phase of the original spins maps to the disordered phase of the dual spins, and the paramagnetic phase maps to the ordered phase of the dual variables.

14.6 Continuum Majorana limit

For $J > 0$ and $h \simeq J$, the gap closes near $k = 0$. Expanding [Eq. \(14.13\)](#) at small k gives

$$\mathcal{H}_{\text{TFIM}}(k) \simeq 2(h - J)\tau_z - 2Jk\tau_y. \quad (14.28)$$

The continuum mass and velocity are therefore

$$m = 2(h - J), \quad v = 2J. \quad (14.29)$$

In real space, the low-energy Hamiltonian may be written in terms of two real Majorana fields χ_R and χ_L as

$$H_{\text{cont}} = \frac{iv}{2} \int dx (\chi_R \partial_x \chi_R - \chi_L \partial_x \chi_L) + im \int dx \chi_R \chi_L. \quad (14.30)$$

The critical point $m = 0$ is the Ising conformal field theory with central charge

$$c = \frac{1}{2}. \quad (14.31)$$

The two signs of m correspond to the two gapped phases. In the fermionic Kitaev-chain description, the sign change of the Majorana mass is the continuum version of the topological transition at $|\mu| = 2|t|$.

14.7 Correlation functions from the Gaussian solution

The fermionic ground state is Gaussian. Therefore all fermionic correlators are obtained from the two-point covariance matrix, and all spin correlators follow from Wick's theorem after expressing Jordan–Wigner strings in terms of Majorana operators.

For example, the transverse magnetization is local in fermions:

$$\sigma_j^z = 1 - 2c_j^\dagger c_j. \quad (14.32)$$

By contrast, the Ising correlator contains a Jordan–Wigner string:

$$\sigma_i^x \sigma_j^x = \left(c_i^\dagger - c_i \right) \left[\prod_{\ell=i+1}^{j-1} (1 - 2n_\ell) \right] \left(c_j + c_j^\dagger \right), \quad i < j. \quad (14.33)$$

In Majorana language this becomes a Pfaffian of a finite antisymmetric matrix. For translation-invariant infinite chains, the Pfaffian often reduces to a Toeplitz determinant. This is the computational bridge between the exact Bogoliubov solution and the exact long-distance order parameter [Eq. \(14.21\)](#).

14.8 Summary

The TFIM is the $\gamma = 1$ reduction of the XY mother model. Its exact solution is controlled by the BdG dispersion

$$E(k) = 2\sqrt{h^2 + J^2 - 2hJ \cos k}. \quad (14.34)$$

For $J, h > 0$, the gap closes at $h = J$, the same point selected by Kramers–Wannier self-duality. The phase $h > J$ is paramagnetic, while $0 \leq h < J$ has Ising ferromagnetic order in the x direction. The continuum critical theory is a massless Majorana fermion, and the two gapped phases are distinguished by the sign of the Majorana mass.

Chapter 15

The XX Chain and Free Fermions

The XX chain is the number-conserving limit of the XY mother model. It is the cleanest example of how a spin chain can become an ordinary tight-binding fermion problem after the Jordan–Wigner transformation. Unlike the TFIM and the generic XY chain, the XX chain has a conserved fermion number, a Fermi sea, and a gapless critical phase with two Fermi points.

15.1 The $\gamma = 0$ limit of the XY chain

Setting $\gamma = 0$ in the transverse-field XY chain gives

$$H_{\text{XX}} = -\frac{J}{2} \sum_{j=1}^L (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y) - h \sum_{j=1}^L \sigma_j^z. \quad (15.1)$$

Equivalently, in terms of spin raising and lowering operators,

$$H_{\text{XX}} = -J \sum_j (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) - h \sum_j \sigma_j^z. \quad (15.2)$$

The first term moves a spin flip from one site to a neighboring site. It does not create or annihilate spin flips in pairs. This is the spin-language origin of fermion-number conservation.

Using the Jordan–Wigner convention of these notes, the bulk fermionic Hamiltonian is

$$H_{\text{XX}}^{\text{bulk}} = -J \sum_j (c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j) - hL + 2h \sum_j n_j. \quad (15.3)$$

There is no pairing term. Therefore

$$[H_{\text{XX}}, N] = 0, \quad N = \sum_j n_j. \quad (15.4)$$

This is stronger than the parity conservation of the generic XY chain.

For a periodic spin chain, the same parity-sector boundary condition as before is present:

$$P_F = p \implies c_{L+1} = -p c_1. \quad (15.5)$$

In the thermodynamic limit this affects only the quantization of allowed momenta.

15.2 Fourier diagonalization

In a fixed boundary sector, use

$$c_j = \frac{1}{\sqrt{L}} \sum_{k \in \mathcal{K}_\vartheta} e^{ikj} c_k, \quad (15.6)$$

where $\vartheta = 0$ or π depending on the sector. The Hamiltonian becomes

$$H_{XX} = -hL + \sum_{k \in \mathcal{K}_\vartheta} \xi(k) c_k^\dagger c_k, \quad (15.7)$$

with single-particle dispersion

$$\xi(k) = 2(h - J \cos k). \quad (15.8)$$

The many-body eigenstates are occupation-number states

$$|\{n_k\}\rangle = \prod_k (c_k^\dagger)^{n_k} |0\rangle, \quad n_k = 0, 1, \quad (15.9)$$

with energy

$$E[\{n_k\}] = -hL + \sum_k \xi(k) n_k. \quad (15.10)$$

The diagonalization is therefore simpler than the BdG problem: no Bogoliubov rotation is required.

15.3 Ground-state Fermi sea

At zero temperature the ground state fills all modes with negative single-particle energy:

$$\xi(k) < 0 \iff \cos k > \frac{h}{J} \quad (J > 0). \quad (15.11)$$

There are three regimes:

$$\begin{aligned} h > J & : \text{ no fermions are occupied,} \\ |h| < J & : \text{ a Fermi sea is present,} \\ h < -J & : \text{ all fermion modes are occupied.} \end{aligned} \quad (15.12)$$

For $|h| < J$, the Fermi points are

$$k = \pm k_F, \quad k_F = \arccos\left(\frac{h}{J}\right). \quad (15.13)$$

The filled region is $|k| < k_F$ in the convention of [Eq. \(15.8\)](#). The fermion density in the thermodynamic limit is

$$n = \langle c_j^\dagger c_j \rangle = \frac{k_F}{\pi}, \quad |h| < J. \quad (15.14)$$

Since $\sigma_j^z = 1 - 2n_j$, the magnetization is

$$m_z = \langle \sigma_j^z \rangle = 1 - \frac{2k_F}{\pi}, \quad k_F = \arccos\left(\frac{h}{J}\right). \quad (15.15)$$

This interpolates from $m_z = -1$ at $h = -J$ to $m_z = +1$ at $h = J$.

The ground-state energy density for $|h| < J$ is

$$e_0 = -h + \frac{1}{2\pi} \int_{-k_F}^{k_F} 2(h - J \cos k) dk = -h + \frac{2hk_F}{\pi} - \frac{2J \sin k_F}{\pi}. \quad (15.16)$$

Outside the partially filled regime this expression must be replaced by the empty or completely filled Fermi sea.

15.4 Low-energy Dirac and Luttinger-liquid limit

In the gapless region $|h| < J$, expand near the two Fermi points:

$$k = \pm k_F + q, \quad |q| \ll 1. \quad (15.17)$$

The dispersion linearizes as

$$\xi(+k_F + q) \simeq v_F q, \quad \xi(-k_F + q) \simeq -v_F q, \quad (15.18)$$

where

$$v_F = \left| \frac{d\xi}{dk} \right|_{k=k_F} = 2J \sin k_F = 2J \sqrt{1 - \left(\frac{h}{J}\right)^2}. \quad (15.19)$$

The continuum fermion is decomposed as

$$c_j \sim e^{ik_F x} \psi_R(x) + e^{-ik_F x} \psi_L(x), \quad x = ja. \quad (15.20)$$

The low-energy Hamiltonian is the massless Dirac Hamiltonian

$$H_{\text{cont}} = -iv_F \int dx \left(\psi_R^\dagger \partial_x \psi_R - \psi_L^\dagger \partial_x \psi_L \right). \quad (15.21)$$

In spin-chain language this is the free-fermion point of a Luttinger liquid. Its central charge is

$$c = 1, \quad (15.22)$$

which distinguishes the XX critical region from the Ising critical point of the TFIM, where $c = 1/2$.

15.5 Correlation functions

Because the Hamiltonian is number conserving and diagonal in momentum space, the basic equal-time fermion correlator is elementary. In the thermodynamic ground state with $|h| < J$,

$$\langle c_i^\dagger c_j \rangle = \frac{1}{2\pi} \int_{-k_F}^{k_F} e^{ik(j-i)} dk = \begin{cases} \frac{k_F}{\pi}, & i = j, \\ \frac{\sin[k_F(j-i)]}{\pi(j-i)}, & i \neq j. \end{cases} \quad (15.23)$$

This $1/r$ decay is the fermionic signature of gaplessness.

The longitudinal spin correlator is local in fermion density:

$$\sigma_j^z = 1 - 2n_j. \quad (15.24)$$

The transverse spin correlators, such as $\langle \sigma_i^x \sigma_j^x \rangle$, contain Jordan–Wigner strings and are computed as determinants or Pfaffians. Their long-distance behavior in the critical region is algebraic, rather than exponential. At half filling ($h = 0$, $k_F = \pi/2$), the leading transverse spin correlations decay as a power law with the characteristic free-fermion/Luttinger-liquid exponent.

15.6 Comparison with the generic XY chain

The XX chain is obtained from the generic XY chain by turning off the pairing amplitude. The distinction can be summarized as follows.

Table 15.1: Generic XY chain versus the XX limit.

Feature	Generic XY chain ($\gamma \neq 0$)	XX chain ($\gamma = 0$)
Fermion number	Not conserved	Conserved
Fermion parity	Conserved	Conserved, with full $U(1)$ enhanced symmetry
Momentum-space block	BdG $(k, -k)$ block	Single-particle dispersion $\xi(k)$
Ground state	BCS paired state	Fermi sea
Generic criticality	Ising lines at $h = \pm J$	Gapless region $ h < J$
Continuum theory	Majorana, $c = 1/2$	Dirac/Luttinger liquid, $c = 1$

Adding a small nonzero anisotropy γ to the gapless XX region introduces a p -wave pairing perturbation. In the fermionic language this pairs the right- and left-moving modes and opens a gap except where the lattice mass also closes at the Ising transition.

15.7 Summary

The XX chain is the $\gamma = 0$ limit of the XY mother model. It is exactly a number-conserving tight-binding model with dispersion

$$\xi(k) = 2(h - J \cos k). \quad (15.25)$$

For $J > 0$, the ground state is empty for $h > J$, completely filled for $h < -J$, and a partially filled Fermi sea for $|h| < J$. The latter regime has two Fermi points, a linear low-energy Dirac description, and central charge $c = 1$. This chapter therefore illustrates the difference between free-fermion solvability by a Fermi sea and free-fermion solvability by BdG pairing.

Chapter 16

The Kitaev Chain as the Fermionic Image of the XY Mother Model

The Kitaev chain is the fermionic image of the transverse-field XY chain. From the spin point of view it is obtained by the Jordan–Wigner transformation; from the fermion point of view it is a one-dimensional spinless p -wave superconductor. This chapter isolates the fermionic interpretation of the XY solution and connects it to Majorana edge modes and the BdG topological invariant.

16.1 Spinless p -wave superconducting chain

We use the convention

$$H_K = -t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - \mu \sum_j \left(n_j - \frac{1}{2} \right) + \Delta \sum_j \left(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger \right), \quad (16.1)$$

where $n_j = c_j^\dagger c_j$. The hopping t and chemical potential μ are ordinary number-conserving terms, while Δ is a nearest-neighbor odd-parity pairing amplitude. For the moment Δ is taken real. A uniform complex phase will be discussed below.

The pairing term breaks fermion-number conservation but preserves fermion parity:

$$[H_K, (-1)^N] = 0, \quad N = \sum_j n_j. \quad (16.2)$$

Thus the Kitaev chain belongs naturally to the same BdG framework used for the XY chain.

16.2 Dictionary from the XY mother model

The bulk Jordan–Wigner image of the XY chain in the conventions of these notes is

$$H_{XY}^{\text{bulk}} = -J \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) - J\gamma \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right) - h \sum_j (1 - 2n_j). \quad (16.3)$$

Using

$$c_{j+1}^\dagger c_j^\dagger = -c_j^\dagger c_{j+1}^\dagger, \quad c_j c_{j+1} = -c_{j+1} c_j, \quad (16.4)$$

comparison between [Eq. \(16.1\)](#) and [Eq. \(16.3\)](#) gives the bulk dictionary

$$t = J, \quad \Delta = J\gamma, \quad \mu = -2h. \quad (16.5)$$

The constant term matches because

$$-\mu \sum_j \left(n_j - \frac{1}{2} \right) = 2h \sum_j n_j - hL \quad (\mu = -2h). \quad (16.6)$$

This dictionary is convention-dependent. If one writes the pairing term as $\Delta_{\text{cr}} \sum_j (c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j)$ instead, then $\Delta_{\text{cr}} = -J\gamma$ in the present Jordan–Wigner convention. The spectrum and phase diagram are unchanged; only the orientation convention for the BdG pairing plane is changed.

For a periodic spin chain, the boundary term is also sector dependent. In a fixed fermion-parity sector,

$$P_F = p \implies c_{L+1} = -p c_1. \quad (16.7)$$

For an intrinsically fermionic Kitaev chain, one may instead specify periodic, anti-periodic, or open boundary conditions directly. The bulk topological phase diagram is the same, but exact finite-size spectra depend on this choice.

16.3 BdG block and spectrum

For a translation-invariant chain, the normal-state dispersion is

$$\xi(k) = -\mu - 2t \cos k. \quad (16.8)$$

With the Fourier and BdG convention used in the earlier chapters, the odd-parity pairing gap for real Δ may be written as

$$\Delta(k) = 2i\Delta \sin k. \quad (16.9)$$

The BdG block is therefore

$$\mathcal{H}_K(k) = \begin{pmatrix} -\mu - 2t \cos k & 2i\Delta \sin k \\ -2i\Delta \sin k & \mu + 2t \cos k \end{pmatrix}. \quad (16.10)$$

Equivalently,

$$\mathcal{H}_K(k) = \mathbf{d}_K(k) \cdot \boldsymbol{\tau}, \quad \mathbf{d}_K(k) = (0, -2\Delta \sin k, -\mu - 2t \cos k). \quad (16.11)$$

The quasiparticle spectrum is

$$E(k) = \sqrt{(-\mu - 2t \cos k)^2 + 4\Delta^2 \sin^2 k}. \quad (16.12)$$

Under [Eq. \(16.5\)](#), this reproduces the XY-chain spectrum

$$E(k) = 2\sqrt{(h - J \cos k)^2 + (J\gamma \sin k)^2}. \quad (16.13)$$

For $\Delta \neq 0$, the bulk gap can close only at $k = 0$ or $k = \pi$. The two conditions are

$$k = 0 : \quad \mu = -2t, \quad k = \pi : \quad \mu = 2t. \quad (16.14)$$

Thus the gapped phases are separated by the lines

$$\mu = \pm 2t. \quad (16.15)$$

16.4 Complex pairing phase and Nambu gauge

Let the pairing amplitude be complex:

$$\Delta = |\Delta| e^{i\theta_\Delta}. \quad (16.16)$$

With the Fourier convention containing a constant phase ϕ ,

$$c_k = \frac{e^{-i\phi}}{\sqrt{L}} \sum_j e^{-ikj} c_j, \quad (16.17)$$

the momentum-space gap becomes

$$\Delta_\phi(k) = 2i |\Delta| e^{i(\theta_\Delta - 2\phi)} \sin k. \quad (16.18)$$

The phase ϕ is a Nambu/Fourier gauge convention. Choosing

$$\phi = \frac{\theta_\Delta}{2} \quad (16.19)$$

removes the uniform superconducting phase from the bulk BdG block. Therefore a spatially uniform pairing phase is not by itself a new bulk phase of matter. It rotates the BdG vector in the pairing plane but leaves $E(k)$ and the topological criterion unchanged.

A phase difference, phase winding, or boundary flux is different. If the pairing phase cannot be removed by a single-valued global gauge transformation, it may represent physical supercurrent or flux data and can affect boundary spectra.

16.5 Majorana representation

Define site Majorana operators using the convention of the covariance-matrix chapter:

$$A_j = c_j + c_j^\dagger, \quad B_j = i(c_j - c_j^\dagger). \quad (16.20)$$

They satisfy

$$A_j^\dagger = A_j, \quad B_j^\dagger = B_j, \quad \{A_i, A_j\} = 2\delta_{ij}, \quad \{B_i, B_j\} = 2\delta_{ij}, \quad \{A_i, B_j\} = 0. \quad (16.21)$$

Also

$$n_j - \frac{1}{2} = -\frac{i}{2} A_j B_j. \quad (16.22)$$

For real Δ , the Hamiltonian becomes

$$H_K = \frac{i}{2} \sum_j [\mu A_j B_j + (t - \Delta) A_j B_{j+1} - (t + \Delta) B_j A_{j+1}], \quad (16.23)$$

up to boundary terms fixed by the chosen boundary condition.

Two limiting cases reveal the phase structure immediately.

Trivial atomic limit. For $t = \Delta = 0$ and $\mu < 0$ or $\mu > 0$, Majoranas on the same site are paired by $i\mu A_j B_j/2$. There are no protected edge Majoranas.

Topological dimerized limit. For $\mu = 0$ and $\Delta = t > 0$,

$$H_K = -it \sum_j B_j A_{j+1}. \quad (16.24)$$

On an open chain this pairs B_j with A_{j+1} on each bond. The operators A_1 and B_L are left unpaired and become zero-energy Majorana edge modes. The two edge Majoranas combine into a nonlocal fermion

$$f_{\text{edge}} = \frac{1}{2}(A_1 + iB_L), \quad (16.25)$$

producing two nearly degenerate ground states of opposite fermion parity in a long open chain.

16.6 Topological invariant

For real parameters, the BdG vector lies in a plane. Define

$$q(k) = d_z(k) + id_y(k) = -\mu - 2t \cos k - 2i\Delta \sin k. \quad (16.26)$$

The integer winding number in the real-pairing BDI convention is

$$\nu = \frac{1}{2\pi i} \int_{-\pi}^{\pi} dk \partial_k \log q(k). \quad (16.27)$$

For $t\Delta \neq 0$, the loop winds around the origin if and only if

$$|\mu| < 2|t|. \quad (16.28)$$

The orientation of the winding depends on the sign of $t\Delta$ and on the chosen definition of $q(k)$, but the distinction between zero and nonzero winding is robust.

If only particle-hole symmetry is kept, the class-D invariant is the corresponding $\mathbb{Z}_2$ quantity. In the present convention it can be written as

$$\mathcal{M} = \text{sgn}\{\xi(0)\xi(\pi)\} = \text{sgn}\{(-\mu - 2t)(-\mu + 2t)\} = \text{sgn}(\mu^2 - 4t^2). \quad (16.29)$$

The topological phase has

$$\mathcal{M} = -1 \iff |\mu| < 2|t|. \quad (16.30)$$

Using the XY dictionary $\mu = -2h$ and $t = J$, this becomes

$$|h| < |J|, \quad (16.31)$$

which is the region between the two Ising gap-closing lines of the XY mother model.

16.7 Relation to the TFIM and the XY chain

The transverse-field Ising model corresponds to the Kitaev chain with

$$\gamma = 1 \iff \Delta = t = J, \quad \mu = -2h. \quad (16.32)$$

For $J, h > 0$, the TFIM transition at $h = J$ is the Kitaev transition at

$$\mu = -2t. \quad (16.33)$$

The spin ferromagnetic phase $h < J$ is the topological superconducting phase $|\mu| < 2|t|$. The spin paramagnet $h > J$ is the trivial superconducting phase.

This identification is conceptually useful but should not be overinterpreted. The fermions in the Kitaev-chain description are Jordan–Wigner fermions, not microscopic electrons of the original spin chain. The Majorana edge modes of the open Kitaev chain correspond, in the spin language, to the near degeneracy of the symmetry-broken Ising phase and to nonlocal string operators.

16.8 Summary

The Kitaev chain is the fermionic form of the XY mother model. The convention used in these notes gives the bulk dictionary

$$t = J, \quad \Delta = J\gamma, \quad \mu = -2h. \quad (16.34)$$

Its BdG spectrum is

$$E(k) = \sqrt{(-\mu - 2t \cos k)^2 + 4|\Delta|^2 \sin^2 k}. \quad (16.35)$$

For nonzero pairing, the bulk transitions occur at $\mu = \pm 2t$. The phase $|\mu| < 2|t|$ is topological and supports Majorana edge modes on an open chain. A uniform complex phase of Δ rotates the BdG pairing plane and can be absorbed by a Nambu gauge choice, while phase winding or boundary flux may carry physical information.

Chapter 17

The SSH Chain and Two-Band Chiral Fermions

The Su–Schrieffer–Heeger (SSH) chain is not a Jordan–Wigner image of the XY mother model. It is nevertheless an essential companion example because it is the simplest number-conserving two-band topological Hamiltonian. The Kitaev chain uses a Nambu particle–hole basis; the SSH chain uses a sublattice basis. Both reduce topology to the winding of a two-component vector around the origin, but the Hilbert-space meaning of the two components is different.

17.1 SSH Hamiltonian

Consider a one-dimensional chain with two sites per unit cell, labelled A and B . Let a_j annihilate a fermion on sublattice A in unit cell j , and let b_j annihilate a fermion on sublattice B . The SSH Hamiltonian is

$$H_{\text{SSH}} = \sum_j \left(t_1 a_j^\dagger b_j + t_2 b_j^\dagger a_{j+1} + \text{h.c.} \right). \quad (17.1)$$

Here t_1 is the intracell hopping and t_2 is the intercell hopping. The model conserves fermion number:

$$[H_{\text{SSH}}, N] = 0, \quad N = \sum_j (a_j^\dagger a_j + b_j^\dagger b_j). \quad (17.2)$$

Thus it is a Bloch-band problem rather than a BdG problem.

The two exactly solvable limits are instructive. If $t_2 = 0$, the chain is a set of isolated intracell dimers. If $t_1 = 0$, the chain is a set of intercell dimers; for an open chain, the leftmost A site and rightmost B site are unpaired edge states. The topological distinction between these two dimerizations is the central content of the SSH model.

17.2 Bloch form

Use the Fourier convention

$$a_j = \frac{1}{\sqrt{L}} \sum_k e^{ikj} a_k, \quad b_j = \frac{1}{\sqrt{L}} \sum_k e^{ikj} b_k. \quad (17.3)$$

In the basis

$$\psi_k = \begin{pmatrix} a_k \\ b_k \end{pmatrix}, \quad (17.4)$$

the Hamiltonian becomes

$$H_{\text{SSH}} = \sum_k \psi_k^\dagger h_{\text{SSH}}(k) \psi_k, \quad (17.5)$$

where

$$h_{\text{SSH}}(k) = \begin{pmatrix} 0 & q(k) \\ q(k)^* & 0 \end{pmatrix}, \quad q(k) = t_1 + t_2 e^{-ik}. \quad (17.6)$$

Equivalently,

$$h_{\text{SSH}}(k) = d_x(k) \sigma_x + d_y(k) \sigma_y, \quad (17.7)$$

with

$$d_x(k) = t_1 + t_2 \cos k, \quad d_y(k) = t_2 \sin k. \quad (17.8)$$

The energy bands are

$$E_{\pm}(k) = \pm |q(k)| = \pm \sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos k}. \quad (17.9)$$

The gap closes when

$$q(k) = 0. \quad (17.10)$$

For real positive hoppings this occurs at

$$t_1 = t_2, \quad k = \pi. \quad (17.11)$$

17.3 Chiral symmetry and winding number

The SSH Bloch Hamiltonian anticommutes with σ_z :

$$\sigma_z h_{\text{SSH}}(k) \sigma_z = -h_{\text{SSH}}(k). \quad (17.12)$$

This is sublattice, or chiral, symmetry. It forces the Hamiltonian to be off-diagonal in the (A, B) basis and pins the spectrum symmetrically around zero energy.

When the bulk is gapped, $q(k)$ never vanishes, so the map

$$k \in S_{\text{BZ}}^1 \mapsto q(k) \in \mathbb{C} \setminus \{0\} \quad (17.13)$$

has an integer winding number. With the off-diagonal convention [Eq. \(17.6\)](#), define

$$\nu_{\text{SSH}} = \frac{1}{2\pi i} \int_{-\pi}^{\pi} dk \partial_k \log q(k). \quad (17.14)$$

For $t_1, t_2 > 0$ and $q(k) = t_1 + t_2 e^{-ik}$, the result is

$$\nu_{\text{SSH}} = \begin{cases} -1, & t_2 > t_1, \\ 0, & t_1 > t_2. \end{cases} \quad (17.15)$$

The sign is an orientation convention coming from the choice e^{-ik} in $q(k)$. Replacing $q(k)$ by $t_1 + t_2 e^{ik}$ reverses the sign. The robust physical distinction is whether the winding is zero or nonzero.

Geometrically, $q(k)$ traces a circle of radius $|t_2|$ centered at t_1 in the complex plane. The nontrivial phase is precisely the case in which this circle encloses the origin:

$$|t_2| > |t_1|. \quad (17.16)$$

17.4 Zak phase

Let

$$q(k) = |q(k)| e^{i\varphi(k)}. \quad (17.17)$$

A normalized occupied-band eigenstate may be chosen locally as

$$|u_-(k)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -e^{i\varphi(k)} \\ 1 \end{pmatrix}. \quad (17.18)$$

Its Berry connection is

$$\mathcal{A}(k) = i \langle u_-(k) | \partial_k u_-(k) \rangle = -\frac{1}{2} \partial_k \varphi(k), \quad (17.19)$$

up to gauge conventions. The Zak phase is therefore

$$\Phi_{\text{Zak}} = \int_{-\pi}^{\pi} \mathcal{A}(k) dk = -\pi \nu_{\text{SSH}} \mod 2\pi. \quad (17.20)$$

Thus the nontrivial SSH phase has Zak phase π modulo 2π , provided the unit-cell convention is fixed. Changing the unit cell shifts the Zak phase by an integer multiple of 2π or changes the representative convention, but it does not remove the protected boundary-state distinction for a specified physical termination.

17.5 Open chain and edge modes

For an open chain with L unit cells, the zero-energy eigenvalue equation at energy $E = 0$ decouples between sublattices. A left edge mode supported on the A sublattice has the form

$$|\psi_L\rangle = \sum_{j=1}^L \alpha_j a_j^\dagger |0\rangle. \quad (17.21)$$

The Schrödinger equation gives the recursion

$$t_1 \alpha_j + t_2 \alpha_{j+1} = 0, \quad (17.22)$$

so that

$$\alpha_j = \left(-\frac{t_1}{t_2} \right)^{j-1} \alpha_1. \quad (17.23)$$

This mode is normalizable at the left edge in the thermodynamic limit if and only if

$$|t_1| < |t_2|. \quad (17.24)$$

The right edge mode is similarly localized on the B sublattice. Thus the bulk winding condition and the edge-state condition agree.

At the exactly dimerized point $t_1 = 0$, the edge modes are simply the unpaired operators $a_1^\dagger$ and $b_L^\dagger$. This is the SSH analogue of the unpaired Majorana operators in the topological limit of the Kitaev chain.

17.6 Relation and distinction from the Kitaev chain

Both the SSH and Kitaev chains are two-band topological models, but the meanings of the two-component spinors are different:

$$\psi_k^{\text{SSH}} = \begin{pmatrix} a_k \\ b_k \end{pmatrix}, \quad \Psi_k^{\text{Kitaev}} = \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix}. \quad (17.25)$$

The SSH Pauli matrices act in sublattice space. The Kitaev Pauli matrices act in Nambu particle-hole space. Consequently:

- the SSH chain conserves fermion number, while the Kitaev chain conserves only fermion parity;
- the SSH spectral symmetry comes from sublattice chiral symmetry, while the Kitaev BdG spectral symmetry comes from particle-hole redundancy;
- SSH edge states are ordinary complex-fermion boundary orbitals, while Kitaev edge states are Majorana zero modes;
- the SSH winding is naturally a chiral-band winding, while the Kitaev winding is a BdG/Nambu winding, reducible to a class-D $\mathbb{Z}_2$ invariant when the real-pairing chiral symmetry is not imposed.

This comparison is useful because the same mathematical form

$$H(k) = d_x(k)\sigma_x + d_y(k)\sigma_y + d_z(k)\sigma_z \quad (17.26)$$

can describe physically distinct objects. A two-band matrix by itself does not say whether the Pauli matrices refer to sublattice, spin, orbital, or Nambu space. The symmetry interpretation must always be specified.

17.7 SSH as a two-band chiral mother example

The SSH chain represents the simplest member of a broader family of one-dimensional chiral two-band Hamiltonians:

$$h(k) = \begin{pmatrix} 0 & q(k) \\ q(k)^* & 0 \end{pmatrix}. \quad (17.27)$$

Every such gapped Hamiltonian has a winding number

$$\nu = \frac{1}{2\pi i} \int_{\text{BZ}} dk \partial_k \log q(k). \quad (17.28)$$

Longer-range hoppings simply replace $q(k) = t_1 + t_2 e^{-ik}$ by a finite Laurent polynomial in e^{ik} . The topology is still controlled by whether and how this complex loop winds around the origin. The SSH model is the minimal case in which this winding can be nonzero.

17.8 Summary

The SSH chain is a number-conserving two-band model with Bloch Hamiltonian

$$h_{\text{SSH}}(k) = \begin{pmatrix} 0 & t_1 + t_2 e^{-ik} \\ t_1 + t_2 e^{ik} & 0 \end{pmatrix}. \quad (17.29)$$

It has chiral symmetry and an integer winding number. For real positive hoppings, the nontrivial phase occurs when $t_2 > t_1$ and supports boundary zero modes on an open chain. It parallels the Kitaev chain at the level of two-band topology, but differs physically because its two-component structure is sublattice space rather than Nambu space.

Part III

Transfer Matrices and Quantum–Classical Correspondence

Chapter 18

The Two-Dimensional Classical Ising Model

The two-dimensional classical Ising model is the first genuinely nontrivial exactly solved lattice model in these notes. It is also the central bridge between two themes developed earlier: transfer matrices and free fermions. In one dimension, the classical Ising transfer matrix is only a 2×2 matrix; in two dimensions, a row-to-row transfer matrix acts on a 2^L -dimensional row Hilbert space. The remarkable fact, discovered by Onsager and reformulated by Kaufman, is that this exponentially large transfer matrix is still exactly solvable because it is a Gaussian operator in Majorana variables.

This chapter has three goals. First, we define the anisotropic zero-field model and construct its row-to-row transfer matrix in the same convention used in the quantum-classical correspondence chapter. Second, we explain why the transfer matrix belongs to a free-fermion algebra. Third, we extract the Onsager critical condition

$$\sinh(2K_x) \sinh(2K_y) = 1 \quad (18.1)$$

from the closing of the transfer-matrix gap.

Remark 18.1 (Zero field is essential). The exact free-fermion solution discussed here applies to the square-lattice Ising model at zero magnetic field. Adding a field term $H \sum_{j,m} s_{j,m}$ destroys the Gaussian structure of the transfer matrix. The two-dimensional Ising model in a nonzero magnetic field is not known to have a comparable closed-form exact solution.

18.1 Classical partition function

Consider an $L \times M$ square lattice with periodic boundary conditions in both directions. A classical spin variable

$$s_{j,m} = \pm 1, \quad j = 1, \dots, L, \quad m = 1, \dots, M, \quad (18.2)$$

lives at each lattice site. We identify

$$s_{L+1,m} = s_{1,m}, \quad s_{j,M+1} = s_{j,1}. \quad (18.3)$$

The anisotropic zero-field Ising partition function is

$$Z_{L,M}(K_x, K_y) = \sum_{\{s_{j,m}=\pm 1\}} \exp \left[K_x \sum_{j,m} s_{j,m} s_{j+1,m} + K_y \sum_{j,m} s_{j,m} s_{j,m+1} \right]. \quad (18.4)$$

Here K_x and K_y are dimensionless couplings. If the physical Hamiltonian is

$$E[s] = -J_x \sum_{j,m} s_{j,m} s_{j+1,m} - J_y \sum_{j,m} s_{j,m} s_{j,m+1}, \quad (18.5)$$

then

$$K_x = \beta J_x, \quad K_y = \beta J_y. \quad (18.6)$$

For most of this chapter we assume $K_x, K_y > 0$, corresponding to the ferromagnetic model. Antiferromagnetic signs can often be moved by sublattice spin flips on a bipartite lattice, but the ferromagnetic case is the canonical one for discussing the thermal order–disorder transition.

Let

$$\mathbf{s}_m = (s_{1,m}, \dots, s_{L,m}) \quad (18.7)$$

denote the m th row configuration. The Boltzmann weight between two adjacent rows can be distributed symmetrically as

$$T_{\mathbf{s}, \mathbf{s}'} = \exp \left[\frac{K_x}{2} \sum_{j=1}^L s_j s_{j+1} + K_y \sum_{j=1}^L s_j s'_j + \frac{K_x}{2} \sum_{j=1}^L s'_j s'_{j+1} \right]. \quad (18.8)$$

Then

$$Z_{L,M}(K_x, K_y) = \text{Tr } T^M. \quad (18.9)$$

The symmetrization in Eq. (18.8) is not physically necessary, but it makes T real symmetric and therefore convenient for spectral analysis.

18.2 Operator form of the row-to-row transfer matrix

We now rewrite T as an operator acting on a row Hilbert space

$$\mathcal{H}_{\text{row}} = (\mathbb{C}^2)^{\otimes L}. \quad (18.10)$$

To remain consistent with the transverse-field Ising convention used earlier in these notes, we label a classical row configuration by eigenvalues of σ_j^x :

$$\sigma_j^x |s_1, \dots, s_L\rangle_x = s_j |s_1, \dots, s_L\rangle_x. \quad (18.11)$$

In this basis, the intra-row interaction is diagonal:

$$\exp \left(\frac{K_x}{2} \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right). \quad (18.12)$$

The vertical one-site Boltzmann matrix is

$$W_{ss'} = e^{K_y ss'} = \begin{pmatrix} e^{K_y} & e^{-K_y} \\ e^{-K_y} & e^{K_y} \end{pmatrix}_{s,s'}. \quad (18.13)$$

Since σ^z flips the σ^x eigenvalue, this matrix can be written as

$$W = A e^{\kappa \sigma^z}, \quad \tanh \kappa = e^{-2K_y}, \quad A = \sqrt{2 \sinh(2K_y)}. \quad (18.14)$$

Indeed,

$$A \cosh \kappa = e^{K_y}, \quad A \sinh \kappa = e^{-K_y}. \quad (18.15)$$

The parameter κ is the dual coupling associated with K_y :

$$e^{-2K_y} = \tanh \kappa \iff \sinh(2K_y) \sinh(2\kappa) = 1. \quad (18.16)$$

Thus the symmetric row-to-row transfer matrix is

$$T = A^L \exp \left(\frac{K_x}{2} \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right) \exp \left(\kappa \sum_{j=1}^L \sigma_j^z \right) \exp \left(\frac{K_x}{2} \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right). \quad (18.17)$$

This is the precise point at which the two-dimensional classical model starts to look like a one-dimensional quantum spin chain: T is an exponential product of operators acting on a one-dimensional row Hilbert space.

Remark 18.2 (Relation to the notation K_τ). In the quantum–classical correspondence chapter, the transfer direction was written as an imaginary-time direction and the vertical coupling was denoted K_τ . In the present chapter we write the two classical directions as x and y . Thus K_y here plays the role of K_τ there.

18.3 Kaufman’s free-fermion structure

At first sight, Eq. (18.17) is not easier than the original classical problem, because it acts on a 2^L -dimensional vector space. The key observation is that both exponential factors are Gaussian after the Jordan–Wigner transformation.

Using the Jordan–Wigner convention fixed earlier in these notes, define Majorana operators

$$a_{2j-1} = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^x, \quad a_{2j} = \left(\prod_{\ell < j} \sigma_\ell^z \right) \sigma_j^y. \quad (18.18)$$

They satisfy

$$a_m^\dagger = a_m, \quad \{a_m, a_n\} = 2\delta_{mn}. \quad (18.19)$$

In the bulk one has the local identities

$$\sigma_j^z = -i a_{2j-1} a_{2j}, \quad \sigma_j^x \sigma_{j+1}^x = -i a_{2j} a_{2j+1}. \quad (18.20)$$

Therefore the two building blocks of T become

$$\exp \left(\kappa \sum_j \sigma_j^z \right) = \exp \left(-i\kappa \sum_j a_{2j-1} a_{2j} \right), \quad (18.21)$$

$$\exp \left(\frac{K_x}{2} \sum_j \sigma_j^x \sigma_{j+1}^x \right) = \exp \left(-\frac{iK_x}{2} \sum_j a_{2j} a_{2j+1} \right), \quad (18.22)$$

up to the usual boundary-sector term for a periodic spin chain.

Equations Eqs. (18.21) and (18.22) show that T is a product of exponentials of Majorana bilinears. Such operators are fermionic Gaussian operators. Their adjoint action maps Majoranas linearly to Majoranas:

$$T^{-1} a_m T = \sum_{n=1}^{2L} R_{mn} a_n, \quad (18.23)$$

where R is a $2L \times 2L$ matrix preserving the Majorana anticommutation relations. The exponentially large transfer-matrix problem is therefore reduced to the diagonalization of a one-particle Majorana transformation.

Remark 18.3 (Boundary sectors). For periodic spin boundary conditions, the boundary term in $\sigma_L^x \sigma_1^x$ contains the fermion-parity operator $P_F = \prod_{j=1}^L \sigma_j^z$. Consequently, the exact finite- L partition function is obtained by combining parity-projected fermion traces with periodic and antiperiodic momenta, in the same spirit as the TFIM parity-sector bookkeeping. In the thermodynamic limit these sector distinctions affect only subleading finite-size terms, not the bulk free energy or the critical curve.

Because the model is translation invariant, the one-particle Majorana problem block-diagonalizes in momentum. For each momentum k , the relevant 2×2 block has eigenvalues $e^{\pm \varepsilon(k)}$, where

$$\cosh \varepsilon(k) = \cosh(2\kappa) \cosh(2K_x) - \sinh(2\kappa) \sinh(2K_x) \cos k. \quad (18.24)$$

The quantity $\varepsilon(k)$ is the transfer-matrix analogue of a quasiparticle energy. More precisely, it is the single-particle gap in the logarithm of the transfer matrix. Its minimum occurs at $k = 0$ for ferromagnetic couplings, and

$$\varepsilon(0) = 2|\kappa - K_x|. \quad (18.25)$$

Thus the transfer-matrix gap closes when

$$\kappa = K_x. \quad (18.26)$$

Using [Eq. \(18.16\)](#), this is equivalent to

$$\boxed{\sinh(2K_x) \sinh(2K_y) = 1.} \quad (18.27)$$

This is the Onsager critical line for the anisotropic two-dimensional Ising model.

18.4 Largest eigenvalue and Onsager free energy

Let

$$\Phi(K_x, K_y) = \lim_{L, M \rightarrow \infty} \frac{1}{LM} \log Z_{L, M}(K_x, K_y) \quad (18.28)$$

be the dimensionless free-energy density with the sign convention appropriate to $\log Z$. The physical free energy per site is

$$f = -\beta^{-1} \Phi \quad (18.29)$$

when $K_\mu = \beta J_\mu$. Since $Z = \text{Tr } T^M$, Φ is the logarithm of the largest transfer-matrix eigenvalue per row site.

From the Gaussian diagonalization one obtains

$$\Phi(K_x, K_y) = \frac{1}{2} \log(2 \sinh(2K_y)) + \frac{1}{2} \int_{-\pi}^{\pi} \frac{dk}{2\pi} \varepsilon(k), \quad (18.30)$$

where $\varepsilon(k)$ is defined by [Eq. \(18.24\)](#). Equivalently,

$$\Phi(K_x, K_y) = \frac{1}{2} \log(2 \sinh(2K_y)) + \frac{1}{2\pi} \int_0^\pi dk \operatorname{arcosh} X(k), \quad (18.31)$$

$$X(k) = \cosh(2\kappa) \cosh(2K_x) - \sinh(2\kappa) \sinh(2K_x) \cos k. \quad (18.32)$$

The appearance of arcosh is the transfer-matrix version of “taking a quasiparticle energy from a one-particle evolution eigenvalue.”

A useful check is the decoupled-row limit $K_x = 0$. Then

$$\varepsilon(k) = 2\kappa, \quad (18.33)$$

and Eq. (18.30) gives

$$\Phi(0, K_y) = \frac{1}{2} \log(2 \sinh(2K_y)) + \kappa = \log(2 \cosh K_y), \quad (18.34)$$

which is exactly the free-energy density of independent vertical Ising chains viewed through the chosen transfer direction.

For the isotropic model $K_x = K_y = K$, the critical point follows from

$$\sinh^2(2K_c) = 1, \quad (18.35)$$

and therefore

$$\boxed{K_c = \frac{1}{2} \log(1 + \sqrt{2})}. \quad (18.36)$$

If $K = \beta J$, this gives

$$k_B T_c = \frac{2J}{\log(1 + \sqrt{2})}. \quad (18.37)$$

18.5 Duality interpretation of the critical curve

The same critical condition follows from Kramers–Wannier duality. The high-temperature expansion writes each bond factor as

$$e^{K s_r s_{r'}} = \cosh K (1 + s_r s_{r'} \tanh K). \quad (18.38)$$

After summing over spins, only closed even-degree bond configurations survive. The low-temperature expansion of a dual Ising model is instead a sum over domain walls. Matching high-temperature loops of the original lattice with domain-wall loops of the dual lattice gives

$$e^{-2K_x^*} = \tanh K_y, \quad e^{-2K_y^*} = \tanh K_x. \quad (18.39)$$

Equivalently,

$$\sinh(2K_x^*) \sinh(2K_y) = 1, \quad \sinh(2K_y^*) \sinh(2K_x) = 1. \quad (18.40)$$

A self-dual point satisfies

$$K_x^* = K_x, \quad K_y^* = K_y, \quad (18.41)$$

and hence obeys

$$\sinh(2K_x) \sinh(2K_y) = 1. \quad (18.42)$$

Duality alone identifies the possible transition curve if there is a unique ferromagnetic critical line. The transfer-matrix solution gives the sharper statement: the one-particle transfer gap actually closes on this curve.

18.6 Order parameter and critical behavior

The finite-volume partition function at zero field has a global $\mathbb{Z}_2$ symmetry $s_{j,m} \mapsto -s_{j,m}$, so the finite-volume magnetization vanishes exactly. The spontaneous magnetization is defined by taking the thermodynamic limit before removing an infinitesimal symmetry-breaking field:

$$M = \lim_{H \rightarrow 0^+} \lim_{L, M \rightarrow \infty} \frac{1}{LM} \sum_{j,m} \langle s_{j,m} \rangle_H. \quad (18.43)$$

Yang's exact result for the anisotropic square-lattice model is

$$M(K_x, K_y) = \begin{cases} \left[1 - \frac{1}{\sinh^2(2K_x) \sinh^2(2K_y)} \right]^{1/8}, & \sinh(2K_x) \sinh(2K_y) > 1, \\ 0, & \sinh(2K_x) \sinh(2K_y) \leq 1. \end{cases} \quad (18.44)$$

For the isotropic model this reduces to

$$M(K) = [1 - \sinh^{-4}(2K)]^{1/8}, \quad K > K_c. \quad (18.45)$$

Thus the magnetization exponent is

$$M \sim (T_c - T)^{1/8}, \quad \beta_{\text{Ising}} = \frac{1}{8}. \quad (18.46)$$

The transfer-matrix dispersion also reveals the divergence of the correlation length. Along the transfer direction, the inverse correlation length is given by the relevant transfer gap. Near the ferromagnetic critical line,

$$\xi_y^{-1} \sim \varepsilon(0) = 2 |\kappa - K_x|. \quad (18.47)$$

Because $\kappa - K_x$ is linear in the distance from the critical curve, the correlation-length exponent is

$$\xi \sim |T - T_c|^{-1}, \quad \nu = 1. \quad (18.48)$$

The full critical behavior of the two-dimensional Ising universality class is summarized by

$$\begin{aligned} \alpha = 0 \quad (\text{logarithmic specific-heat divergence}), \quad \beta = \frac{1}{8}, \quad \gamma = \frac{7}{4}, \\ \nu = 1, \quad \eta = \frac{1}{4}, \quad \delta = 15. \end{aligned} \quad (18.49)$$

In these notes, the most important point is not the list of exponents itself, but their origin: the classical thermal transition is controlled by a massless free-fermion transfer-matrix theory at criticality.

18.7 Connection to the one-dimensional TFIM

The transfer matrix in [Eq. \(18.17\)](#) becomes the imaginary-time evolution operator of the one-dimensional TFIM in a strongly anisotropic limit. Set

$$K_x = \delta\tau J, \quad \kappa = \delta\tau h, \quad \tanh(\delta\tau h) = e^{-2K_y}. \quad (18.50)$$

Then, using the symmetric Trotter formula,

$$T = A^L \exp \left[-\delta\tau H_{\text{TFIM}} + O(\delta\tau^3) \right], \quad (18.51)$$

with

$$H_{\text{TFIM}} = -J \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^L \sigma_j^z. \quad (18.52)$$

The Onsager condition becomes

$$\sinh(2\delta\tau J) \sinh(2K_y) = 1. \quad (18.53)$$

Using

$$e^{-2K_y} = \tanh(\delta\tau h) = \delta\tau h + O(\delta\tau^3), \quad (18.54)$$

one finds

$$h = J. \quad (18.55)$$

Thus the thermal critical point of the anisotropic two-dimensional classical Ising model maps to the zero-temperature quantum critical point of the one-dimensional transverse-field Ising model.

Classical 2D Ising object	Transfer-matrix object	1D quantum interpretation
Row configuration $\mathbf{s}_m$	Basis vector in $\mathcal{H}_{\text{row}}$	Spin-chain configuration
Row-to-row transfer matrix T	Positive operator on $\mathcal{H}_{\text{row}}$	Imaginary-time step $e^{-\delta\tau H}$
Largest eigenvalue of T	Ground-state sector of $-\log T$	Free-energy density / ground-state energy density
Transfer gap $\varepsilon(0)$	Mass gap of $-\log T$	Quantum excitation gap
Onsager curve	Gapless transfer matrix	TFIM critical point $h = J$

18.8 Summary

The two-dimensional zero-field Ising model is exactly solvable because its row-to-row transfer matrix is a fermionic Gaussian operator. The essential chain of ideas is

$$\begin{aligned} \text{2D classical partition function} &\longrightarrow \text{row transfer matrix} \longrightarrow \text{Majorana Gaussian operator} \\ &\longrightarrow \text{one-particle dispersion } \varepsilon(k). \end{aligned} \quad (18.56)$$

The Onsager critical point is the point where the transfer-matrix quasiparticle gap closes:

$$\varepsilon(0) = 0 \iff \kappa = K_x \iff \sinh(2K_x) \sinh(2K_y) = 1. \quad (18.57)$$

This is why the model belongs naturally in a lecture note on solvable quantum many-body systems: its exact solution is not merely a classical combinatorial miracle, but a free-fermion solution in transfer-matrix language.

Chapter 19

From 2D Classical Ising to 1D Quantum TFIM

The previous chapter solved the square-lattice Ising model by rewriting its row-to-row transfer matrix as a free-fermion operator. We now sharpen one particular consequence of that construction: in a strongly anisotropic limit, the two-dimensional classical Ising transfer matrix becomes the short imaginary-time evolution operator of the one-dimensional transverse-field Ising model (TFIM). This is the cleanest example of the general principle

$$d\text{-dimensional quantum system} \longleftrightarrow (d+1)\text{-dimensional classical statistical system.} \quad (19.1)$$

For the present case, the correspondence is

$$1\text{D quantum TFIM} \longleftrightarrow \text{anisotropic 2D classical Ising model.} \quad (19.2)$$

The additional classical direction is Euclidean imaginary time.

The discussion in this chapter has three complementary goals. First, we derive precisely how the row-to-row transfer matrix becomes $e^{-\delta\tau H_{\text{TFIM}}}$. Second, we give the explicit dictionary between classical couplings and quantum Hamiltonian parameters in the conventions of these notes. Third, we explain the physical interpretation of the map: spatial Ising bonds become the quantum Ising interaction, while rare domain walls in the transfer direction become transverse-field spin flips.

19.1 Row transfer matrix and basis convention

Consider the anisotropic zero-field classical Ising model on an $L \times M$ square lattice,

$$Z_{L,M}(K_x, K_y) = \sum_{\{s_{j,m}=\pm 1\}} \exp \left[K_x \sum_{j,m} s_{j,m} s_{j+1,m} + K_y \sum_{j,m} s_{j,m} s_{j,m+1} \right], \quad (19.3)$$

with periodic boundary conditions in both directions. The index j labels the one-dimensional spatial direction, while m labels the transfer direction. In the quantum interpretation, m will become a discretized imaginary-time index.

Following the convention used throughout these notes, a classical row configuration is identified with a simultaneous eigenstate of the Pauli operators σ_j^x :

$$\sigma_j^x |s_1, \dots, s_L\rangle_x = s_j |s_1, \dots, s_L\rangle_x, \quad s_j = \pm 1. \quad (19.4)$$

This convention is chosen so that the resulting quantum Hamiltonian is written in the same form as the TFIM chapter,

$$H_{\text{TFIM}} = -J \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x - h \sum_{j=1}^L \sigma_j^z. \quad (19.5)$$

Many books use the rotated convention $-J \sum_j \sigma_j^z \sigma_{j+1}^z - \Gamma \sum_j \sigma_j^x$. The two are related by a global spin rotation. In these notes the x -basis convention is more natural because the Jordan–Wigner occupation convention uses $n_j = (1 - \sigma_j^z)/2$.

The symmetrized row-to-row transfer matrix is

$$T = A_y^L \exp \left(\frac{K_x}{2} \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right) \exp \left(\kappa_y \sum_{j=1}^L \sigma_j^z \right) \exp \left(\frac{K_x}{2} \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x \right), \quad (19.6)$$

where

$$\tanh \kappa_y = e^{-2K_y}, \quad A_y = \sqrt{2 \sinh(2K_y)}. \quad (19.7)$$

Equivalently,

$$\kappa_y = \operatorname{arctanh}(e^{-2K_y}) = \frac{1}{2} \log \coth K_y. \quad (19.8)$$

The normalization A_y^L is important for the absolute classical free energy, but it becomes an additive constant in the quantum Hamiltonian and can be ignored when discussing normalized correlators and excitation gaps.

Remark 19.1 (Why σ^z appears in the vertical transfer operator). In the σ^x basis, σ^z flips the sign of s_j . Therefore the one-site matrix

$$W_{ss'} = e^{K_y ss'} \quad (19.9)$$

can be represented as $W = A_y e^{\kappa_y \sigma^z}$. A temporal mismatch $s \neq s'$ is a domain wall in the transfer direction; in the quantum model it is generated by the transverse-field operator σ^z .

19.2 Anisotropic Hamiltonian limit

The classical transfer matrix defines an exact operator

$$H_{\text{eff}} = -\frac{1}{a_\tau} \log T, \quad (19.10)$$

where a_τ is an arbitrary microscopic time spacing. For generic finite K_x and K_y , however, H_{eff} is not simply the local nearest-neighbour TFIM Hamiltonian. The logarithm of a product of noncommuting many-body operators generally contains longer-range and higher-body terms.

The local TFIM is obtained in the *strongly anisotropic limit*. Introduce a small imaginary-time lattice spacing $\delta\tau$ and set

$$K_x = \delta\tau J, \quad \kappa_y = \delta\tau h, \quad \tanh(\delta\tau h) = e^{-2K_y}. \quad (19.11)$$

Thus

$$K_x \rightarrow 0, \quad K_y \rightarrow \infty, \quad K_x e^{2K_y} \text{ fixed}. \quad (19.12)$$

Indeed, for small $\delta\tau$,

$$e^{-2K_y} = \tanh(\delta\tau h) = \delta\tau h + O(\delta\tau^3), \quad K_y = -\frac{1}{2} \log(\delta\tau h) + O(\delta\tau^2). \quad (19.13)$$

The horizontal classical coupling becomes weak, while the vertical coupling becomes strong. This is exactly what one expects from an imaginary-time path integral: neighbouring time slices are almost identical, and spin flips in time are rare events with probability proportional to $\delta\tau h$.

Define

$$X = J \sum_{j=1}^L \sigma_j^x \sigma_{j+1}^x, \quad Z = h \sum_{j=1}^L \sigma_j^z. \quad (19.14)$$

Then Eq. (19.6) becomes

$$T = A_y^L e^{\frac{\delta\tau}{2} X} e^{\delta\tau Z} e^{\frac{\delta\tau}{2} X}. \quad (19.15)$$

The symmetric Trotter formula gives

$$e^{\frac{\delta\tau}{2} X} e^{\delta\tau Z} e^{\frac{\delta\tau}{2} X} = \exp \left[\delta\tau (X + Z) + O(\delta\tau^3) \right]. \quad (19.16)$$

Since $X + Z = -H_{\text{TFIM}}$ in the convention of Eq. (19.5), we obtain

$$T = A_y^L \exp \left[-\delta\tau H_{\text{TFIM}} + O(\delta\tau^3) \right]. \quad (19.17)$$

Equivalently,

$$-\frac{1}{\delta\tau} \log T = H_{\text{TFIM}} - \frac{L}{\delta\tau} \log A_y \mathbf{1} + O(\delta\tau^2). \quad (19.18)$$

The second term is an additive constant. It affects the absolute free energy but not the eigenvectors, normalized expectation values, or excitation gaps.

Proposition 19.1 (Hamiltonian limit of the anisotropic Ising transfer matrix). *With the scaling*

$$K_x = \delta\tau J, \quad \tanh(\delta\tau h) = e^{-2K_y}, \quad (19.19)$$

the row-to-row transfer matrix of the anisotropic two-dimensional classical Ising model satisfies

$$T = A_y^L e^{-\delta\tau H_{\text{TFIM}} + O(\delta\tau^3)}, \quad (19.20)$$

where

$$H_{\text{TFIM}} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z. \quad (19.21)$$

Therefore the local quantum Hamiltonian is recovered from the logarithm of the classical transfer matrix up to an additive constant and errors of order $O(\delta\tau^2)$ in the Hamiltonian density.

19.3 Partition functions and the continuum-time limit

The classical partition function is

$$Z_{L,M}(K_x, K_y) = \text{Tr } T^M. \quad (19.22)$$

Using Eq. (19.17), one obtains

$$Z_{L,M}(K_x, K_y) = A_y^{LM} \text{Tr} \exp \left[-M\delta\tau H_{\text{TFIM}} + O(M\delta\tau^3) \right]. \quad (19.23)$$

If

$$\beta = M\delta\tau \quad (19.24)$$

is held fixed while $M \rightarrow \infty$ and $\delta\tau \rightarrow 0$, then

$$Z_{L,M}(K_x, K_y) = A_y^{LM} Z_{\text{TFIM}}(\beta) [1 + \text{controlled Trotter corrections}], \quad (19.25)$$

where

$$Z_{\text{TFIM}}(\beta) = \text{Tr} e^{-\beta H_{\text{TFIM}}}. \quad (19.26)$$

For local thermodynamic quantities, the logarithm of the correction is of order $O(\beta L \delta\tau^2)$ for the symmetric decomposition. Thus the limit is controlled at fixed system size and fixed inverse temperature, and remains controlled for thermodynamic densities when the usual local Trotter estimates are used.

The normalization factor A_y^{LM} gives an additive contribution to the classical free energy:

$$-\log Z_{L,M} = -LM \log A_y - \log Z_{\text{TFIM}}(\beta) + \dots \quad (19.27)$$

This term has no effect on normalized quantum correlation functions. It is, however, part of the absolute classical free energy. One should therefore keep it when comparing the full classical free-energy density, but drop it when extracting the quantum Hamiltonian spectrum.

19.4 Dictionary of couplings

There are two equivalent ways to present the coupling dictionary.

19.4.1 From quantum TFIM to classical Ising

Given a TFIM Hamiltonian

$$H_{\text{TFIM}} = -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z, \quad (19.28)$$

and a small imaginary-time spacing $\delta\tau$, the corresponding classical couplings are

$$\boxed{K_x = \delta\tau J, \quad K_y = -\frac{1}{2} \log \tanh(\delta\tau h)}. \quad (19.29)$$

Equivalently,

$$\kappa_y = \delta\tau h, \quad \tanh \kappa_y = e^{-2K_y}. \quad (19.30)$$

The classical lattice has $M = \beta/\delta\tau$ rows in the imaginary-time direction when representing a quantum thermal partition function at inverse temperature β .

19.4.2 From classical Ising to quantum TFIM

Given classical couplings K_x and K_y , define the dual transfer-direction coupling

$$\kappa_y = \frac{1}{2} \log \coth K_y. \quad (19.31)$$

If the transfer matrix is close to the identity in the sense that $K_x \ll 1$ and $\kappa_y \ll 1$, then it is naturally interpreted as a short-time quantum evolution operator. Choosing an imaginary-time spacing $\delta\tau$,

$$\boxed{J = \frac{K_x}{\delta\tau}, \quad h = \frac{\kappa_y}{\delta\tau} = \frac{1}{2\delta\tau} \log \coth K_y.} \quad (19.32)$$

Only the dimensionless ratio

$$\boxed{\frac{h}{J} = \frac{\kappa_y}{K_x} = \frac{1}{2K_x} \log \coth K_y} \quad (19.33)$$

is independent of the arbitrary choice of $\delta\tau$. The overall scale of J and h is fixed only after one decides how the classical transfer step is to be measured in imaginary-time units.

Object	Classical Ising transfer-matrix language	Quantum TFIM language
Spatial coordinate	Column index j	Spin-chain site j
Transfer coordinate	Row index m	Imaginary time $\tau = m\delta\tau$
Row spin $s_{j,m}$	Eigenvalue of σ_j^x	Order-parameter basis variable
Horizontal coupling	$K_x = \delta\tau J$	Ising exchange J
Vertical coupling	$\tanh \kappa_y = e^{-2K_y}$	Transverse field $h = \kappa_y/\delta\tau$
Transfer matrix	T	$A_y^L e^{-\delta\tau H_{\text{TFIM}}}$
Vertical domain wall	$s_{j,m} \neq s_{j,m+1}$	Spin flip generated by σ_j^z

19.5 Critical line and self-duality

The anisotropic square-lattice Ising model is critical on the Onsager curve

$$\sinh(2K_x) \sinh(2K_y) = 1. \quad (19.34)$$

Using

$$\sinh(2K_y) \sinh(2\kappa_y) = 1, \quad (19.35)$$

this condition is equivalent, for positive couplings, to

$$\kappa_y = K_x. \quad (19.36)$$

Under the Hamiltonian-limit dictionary $K_x = \delta\tau J$ and $\kappa_y = \delta\tau h$, this becomes

$$h = J. \quad (19.37)$$

Thus the thermal critical line of the anisotropic two-dimensional classical Ising model reduces to the zero-temperature quantum critical point of the one-dimensional TFIM.

This result is also the transfer-matrix form of Kramers–Wannier duality. The coupling κ_y is the dual of K_y , and the critical line is the self-dual condition

$$K_x = \kappa_y. \quad (19.38)$$

The TFIM inherits this as the quantum self-dual condition $J = h$. On one side of this point, $J > h$, the quantum ground state is ferromagnetically ordered in the x direction. On the other side, $h > J$, the ground state is paramagnetic. The classical high-temperature and low-temperature phases become the two quantum phases separated by a zero-temperature transition.

Remark 19.2 (Thermal versus quantum criticality). The two-dimensional classical transition is driven by thermal fluctuations. The one-dimensional TFIM transition at $T = 0$ is driven by quantum fluctuations from the transverse field. The transfer-matrix correspondence identifies their Euclidean statistical weights, and therefore their universal critical theory, not their microscopic dynamical origin.

19.6 Transfer-matrix spectrum and the TFIM dispersion

The correspondence is sharper than a relation between critical points. It maps the full transfer-matrix quasiparticle spectrum to the quantum excitation spectrum.

In the free-fermion solution of the two-dimensional Ising transfer matrix, the one-particle transfer dispersion may be written as

$$\cosh \varepsilon_{\text{tr}}(k) = \cosh(2K_x) \cosh(2\kappa_y) - \sinh(2K_x) \sinh(2\kappa_y) \cos k. \quad (19.39)$$

The transfer gap closes at $k = 0$ when $K_x = \kappa_y$, reproducing the critical condition above.

Now impose the Hamiltonian-limit scaling

$$K_x = \delta\tau J, \quad \kappa_y = \delta\tau h. \quad (19.40)$$

For small $\delta\tau$,

$$\cosh \varepsilon_{\text{tr}}(k) = 1 + 2\delta\tau^2 (J^2 + h^2 - 2Jh \cos k) + O(\delta\tau^4). \quad (19.41)$$

Therefore

$$\varepsilon_{\text{tr}}(k) = \delta\tau E_{\text{TFIM}}(k) + O(\delta\tau^3), \quad (19.42)$$

where

$$E_{\text{TFIM}}(k) = 2\sqrt{h^2 + J^2 - 2Jh \cos k} = 2\sqrt{(h - J \cos k)^2 + J^2 \sin^2 k} \quad (19.43)$$

is exactly the Bogoliubov quasiparticle dispersion of the TFIM in the convention of [Eq. \(19.5\)](#).

Thus transfer-matrix eigenvalue ratios become quantum energy gaps:

$$-\log \frac{\lambda_n}{\lambda_0} = \delta\tau (E_n - E_0) + O(\delta\tau^3). \quad (19.44)$$

At the ferromagnetic critical point $h = J > 0$,

$$E_{\text{TFIM}}(k) = 4J |\sin(k/2)| \simeq 2J |k| \quad (k \rightarrow 0). \quad (19.45)$$

The continuum limit is therefore a massless Majorana theory. Away from the critical point, the quantum gap is

$$\Delta = E_{\text{TFIM}}(0) = 2|h - J| \quad (19.46)$$

for $J, h > 0$.

19.7 Operator dictionary and Euclidean correlators

The transfer-matrix map also identifies correlation functions. A classical spin on row m becomes the quantum operator σ_j^x inserted at imaginary time

$$\tau = m\delta\tau. \quad (19.47)$$

Thus, in the Hamiltonian limit,

$$\langle s_{j,m} s_{j',0} \rangle_{\text{cl}} \longrightarrow \frac{1}{Z_{\text{TFIM}}} \text{Tr} \left[e^{-\beta H_{\text{TFIM}}} \sigma_j^x(\tau) \sigma_{j'}^x(0) \right], \quad (19.48)$$

where

$$\sigma_j^x(\tau) = e^{\tau H_{\text{TFIM}}} \sigma_j^x e^{-\tau H_{\text{TFIM}}}. \quad (19.49)$$

At zero quantum temperature, $\beta \rightarrow \infty$, the trace becomes a ground-state expectation value:

$$\langle s_{j,m} s_{j',0} \rangle_{\text{cl}} \longrightarrow \langle 0 | \sigma_j^x(\tau) \sigma_{j'}^x(0) | 0 \rangle. \quad (19.50)$$

Spatial classical bonds map to the Ising energy operator,

$$s_{j,m} s_{j+1,m} \longleftrightarrow \sigma_j^x \sigma_{j+1}^x. \quad (19.51)$$

Temporal bonds are represented by insertions obtained by differentiating the vertical transfer matrix with respect to K_y . Since

$$W = A_y e^{\kappa_y \sigma^z}, \quad (19.52)$$

vertical-bond observables contain an identity contribution and a contribution proportional to σ^z . In the quantum Hamiltonian limit, the nontrivial operator associated with temporal fluctuations is therefore the transverse-field operator

$$\text{temporal domain-wall fluctuations} \longleftrightarrow \sigma_j^z. \quad (19.53)$$

This is the operator-level reason why the vertical classical coupling becomes the transverse field rather than another Ising exchange term.

19.8 Physical interpretation of the anisotropic limit

The anisotropic limit has a simple worldline interpretation. In the σ^x basis, a quantum spin configuration at imaginary time τ is a row configuration

$$\mathbf{s}(\tau) = (s_1(\tau), \dots, s_L(\tau)). \quad (19.54)$$

The diagonal part of the Hamiltonian,

$$-J \sum_j \sigma_j^x \sigma_{j+1}^x, \quad (19.55)$$

assigns an energy to each row configuration. Over a short imaginary-time interval $\delta\tau$, it contributes the Boltzmann factor

$$\exp \left[\delta\tau J \sum_j s_j(\tau) s_{j+1}(\tau) \right], \quad (19.56)$$

which is precisely the horizontal classical Ising weight with $K_x = \delta\tau J$.

The off-diagonal part of the Hamiltonian,

$$-h \sum_j \sigma_j^z, \quad (19.57)$$

flips σ^x eigenvalues. For one spin,

$$\langle s' |_x e^{\delta\tau h \sigma^z} | s \rangle_x = \begin{cases} \cosh(\delta\tau h), & s' = s, \\ \sinh(\delta\tau h), & s' = -s. \end{cases} \quad (19.58)$$

The relative weight of a flip is therefore

$$\frac{\sinh(\delta\tau h)}{\cosh(\delta\tau h)} = \tanh(\delta\tau h) = e^{-2K_y}. \quad (19.59)$$

A vertical classical domain wall costs a factor e^{-2K_y} relative to a non-domain-wall segment. Hence the transverse field is the rate per unit imaginary time at which a worldline flips:

$$e^{-2K_y} \simeq \delta\tau h. \quad (19.60)$$

This is the microscopic origin of the dictionary.

The ordered and disordered phases can be read from this worldline picture:

- If $J \gg h$, spatial alignment dominates and spin worldlines remain ordered in the x direction. This is the ferromagnetic phase of the TFIM.
- If $h \gg J$, spin flips proliferate in imaginary time and destroy long-range x order. This is the quantum paramagnet.
- At $h = J$, domain-wall proliferation is critical. The transfer-matrix gap closes and the Euclidean theory becomes scale invariant.

19.9 Finite temperature, finite size, and the extra dimension

The number of rows M in the transfer direction determines the quantum inverse temperature:

$$\beta = M\delta\tau. \quad (19.61)$$

Therefore:

- A classical cylinder with finite circumference M in the transfer direction represents a quantum TFIM at finite temperature $T_q = 1/\beta$.
- The limit $M \rightarrow \infty$ at fixed $\delta\tau$ projects onto the largest eigenvalue of the transfer matrix. In the quantum language this is the zero-temperature or ground-state limit.
- The continuum-time quantum limit requires $M \rightarrow \infty$ and $\delta\tau \rightarrow 0$ with $\beta = M\delta\tau$ fixed.

The spatial size L is the number of sites in the quantum spin chain. Thus the classical thermodynamic limit $L, M \rightarrow \infty$ contains both the quantum thermodynamic limit and the zero-temperature limit.

The transfer-matrix correspondence also explains why a quantum critical point in d spatial dimensions is naturally described by a $(d + 1)$ -dimensional Euclidean field theory. The quantum partition function

$$Z_q = \text{Tr} e^{-\beta H} \quad (19.62)$$

is a sum over histories in imaginary time. When the quantum gap closes, the correlation length in space and the correlation length in imaginary time both diverge. The relative scaling of the two is controlled by the dynamical exponent z . For the TFIM critical point, $z = 1$, and the long-distance continuum theory is effectively isotropic after a nonuniversal rescaling of time.

19.10 Common pitfalls

There are several points that are easy to misstate.

- (i) **The generic transfer matrix is not automatically a local TFIM Hamiltonian.** For finite K_x and K_y , the exact logarithm $-\log T$ is a well-defined free-fermion Hamiltonian, but it is generally not the nearest-neighbour TFIM. Locality in the simple TFIM form emerges in the strongly anisotropic limit.
- (ii) **The normalization A_y^L is not a physical transverse field.** It shifts the absolute free energy or the zero of quantum energy. The transverse field is encoded in κ_y , not in A_y .
- (iii) **The choice of spin basis matters.** In these notes the classical Ising variable is the eigenvalue of σ^x . If one instead labels classical spins by σ^z , the same physics is obtained after a global spin rotation, but the symbols in the Hamiltonian are interchanged.
- (iv) **Classical temperature and quantum temperature are different concepts.** The classical couplings K_x, K_y already include the classical inverse temperature. The quantum inverse temperature is the length of the transfer direction, $\beta = M\delta\tau$.
- (v) **Boundary conditions introduce fermion parity sectors.** After Jordan–Wigner transformation, periodic spin boundary conditions split the free-fermion solution into parity-dependent boundary conditions. This sector bookkeeping is essential for exact finite-size spectra, although it does not change the local Hamiltonian-limit dictionary above.

19.11 Summary

The anisotropic two-dimensional classical Ising model contains the one-dimensional quantum TFIM as its Hamiltonian limit. The essential chain of relations is

$$\begin{aligned} Z_{\text{2D Ising}} &= \text{Tr} T^M, & T &= A_y^L e^{\frac{K_x}{2} \sum_j \sigma_j^x \sigma_{j+1}^x} e^{\kappa_y \sum_j \sigma_j^z} e^{\frac{K_x}{2} \sum_j \sigma_j^x \sigma_{j+1}^x}, \\ K_x &= \delta\tau J, & \kappa_y &= \delta\tau h, & \tanh \kappa_y &= e^{-2K_y}, \\ T &= A_y^L e^{-\delta\tau H_{\text{TFIM}} + O(\delta\tau^3)}, & H_{\text{TFIM}} &= -J \sum_j \sigma_j^x \sigma_{j+1}^x - h \sum_j \sigma_j^z. \end{aligned} \quad (19.63)$$

The coupling ratio is

$$\frac{h}{J} = \frac{\kappa_y}{K_x} = \frac{1}{2K_x} \log \coth K_y. \quad (19.64)$$

The Onsager critical curve becomes the TFIM quantum critical point:

$$\sinh(2K_x) \sinh(2K_y) = 1 \quad \Longleftrightarrow \quad K_x = \kappa_y \quad \Longleftrightarrow \quad J = h. \quad (19.65)$$

At the spectral level,

$$\varepsilon_{\text{tr}}(k) = \delta\tau \, 2\sqrt{h^2 + J^2 - 2Jh \cos k} + O(\delta\tau^3), \quad (19.66)$$

so the transfer-matrix gap is the quantum excitation gap measured in imaginary- time units. This chapter therefore gives the canonical solvable example of how classical transfer matrices, quantum Hamiltonians, duality, and free-fermion criticality are different representations of the same underlying structure.

Chapter 20

Six-Vertex Model and the XXZ Chain

The two-dimensional Ising model was solvable because its transfer matrix could be rewritten in terms of free fermions. The six-vertex model is solvable for a different reason. It is generally not a free-fermion model. Its solvability comes from the existence of an R -matrix satisfying the Yang–Baxter equation. This equation implies a commuting family of transfer matrices, and the logarithmic derivative of this family produces the one-dimensional XXZ quantum spin chain.

This chapter has three goals:

- (i) define the six allowed local vertices and the row-to-row transfer matrix;
- (ii) introduce the six-vertex R -matrix and explain how the Yang–Baxter equation gives commuting transfer matrices; and
- (iii) take the Hamiltonian limit and obtain the XXZ chain with anisotropy parameter Δ .

The coordinate and algebraic Bethe-ansatz diagonalizations are postponed to the following chapters. Here the main point is the exact map

six-vertex transfer matrix $\implies$ commuting transfer matrices $\implies$ XXZ Hamiltonian.

20.1 Arrows, the ice rule, and six local weights

The six-vertex model is a classical statistical model on a square lattice. The degrees of freedom are arrows living on the edges. At each vertex four arrows meet. The allowed configurations obey the *ice rule*: exactly two arrows point into the vertex and exactly two arrows point out of the vertex.

Equivalently, each local vertex conserves a two-state charge. We identify an arrow state with a spin-1/2 basis vector

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad V \simeq \mathbb{C}^2. \quad (20.1)$$

A local vertex weight is then an operator on $V \otimes V$. In the ordered basis

$$|\uparrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle, \quad (20.2)$$

the most common six-vertex weight matrix has the block form

$$R = \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & b & c & 0 \\ 0 & c & b & 0 \\ 0 & 0 & 0 & a \end{pmatrix}. \quad (20.3)$$

The six nonzero matrix elements are

$$\begin{aligned} |\uparrow\uparrow\rangle &\longrightarrow |\uparrow\uparrow\rangle && \text{with weight } a, \\ |\downarrow\downarrow\rangle &\longrightarrow |\downarrow\downarrow\rangle && \text{with weight } a, \\ |\uparrow\downarrow\rangle &\longrightarrow |\uparrow\downarrow\rangle && \text{with weight } b, \\ |\downarrow\uparrow\rangle &\longrightarrow |\downarrow\uparrow\rangle && \text{with weight } b, \\ |\uparrow\downarrow\rangle &\longrightarrow |\downarrow\uparrow\rangle && \text{with weight } c, \\ |\downarrow\uparrow\rangle &\longrightarrow |\uparrow\downarrow\rangle && \text{with weight } c. \end{aligned} \quad (20.4)$$

All other matrix elements vanish. Thus R conserves the total z -spin on the two lines:

$$[R, \sigma^z \otimes \mathbf{1} + \mathbf{1} \otimes \sigma^z] = 0. \quad (20.5)$$

This is the algebraic form of the ice rule.

Definition 20.1 (Six-vertex anisotropy parameter). For nonzero a and b , the standard six-vertex anisotropy parameter is

$$\Delta = \frac{a^2 + b^2 - c^2}{2ab}. \quad (20.6)$$

Multiplying all weights by a common scalar changes the normalization of the partition function but not Δ .

The parameter Δ is the same anisotropy parameter that appears in the XXZ spin chain obtained from the Hamiltonian limit. The local weights (a, b, c) control a two-dimensional classical model, whereas Δ controls the relative strength of the z interaction in the associated one-dimensional quantum chain.

Remark 20.1 (Why there are six vertices). Without the ice rule there would be $2^4 = 16$ possible arrow configurations at a four-valent vertex. The constraint “two in and two out” leaves six. In the spin notation this is exactly the statement that the local weight preserves the number of up spins, or equivalently the total S^z charge, on the two lines.

20.2 Row-to-row transfer matrix

Consider a square lattice with L columns and M rows. We impose periodic boundary conditions in the horizontal direction. A row configuration is specified by the L vertical arrows crossing a horizontal cut, hence the row Hilbert space is

$$\mathcal{H}_L = V_1 \otimes V_2 \otimes \cdots \otimes V_L, \quad V_j \simeq \mathbb{C}^2. \quad (20.7)$$

A row-to-row transfer matrix maps one row of vertical arrows to the next row, summing over the internal horizontal arrows.

The efficient way to write this object is to introduce an auxiliary space $V_a \simeq \mathbb{C}^2$, which represents the horizontal arrow passing through the row. Let $R_{aj}(u)$ act on the auxiliary space a and the quantum space j . The monodromy matrix is

$$M_a(u) = R_{aL}(u)R_{a,L-1}(u) \cdots R_{a1}(u), \quad (20.8)$$

and the transfer matrix is the auxiliary trace

$$\tau(u) = \text{Tr}_a M_a(u) = \text{Tr}_a [R_{aL}(u)R_{a,L-1}(u) \cdots R_{a1}(u)]. \quad (20.9)$$

The trace over a closes the horizontal arrow into a periodic loop. The partition function on an $L \times M$ torus is then

$$Z_{L,M}(u) = \text{Tr}_{\mathcal{H}_L} \tau(u)^M. \quad (20.10)$$

Thus the thermodynamics of the two-dimensional classical model is controlled by the spectrum of $\tau(u)$.

Remark 20.2 (Two meanings of the same space). In the classical model, $\mathcal{H}_L$ is a convenient vector space for row configurations. In the quantum Hamiltonian limit, the same tensor product becomes the Hilbert space of a length- L spin chain. This is the same conceptual move as in the Ising transfer-matrix construction: a classical transfer direction becomes imaginary time for a quantum problem.

The ice rule implies a global $U(1)$ symmetry of the transfer matrix. Let

$$S_{\text{tot}}^z = \frac{1}{2} \sum_{j=1}^L \sigma_j^z. \quad (20.11)$$

Since every local $R_{aj}(u)$ conserves the sum of the auxiliary and quantum z -spins, tracing over the auxiliary space gives

$$[\tau(u), S_{\text{tot}}^z] = 0. \quad (20.12)$$

The corresponding XXZ Hamiltonian therefore conserves total magnetization.

20.3 The trigonometric six-vertex R -matrix

A single transfer matrix is not yet enough to imply exact solvability. The integrable structure appears when the vertex weights are embedded in a spectral-parameter family $R(u)$ satisfying the Yang–Baxter equation. A standard hyperbolic parametrization is

$$R(u) = \begin{pmatrix} \sinh(u + \eta) & 0 & 0 & 0 \\ 0 & \sinh u & \sinh \eta & 0 \\ 0 & \sinh \eta & \sinh u & 0 \\ 0 & 0 & 0 & \sinh(u + \eta) \end{pmatrix}. \quad (20.13)$$

Here u is the spectral parameter and η is the anisotropy parameter. In this convention

$$a(u) = \sinh(u + \eta), \quad b(u) = \sinh u, \quad c(u) = \sinh \eta, \quad (20.14)$$

and Eq. (20.6) gives

$$\Delta = \cosh \eta. \quad (20.15)$$

For the critical regime one often uses the trigonometric parametrization

$$R(u) = \begin{pmatrix} \sin(u + \gamma) & 0 & 0 & 0 \\ 0 & \sin u & \sin \gamma & 0 \\ 0 & \sin \gamma & \sin u & 0 \\ 0 & 0 & 0 & \sin(u + \gamma) \end{pmatrix}, \quad \Delta = \cos \gamma. \quad (20.16)$$

The two forms are related by analytic continuation $\eta \leftrightarrow i\gamma$, up to harmless scalar and phase normalizations. The hyperbolic form is natural for $\Delta > 1$, while the trigonometric form is natural for $|\Delta| < 1$.

Remark 20.3 (Scalar normalization). The transformation

$$R(u) \mapsto f(u)R(u) \quad (20.17)$$

with scalar $f(u) \neq 0$ preserves the Yang–Baxter equation. It multiplies the transfer matrix by $f(u)^L$ and therefore changes only the scalar part of $\log \tau(u)$. In the Hamiltonian limit this produces an additive constant in H , not a change of the interacting physics.

The R -matrix in [Eq. \(20.13\)](#) is *regular*:

$$R(0) = \sinh \eta P, \quad (20.18)$$

where $P : V \otimes V \rightarrow V \otimes V$ is the permutation operator

$$P(|\alpha\rangle \otimes |\beta\rangle) = |\beta\rangle \otimes |\alpha\rangle. \quad (20.19)$$

In Pauli matrices,

$$P = \frac{1}{2} (\mathbf{1} \otimes \mathbf{1} + \sigma^x \otimes \sigma^x + \sigma^y \otimes \sigma^y + \sigma^z \otimes \sigma^z). \quad (20.20)$$

Regularity is the key property that makes the logarithmic derivative of $\tau(u)$ local.

20.4 Yang–Baxter equation and commuting transfer matrices

Let $R_{ij}(u)$ denote $R(u)$ acting on the i th and j th tensor factors and as identity on all others. The Yang–Baxter equation is

$$R_{12}(u - v)R_{13}(u)R_{23}(v) = R_{23}(v)R_{13}(u)R_{12}(u - v). \quad (20.21)$$

It is an operator identity on $V_1 \otimes V_2 \otimes V_3$. The six-vertex R -matrix [Eq. \(20.13\)](#) satisfies this equation. In the classical vertex-model language, [Eq. \(20.21\)](#) is a local star–triangle move. In the quantum scattering language, it is the consistency condition for factorized three-body scattering.

Theorem 20.1 (Commuting six-vertex transfer matrices). *Let $R(u)$ satisfy the Yang–Baxter equation [Eq. \(20.21\)](#). Define $\tau(u)$ by [Eq. \(20.9\)](#). Then*

$$[\tau(u), \tau(v)] = 0, \quad \text{for all } u, v. \quad (20.22)$$

Proof. Introduce two auxiliary spaces a and b . The Yang–Baxter equation implies the local relation

$$R_{ab}(u-v)R_{aj}(u)R_{bj}(v) = R_{bj}(v)R_{aj}(u)R_{ab}(u-v) \quad (20.23)$$

for each quantum site j . Multiplying this identity over all sites gives the RTT relation

$$R_{ab}(u-v)M_a(u)M_b(v) = M_b(v)M_a(u)R_{ab}(u-v). \quad (20.24)$$

Taking $\text{Tr}_a \text{Tr}_b$ and using cyclicity of the trace in the auxiliary spaces, $R_{ab}(u-v)$ can be moved through the trace and canceled. The result is

$$\text{Tr}_a M_a(u) \text{Tr}_b M_b(v) = \text{Tr}_b M_b(v) \text{Tr}_a M_a(u), \quad (20.25)$$

which is exactly $\tau(u)\tau(v) = \tau(v)\tau(u)$. $\square$

The implication is substantial. Expanding the logarithm of $\tau(u)$ around a regular point gives an infinite sequence of mutually commuting conserved charges:

$$\log \tau(u) = \sum_{n=0}^{\infty} (u - u_0)^n Q_n, \quad [Q_m, Q_n] = 0. \quad (20.26)$$

The first nontrivial local charge is the XXZ Hamiltonian. Higher charges have larger spatial range and are responsible for integrability.

Principle 20.1 (Yang–Baxter integrability). The Yang–Baxter equation is the local algebraic identity that upgrades one transfer matrix into a commuting one-parameter family. The logarithm of this family generates the Hamiltonian and the higher conserved charges.

20.5 Hamiltonian limit and the XXZ chain

We now derive the quantum Hamiltonian associated with the regular six-vertex transfer matrix. For a homogeneous periodic chain, regularity gives

$$\tau(0) = (\sinh \eta)^L U, \quad (20.27)$$

where U is the one-site translation operator on the spin chain. Taking the logarithmic derivative removes this translation and produces a nearest-neighbor operator:

$$\left. \frac{d}{du} \log \tau(u) \right|_{u=0} = \sum_{j=1}^L \frac{1}{\sinh \eta} P_{j,j+1} R'_{j,j+1}(0), \quad (20.28)$$

with $j+1$ understood modulo L . The local operator can be computed explicitly. From Eq. (20.13),

$$R'(0) = \begin{pmatrix} \cosh \eta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cosh \eta \end{pmatrix}. \quad (20.29)$$

Therefore

$$\frac{1}{\sinh \eta} P R'(0) = \frac{1}{2 \sinh \eta} (\sigma^x \otimes \sigma^x + \sigma^y \otimes \sigma^y + \cosh \eta \sigma^z \otimes \sigma^z + \cosh \eta \mathbf{1} \otimes \mathbf{1}). \quad (20.30)$$

Multiplying the logarithmic derivative by $2 \sinh \eta$ and subtracting an additive constant gives

$$H_{\text{XXZ}}^{(+)} = J \sum_{j=1}^L (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1)), \quad \Delta = \cosh \eta. \quad (20.31)$$

Equivalently,

$$H_{\text{XXZ}}^{(+)} = 2J \sinh \eta \left. \frac{d}{du} \log \tau(u) \right|_{u=0} - 2J\Delta L. \quad (20.32)$$

The superscript (+) indicates the standard antiferromagnetic integrability normalization for $J > 0$ and $\Delta > 0$. The constant $-2J\Delta L$ fixes the energy of the fully polarized reference state to zero; it has no effect on eigenvectors, gaps, or correlation functions.

In spin-operator normalization, $S_j^\alpha = \sigma_j^\alpha/2$, the same Hamiltonian is

$$H_{\text{XXZ}}^{(+)} = 4J \sum_{j=1}^L (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z) - J\Delta L. \quad (20.33)$$

Only the overall energy scale and additive constant differ from other common XXZ conventions.

Remark 20.4 (Relation to the earlier Pauli convention). Earlier chapters used the ferromagnetic Pauli convention

$$H_{\text{XXZ}}^{\text{ferro}} = -\frac{J_f}{2} \sum_j (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta \sigma_j^z \sigma_{j+1}^z) - h \sum_j \sigma_j^z. \quad (20.34)$$

The Hamiltonian generated directly by the six-vertex R -matrix is instead the integrability convention [Eq. \(20.31\)](#). The two are related by an overall sign and energy-scale choice, plus an additive constant. Whenever Bethe-ansatz formulas are quoted, the sign convention must be stated explicitly.

The trigonometric R -matrix gives the same formula with

$$\Delta = \cos \gamma, \quad \sinh \eta \longrightarrow \sin \gamma. \quad (20.35)$$

Thus the critical XXZ regime $|\Delta| < 1$ is naturally parametrized by $0 < \gamma < \pi$.

20.6 Why the Hamiltonian is local

It is useful to isolate the mechanism behind locality. The transfer matrix

$$\tau(u) = \text{Tr}_a R_{aL}(u) \cdots R_{a1}(u) \quad (20.36)$$

is a nonlocal operator on the whole chain. Nevertheless, at the regular point $u = 0$ every $R_{aj}(0)$ is a permutation. The product of permutations simply moves the auxiliary spin through the chain and becomes the translation operator after the auxiliary trace. Differentiating with respect to u replaces exactly one local permutation by $R'(0)$, while all other factors remain permutations. After multiplying by $\tau(0)^{-1}$, this single defect becomes a local nearest-neighbor density.

This is the vertex-model analogue of the Trotter logic in the Ising–TFIM correspondence. The difference is that the present transfer matrix is not merely one short-time evolution operator. It belongs to a commuting spectral-parameter family, and this additional structure produces infinitely many conserved charges.

Proposition 20.1 (Local charge from regularity). *Let $R(u)$ satisfy $R(0) = \rho P$ with $\rho \neq 0$, and define the homogeneous periodic transfer matrix by Eq. (20.9). Then*

$$\left. \frac{d}{du} \log \tau(u) \right|_{u=0} = \sum_{j=1}^L h_{j,j+1}, \quad h_{j,j+1} = \rho^{-1} P_{j,j+1} R'_{j,j+1}(0), \quad (20.37)$$

up to the conventional orientation of the one-site translation operator.

20.7 Conserved magnetization and interacting character

The XXZ Hamiltonian obtained above satisfies

$$[H_{\text{XXZ}}^{(+)}, S_{\text{tot}}^z] = 0. \quad (20.38)$$

Using $\sigma_j^\pm = (\sigma_j^x \pm i\sigma_j^y)/2$, one may rewrite the exchange part as

$$\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y = 2(\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+). \quad (20.39)$$

Thus the XY part hops a down spin through an up-spin background, while the $\Delta \sigma_j^z \sigma_{j+1}^z$ term gives a nearest-neighbor interaction between magnons. At $\Delta = 0$ the model is the XX chain and becomes free after the Jordan–Wigner transformation. At generic Δ , the Jordan–Wigner image contains a density–density interaction and is not a quadratic fermion model.

Explicitly, under Jordan–Wigner fermionization, the XY part becomes hopping, whereas

$$\sigma_j^z \sigma_{j+1}^z = (1 - 2n_j)(1 - 2n_{j+1}) = 1 - 2(n_j + n_{j+1}) + 4n_j n_{j+1}. \quad (20.40)$$

The quartic term $n_j n_{j+1}$ is the interacting part. Therefore the XXZ chain is a genuinely interacting integrable model, except at the free-fermion point $\Delta = 0$.

Remark 20.5 (Why this chapter leaves the spectrum unfinished). For the Ising, XY, Kitaev, and SSH chains, exact solvability meant reducing the Hamiltonian to independent BdG blocks. For the XXZ chain this is impossible at generic Δ because of the quartic Jordan–Wigner interaction. Its solution requires Bethe ansatz: many-body eigenstates are built from interacting magnons whose scattering factorizes. The six-vertex transfer matrix supplies the Yang–Baxter reason why that factorization is consistent.

20.8 Anisotropy regimes

For the antiferromagnetic XXZ convention

$$H_{\text{XXZ}}^S = \mathcal{J} \sum_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z), \quad \mathcal{J} > 0, \quad (20.41)$$

the familiar zero-field phase structure is summarized as follows:

Parameter regime	Useful parametrization	Quantum-chain behavior
$\Delta > 1$	$\Delta = \cosh \eta$	gapped antiferromagnetic, or Néel, regime
$-1 < \Delta < 1$	$\Delta = \cos \gamma$	gapless critical Luttinger-liquid regime
$\Delta = 1$	isotropic XXX point	critical with full $SU(2)$ symmetry and marginal corrections
$\Delta < -1$	analytic continuation or sign-twisted convention	gapped ferromagnetic regime

For the classical six-vertex model with positive weights, one also classifies regions using

$$\Delta = \frac{a^2 + b^2 - c^2}{2ab}. \quad (20.42)$$

The regime $|\Delta| < 1$ is the critical disordered phase of the classical vertex model. The regions $\Delta > 1$ and $\Delta < -1$ correspond to ordered ferroelectric and antiferroelectric regimes, respectively, depending on which weights dominate. The exact terminology is classical-model dependent, but the same invariant Δ controls the integrable family.

Remark 20.6 (Classical phases versus quantum phases). The two-dimensional six-vertex partition function at a fixed spectral parameter and the one-dimensional XXZ Hamiltonian obtained by a logarithmic derivative are not the same physical object. They are two different uses of the same commuting transfer-matrix family. The shared parameter Δ explains why their phase diagrams are closely related, but one should not identify a classical ordered region with a quantum ground-state phase without specifying the Hamiltonian limit and the sign convention.

20.9 What the transfer matrix solution will become

The six-vertex transfer matrix is diagonalized by Bethe ansatz. In the algebraic form, one writes the monodromy matrix as a 2×2 matrix in auxiliary space,

$$M_a(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix}_a, \quad \tau(u) = A(u) + D(u). \quad (20.43)$$

The operator $B(u)$ creates Bethe quasiparticles above a reference state, and the RTT relation generated by the Yang–Baxter equation gives the algebra needed to move $A(u)$ and $D(u)$ through a product of B ’s. Requiring unwanted terms to vanish gives the Bethe equations.

In the coordinate form, the same result appears as factorized magnon scattering. In a sector with M down spins, a wave function is a sum over plane waves in each ordered coordinate region,

$$\Psi(x_1, \dots, x_M) = \sum_{P \in S_M} A(P) \exp \left(i \sum_{a=1}^M k_{P_a} x_a \right), \quad x_1 < \dots < x_M. \quad (20.44)$$

The two-body scattering phase determines how the amplitudes $A(P)$ change when neighboring momenta are exchanged. Periodic boundary conditions then lead to Bethe equations. The following chapters derive these statements explicitly.

20.10 Summary

The essential chain of ideas is

$$\begin{array}{l}
 \text{ice-rule vertex weights} \implies R(u) \text{ with } U(1) \text{ conservation} \\
 \implies \text{Yang-Baxter equation} \implies [\tau(u), \tau(v)] = 0 \\
 \implies H_{\text{XXZ}} \propto \left. \frac{d}{du} \log \tau(u) \right|_{u=0} \implies \text{interacting Bethe-ansatz integrability.}
 \end{array} \tag{20.45}$$

The six-vertex model is therefore the canonical bridge between two-dimensional classical vertex models and one-dimensional interacting quantum spin chains. Unlike the Ising–TFIM correspondence, the resulting quantum chain is not solved by free fermions at generic anisotropy. Its solvability is instead controlled by Yang–Baxter integrability and factorized scattering.

Chapter 21

Eight-Vertex Model and the XYZ Chain

The six-vertex model is the vertex-model counterpart of the XXZ chain. Its ice rule gives a local $U(1)$ conservation law, and the trigonometric R -matrix gives an interacting but integrable quantum spin chain. The eight-vertex model is the next member in the same hierarchy. It relaxes the ice rule while preserving an even-arrow, or $\mathbb{Z}_2$, constraint. Algebraically this replaces the six-vertex R -matrix by an elliptic R -matrix. The Hamiltonian limit is no longer XXZ but the fully anisotropic XYZ spin chain.

The exact correspondence is

$$\begin{aligned} \text{eight-vertex transfer matrix} &\implies \text{elliptic Yang-Baxter equation} \\ &\implies \text{XYZ Hamiltonian.} \end{aligned}$$

The important conceptual change from the previous chapter is that the model is still Yang-Baxter integrable, but it is no longer organized by total magnetization sectors. The missing $U(1)$ symmetry is the main reason why the solution of the eight-vertex model is technically harder than the six-vertex case.

This chapter does not derive Baxter's full solution. Its goal is to set up the model, state the elliptic R -matrix structure, and derive the Hamiltonian limit that produces the XYZ chain. Baxter's TQ relation and the full spectral theory belong to a more advanced treatment of elliptic integrability.

21.1 Eight allowed vertices and local weights

The eight-vertex model is a classical statistical model on the square lattice. As in the six-vertex model, arrows live on edges and a Boltzmann weight is assigned to each vertex. The local constraint is weaker than the ice rule:

$$\text{an allowed vertex has an even number of arrows pointing in.} \quad (21.1)$$

Equivalently, the allowed numbers of incoming arrows are 0, 2, and 4. There are therefore eight allowed local configurations. The six ice-rule vertices are included, and the two additional vertices have all arrows pointing in or all arrows pointing out.

In spin notation a local vertex weight is an operator on $V \otimes V$, where $V \simeq \mathbb{C}^2$. We use the same ordered basis as in the six-vertex chapter,

$$|\uparrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle. \quad (21.2)$$

A standard symmetric eight-vertex weight matrix has the form

$$R = \begin{pmatrix} a & 0 & 0 & d \\ 0 & b & c & 0 \\ 0 & c & b & 0 \\ d & 0 & 0 & a \end{pmatrix}. \quad (21.3)$$

The six-vertex model is recovered when $d = 0$. The extra d -terms are the new eight-vertex processes

$$|\uparrow\uparrow\rangle \longleftrightarrow |\downarrow\downarrow\rangle. \quad (21.4)$$

They change the total S^z of the two lines by two units. Thus the model no longer conserves total S^z , but it conserves its parity.

Proposition 21.1 ($\mathbb{Z}_2$ conservation). *Let*

$$Q_2 = \sigma^z \otimes \sigma^z. \quad (21.5)$$

For the eight-vertex weight Eq. (21.3),

$$[R, Q_2] = 0. \quad (21.6)$$

However, for $d \neq 0$,

$$[R, \sigma^z \otimes \mathbf{1} + \mathbf{1} \otimes \sigma^z] \neq 0. \quad (21.7)$$

Thus the local $U(1)$ conservation of the six-vertex model is reduced to a $\mathbb{Z}_2$ parity conservation.

Proof. The diagonal a and b processes clearly preserve both the total z -spin and its parity. The c processes exchange $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$, so they also preserve total z -spin. The d processes exchange $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$, changing total S^z by ± 2 but preserving the eigenvalue of $\sigma^z \otimes \sigma^z$. This is exactly Eq. (21.6). $\square$

Remark 21.1 (Why the eight-vertex model is harder). The six-vertex transfer matrix decomposes into fixed-magnetization sectors. This sector decomposition is the basic input behind the usual algebraic Bethe ansatz with a simple ferromagnetic reference state. In the eight-vertex model the d -processes destroy S^z conservation, so the standard highest-weight construction no longer applies directly. Baxter's solution replaces it with an elliptic functional-relation method.

21.2 Row-to-row transfer matrix

The transfer-matrix construction is formally the same as in the six-vertex model. On a row of length L , the row space is

$$\mathcal{H}_L = V_1 \otimes V_2 \otimes \cdots \otimes V_L, \quad V_j \simeq \mathbb{C}^2. \quad (21.8)$$

Introduce an auxiliary space $V_a \simeq \mathbb{C}^2$. Given a spectral-parameter family $R(u)$ of eight-vertex weights, define the monodromy matrix

$$M_a(u) = R_{aL}(u) R_{a,L-1}(u) \cdots R_{a1}(u), \quad (21.9)$$

and the row-to-row transfer matrix

$$\tau(u) = \text{Tr}_a M_a(u) = \text{Tr}_a [R_{aL}(u) R_{a,L-1}(u) \cdots R_{a1}(u)]. \quad (21.10)$$

For an $L \times M$ torus,

$$Z_{L,M}(u) = \text{Tr}_{\mathcal{H}_L} \tau(u)^M. \quad (21.11)$$

The spectrum of $\tau(u)$ controls the thermodynamic limit of the classical model.

Because of the local parity conservation in Eq. (21.6), the transfer matrix commutes with the global spin-parity operator

$$\mathcal{P} = \prod_{j=1}^L \sigma_j^z, \quad [\tau(u), \mathcal{P}] = 0. \quad (21.12)$$

For $d \neq 0$ one does not generally have

$$[\tau(u), S_{\text{tot}}^z] = 0, \quad S_{\text{tot}}^z = \frac{1}{2} \sum_{j=1}^L \sigma_j^z. \quad (21.13)$$

This is the transfer-matrix reflection of the fact that the corresponding XYZ Hamiltonian has only a discrete spin-flip/parity symmetry, not a continuous $U(1)$ symmetry.

21.3 Baxter's elliptic parametrization

A generic matrix of the form Eq. (21.3) does not automatically produce commuting transfer matrices. Integrability requires that the weights be embedded in an elliptic spectral-parameter family satisfying the Yang–Baxter equation.

Let $\vartheta_1(u)$ and $\vartheta_4(u)$ denote Jacobi theta functions with a fixed nome q , $|q| < 1$. Let η be the crossing parameter. One common Baxter parametrization, shifted so that the regular point is at $u = 0$, is

$$\begin{aligned} a(u) &= \rho \vartheta_4(2\eta) \vartheta_4(u) \vartheta_1(u + 2\eta), \\ b(u) &= \rho \vartheta_4(2\eta) \vartheta_1(u) \vartheta_4(u + 2\eta), \\ c(u) &= \rho \vartheta_1(2\eta) \vartheta_4(u) \vartheta_4(u + 2\eta), \\ d(u) &= \rho \vartheta_1(2\eta) \vartheta_1(u) \vartheta_1(u + 2\eta). \end{aligned} \quad (21.14)$$

Here ρ is an arbitrary nonzero scalar normalization. The corresponding R -matrix is

$$R(u) = \begin{pmatrix} a(u) & 0 & 0 & d(u) \\ 0 & b(u) & c(u) & 0 \\ 0 & c(u) & b(u) & 0 \\ d(u) & 0 & 0 & a(u) \end{pmatrix}. \quad (21.15)$$

Since $\vartheta_1(0) = 0$ and $\vartheta_4'(0) = 0$, the regular point satisfies

$$b(0) = d(0) = 0, \quad a(0) = c(0) = \rho_0, \quad (21.16)$$

where

$$\rho_0 = \rho \vartheta_4(0) \vartheta_4(2\eta) \vartheta_1(2\eta). \quad (21.17)$$

Therefore

$$R(0) = \rho_0 P, \quad (21.18)$$

where P is the two-spin permutation operator,

$$P |\alpha\rangle \otimes |\beta\rangle = |\beta\rangle \otimes |\alpha\rangle. \quad (21.19)$$

Remark 21.2 (Normalization and shift conventions). The eight-vertex literature contains several equivalent theta-function conventions. Some place the regular point at $u = \eta$ rather than $u = 0$, and some use theta functions denoted by H and Θ . The convention in Eq. (21.14) is chosen to match the Hamiltonian-limit convention used in these notes: $R(0)$ is proportional to the permutation operator.

For the family Eq. (21.15), the Yang–Baxter equation takes the same formal form as in the six-vertex case:

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \quad (21.20)$$

This identity is a nontrivial theta-function identity. Its consequence is the commutativity of transfer matrices:

$$[\tau(u), \tau(v)] = 0 \quad \text{for all } u, v. \quad (21.21)$$

Thus the eight-vertex model is integrable in the same structural sense as the six-vertex model: there is a one-parameter commuting family of transfer matrices.

Principle 21.1 (Elliptic integrability). The hierarchy

$$\text{rational } R\text{-matrix} \longrightarrow \text{trigonometric } R\text{-matrix} \longrightarrow \text{elliptic } R\text{-matrix} \quad (21.22)$$

corresponds, respectively, to the XXX, XXZ, and XYZ spin chains. The eight-vertex model is the elliptic vertex model whose Hamiltonian limit is the XYZ chain.

A useful way to see the difference from the six-vertex model is the Pauli-basis form. Since

$$\mathbf{1} \otimes \mathbf{1} \mapsto \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (21.23)$$

$$\sigma^x \otimes \sigma^x \mapsto \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad (21.24)$$

$$\sigma^y \otimes \sigma^y \mapsto \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad (21.25)$$

$$\sigma^z \otimes \sigma^z \mapsto \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (21.26)$$

we may rewrite Eq. (21.15) as

$$R(u) = A(u) \mathbf{1} \otimes \mathbf{1} + B(u) \sigma^x \otimes \sigma^x + C(u) \sigma^y \otimes \sigma^y + D(u) \sigma^z \otimes \sigma^z, \quad (21.27)$$

with

$$A(u) = \frac{a(u) + b(u)}{2}, \quad D(u) = \frac{a(u) - b(u)}{2}, \quad (21.28)$$

$$B(u) = \frac{c(u) + d(u)}{2}, \quad C(u) = \frac{c(u) - d(u)}{2}. \quad (21.29)$$

The distinction between $\sigma^x \sigma^x$ and $\sigma^y \sigma^y$ is controlled by the d -weight. When $d = 0$, the coefficients of $\sigma^x \otimes \sigma^x$ and $\sigma^y \otimes \sigma^y$ coincide, and the model reduces to the six-vertex/XXZ type.

21.4 Hamiltonian limit and the XYZ chain

The Hamiltonian limit is obtained from the logarithmic derivative of the transfer matrix at the regular point. Because $R(0) = \rho_0 P$, the transfer matrix at $u = 0$ is proportional to the one-site translation operator. The first derivative therefore gives a sum of local nearest-neighbour terms.

Proposition 21.2 (Hamiltonian from the eight-vertex transfer matrix). *Let $\tau(u)$ be the periodic transfer matrix built from the regular R -matrix [Eq. \(21.15\)](#). Define*

$$H_{\text{XYZ}} = \left. \frac{d}{du} \log \tau(u) \right|_{u=0}. \quad (21.30)$$

After dropping an additive scalar multiple of the identity and allowing an overall energy-scale normalization, one obtains

$$H_{\text{XYZ}} = \sum_{j=1}^L (J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y + J_z \sigma_j^z \sigma_{j+1}^z), \quad \sigma_{L+1}^\alpha \equiv \sigma_1^\alpha. \quad (21.31)$$

The three couplings are determined by the first derivatives of the vertex weights at the regular point:

$$J_x = \frac{b'(0) + d'(0)}{2\rho_0}, \quad J_y = \frac{b'(0) - d'(0)}{2\rho_0}, \quad J_z = \frac{a'(0) - c'(0)}{2\rho_0}. \quad (21.32)$$

The discarded additive constant is

$$E_0 = \frac{a'(0) + c'(0)}{2\rho_0} \quad (21.33)$$

per bond.

Proof. By regularity, $R(0) = \rho_0 P$. The standard transfer-matrix argument gives a local logarithmic derivative

$$h_{j,j+1} = \frac{1}{\rho_0} P_{j,j+1} R'_{j,j+1}(0). \quad (21.34)$$

For the matrix form [Eq. \(21.15\)](#),

$$\frac{1}{\rho_0} P R'(0) = \frac{1}{\rho_0} \begin{pmatrix} a'(0) & 0 & 0 & d'(0) \\ 0 & c'(0) & b'(0) & 0 \\ 0 & b'(0) & c'(0) & 0 \\ d'(0) & 0 & 0 & a'(0) \end{pmatrix}. \quad (21.35)$$

Comparing this matrix with

$$E_0 \mathbf{1} + J_x \sigma^x \otimes \sigma^x + J_y \sigma^y \otimes \sigma^y + J_z \sigma^z \otimes \sigma^z \quad (21.36)$$

gives exactly [Eq. \(21.32\)](#) and [Eq. \(21.33\)](#). $\square$

For the theta-function weights in Eq. (21.14), these formulas give

$$J_x = \frac{\vartheta'_1(0)}{2\vartheta_4(0)\vartheta_4(2\eta)\vartheta_1(2\eta)} [\vartheta_4(2\eta)^2 + \vartheta_1(2\eta)^2], \quad (21.37)$$

$$J_y = \frac{\vartheta'_1(0)}{2\vartheta_4(0)\vartheta_4(2\eta)\vartheta_1(2\eta)} [\vartheta_4(2\eta)^2 - \vartheta_1(2\eta)^2], \quad (21.38)$$

and

$$J_z = \frac{1}{2} \left[\frac{\vartheta'_1(2\eta)}{\vartheta_1(2\eta)} - \frac{\vartheta'_4(2\eta)}{\vartheta_4(2\eta)} \right]. \quad (21.39)$$

These expressions depend on the chosen normalization of the spectral parameter. A rescaling $u \mapsto \lambda u$ rescales all three couplings by the common factor λ . The physically meaningful anisotropies are the coupling ratios.

Remark 21.3 (Sign convention). Some condensed-matter texts write

$$H = - \sum_j (J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y + J_z \sigma_j^z \sigma_{j+1}^z). \quad (21.40)$$

This differs from Eq. (21.31) by redefining $J_\alpha \mapsto -J_\alpha$. The integrable structure is unaffected by this overall convention.

21.5 Relation to the XXZ chain and special limits

The XYZ chain is the most general nearest-neighbour spin-1/2 chain with diagonal exchange couplings and no magnetic field:

$$H_{\text{XYZ}} = \sum_j (J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y + J_z \sigma_j^z \sigma_{j+1}^z). \quad (21.41)$$

The XXZ chain is the special case

$$J_x = J_y, \quad J_z = \Delta J_x. \quad (21.42)$$

In the vertex-model language this corresponds to the degeneration

$$d(u) \rightarrow 0, \quad (21.43)$$

so that the eight-vertex R -matrix becomes a six-vertex R -matrix and the elliptic parametrization degenerates to a trigonometric one. In theta-function language this occurs in the trigonometric limit of the elliptic nome. In that limit the d -weight disappears, $J_x - J_y \rightarrow 0$, and the $U(1)$ symmetry of the XXZ chain is restored.

Vertex model	Quantum chain	Symmetry
Six-vertex	XXZ	$U(1)$ generated by S_{tot}^z
Eight-vertex	XYZ	$\mathbb{Z}_2$ generated by $\prod_j \sigma_j^z$
Trigonometric R	XXZ anisotropy	$J_x = J_y$
Elliptic R	XYZ anisotropy	$J_x \neq J_y \neq J_z$ generically

The transverse-field Ising and XY chains studied earlier in these notes are also anisotropic spin chains, but their solvability mechanism is different. They become quadratic fermion models after the Jordan–Wigner transformation. The generic XYZ chain does not become a free-fermion Hamiltonian under Jordan–Wigner; it becomes an interacting fermion model. Its exact solvability comes instead from the commuting transfer matrices inherited from the elliptic Yang–Baxter equation.

21.6 Fermionic image and loss of free-fermion structure

It is useful to compare the XYZ chain with the XY mother model. Write

$$\begin{aligned} J_x \sigma_j^x \sigma_{j+1}^x + J_y \sigma_j^y \sigma_{j+1}^y &= (J_x + J_y) (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) \\ &\quad + (J_x - J_y) (\sigma_j^+ \sigma_{j+1}^+ + \sigma_j^- \sigma_{j+1}^-). \end{aligned} \quad (21.44)$$

The first line gives fermion hopping after Jordan–Wigner, while the second line gives fermion pairing. The additional J_z term gives

$$\sigma_j^z \sigma_{j+1}^z = (1 - 2n_j)(1 - 2n_{j+1}) = 1 - 2(n_j + n_{j+1}) + 4n_j n_{j+1}, \quad (21.45)$$

up to the convention for which spin state is identified with an occupied fermion. Therefore the XYZ chain maps schematically to

$$H_{\text{XYZ}} \longmapsto H_{\text{hopping}} + H_{\text{pairing}} + 4J_z \sum_j n_j n_{j+1} + \text{chemical-potential and boundary terms.} \quad (21.46)$$

The density-density interaction is absent only when $J_z = 0$. Hence the generic XYZ chain is not a quadratic BdG problem.

Remark 21.4 (Two different meanings of anisotropy). In the free-fermion XY chain, anisotropy between x and y exchange produces pairing but does not destroy quadratic solvability. In the XYZ chain, the $J_z \sigma^z \sigma^z$ term produces a genuine fermion interaction. Therefore “anisotropic spin chain” does not by itself specify the solution method.

21.7 Baxter functional relations

The six-vertex model can be diagonalized by coordinate or algebraic Bethe ansatz because the transfer matrix has a simple reference state, such as the all-up state. The eight-vertex transfer matrix does not preserve the number of down spins, so that reference-state construction is not directly available. Baxter’s solution instead uses functional relations.

The central object is a second commuting family of operators $Q(u)$ satisfying a relation of the schematic form

$$\tau(u)Q(u) = \phi_+(u)Q(u - 2\eta) + \phi_-(u)Q(u + 2\eta), \quad (21.47)$$

where $\phi_{\pm}(u)$ are known scalar functions determined by the elliptic weights and the system size. On a common eigenvector of $\tau(u)$ and $Q(u)$, this becomes a scalar finite-difference equation for the transfer-matrix eigenvalue.

Principle 21.2 (Baxter’s replacement for the Bethe ansatz). For the eight-vertex model, the role of Bethe roots is played by zeros of the Q -eigenvalue. The Bethe equations arise from requiring the apparent poles or zeros in the TQ relation to be consistent. Thus the solution is still an exact spectral solution, but it is naturally formulated as an elliptic functional relation rather than as a magnon scattering problem.

This is why the eight-vertex and XYZ models are often regarded as more advanced than the six-vertex and XXZ models. They are not merely more anisotropic; they belong to the elliptic level of Yang–Baxter integrability.

21.8 Summary of the model hierarchy

The hierarchy developed in the last two chapters can be summarized as follows:

$$\begin{array}{llll}
 \text{XXX chain} & \longleftrightarrow & \text{rational } R\text{-matrix} & \longleftrightarrow & \Delta = 1, \\
 \text{XXZ chain} & \longleftrightarrow & \text{trigonometric } R\text{-matrix} & \longleftrightarrow & J_x = J_y \neq J_z, \\
 \text{XYZ chain} & \longleftrightarrow & \text{elliptic } R\text{-matrix} & \longleftrightarrow & J_x \neq J_y \neq J_z.
 \end{array} \tag{21.48}$$

The same progression appears on the classical two-dimensional side:

$$\text{six-vertex model} \subset \text{eight-vertex model}, \tag{21.49}$$

where the inclusion is obtained by setting the pair-flip weight d to zero.

The eight-vertex model is the elliptic vertex model whose regular transfer matrix generates the XYZ chain. Its solvability is not free-fermionic and not based on $U(1)$ magnon-number conservation. It is Yang–Baxter integrability at the elliptic level, realized through commuting transfer matrices and Baxter functional relations.

Part IV

**Interacting Integrable Quantum
Chains**

Chapter 22

Coordinate Bethe Ansatz for the XXZ Chain

The preceding chapters introduced the XXZ chain as the Hamiltonian limit of the six-vertex transfer matrix. We now solve the same quantum chain directly in real space. This is the coordinate Bethe ansatz. Its logic is deliberately close to ordinary few-body quantum mechanics: write a plane-wave superposition in each ordered coordinate sector, impose the Schrodinger equation at particle contact, and finally impose periodic boundary conditions.

Throughout this chapter we use the integrability normalization obtained from the six-vertex logarithmic derivative,

$$H_{\text{XXZ}}^{(+)} = J \sum_{j=1}^L (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1)), \quad (22.1)$$

with periodic boundary conditions $\sigma_{L+1}^\alpha = \sigma_1^\alpha$. The fully polarized state

$$|0\rangle = |\uparrow\uparrow \cdots \uparrow\rangle \quad (22.2)$$

is an eigenstate with zero energy. It is not necessarily the ground state of the antiferromagnetic chain, but it is the natural pseudovacuum for Bethe ansatz. Down spins above this reference state will be called magnons.

The three goals of the chapter are:

- (i) solve the one-magnon problem and identify the bare dispersion;
- (ii) derive the two-magnon scattering amplitude from the contact equation;
- (iii) assemble the M -magnon Bethe wave function and obtain the Bethe equations.

The result is the first genuinely interacting exact solution in these notes: the energy is additive in Bethe momenta, while the allowed momenta are determined by many-body quantization conditions containing all two-body phase shifts.

22.1 Conserved magnon number and coordinate basis

Using

$$\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y = 2 (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+), \quad (22.3)$$

one sees that the XY part only exchanges neighboring up and down spins. The total magnetization

$$S_{\text{tot}}^z = \frac{1}{2} \sum_{j=1}^L \sigma_j^z \quad (22.4)$$

therefore commutes with $H_{\text{XXZ}}^{(+)}$. Equivalently, the number of down spins is conserved.

In the sector with M down spins, a convenient basis is

$$|x_1, \dots, x_M\rangle = \sigma_{x_1}^- \cdots \sigma_{x_M}^- |0\rangle, \quad 1 \leq x_1 < x_2 < \cdots < x_M \leq L. \quad (22.5)$$

A general state in this sector is

$$|\Psi_M\rangle = \sum_{1 \leq x_1 < \cdots < x_M \leq L} \Psi(x_1, \dots, x_M) |x_1, \dots, x_M\rangle. \quad (22.6)$$

The coordinate Bethe ansatz is an ansatz for the amplitudes $\Psi(x_1, \dots, x_M)$.

It is useful to separate two kinds of equations. When no two magnons are neighbors, the Schrodinger equation is just a discrete free-particle equation. When $x_{a+1} = x_a + 1$, the two magnons touch, and the nearest-neighbor Ising term produces a contact condition. This contact condition is the source of the scattering phase.

22.2 One-magnon sector

A one-magnon state has the form

$$|\Psi_1\rangle = \sum_{x=1}^L \psi(x) |x\rangle, \quad |x\rangle = \sigma_x^- |0\rangle. \quad (22.7)$$

Acting with [Eq. \(22.1\)](#), one obtains

$$E\psi(x) = 2J\psi(x-1) + 2J\psi(x+1) - 4J\Delta\psi(x), \quad (22.8)$$

where x is understood modulo L . The hopping amplitude is $2J$, and the potential term $-4J\Delta$ comes from the two domain walls adjacent to the down spin.

The plane wave

$$\psi_k(x) = e^{ikx} \quad (22.9)$$

is therefore an eigenfunction with dispersion

$$\varepsilon(k) = 4J(\cos k - \Delta). \quad (22.10)$$

Periodic boundary conditions give

$$e^{ikL} = 1, \quad k = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}. \quad (22.11)$$

Remark 22.1 (Energy convention). The dispersion in [Eq. \(22.10\)](#) is written relative to the fully polarized reference state, whose energy was fixed to zero by the constant in [Eq. \(22.1\)](#). Reversing the overall sign of the Hamiltonian or removing the constant shifts and rescales the formulas, but does not change the Bethe wave functions or the scattering phase.

22.3 Two-magnon scattering

The first nontrivial part of the Bethe ansatz appears for two magnons. For coordinates $x_1 < x_2$, take

$$\Psi(x_1, x_2) = A_{12} e^{i(k_1 x_1 + k_2 x_2)} + A_{21} e^{i(k_2 x_1 + k_1 x_2)}. \quad (22.12)$$

The two terms describe the same two momenta assigned to the two ordered coordinates in two different ways. Away from contact, namely for $x_2 > x_1 + 1$, the wave function satisfies the free difference equation and has energy

$$E(k_1, k_2) = 4J(\cos k_1 + \cos k_2 - 2\Delta). \quad (22.13)$$

The energy is additive in the two bare one-magnon energies.

At contact, $x_2 = x_1 + 1$, two of the four domain walls of separated magnons have annihilated. The Schrodinger equation becomes

$$(E + 4J\Delta)\Psi(x, x + 1) = 2J\Psi(x - 1, x + 1) + 2J\Psi(x, x + 2). \quad (22.14)$$

Substituting Eq. (22.12) into Eq. (22.14) fixes the ratio of amplitudes:

$$\frac{A_{21}}{A_{12}} = S(k_1, k_2) = -\frac{1 + e^{i(k_1 + k_2)} - 2\Delta e^{ik_1}}{1 + e^{i(k_1 + k_2)} - 2\Delta e^{ik_2}}. \quad (22.15)$$

This is the two-magnon scattering amplitude of the XXZ chain in the convention of Eq. (22.1).

Several checks are immediate. First,

$$S(k_1, k_2)S(k_2, k_1) = 1, \quad (22.16)$$

so exchanging two magnons twice gives back the original amplitude. Second, at $\Delta = 0$,

$$S(k_1, k_2) = -1, \quad (22.17)$$

which is the free-fermion sign of the XX chain. Thus the XX point is recovered as the noninteracting limit of the coordinate Bethe ansatz.

Principle 22.1 (Interaction as a boundary condition). In the coordinate Bethe ansatz, the interaction does not destroy plane waves in each ordered region. Instead, it appears as a matching condition at contact, which relates different plane-wave amplitudes by a two-body scattering phase.

22.4 The many-magnon coordinate Bethe ansatz

For M magnons, the ordered coordinate region is

$$1 \leq x_1 < x_2 < \cdots < x_M \leq L. \quad (22.18)$$

The coordinate Bethe ansatz is

$$\Psi(x_1, \dots, x_M) = \sum_{P \in S_M} A(P) \exp \left(i \sum_{a=1}^M k_{P_a} x_a \right), \quad (22.19)$$

where S_M is the permutation group of M objects. The amplitude $A(P)$ is attached to the assignment of momentum k_{P_a} to the ordered coordinate x_a .

Whenever all magnons are separated, the Schrodinger equation gives the additive energy

$$E(k_1, \dots, k_M) = \sum_{a=1}^M \varepsilon(k_a) = 4J \sum_{a=1}^M (\cos k_a - \Delta). \quad (22.20)$$

Whenever two adjacent coordinates touch, the same two-body contact equation as in the two-magnon problem applies. Therefore adjacent permutations obey

$$A(Ps_a) = S(k_{P_a}, k_{P_{a+1}})A(P), \quad a = 1, \dots, M-1, \quad (22.21)$$

where s_a exchanges the entries in positions a and $a+1$.

Because the scattering amplitude is a scalar phase, consistency of different ways of reducing a permutation follows from ordinary multiplication. In models with internal degrees of freedom, the analogous consistency condition becomes the Yang–Baxter equation. This is why the coordinate Bethe ansatz here is the real-space shadow of the six-vertex Yang–Baxter structure.

22.5 Periodic boundary conditions and Bethe equations

The remaining step is to impose periodic boundary conditions. Move magnon j once around the ring. It accumulates a free propagation phase $e^{ik_j L}$ and scatters once with every other magnon. With the scattering convention of Eq. (22.15), the result is

$$e^{ik_j L} = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M S(k_j, k_\ell), \quad j = 1, \dots, M. \quad (22.22)$$

Together with the energy formula Eq. (22.20), these equations solve the finite periodic XXZ chain in each fixed-magnon sector, up to the usual questions of completeness and string organization.

Writing

$$S(k_j, k_\ell) = e^{i\theta(k_j, k_\ell)}, \quad (22.23)$$

one obtains the logarithmic form

$$Lk_j = 2\pi I_j + \sum_{\substack{\ell=1 \\ \ell \neq j}}^M \theta(k_j, k_\ell), \quad I_j \in \mathbb{Z} \text{ or } \mathbb{Z} + \frac{1}{2}. \quad (22.24)$$

The integer versus half-integer choice depends on M , L , the branch choice of θ , and the boundary-condition convention. The exponential form Eq. (22.22) is the unambiguous statement.

Remark 22.2 (What is solved by the Bethe equations). The Bethe equations do not say that the interacting chain has free momenta. They say that eigenstates are still labelled by rapidities or momenta, but those labels are quantized collectively. Each k_j is shifted by the phase shifts against all other magnons.

22.6 Rapidity parametrizations

The momentum-space scattering amplitude is explicit but not the most useful form for analysis. The standard Bethe equations become simpler after introducing a rapidity variable. The appropriate parametrization depends on the anisotropy regime.

Gapless regime: $-1 < \Delta < 1$

Let

$$\Delta = \cos \gamma, \quad 0 < \gamma < \pi. \quad (22.25)$$

Define the rapidity λ by

$$e^{ik(\lambda)} = \frac{\sinh(\lambda + i\gamma/2)}{\sinh(\lambda - i\gamma/2)}. \quad (22.26)$$

Then Eq. (22.22) becomes

$$\left[\frac{\sinh(\lambda_j + i\gamma/2)}{\sinh(\lambda_j - i\gamma/2)} \right]^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\sinh(\lambda_j - \lambda_\ell + i\gamma)}{\sinh(\lambda_j - \lambda_\ell - i\gamma)}. \quad (22.27)$$

The energy becomes

$$E = -4J \sum_{j=1}^M \frac{\sin^2 \gamma}{\cosh(2\lambda_j) - \cos \gamma}. \quad (22.28)$$

The total lattice momentum is

$$P = \sum_{j=1}^M k(\lambda_j) \pmod{2\pi}. \quad (22.29)$$

Massive antiferromagnetic regime: $\Delta > 1$

Let

$$\Delta = \cosh \eta, \quad \eta > 0. \quad (22.30)$$

A convenient rapidity map is

$$e^{ik(\lambda)} = \frac{\sin(\lambda + i\eta/2)}{\sin(\lambda - i\eta/2)}. \quad (22.31)$$

The Bethe equations are

$$\left[\frac{\sin(\lambda_j + i\eta/2)}{\sin(\lambda_j - i\eta/2)} \right]^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\sin(\lambda_j - \lambda_\ell + i\eta)}{\sin(\lambda_j - \lambda_\ell - i\eta)}. \quad (22.32)$$

The energy is

$$E = -4J \sum_{j=1}^M \frac{\sinh^2 \eta}{\cosh \eta - \cos(2\lambda_j)}. \quad (22.33)$$

In this regime complex rapidities organize into string patterns in the thermodynamic limit. Strings describe bound states of magnons and are essential for thermodynamics, but the finite-size Bethe equations above are the starting point.

Isotropic XXX point

At $\Delta = 1$, one may use the rational rapidity variable

$$e^{ik(\lambda)} = \frac{\lambda + i/2}{\lambda - i/2}. \quad (22.34)$$

The Bethe equations reduce to

$$\left(\frac{\lambda_j + i/2}{\lambda_j - i/2} \right)^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\lambda_j - \lambda_\ell + i}{\lambda_j - \lambda_\ell - i}, \quad (22.35)$$

and the energy is

$$E = -2J \sum_{j=1}^M \frac{1}{\lambda_j^2 + 1/4}. \quad (22.36)$$

This is the rational limit of the trigonometric XXZ equations.

22.7 Special limits and physical interpretation

Example 22.1 (XX point). At $\Delta = 0$, the scattering amplitude becomes $S = -1$. The Bethe equations are therefore

$$e^{ik_j L} = (-1)^{M-1}. \quad (22.37)$$

This is precisely the parity-dependent boundary condition of Jordan–Wigner fermions in the M -particle sector. Thus the coordinate Bethe ansatz reduces to free fermions at the XX point.

Example 22.2 (Two-magnon bound states). For $\Delta > 1$, the scattering amplitude has poles in the complex momentum or rapidity plane. Normalizable two-magnon wave functions can then have complex conjugate momenta, producing an exponentially localized relative coordinate. These states are the simplest examples of Bethe strings. In the many-body problem, strings encode bound-state quasiparticles.

Remark 22.3 (Pseudovacuum versus physical ground state). The coordinate Bethe ansatz above builds states by flipping spins above $|0\rangle = |\uparrow \cdots \uparrow\rangle$. For the antiferromagnetic convention with $J > 0$ and $\Delta > 0$, this reference state is not the physical ground state in the zero-magnetization sector. The true ground state is obtained by solving the Bethe equations with an extensive number of rapidities. The pseudovacuum is used because it makes the scattering construction algebraically simple.

22.8 Relation to the algebraic Bethe ansatz

The coordinate Bethe ansatz diagonalizes the Hamiltonian by solving the real-space Schrodinger equation. The algebraic Bethe ansatz diagonalizes the whole transfer matrix family using the R -matrix algebra. The two methods produce the same Bethe equations, but they organize the calculation differently:

Method	Basic object	Origin of Bethe equations
Coordinate Bethe ansatz	Wave function $\Psi(x_1, \dots, x_M)$	Periodic boundary conditions plus two-body scattering
Algebraic Bethe ansatz	Bethe vector $B(\lambda_1) \cdots B(\lambda_M) 0\rangle$	Cancellation of unwanted terms in the RTT algebra

The coordinate method is more physically transparent: it shows directly that the XXZ chain is solvable because many-body scattering factorizes into two-body phases. The algebraic method is more structural: it explains why this factorization is guaranteed by the Yang–Baxter equation and gives direct access to transfer-matrix eigenvalues and higher conserved charges.

22.9 Summary

The coordinate Bethe ansatz for the XXZ chain can be summarized as

$$\begin{aligned}
H_{\text{XXZ}}^{(+)} &\implies M\text{-magnon coordinate sectors} \\
\implies \Psi(x_1, \dots, x_M) &= \sum_{P \in S_M} A(P) \exp \left(i \sum_a k_{P_a} x_a \right) \\
&\implies A(P s_a) = S(k_{P_a}, k_{P_{a+1}}) A(P) \\
\implies e^{ik_j L} &= \prod_{\ell \neq j} S(k_j, k_\ell), \quad E = 4J \sum_j (\cos k_j - \Delta).
\end{aligned} \tag{22.38}$$

The one-magnon sector gives the bare dispersion. The two-magnon contact equation gives the scattering phase. The many-magnon solution works because all collisions factorize into the same two-body scattering amplitude. This is the coordinate-space form of the Yang–Baxter integrability already encountered in the six-vertex model.

Chapter 23

Algebraic Bethe Ansatz

The coordinate Bethe ansatz solves the XXZ chain by constructing wave functions in ordered magnon-coordinate sectors. The algebraic Bethe ansatz solves the same problem from the transfer-matrix side. Instead of writing a many-body wave function explicitly, one uses the Yang–Baxter equation to build an operator algebra. Bethe states are then obtained by applying a creation operator $B(u)$, extracted from the monodromy matrix, to a simple reference state.

The conceptual chain is

$$\begin{aligned} R\text{-matrix satisfying YBE} &\implies \text{RTT algebra for } A(u), B(u), C(u), D(u) \\ &\implies \text{Bethe vectors } B(\lambda_1) \cdots B(\lambda_M) |0\rangle \\ &\implies \text{transfer-matrix eigenvalues and Bethe equations.} \end{aligned}$$

Thus the algebraic Bethe ansatz diagonalizes not only one Hamiltonian, but the whole commuting transfer-matrix family. The Hamiltonian eigenvalues are obtained afterwards by taking the logarithmic derivative of the transfer-matrix eigenvalue.

Throughout this chapter we use the homogeneous six-vertex convention of [Chapter 20](#). The local R -matrix is

$$R(u) = \begin{pmatrix} \sinh(u + \eta) & 0 & 0 & 0 \\ 0 & \sinh u & \sinh \eta & 0 \\ 0 & \sinh \eta & \sinh u & 0 \\ 0 & 0 & 0 & \sinh(u + \eta) \end{pmatrix}, \quad \Delta = \cosh \eta. \quad (23.1)$$

The trigonometric critical regime follows by analytic continuation $\eta \mapsto i\gamma$, giving $\Delta = \cos \gamma$. We keep the hyperbolic notation because it makes the algebraic formulas compact.

23.1 Monodromy matrix

Let $V_a \simeq \mathbb{C}^2$ be an auxiliary spin-1/2 space and

$$\mathcal{H}_L = V_1 \otimes \cdots \otimes V_L, \quad V_j \simeq \mathbb{C}^2 \quad (23.2)$$

be the quantum spin-chain Hilbert space. The monodromy matrix is the ordered product

$$M_a(u) = R_{aL}(u) R_{a,L-1}(u) \cdots R_{a1}(u). \quad (23.3)$$

It is a 2×2 matrix in the auxiliary space, whose entries are operators on $\mathcal{H}_L$:

$$M_a(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix}_a. \quad (23.4)$$

The transfer matrix is the auxiliary trace

$$\tau(u) = \text{Tr}_a M_a(u) = A(u) + D(u). \quad (23.5)$$

The entries A, B, C, D are not four independent Hamiltonians. They are noncommuting operator-valued functions constrained by the RTT algebra below.

It is useful to write the local R -operator as a matrix in the auxiliary space with quantum-site operators as entries. In the convention of Eq. (23.1),

$$R_{aj}(u) = \begin{pmatrix} a_R(u)e_j^{11} + b_R(u)e_j^{22} & c_R e_j^{21} \\ c_R e_j^{12} & b_R(u)e_j^{11} + a_R(u)e_j^{22} \end{pmatrix}_a, \quad (23.6)$$

where

$$a_R(u) = \sinh(u + \eta), \quad b_R(u) = \sinh u, \quad c_R = \sinh \eta, \quad (23.7)$$

and $e_j^{\alpha\beta} = |\alpha\rangle_j \langle\beta|_j$. With $|\uparrow\rangle = |1\rangle$ and $|\downarrow\rangle = |2\rangle$, one has $e^{21} = \sigma^-$ and $e^{12} = \sigma^+$.

Choose the ferromagnetic reference state

$$|0\rangle = |\uparrow\uparrow \cdots \uparrow\rangle. \quad (23.8)$$

On this state every local $R_{aj}(u)$ becomes upper triangular in auxiliary space:

$$R_{aj}(u) |\uparrow\rangle_j = \begin{pmatrix} a_R(u) |\uparrow\rangle_j & c_R |\downarrow\rangle_j \\ 0 & b_R(u) |\uparrow\rangle_j \end{pmatrix}_a. \quad (23.9)$$

Multiplying over all sites preserves triangularity. Hence

$$C(u) |0\rangle = 0, \quad A(u) |0\rangle = \alpha(u) |0\rangle, \quad D(u) |0\rangle = \delta(u) |0\rangle, \quad (23.10)$$

with vacuum eigenvalues

$$\alpha(u) = a_R(u)^L = \sinh^L(u + \eta), \quad \delta(u) = b_R(u)^L = \sinh^L u. \quad (23.11)$$

The operator $B(u)$ lowers one or more spins when expanded as an operator on the chain. Its leading action on the reference state is a one-magnon superposition, so $B(u)$ is the algebraic creation operator for Bethe quasiparticles.

Remark 23.1 (Pseudovacuum). The word “vacuum” here is algebraic. For the antiferromagnetic XXZ chain the physical ground state is usually not $|0\rangle$. The state $|0\rangle$ is used because it is a highest-weight state for the RTT algebra: $C(u)$ annihilates it, while $A(u)$ and $D(u)$ act diagonally.

23.2 RTT relation and commuting transfer matrices

The Yang–Baxter equation for the six-vertex R -matrix implies the local intertwining relation

$$R_{ab}(u-v)R_{aj}(u)R_{bj}(v) = R_{bj}(v)R_{aj}(u)R_{ab}(u-v). \quad (23.12)$$

Multiplying this identity over all sites gives the RTT relation

$$R_{ab}(u-v)M_a(u)M_b(v) = M_b(v)M_a(u)R_{ab}(u-v). \quad (23.13)$$

This is the defining algebraic relation of the algebraic Bethe ansatz. It is an operator identity on

$$V_a \otimes V_b \otimes \mathcal{H}_L. \quad (23.14)$$

The two auxiliary spaces a and b are bookkeeping devices; the nontrivial operators act on the same quantum chain $\mathcal{H}_L$.

Theorem 23.1 (Commuting transfer matrices). *The RTT relation implies*

$$[\tau(u), \tau(v)] = 0, \quad \text{for all } u, v. \quad (23.15)$$

Proof. Take $\text{Tr}_a \text{Tr}_b$ of Eq. (23.13). By cyclicity of the trace in the auxiliary spaces, $R_{ab}(u-v)$ can be moved through the trace and canceled. This gives

$$\text{Tr}_a M_a(u) \text{Tr}_b M_b(v) = \text{Tr}_b M_b(v) \text{Tr}_a M_a(u), \quad (23.16)$$

which is exactly Eq. (23.15). $\square$

The commutativity is the spectral reason for integrability. Expanding around the regular point gives mutually commuting charges,

$$\log \tau(u) = \sum_{n \geq 0} u^n Q_n, \quad [Q_m, Q_n] = 0. \quad (23.17)$$

The XXZ Hamiltonian is one of these charges:

$$H_{\text{XXZ}}^{(+)} = 2J \sinh \eta \left. \frac{d}{du} \log \tau(u) \right|_{u=0} - 2J \cosh \eta L, \quad (23.18)$$

which is the same normalization as

$$H_{\text{XXZ}}^{(+)} = J \sum_{j=1}^L (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1)), \quad \Delta = \cosh \eta. \quad (23.19)$$

The constant fixes the energy of the fully polarized reference state to zero.

23.3 The fundamental commutation relations

The RTT relation contains many component equations. The algebraic Bethe ansatz only needs the relations that move $A(u)$ and $D(u)$ through products of B 's. Introduce

$$f(u) = \frac{\sinh(u+\eta)}{\sinh u}, \quad g(u) = \frac{\sinh \eta}{\sinh u}. \quad (23.20)$$

With the monodromy order [Eq. \(23.3\)](#), the needed RTT relations are

$$B(u)B(v) = B(v)B(u), \quad (23.21)$$

$$A(u)B(v) = f(v-u)B(v)A(u) - g(v-u)B(u)A(v), \quad (23.22)$$

$$D(u)B(v) = f(u-v)B(v)D(u) - g(u-v)B(u)D(v). \quad (23.23)$$

The first identity says that Bethe creation operators commute. The second and third identities have two parts. The first terms preserve the set of Bethe rapidities and will produce the transfer-matrix eigenvalue. The second terms replace one rapidity v by the spectral parameter u ; these are the unwanted terms whose cancellation gives the Bethe equations.

Remark 23.2 (Convention dependence). If one reverses the monodromy ordering, changes $R(u-v)$ to $R(v-u)$, or shifts the spectral parameter by $\eta/2$, the displayed formulas may be inverted or shifted. The invariant content is the same: the ratio of vacuum eigenvalues is balanced against the product of two-body scattering factors.

23.4 Bethe vectors

For rapidities $\lambda_1, \dots, \lambda_M$, define the Bethe vector

$$|\lambda_1, \dots, \lambda_M\rangle = B(\lambda_1)B(\lambda_2) \cdots B(\lambda_M) |0\rangle. \quad (23.24)$$

Because the B 's commute, this vector is symmetric under permutations of the rapidities. It lies in the sector with M down spins:

$$S_{\text{tot}}^z |\lambda_1, \dots, \lambda_M\rangle = \left(\frac{L}{2} - M\right) |\lambda_1, \dots, \lambda_M\rangle. \quad (23.25)$$

This matches the coordinate Bethe ansatz, where M was the number of magnons.

Example 23.1 (One Bethe excitation). For $M = 1$,

$$|\lambda\rangle = B(\lambda) |0\rangle. \quad (23.26)$$

Using [Eqs. \(23.22\)](#) and [\(23.23\)](#),

$$\begin{aligned} \tau(u)B(\lambda) |0\rangle &= [\alpha(u)f(\lambda-u) + \delta(u)f(u-\lambda)] B(\lambda) |0\rangle \\ &\quad + g(\lambda-u) [\delta(\lambda) - \alpha(\lambda)] B(u) |0\rangle. \end{aligned} \quad (23.27)$$

The first line is proportional to the original vector. The second line is unwanted. It vanishes exactly when

$$\alpha(\lambda) = \delta(\lambda), \quad \left[\frac{\sinh(\lambda + \eta)}{\sinh \lambda} \right]^L = 1. \quad (23.28)$$

This is the one-magnon momentum quantization condition in algebraic form.

The many-particle calculation is the repeated version of the same mechanism. Move $A(u)$ and $D(u)$ through the string of B 's until they reach the pseudovacuum. The wanted part keeps all rapidities fixed. Each unwanted term replaces one λ_j by u .

23.5 Transfer-matrix eigenvalue and Bethe equations

Acting with $\tau(u) = A(u) + D(u)$ on the Bethe vector gives the off-shell form

$$\tau(u) |\lambda_1, \dots, \lambda_M\rangle = \Lambda(u | \boldsymbol{\lambda}) |\lambda_1, \dots, \lambda_M\rangle + \sum_{j=1}^M \mathcal{F}_j(u | \boldsymbol{\lambda}) B(u) \prod_{\ell \neq j} B(\lambda_\ell) |0\rangle. \quad (23.29)$$

Here the wanted coefficient is

$$\begin{aligned} \Lambda(u | \boldsymbol{\lambda}) &= \alpha(u) \prod_{j=1}^M f(\lambda_j - u) + \delta(u) \prod_{j=1}^M f(u - \lambda_j) \\ &= \sinh^L(u + \eta) \prod_{j=1}^M \frac{\sinh(\lambda_j - u + \eta)}{\sinh(\lambda_j - u)} \\ &\quad + \sinh^L u \prod_{j=1}^M \frac{\sinh(u - \lambda_j + \eta)}{\sinh(u - \lambda_j)}. \end{aligned} \quad (23.30)$$

Up to a nonzero scalar factor, the unwanted coefficient is

$$\mathcal{F}_j(u | \boldsymbol{\lambda}) \propto \delta(\lambda_j) \prod_{\ell \neq j} f(\lambda_j - \lambda_\ell) - \alpha(\lambda_j) \prod_{\ell \neq j} f(\lambda_\ell - \lambda_j). \quad (23.31)$$

Therefore the Bethe vector is an eigenvector of $\tau(u)$ for all u if

$$\frac{\alpha(\lambda_j)}{\delta(\lambda_j)} = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{f(\lambda_j - \lambda_\ell)}{f(\lambda_\ell - \lambda_j)}, \quad j = 1, \dots, M. \quad (23.32)$$

Substituting the six-vertex functions gives the algebraic Bethe equations

$$\left[\frac{\sinh(\lambda_j + \eta)}{\sinh \lambda_j} \right]^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\sinh(\lambda_j - \lambda_\ell + \eta)}{\sinh(\lambda_j - \lambda_\ell - \eta)}, \quad j = 1, \dots, M. \quad (23.33)$$

When these equations hold, all unwanted terms in [Eq. \(23.29\)](#) vanish, and

$$\tau(u) |\lambda_1, \dots, \lambda_M\rangle = \Lambda(u | \boldsymbol{\lambda}) |\lambda_1, \dots, \lambda_M\rangle. \quad (23.34)$$

Principle 23.1 (Bethe equations as analyticity conditions). The transfer-matrix eigenvalue [Eq. \(23.30\)](#) appears to have simple poles at $u = \lambda_j$. The Bethe equations are exactly the conditions that the residues cancel. Thus the same equations can be read either as cancellation of unwanted vectors or as analyticity of the transfer-matrix eigenvalue.

23.6 Hamiltonian eigenvalues

The transfer-matrix eigenvalue contains the eigenvalues of all commuting charges. Using [Eq. \(23.18\)](#), the XXZ energy is

$$E(\boldsymbol{\lambda}) = 2J \sinh \eta \left. \frac{d}{du} \log \Lambda(u | \boldsymbol{\lambda}) \right|_{u=0} - 2J \cosh \eta L. \quad (23.35)$$

For the eigenvalue Eq. (23.30), this gives

$$E(\boldsymbol{\lambda}) = 2J \sinh^2 \eta \sum_{j=1}^M \frac{1}{\sinh \lambda_j \sinh(\lambda_j + \eta)}. \quad (23.36)$$

The empty Bethe state $M = 0$ has $E = 0$, as required by the constant chosen in Eq. (23.19).

The lattice momentum is obtained from the regular-point transfer matrix. Since $R(0) = \sinh \eta P$, one has

$$\tau(0) = (\sinh \eta)^L U, \quad (23.37)$$

where U is the one-site translation operator, with the orientation determined by the monodromy ordering. Therefore

$$U |\boldsymbol{\lambda}\rangle = \left[\prod_{j=1}^M \frac{\sinh(\lambda_j + \eta)}{\sinh \lambda_j} \right] |\boldsymbol{\lambda}\rangle. \quad (23.38)$$

Depending on whether one defines U as a left or right shift, this eigenvalue is e^{iP} or e^{-iP} . This is only an orientation convention.

If one introduces a one-particle momentum by

$$e^{ik(\lambda)} = \frac{\sinh(\lambda + \eta)}{\sinh \lambda}, \quad (23.39)$$

then

$$\frac{2J \sinh^2 \eta}{\sinh \lambda \sinh(\lambda + \eta)} = 4J (\cos k(\lambda) - \cosh \eta), \quad (23.40)$$

so Eq. (23.36) reproduces the coordinate Bethe-ansatz energy

$$E = 4J \sum_{j=1}^M (\cos k_j - \Delta). \quad (23.41)$$

If the opposite translation orientation is used, k is replaced by $-k$ and the energy is unchanged.

23.7 Symmetric rapidities and comparison with the coordinate Bethe ansatz

The rapidity in Eq. (23.33) is tied to the unshifted six-vertex R -matrix. A more symmetric variable is

$$\mu = \lambda + \frac{\eta}{2}. \quad (23.42)$$

Then

$$e^{ik(\mu)} = \frac{\sinh(\mu + \eta/2)}{\sinh(\mu - \eta/2)}, \quad (23.43)$$

and the Bethe equations become

$$\left[\frac{\sinh(\mu_j + \eta/2)}{\sinh(\mu_j - \eta/2)} \right]^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\sinh(\mu_j - \mu_\ell + \eta)}{\sinh(\mu_j - \mu_\ell - \eta)}. \quad (23.44)$$

After analytic continuation $\eta = i\gamma$, this is the standard critical XXZ form

$$\left[\frac{\sinh(\mu_j + i\gamma/2)}{\sinh(\mu_j - i\gamma/2)} \right]^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\sinh(\mu_j - \mu_\ell + i\gamma)}{\sinh(\mu_j - \mu_\ell - i\gamma)}. \quad (23.45)$$

This is the same equation obtained in the coordinate Bethe ansatz, up to the orientation convention for lattice momentum.

The two approaches differ in what they make manifest:

Method	Main calculation	Output
Coordinate Bethe ansatz	Solve contact conditions in real-space wave functions	Two-body scattering phase and momentum quantization
Algebraic Bethe ansatz	Commute $A(u)$ and $D(u)$ through products of B 's	Transfer-matrix eigenvalue and Bethe equations

The coordinate method emphasizes factorized scattering. The algebraic method explains why factorized scattering exists: the RTT relation is the operator form of the Yang–Baxter equation.

23.8 Higher conserved charges

Because the Bethe vector is an eigenvector of $\tau(u)$ for all u , it is also a simultaneous eigenvector of all charges generated by $\log \tau(u)$. If

$$\log \tau(u) = \sum_{n \geq 0} u^n Q_n, \quad (23.46)$$

then

$$Q_n |\boldsymbol{\lambda}\rangle = q_n(\boldsymbol{\lambda}) |\boldsymbol{\lambda}\rangle, \quad q_n(\boldsymbol{\lambda}) = \frac{1}{n!} \frac{d^n}{du^n} \log \Lambda(u | \boldsymbol{\lambda}) \Big|_{u=0} \quad (23.47)$$

up to the same normalization convention used in defining the charges. The Hamiltonian is the first nontrivial local charge. Higher Q_n have larger spatial support and provide the extensive set of conservation laws characteristic of an integrable chain.

Remark 23.3 (Why this is stronger than diagonalizing one Hamiltonian). A generic many-body Hamiltonian may be diagonalized in principle, but its eigenvectors do not usually diagonalize a whole analytic family of local and quasilocal conserved charges. The algebraic Bethe ansatz gives simultaneous eigenvectors of the transfer matrix for all spectral parameters. This is the finite-size algebraic signature of integrability.

23.9 Trigonometric and rational limits

For $|\Delta| < 1$, set

$$\eta = i\gamma, \quad 0 < \gamma < \pi, \quad \Delta = \cos \gamma. \quad (23.48)$$

The same algebraic derivation applies, with $\sinh$ functions analytically continued. In the symmetric rapidity convention, the Bethe equations are exactly [Eq. \(23.45\)](#).

The isotropic XXX chain is the rational degeneration. Scaling the spectral parameter and anisotropy together produces the rational R -matrix

$$R_{\text{XXX}}(u) = u \mathbf{1} + iP, \quad \text{up to an overall scalar normalization.} \quad (23.49)$$

In the usual XXX rapidity variable, the Bethe equations become

$$\left(\frac{\lambda_j + i/2}{\lambda_j - i/2} \right)^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\lambda_j - \lambda_\ell + i}{\lambda_j - \lambda_\ell - i}. \quad (23.50)$$

This rational limit will reappear when the XXX chain is discussed as a special case of the XXZ chain.

23.10 Common pitfalls

Remark 23.4 (The operators A, B, C, D are nonlocal). Although $R_{aj}(u)$ is local in the quantum site j , the monodromy entries $A(u), B(u), C(u), D(u)$ are products over the whole chain. The operator $B(u)$ is therefore not a local spin-lowering operator. It is a spectral-parameter dependent, nonlocal creation operator for Bethe quasiparticles.

Remark 23.5 (Bethe roots cannot sit at generic poles). The formulas above contain denominators such as $\sinh \lambda_j$ and $\sinh(\lambda_j - \lambda_\ell)$. Ordinary Bethe roots are assumed to avoid these poles. Singular solutions can occur at special points and require a limiting procedure. They are not part of the basic algebraic derivation.

Remark 23.6 (Completeness is a separate question). The algebraic Bethe ansatz constructs eigenvectors whenever the Bethe equations hold and the resulting vector is nonzero. Showing that all eigenvectors are obtained requires a separate completeness analysis. For the finite XXZ chain this is well understood, but it is conceptually distinct from deriving the Bethe equations.

23.11 Summary

The algebraic Bethe ansatz for the XXZ chain can be compressed into the following sequence:

$$\boxed{\begin{aligned} R(u) \text{ satisfies YBE} &\implies R_{ab}(u-v)M_a(u)M_b(v) = M_b(v)M_a(u)R_{ab}(u-v), \\ M_a(u) &= \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix}_a, \quad \tau(u) = A(u) + D(u), \\ |\boldsymbol{\lambda}\rangle &= B(\lambda_1) \cdots B(\lambda_M) |0\rangle, \\ \tau(u) |\boldsymbol{\lambda}\rangle &= \Lambda(u | \boldsymbol{\lambda}) |\boldsymbol{\lambda}\rangle \quad \text{if } \boldsymbol{\lambda} \text{ satisfies the Bethe equations,} \\ \left[\frac{\sinh(\lambda_j + \eta)}{\sinh \lambda_j} \right]^L &= \prod_{\ell \neq j} \frac{\sinh(\lambda_j - \lambda_\ell + \eta)}{\sinh(\lambda_j - \lambda_\ell - \eta)}. \end{aligned}} \quad (23.51)$$

The Bethe equations arise from cancellation of unwanted terms in the RTT algebra. The transfer-matrix eigenvalue then gives the complete set of commuting-charge eigenvalues, including the XXZ Hamiltonian energy. This is the algebraic counterpart of the factorized scattering picture developed in the coordinate Bethe-ansatz chapter.

Chapter 24

Further Integrable Models

The previous two chapters solved the XXZ chain by two complementary methods. The coordinate Bethe ansatz emphasized factorized scattering of magnons, while the algebraic Bethe ansatz emphasized the Yang–Baxter algebra of the six-vertex transfer matrix. This chapter explains how the same ideas extend beyond the nearest-neighbor XXZ chain. The goal is not to give a second complete solution of every model. Instead, the purpose is to identify the structural data that make each model solvable and to locate it within the mother-model organization of these notes.

We discuss three classes of examples:

- (i) the isotropic XXX chain, obtained as the rational limit of the XXZ chain;
- (ii) the one-dimensional Hubbard model, whose exact solution requires a nested Bethe ansatz because particles carry both charge and spin;
- (iii) long-range integrable models, represented by the Haldane–Shastry spin chain and the Calogero–Sutherland model.

The recurring theme is that integrability is not synonymous with free fermions. In these models the many-body spectrum is controlled by nontrivial quantization conditions, hidden symmetries, or special long-range interactions. The excitations are still strongly constrained, but they need not be free Bogoliubov quasiparticles.

24.1 The XXX limit

The XXZ Hamiltonian used in [Chapters 22](#) and [23](#) was

$$H_{\text{XXZ}}^{(+)} = J \sum_{j=1}^L \left(\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1) \right), \quad (24.1)$$

with periodic boundary conditions. The isotropic XXX chain is the special point

$$\Delta \rightarrow 1, \quad H_{\text{XXX}}^{(+)} = J \sum_{j=1}^L (\boldsymbol{\sigma}_j \cdot \boldsymbol{\sigma}_{j+1} - 1). \quad (24.2)$$

Equivalently, in terms of spin operators $\mathbf{S}_j = \boldsymbol{\sigma}_j/2$,

$$H_{\text{XXX}}^{(+)} = 4J \sum_{j=1}^L \left(\mathbf{S}_j \cdot \mathbf{S}_{j+1} - \frac{1}{4} \right). \quad (24.3)$$

The constant is chosen so that the fully polarized reference state has zero energy, as in the XXZ Bethe ansatz chapters.

Remark 24.1 (Sign convention). With $J > 0$ in Eq. (24.2), a one-magnon state has $\varepsilon(k) = 4J(\cos k - 1) \leq 0$ relative to the ferromagnetic reference state. Many condensed-matter texts instead write the antiferromagnetic XXX Hamiltonian as $J_{\text{AF}} \sum_j \mathbf{S}_j \cdot \mathbf{S}_{j+1}$ with $J_{\text{AF}} > 0$. The two conventions differ by an overall sign and an additive constant. The Bethe equations are convention-independent; only the energy formula changes by this normalization.

24.1.1 Rational R-matrix

The XXX chain is also obtained from the six-vertex R -matrix by taking the rational limit. In the hyperbolic convention of the algebraic Bethe ansatz chapter,

$$R_{\text{XXZ}}(u) = \begin{pmatrix} \sinh(u + \eta) & 0 & 0 & 0 \\ 0 & \sinh u & \sinh \eta & 0 \\ 0 & \sinh \eta & \sinh u & 0 \\ 0 & 0 & 0 & \sinh(u + \eta) \end{pmatrix}, \quad \Delta = \cosh \eta. \quad (24.4)$$

Sending $u \mapsto \eta u$ and then taking $\eta \rightarrow 0$ gives, up to an irrelevant scalar normalization,

$$R_{\text{XXX}}(u) = u \mathbf{1} + P, \quad (24.5)$$

where P permutes the two spin-1/2 spaces:

$$P |\alpha\rangle \otimes |\beta\rangle = |\beta\rangle \otimes |\alpha\rangle. \quad (24.6)$$

This rational R -matrix satisfies the Yang–Baxter equation

$$R_{12}(u - v) R_{13}(u - w) R_{23}(v - w) = R_{23}(v - w) R_{13}(u - w) R_{12}(u - v). \quad (24.7)$$

It is regular because

$$R_{\text{XXX}}(0) = P. \quad (24.8)$$

Therefore the logarithmic derivative of the rational transfer matrix gives a nearest-neighbor Hamiltonian. Using

$$P_{j,j+1} = \frac{1}{2} (\mathbf{1} + \boldsymbol{\sigma}_j \cdot \boldsymbol{\sigma}_{j+1}), \quad (24.9)$$

one obtains the isotropic exchange interaction.

Principle 24.1 (Trigonometric versus rational integrability). The XXZ chain is controlled by the trigonometric six-vertex R -matrix. The XXX chain is controlled by its rational degeneration. The solution method is unchanged: monodromy matrix, transfer matrix, RTT algebra, Bethe vectors, Bethe equations. What changes is the analytic form of the scattering phase.

24.1.2 Bethe equations in rapidity form

The rational rapidity variable λ is related to the one-magnon momentum by

$$e^{ik} = \frac{\lambda + \frac{i}{2}}{\lambda - \frac{i}{2}}. \quad (24.10)$$

For M magnons, the XXX Bethe equations are

$$\left(\frac{\lambda_j + \frac{i}{2}}{\lambda_j - \frac{i}{2}}\right)^L = \prod_{\substack{\ell=1 \\ \ell \neq j}}^M \frac{\lambda_j - \lambda_\ell + i}{\lambda_j - \lambda_\ell - i}, \quad j = 1, \dots, M. \quad (24.11)$$

The corresponding two-body scattering amplitude is

$$S_{\text{XXX}}(\lambda_j - \lambda_\ell) = \frac{\lambda_j - \lambda_\ell + i}{\lambda_j - \lambda_\ell - i}. \quad (24.12)$$

Thus the many-body quantization condition still has the universal Bethe form

$$e^{ik_j L} = \prod_{\ell \neq j} S(k_j, k_\ell), \quad (24.13)$$

but the scattering phase has become rational rather than trigonometric.

Using Eq. (24.10), one finds

$$\cos k = \frac{\lambda^2 - \frac{1}{4}}{\lambda^2 + \frac{1}{4}}. \quad (24.14)$$

Therefore the energy in the normalization of Eq. (24.2) is

$$E = 4J \sum_{j=1}^M (\cos k_j - 1) = -2J \sum_{j=1}^M \frac{1}{\lambda_j^2 + \frac{1}{4}}. \quad (24.15)$$

The total lattice momentum is

$$e^{iP} = \prod_{j=1}^M \frac{\lambda_j + \frac{i}{2}}{\lambda_j - \frac{i}{2}}. \quad (24.16)$$

24.1.3 Enhanced $SU(2)$ symmetry

For generic Δ , the XXZ chain conserves only the $U(1)$ charge S_{tot}^z . At $\Delta = 1$, the symmetry is enhanced to full spin rotation symmetry:

$$[H_{\text{XXX}}^{(+)}, S_{\text{tot}}^\alpha] = 0, \quad S_{\text{tot}}^\alpha = \frac{1}{2} \sum_{j=1}^L \sigma_j^\alpha, \quad \alpha = x, y, z. \quad (24.17)$$

This enhancement has an important consequence. Bethe states constructed above the all-up pseudovacuum are usually highest-weight states of $SU(2)$ multiplets. The remaining members of the multiplet are obtained by acting with S_{tot}^- . This is why completeness questions in the XXX chain are slightly more delicate than merely counting finite rapidity solutions of Eq. (24.11).

Remark 24.2 (Infinite rapidities). In the XXX chain, acting with the global spin-lowering operator can be viewed as adding a Bethe root at infinity. This is a useful algebraic way to understand how $SU(2)$ descendants fit into the Bethe-ansatz framework.

24.2 Hubbard model and nested Bethe ansatz

The one-dimensional Hubbard model is the canonical example of an interacting fermion model that is exactly solvable but not reducible to a single-species magnon Bethe ansatz. Its local degrees of freedom are

$$|0\rangle, \quad |\uparrow\rangle, \quad |\downarrow\rangle, \quad |\uparrow\downarrow\rangle, \quad (24.18)$$

so each site carries both charge and spin. The Hamiltonian on a periodic chain is

$$H_{\text{Hub}} = -t \sum_{j=1}^L \sum_{\sigma=\uparrow,\downarrow} \left(c_{j\sigma}^\dagger c_{j+1,\sigma} + c_{j+1,\sigma}^\dagger c_{j\sigma} \right) + U \sum_{j=1}^L n_{j\uparrow} n_{j\downarrow}. \quad (24.19)$$

Here

$$n_{j\sigma} = c_{j\sigma}^\dagger c_{j\sigma}, \quad c_{L+1,\sigma} = c_{1\sigma}. \quad (24.20)$$

The hopping amplitude is t , and U is the on-site interaction.

24.2.1 Why a nested ansatz is needed

In the XXZ chain, the Bethe particles are down spins moving in an all-up reference state. There is only one species of quasiparticle, so every collision is described by a scalar two-body phase. In the Hubbard model, a particle carries charge momentum and spin. When two fermions scatter, their spin labels can be rearranged. The two-body scattering matrix therefore acts on an internal spin space. Diagonalizing the charge problem leaves a second spin problem to be solved. This second step is the nested Bethe ansatz.

The exact solution is parametrized by two sets of variables:

$$\{k_j\}_{j=1}^N, \quad \{\lambda_\alpha\}_{\alpha=1}^M, \quad (24.21)$$

where N is the total number of electrons and M is the number of down-spin electrons. The k_j are charge momenta, while the λ_α are spin rapidities.

Principle 24.2 (Nested Bethe ansatz). When the two-body scattering matrix carries an internal index, Bethe ansatz becomes a two-level problem. First one writes a coordinate Bethe ansatz for charge motion. Then one diagonalizes the internal spin scattering problem by a second Bethe ansatz. The final answer contains coupled Bethe equations for charge and spin rapidities.

24.2.2 Lieb–Wu equations

Introduce the dimensionless interaction parameter

$$u = \frac{U}{4t}. \quad (24.22)$$

For the Hamiltonian Eq. (24.19), the Lieb–Wu Bethe equations are

$$e^{ik_j L} = \prod_{\alpha=1}^M \frac{\sin k_j - \lambda_\alpha + iu}{\sin k_j - \lambda_\alpha - iu}, \quad j = 1, \dots, N, \quad (24.23)$$

$$\prod_{j=1}^N \frac{\lambda_\alpha - \sin k_j + iu}{\lambda_\alpha - \sin k_j - iu} = \prod_{\substack{\beta=1 \\ \beta \neq \alpha}}^M \frac{\lambda_\alpha - \lambda_\beta + 2iu}{\lambda_\alpha - \lambda_\beta - 2iu}, \quad \alpha = 1, \dots, M. \quad (24.24)$$

The energy and total momentum are

$$E = -2t \sum_{j=1}^N \cos k_j, \quad P = \sum_{j=1}^N k_j \pmod{2\pi}, \quad (24.25)$$

for the unshifted convention in Eq. (24.19). If one uses the particle-hole symmetric Hamiltonian

$$H'_{\text{Hub}} = -t \sum_{j,\sigma} \left(c_{j\sigma}^\dagger c_{j+1,\sigma} + c_{j+1,\sigma}^\dagger c_{j\sigma} \right) + U \sum_j \left(n_{j\uparrow} - \frac{1}{2} \right) \left(n_{j\downarrow} - \frac{1}{2} \right), \quad (24.26)$$

then the same Bethe roots give the shifted energy

$$E' = E - \frac{U}{2}N + \frac{U}{4}L. \quad (24.27)$$

Remark 24.3 (Why $\sin k$ appears). The Hubbard equations are not obtained by simply replacing the XXZ rapidity by k . The charge dispersion is that of a tight-binding lattice fermion, $-2t \cos k$, while the spin scattering depends on $\sin k$. This mixture is a characteristic feature of the Hubbard nested solution.

24.2.3 Half filling and the Mott gap

At half filling, $N = L$. For repulsive interaction $U > 0$, the exact solution shows that the charge sector is gapped while the spin sector remains gapless. This is the one-dimensional Mott-insulating mechanism in its cleanest exact form. The Hilbert space still allows charge fluctuations, but the low-energy sector separates into

$$\text{gapped charge excitations} \quad \text{and} \quad \text{gapless spin excitations.} \quad (24.28)$$

For large positive U/t , virtual hopping produces the effective antiferromagnetic Heisenberg chain

$$H_{\text{eff}} = J_{\text{ex}} \sum_{j=1}^L \left(\mathbf{S}_j \cdot \mathbf{S}_{j+1} - \frac{1}{4} \right), \quad J_{\text{ex}} = \frac{4t^2}{U}, \quad (24.29)$$

inside the subspace with one electron per site. Thus the XXX chain appears as the spin-sector low-energy limit of the repulsive half-filled Hubbard model.

Example 24.1 (From Hubbard to Heisenberg). Consider two neighboring sites with one electron on each site. A hopping event creates a virtual doubly occupied site and costs energy U . A second hopping event returns to the singly occupied subspace. Second-order perturbation theory therefore gives an exchange scale t^2/U . Fermionic statistics select the antiferromagnetic exchange, producing $J_{\text{ex}} = 4t^2/U$.

24.2.4 Integrability structure

The Hubbard model is integrable, but its integrability is more subtle than that of the six-vertex model. The relevant R -matrix is not the elementary six-vertex R -matrix; it is a higher structure compatible with two coupled spin species. Equivalently, one may view the model as having two intertwined XX-type hopping problems connected by the on-site interaction. The important lesson for these notes is the following:

The Hubbard model is solvable not because it is quadratic, but because its many-body scattering is still factorized after accounting for the nested charge and spin structure.

24.3 Long-range integrable models

The nearest-neighbor XXX, XXZ, and XYZ chains teach that local Hamiltonians can be integrable when they arise from logarithmic derivatives of commuting transfer matrices. There is another important class of integrable models whose defining feature is not nearest-neighbor locality, but a very special long-range interaction. The two standard examples are the Haldane–Shastry spin chain and the Calogero–Sutherland particle model.

24.3.1 Haldane–Shastry chain

Place L spin-1/2 degrees of freedom at equally spaced points on the unit circle,

$$z_j = e^{2\pi i j/L}, \quad j = 1, \dots, L. \quad (24.30)$$

The Haldane–Shastry Hamiltonian is

$$H_{\text{HS}} = J \left(\frac{\pi}{L} \right)^2 \sum_{1 \leq j < \ell \leq L} \frac{\mathbf{S}_j \cdot \mathbf{S}_\ell - \frac{1}{4}}{\sin^2 \left(\frac{\pi(j-\ell)}{L} \right)}. \quad (24.31)$$

The interaction decays as the inverse square of the chord distance. Although this Hamiltonian is long-ranged, it is more exactly solvable than a generic nearest-neighbor interacting spin chain.

At half filling of down spins, an exact ground-state wave function can be written in terms of the down-spin coordinates $z_1, \dots, z_{L/2}$ as a lattice Jastrow state,

$$\Psi_{\text{HS}}(z_1, \dots, z_{L/2}) = \prod_{a < b} (z_a - z_b)^2 \prod_{a=1}^{L/2} z_a. \quad (24.32)$$

This is the spin-chain analogue of a Laughlin-type wave function. The square Jastrow factor enforces strong correlations between spin flips.

The elementary excitations are spinons. They carry spin 1/2 rather than spin 1, and they obey fractional exclusion statistics. This is qualitatively different from the magnons of the ferromagnetic Bethe ansatz, which are spin-flip particles above a polarized vacuum.

Remark 24.4 (Yangian symmetry). The hidden symmetry of the Haldane–Shastry chain is a Yangian extension of $SU(2)$. This enlarged symmetry is responsible for large spectral degeneracies and for the unusually simple spinon structure. It is the long-range analogue of the algebraic constraints imposed by the Yang–Baxter equation in nearest-neighbor chains.

24.3.2 Calogero–Sutherland model

The Calogero–Sutherland model is a continuum many-body system on a circle with inverse-square interactions. A common normalization is

$$H_{\text{CS}} = - \sum_{j=1}^N \frac{\partial^2}{\partial x_j^2} + g(g-1) \left(\frac{\pi}{L} \right)^2 \sum_{1 \leq j < \ell \leq N} \frac{1}{\sin^2 \left(\frac{\pi(x_j - x_\ell)}{L} \right)}. \quad (24.33)$$

The exact ground state has the Jastrow form

$$\Psi_0(x_1, \dots, x_N) = \prod_{j < \ell} \left| \sin \frac{\pi(x_j - x_\ell)}{L} \right|^g. \quad (24.34)$$

The excited states are obtained by multiplying this ground state by symmetric polynomials. More precisely, Jack polynomials provide the natural eigenbasis. Thus the Calogero–Sutherland model is solvable by special-function and representation-theoretic methods rather than by a nearest-neighbor coordinate Bethe ansatz.

The coupling g controls the statistical interaction between particles. At $g = 0$ or $g = 1$, the inverse-square potential vanishes, corresponding to free bosonic or fermionic endpoints depending on the imposed symmetry. For general g , the model realizes fractional exclusion statistics in a continuum setting.

24.3.3 Freezing trick and relation to spin chains

The Haldane–Shastry chain and the Calogero–Sutherland model are not unrelated. In the strong-coupling limit of spin Calogero–Sutherland-type models, the particle coordinates freeze near equally spaced classical equilibrium positions on the circle. The remaining low-energy dynamics is carried by the internal spin variables. This limiting procedure produces inverse-square spin-chain Hamiltonians of Haldane–Shastry type.

Schematically,

$$\text{spin Calogero–Sutherland particles} \xrightarrow[\text{coordinates freeze}]{\text{large coupling}} \text{Haldane–Shastry spin chain.} \quad (24.35)$$

This relation is called the freezing trick. It is another example of the mother-model principle: a continuum integrable particle model contains a lattice integrable spin chain as a controlled limit.

24.4 Comparison of integrable structures

The examples in this chapter illustrate several distinct meanings of exact solvability. They are summarized in [Table 24.1](#).

The following hierarchy is useful:

$$\begin{aligned} \text{free fermion solvability} &\subset \text{Bethe/Yang–Baxter integrability} \subset \text{broader exact solvability,} \\ &\text{but none of these inclusions should be interpreted as equality.} \end{aligned} \quad (24.36)$$

Free-fermion models are solvable because the Hamiltonian becomes quadratic. Bethe-ansatz models are solvable because many-body scattering factorizes. Long-range inverse-square models are solvable because the interaction is tuned to special symmetry and polynomial structures. Commuting-projector models, which appear in the next part of the notes, are solvable for yet another reason: the Hamiltonian decomposes into mutually compatible local constraints.

24.5 What should be remembered

The main points of this chapter are:

- (i) The XXX chain is the rational limit of the XXZ chain. Its Bethe equations are obtained by replacing trigonometric scattering data by rational scattering data.
- (ii) The XXX point has enhanced $SU(2)$ symmetry. Bethe roots at infinity and highest-weight multiplets are natural consequences of this symmetry.

Model	Solvability mechanism	Main spectral data
XXX chain	Rational R -matrix; Bethe ansatz; enhanced $SU(2)$ symmetry	Rapidities satisfying rational Bethe equations
XXZ chain	Trigonometric six-vertex R -matrix; coordinate/algebraic Bethe ansatz	Rapidities satisfying trigonometric Bethe equations
XYZ chain	Elliptic eight-vertex R -matrix; Baxter TQ relation	Elliptic spectral parameter and functional equations
Hubbard chain	Nested Bethe ansatz; factorized scattering with charge and spin variables	Charge momenta k_j and spin rapidities λ_α satisfying Lieb–Wu equations
Haldane–Shastry chain	Inverse-square interaction; Yangian symmetry; spinon basis	Fractional-statistics spinons and exact Jastrow ground state
Calogero–Sutherland model	Inverse-square continuum interaction; Jack-polynomial eigenfunctions	Pseudomomenta and Jack-polynomial labels

Table 24.1: Representative integrable models beyond the basic XXZ coordinate Bethe ansatz. The word “integrable” refers to different but related structures: rational, trigonometric, or elliptic R -matrices; nested Bethe ansatz; Yangian symmetry; or inverse-square special-function solvability.

- (iii) The Hubbard model requires a nested Bethe ansatz because the particles carry both charge and spin. Its solution is encoded in the coupled Lieb–Wu equations.
- (iv) At half filling and repulsive interaction, the Hubbard model has a gapped charge sector and a gapless spin sector. In the large- U limit, the low-energy spin dynamics reduces to an antiferromagnetic Heisenberg chain.
- (v) Long-range inverse-square models such as the Haldane–Shastry and Calogero–Sutherland models are integrable for reasons different from the nearest-neighbor transfer-matrix construction. Their exact wave functions and spectra are governed by Jastrow factors, Yangian symmetry, and special polynomials.

The common lesson is that exact solvability is a structural property. Quadratic fermions, Yang–Baxter algebras, nested scattering, inverse-square interactions, and commuting projectors are different mechanisms, but each replaces the generic many-body problem by a constrained algebraic or analytic problem.

Part V

**Two-Dimensional Quantum Exact
Solvability**

Chapter 25

Commuting-Projector Models and the Toric Code

The previous part studied interacting one-dimensional quantum chains whose exact solvability comes from factorized scattering and the Yang–Baxter equation. In this part we move to two spatial dimensions. The first solvable structure is very different from Bethe-ansatz integrability: the Hamiltonian is a sum of local terms that commute with one another. Such models are exactly solvable because all local conserved projectors can be diagonalized simultaneously.

The toric code is the canonical example. It is simultaneously

- (i) a stabilizer Hamiltonian;
- (ii) a commuting-projector model;
- (iii) a lattice $\mathbb{Z}_2$ gauge theory at zero gauge coupling;
- (iv) the simplest fixed-point model of intrinsic topological order.

It is therefore an ideal first model for two-dimensional quantum exact solvability. Its solution does not require Fourier transforms, Bogoliubov rotations, or Bethe rapidities. Instead, the spectrum follows from a finite set of mutually commuting $\mathbb{Z}_2$ -valued operators, while its physical content is encoded in nonlocal string operators.

25.1 Commuting-projector solvability

Let $\mathcal{H}$ be a many-body Hilbert space and suppose that a Hamiltonian can be written as

$$H = \sum_{\alpha} E_{\alpha} P_{\alpha}, \quad P_{\alpha}^2 = P_{\alpha}, \quad [P_{\alpha}, P_{\beta}] = 0. \quad (25.1)$$

Then all P_{α} can be diagonalized simultaneously. The many-body spectrum is reduced to the problem of finding which binary eigenvalue patterns $p_{\alpha} \in \{0, 1\}$ are compatible with the algebraic constraints among the projectors.

Equivalently, if each local term is written in terms of an involution $Q_{\alpha}^2 = 1$,

$$P_{\alpha} = \frac{1 - Q_{\alpha}}{2}, \quad Q_{\alpha} = 1 - 2P_{\alpha}, \quad (25.2)$$

then exact solvability is the statement that the commuting stabilizers Q_{α} admit simultaneous eigenvalues $q_{\alpha} = \pm 1$.

Definition 25.1 (Commuting-projector Hamiltonian). A local Hamiltonian is called a commuting-projector Hamiltonian if it can be written as a sum of local projectors satisfying Eq. (25.1). If the ground state minimizes every local projector separately, the Hamiltonian is called frustration-free.

Remark 25.1 (Exact solvability versus integrability). A commuting-projector model has an extensive set of strictly local conserved operators. This is stronger than ordinary quantum integrability in one dimension, where conserved charges are usually quasilocal or generated by a transfer matrix. However, the strength of the solvability is also more specialized: a generic perturbation that destroys commutativity usually cannot be solved by the same method. The toric code should therefore be viewed as an exactly solvable fixed point, not as a representative of a continuously soluble family in the Bethe-ansatz sense.

The simplest nontrivial commuting-projector models are stabilizer Hamiltonians. There one has N qubits and an Abelian subgroup $\mathcal{S}$ of the Pauli group, generated by mutually commuting Pauli strings S_a . The common $+1$ eigenspace

$$\mathcal{H}_{\text{code}} = \{|\psi\rangle \in (\mathbb{C}^2)^{\otimes N} : S_a |\psi\rangle = |\psi\rangle \text{ for all } a\} \quad (25.3)$$

is the ground space of a positive-semidefinite Hamiltonian

$$H_{\text{stab}} = \sum_a \frac{1 - S_a}{2}. \quad (25.4)$$

If r of the stabilizer generators are independent, then

$$\dim \mathcal{H}_{\text{code}} = 2^{N-r}. \quad (25.5)$$

This counting formula will give the toric-code ground-state degeneracy on a closed surface.

25.2 Lattice, Hilbert space, and Hamiltonian

Consider an oriented square lattice embedded on a closed orientable surface Σ_g of genus g . The set of vertices, edges, and plaquettes is denoted by

$$V, \quad E, \quad P, \quad (25.6)$$

with cardinalities $N_v = |V|$, $N_e = |E|$, and $N_p = |P|$. A spin-1/2 degree of freedom lives on every edge $\ell \in E$. Thus

$$\mathcal{H}_{\text{TC}} = \bigotimes_{\ell \in E} \mathbb{C}_\ell^2, \quad \dim \mathcal{H}_{\text{TC}} = 2^{N_e}. \quad (25.7)$$

We write Pauli operators on edge ℓ as σ_ℓ^x , σ_ℓ^y , and σ_ℓ^z .

For each vertex $s \in V$, define the star operator

$$A_s = \prod_{\ell \ni s} \sigma_\ell^x, \quad (25.8)$$

where the product is over the edges incident on s . For each plaquette $p \in P$, define the plaquette operator

$$B_p = \prod_{\ell \in \partial p} \sigma_\ell^z, \quad (25.9)$$

where the product is over the boundary edges of p . On a square lattice each star and plaquette contains four edges, but the algebraic construction only requires that the lattice be a cellulation of a closed oriented surface.

The toric-code Hamiltonian is

$$H_{\text{TC}} = -J_e \sum_{s \in V} A_s - J_m \sum_{p \in P} B_p, \quad J_e, J_m > 0. \quad (25.10)$$

Equivalently, up to an additive constant,

$$H_{\text{TC}} = 2J_e \sum_{s \in V} \frac{1 - A_s}{2} + 2J_m \sum_{p \in P} \frac{1 - B_p}{2} - (J_e N_v + J_m N_p). \quad (25.11)$$

The terms $(1 - A_s)/2$ and $(1 - B_p)/2$ are projectors onto violated star and plaquette constraints.

Remark 25.2 (Naming convention). The notation J_e and J_m anticipates the excitation content. A violated star, $A_s = -1$, is called an electric charge or e particle. A violated plaquette, $B_p = -1$, is called a magnetic flux or m particle. The model is self-dual under exchange of direct and dual lattices together with $A_s \leftrightarrow B_p$, $J_e \leftrightarrow J_m$, and $\sigma^x \leftrightarrow \sigma^z$.

25.3 Commutativity and simultaneous diagonalization

The toric code is exactly solvable because every star and every plaquette term commutes.

Proposition 25.1 (Toric-code stabilizer algebra). *The operators A_s and B_p satisfy*

$$A_s^2 = B_p^2 = \mathbf{1}, \quad [A_s, A_{s'}] = 0, \quad [B_p, B_{p'}] = 0, \quad [A_s, B_p] = 0. \quad (25.12)$$

Proof. The first identity follows from $(\sigma^x)^2 = (\sigma^z)^2 = \mathbf{1}$. Two star operators contain only σ^x operators, hence commute. Two plaquette operators contain only σ^z operators, hence commute.

It remains to check $[A_s, B_p] = 0$. The operators A_s and B_p can only fail to commute on edges shared by the star s and the plaquette p . On a shared edge,

$$\sigma_\ell^x \sigma_\ell^z = -\sigma_\ell^z \sigma_\ell^x. \quad (25.13)$$

On a square lattice embedded without boundary, a star and a plaquette share either zero or two edges. Therefore moving A_s through B_p produces the factor $(-1)^n$, where $n \in \{0, 2\}$. Since $(-1)^n = +1$, the two operators commute. $\square$

Because of [Eq. \(25.12\)](#), the Hamiltonian can be diagonalized by simultaneous eigenstates

$$A_s |\psi\rangle = a_s |\psi\rangle, \quad B_p |\psi\rangle = b_p |\psi\rangle, \quad a_s, b_p \in \{+1, -1\}. \quad (25.14)$$

The energy of such a state is

$$E[\{a_s\}, \{b_p\}] = -J_e \sum_{s \in V} a_s - J_m \sum_{p \in P} b_p. \quad (25.15)$$

Thus the full spectrum is determined once we know the allowed stabilizer eigenvalue patterns and their degeneracies.

On a closed surface there are two global relations:

$$\prod_{s \in V} A_s = 1, \quad \prod_{p \in P} B_p = 1. \quad (25.16)$$

The first relation holds because every edge touches exactly two vertices and therefore appears twice in $\prod_s A_s$. The second holds because every edge belongs to exactly two plaquette boundaries and therefore appears twice in $\prod_p B_p$.

It follows that the number of star violations and the number of plaquette violations must both be even:

$$\prod_{s \in V} a_s = 1, \quad \prod_{p \in P} b_p = 1. \quad (25.17)$$

In particular, on a closed surface a single isolated e or m excitation cannot be created by a finite local operator. Excitations are created in pairs, or else one must use open boundaries or punctures.

25.4 Ground states as flat $\mathbb{Z}_2$ gauge fields

The ground-state conditions are

$$A_s |\Omega\rangle = |\Omega\rangle, \quad B_p |\Omega\rangle = |\Omega\rangle \quad (25.18)$$

for all vertices s and plaquettes p . These equations have a useful gauge theory interpretation. The σ^z -basis assigns a $\mathbb{Z}_2$ variable to each edge. The plaquette constraint $B_p = +1$ says that the $\mathbb{Z}_2$ flux through every plaquette is trivial. The star operator A_s flips all edge variables adjacent to a vertex; it acts as a local $\mathbb{Z}_2$ gauge transformation. Thus the toric-code ground states are gauge-invariant superpositions of flat $\mathbb{Z}_2$ connections.

A concrete construction is obtained as follows. Let $|0_z\rangle$ be the state with $\sigma_\ell^z = +1$ for all edges. It already satisfies $B_p |0_z\rangle = |0_z\rangle$ for every plaquette. The projector onto the +1-eigenspace of all star operators is

$$\Pi_A = \prod_{s \in V} \frac{1 + A_s}{2}. \quad (25.19)$$

Therefore

$$|\Omega_0\rangle \propto \Pi_A |0_z\rangle \quad (25.20)$$

is a ground state in the topologically trivial sector. In the σ^z basis, this state is an equal-amplitude superposition of closed-loop configurations. The word “closed” means that every vertex is touched by an even number of flipped edges; this is precisely the no-charge condition $A_s = +1$.

On a torus there are additional sectors distinguished by noncontractible $\mathbb{Z}_2$ fluxes. Local star operations cannot change these global fluxes, so one obtains four independent ground states rather than one. We derive this counting in [Section 25.7](#).

25.5 String operators and anyon excitations

The elementary excitations are defects of the stabilizer constraints.

Definition 25.2 (Electric and magnetic excitations). A vertex s with $A_s = -1$ is an electric charge, denoted e . A plaquette p with $B_p = -1$ is a magnetic flux, denoted m . The composite excitation $\epsilon = e \times m$ is obtained by placing an e and an m at the same position in the continuum topological description.

The energy cost of one violated star is $2J_e$, because its contribution to Eq. (25.10) changes from $-J_e$ to $+J_e$. Similarly, one violated plaquette costs $2J_m$. On a closed surface the minimal locally created excitations are pairs, with energies

$$\Delta E_{e\text{-pair}} = 4J_e, \quad \Delta E_{m\text{-pair}} = 4J_m. \quad (25.21)$$

25.5.1 Electric strings

Let γ be a path on the direct lattice, i.e. a sequence of edges. Define the electric string operator

$$W_e(\gamma) = \prod_{\ell \in \gamma} \sigma_\ell^z. \quad (25.22)$$

Since $W_e(\gamma)$ is made of σ^z operators, it commutes with all plaquette operators B_p . It anticommutes with a star operator A_s if and only if s is an endpoint of γ . Therefore an open electric string creates or moves a pair of e excitations at its endpoints:

$$A_s W_e(\gamma) = \begin{cases} -W_e(\gamma) A_s, & s \in \partial\gamma, \\ W_e(\gamma) A_s, & s \notin \partial\gamma. \end{cases} \quad (25.23)$$

For a closed path, $\partial\gamma = \emptyset$, the string operator commutes with the Hamiltonian.

25.5.2 Magnetic strings

Let γ^* be a path on the dual lattice. It crosses a set of direct lattice edges. Define

$$W_m(\gamma^*) = \prod_{\ell \perp \gamma^*} \sigma_\ell^x. \quad (25.24)$$

This operator commutes with all star operators A_s , but it anticommutes with the two plaquette operators at the endpoints of γ^* . Thus an open magnetic string creates or moves a pair of m fluxes.

25.5.3 Fusion rules

Because e and m excitations are $\mathbb{Z}_2$ -valued defects, bringing two identical excitations together annihilates them. The fusion rules are

$$e \times e = 1, \quad m \times m = 1, \quad e \times m = \epsilon, \quad \epsilon \times \epsilon = 1. \quad (25.25)$$

The set of anyon types is therefore

$$\{1, e, m, \epsilon\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \quad (25.26)$$

under fusion.

25.6 Mutual statistics

The nontrivial topological content of the toric code is not the energy of the excitations, which is completely local, but their braiding statistics. Let γ be a direct-lattice path and γ^* a dual-lattice path. The corresponding string operators obey

$$W_e(\gamma)W_m(\gamma^*) = (-1)^{I(\gamma, \gamma^*)}W_m(\gamma^*)W_e(\gamma), \quad (25.27)$$

where $I(\gamma, \gamma^*)$ is the number of intersections modulo 2. The reason is local: at each crossing the two strings act on the same edge by σ^z and σ^x , which anticommute.

If an e particle is adiabatically moved once around an m particle, the worldline process is represented algebraically by a closed electric string that intersects a magnetic string once. Hence the wave function acquires the phase

$$\theta_{em} = -1. \quad (25.28)$$

Thus e and m are mutual semions.

The individual self-statistics are bosonic,

$$\theta_e = +1, \quad \theta_m = +1, \quad (25.29)$$

because exchanging two identical e 's or two identical m 's can be represented without producing a nontrivial string crossing. Their composite $\epsilon = e \times m$ has fermionic self-statistics,

$$\theta_\epsilon = -1. \quad (25.30)$$

This is one of the simplest examples where a purely spin Hamiltonian supports an emergent fermionic quasiparticle.

Remark 25.3 (Why this is topological). The phase in Eq. (25.27) depends only on the intersection number modulo 2, not on the metric length or detailed shape of the paths. This is the operational meaning of topological statistics in the toric-code fixed-point model.

25.7 Topological degeneracy on closed surfaces

We now count the ground-state degeneracy on a closed orientable surface Σ_g . The torus corresponds to $g = 1$.

The toric code has N_e physical qubits. The stabilizer group is generated by N_v star operators and N_p plaquette operators, but the two relations in Eq. (25.16) remove two independent generators. Hence the number of independent stabilizers is

$$r = N_v + N_p - 2. \quad (25.31)$$

By the stabilizer counting formula Eq. (25.5),

$$\dim \mathcal{H}_{\text{GS}} = 2^{N_e - r} = 2^{N_e - N_v - N_p + 2}. \quad (25.32)$$

Using the Euler characteristic

$$N_v - N_e + N_p = 2 - 2g, \quad (25.33)$$

we obtain

$$\boxed{\dim \mathcal{H}_{\text{GS}}(\Sigma_g) = 2^{2g}.} \quad (25.34)$$

On a torus, $g = 1$, this gives

$$\dim \mathcal{H}_{\text{GS}}(T^2) = 4. \quad (25.35)$$

Theorem 25.1 (Toric-code topological degeneracy). *On a closed orientable surface of genus g , the toric-code Hamiltonian has 2^{2g} exactly degenerate ground states. In particular, on the torus the ground-state degeneracy is four.*

Remark 25.4 (Why the degeneracy is not symmetry breaking). The degeneracy in Eq. (25.34) is not caused by a local order parameter. It is tied to noncontractible cycles of the spatial manifold. On a sphere, $g = 0$, the ground state is unique. On a torus, the four ground states cannot be distinguished by any strictly local operator in the thermodynamic limit; they are distinguished by nonlocal Wilson loop operators.

25.8 Logical Wilson loops on the torus

Let α and β be the two noncontractible cycles of a torus. Choose direct-lattice cycles $\gamma_\alpha, \gamma_\beta$ and dual-lattice cycles $\gamma_\alpha^*, \gamma_\beta^*$. The corresponding closed string operators are

$$\bar{Z}_\alpha = W_e(\gamma_\alpha), \quad \bar{Z}_\beta = W_e(\gamma_\beta), \quad \bar{X}_\alpha = W_m(\gamma_\alpha^*), \quad \bar{X}_\beta = W_m(\gamma_\beta^*). \quad (25.36)$$

All four operators commute with H_{TC} , because their strings are closed. However, they do not all commute with one another. Their algebra is determined by intersection numbers:

$$\bar{Z}_\alpha \bar{X}_\beta = -\bar{X}_\beta \bar{Z}_\alpha, \quad \bar{Z}_\beta \bar{X}_\alpha = -\bar{X}_\alpha \bar{Z}_\beta, \quad (25.37)$$

while pairs with zero intersection commute. Thus the ground-state space on the torus carries two encoded qubits. One may choose $\bar{Z}_\alpha$ and $\bar{Z}_\beta$ as a commuting set of labels, while the corresponding $\bar{X}$ -loops act as logical bit flips.

This is the stabilizer-code interpretation of the toric code. Local errors create anyon pairs, while noncontractible error strings implement logical operators. The code distance is set by the shortest noncontractible cycle of the lattice.

25.9 Local indistinguishability and robustness

Let $|\Omega_a\rangle$ and $|\Omega_b\rangle$ be two different ground states on a large torus, distinguished by Wilson-loop eigenvalues. A local operator O_{loc} supported on a contractible region cannot measure a noncontractible flux. Within the exactly solvable fixed-point model, this gives the selection rule

$$\langle \Omega_a | O_{\text{loc}} | \Omega_b \rangle = C(O_{\text{loc}}) \delta_{ab}. \quad (25.38)$$

This holds for operators whose support is small compared with the system size, up to finite-size corrections when perturbations are added.

Physically, this is why the toric-code degeneracy is stable against weak local perturbations in the thermodynamic limit. A perturbation can split the ground states only through virtual processes that create an anyon pair, move one anyon around a noncontractible cycle, and annihilate the pair. The amplitude of such a process is exponentially small in the linear system size in the gapped phase.

Principle 25.1 (Exact fixed point of topological order). The toric code is not merely a solvable Hamiltonian. It is a fixed-point representative of a stable gapped phase. Its universal data are the anyon types $\{1, e, m, \epsilon\}$, their fusion rules, their mutual statistics, and the genus-dependent ground-state degeneracy 2^{2g} .

25.10 Summary and relation to later chapters

The exact solution of the toric code is based on four elementary facts:

- (i) the Hamiltonian is a sum of commuting stabilizers;
- (ii) star and plaquette violations define e and m excitations;
- (iii) open strings create anyon pairs, while closed noncontractible strings act within the ground-state space;
- (iv) the ground-state degeneracy on Σ_g is 2^{2g} .

This gives the simplest example of intrinsic topological order in these notes. The following chapters generalize the same structural ideas in two directions. Quantum double models replace the group $\mathbb{Z}_2$ by a finite group G , leading in general to non-Abelian anyons. The Kitaev honeycomb model keeps spin-1/2 degrees of freedom in two dimensions but solves them by mapping to Majorana fermions moving in a static $\mathbb{Z}_2$ gauge field. Thus the toric code is the commuting-projector endpoint of a broader family of exactly solvable spin-liquid models.

Chapter 26

Kitaev Quantum Double Models

The toric code is the quantum double model for the finite group $G = \mathbb{Z}_2$. The purpose of this chapter is to explain the general finite-group construction. The resulting models, usually denoted by $D(G)$, are commuting-projector Hamiltonians whose local Hilbert spaces are group algebras rather than qubits. They are exactly solvable for every finite group G , and for non-Abelian G they give fixed-point lattice models with non-Abelian anyon excitations.

There are two guiding ideas.

- (i) The toric-code star operator is a local $\mathbb{Z}_2$ gauge transformation. For a general finite group G , the vertex term is the projector onto local gauge-invariant states.
- (ii) The toric-code plaquette operator enforces zero $\mathbb{Z}_2$ flux. For a general finite group G , the plaquette term projects onto states with trivial group holonomy around each plaquette.

The exact solvability follows from the same mechanism as in the toric code: local gauge projectors and flatness projectors commute.

26.1 From $\mathbb{Z}_2$ to finite groups

Let G be a finite group with identity element e . The local Hilbert space on each oriented edge ℓ is the group algebra

$$\mathcal{H}_\ell = \mathbb{C}[G] = \text{span}\{|g\rangle : g \in G\}, \quad \langle g|g'\rangle = \delta_{g,g'}. \quad (26.1)$$

The total Hilbert space on a lattice $\Gamma = (V, E, P)$, with vertices V , edges E , and plaquettes P , is

$$\mathcal{H}_G = \bigotimes_{\ell \in E} \mathbb{C}[G]_\ell, \quad \dim \mathcal{H}_G = |G|^{|E|}. \quad (26.2)$$

A basis state is a group-valued lattice gauge field

$$|\{g_\ell\}\rangle = \bigotimes_{\ell \in E} |g_\ell\rangle_\ell. \quad (26.3)$$

Each edge is assigned an arbitrary reference orientation. If a path traverses an edge ℓ in the reference direction, it contributes g_ℓ ; if it traverses ℓ in the opposite direction, it contributes g_ℓ^{-1} . This is the lattice analogue of parallel transport by a gauge connection.

For every $h \in G$, define left and right multiplication operators on $\mathbb{C}[G]$ by

$$L^h |g\rangle = |hg\rangle, \quad R^h |g\rangle = |gh^{-1}\rangle. \quad (26.4)$$

The inverse in the definition of R^h is convenient because both L^h and R^h define honest group actions:

$$L^h L^k = L^{hk}, \quad R^h R^k = R^{hk}, \quad [L^h, R^k] = 0. \quad (26.5)$$

Remark 26.1 (Orientation convention). A local gauge transformation at a vertex v should multiply an outgoing edge variable from the left and an incoming edge variable from the right. With the convention Eq. (26.4), this means

$$g_\ell \mapsto \begin{cases} hg_\ell, & \ell \text{ starts at } v, \\ g_\ell h^{-1}, & \ell \text{ ends at } v. \end{cases} \quad (26.6)$$

This is the lattice version of $U_{ij} \mapsto h_i U_{ij} h_j^{-1}$.

26.2 Vertex gauge transformations

For a vertex $v \in V$ and a group element $h \in G$, define

$$A_v^h = \prod_{\ell: s(\ell)=v} L_\ell^h \prod_{\ell: t(\ell)=v} R_\ell^h, \quad (26.7)$$

where $s(\ell)$ and $t(\ell)$ denote the source and target of the oriented edge ℓ . Thus A_v^h implements a gauge transformation by h at the single vertex v . The vertex projector is the group average

$$A_v = \frac{1}{|G|} \sum_{h \in G} A_v^h. \quad (26.8)$$

Proposition 26.1 (Vertex projectors). *The operators A_v are mutually commuting projectors:*

$$A_v^2 = A_v, \quad [A_v, A_w] = 0. \quad (26.9)$$

Proof. For a fixed vertex v , the map $h \mapsto A_v^h$ is a representation of G , so

$$A_v^2 = \frac{1}{|G|^2} \sum_{h,k \in G} A_v^{hk} = \frac{1}{|G|} \sum_{q \in G} A_v^q = A_v. \quad (26.10)$$

For $v \neq w$, the only possible overlap is an edge connecting v and w . On such an edge one operator acts by left multiplication and the other by right multiplication. These commute by Eq. (26.5). Therefore $[A_v^h, A_w^k] = 0$ for all h, k , and hence $[A_v, A_w] = 0$. $\square$

The condition

$$A_v |\psi\rangle = |\psi\rangle \quad (26.11)$$

is the lattice Gauss-law constraint. Thus the vertex terms impose local gauge invariance.

26.3 Plaquette holonomy and flatness projectors

Choose an orientation and a base vertex $v \in \partial p$ for each plaquette p . Let $\partial p = (\ell_1, \dots, \ell_n)$ be the ordered boundary path starting and ending at v . Define the plaquette holonomy

$$H_{p,v}(\{g_\ell\}) = g_{\ell_1}^{\epsilon_1} g_{\ell_2}^{\epsilon_2} \cdots g_{\ell_n}^{\epsilon_n}, \quad \epsilon_j = \begin{cases} +1, & \ell_j \text{ is traversed with the reference orientation,} \\ -1, & \ell_j \text{ is traversed against it.} \end{cases} \quad (26.12)$$

For $k \in G$, define a plaquette flux projector by

$$B_p^k |\{g_\ell\}\rangle = \delta_{H_{p,v}(\{g_\ell\}), k} |\{g_\ell\}\rangle. \quad (26.13)$$

The Hamiltonian uses the trivial-flux projector

$$B_p := B_p^e. \quad (26.14)$$

Remark 26.2 (Basepoint dependence). For non-Abelian G , the holonomy around a plaquette depends on the basepoint by conjugation. If the starting point is changed, $H_{p,v}$ is replaced by $xH_{p,v}x^{-1}$ for some boundary transport x . Therefore the condition $H_{p,v} = e$ is basepoint-independent, while the condition $H_{p,v} = k$ is not basepoint-independent unless it is stated up to conjugacy.

The plaquette projectors are diagonal in the group basis and obey

$$(B_p^k)^2 = B_p^k, \quad B_p^k B_p^{k'} = \delta_{k,k'} B_p^k, \quad \sum_{k \in G} B_p^k = \mathbf{1}. \quad (26.15)$$

The physical plaquette term $B_p = B_p^e$ enforces flatness of the lattice gauge field at p .

26.4 The quantum double Hamiltonian

The Kitaev quantum double Hamiltonian is

$$H_{D(G)} = -J_e \sum_{v \in V} A_v - J_m \sum_{p \in P} B_p, \quad J_e, J_m > 0. \quad (26.16)$$

Equivalently, up to an additive constant,

$$H_{D(G)} = J_e \sum_{v \in V} (\mathbf{1} - A_v) + J_m \sum_{p \in P} (\mathbf{1} - B_p) - (J_e |V| + J_m |P|). \quad (26.17)$$

The exact ground states are those satisfying

$$A_v |\psi\rangle = |\psi\rangle, \quad B_p |\psi\rangle = |\psi\rangle, \quad \forall v \in V, p \in P. \quad (26.18)$$

Thus the ground space consists of gauge-invariant wavefunctions supported on flat G -connections.

Theorem 26.1 (Commuting-projector structure). *The terms of $H_{D(G)}$ commute:*

$$[A_v, A_w] = 0, \quad [B_p, B_q] = 0, \quad [A_v, B_p] = 0. \quad (26.19)$$

Therefore $H_{D(G)}$ is an exactly solvable commuting-projector Hamiltonian.

Proof. The first equality was proved in [Section 26.2](#). The second equality follows because all B_p are diagonal multiplication operators in the group basis.

It remains to check $[A_v, B_p] = 0$. If $v \notin \partial p$, the operators act on disjoint local data or act in a way that does not change the plaquette holonomy, so they commute. If $v \in \partial p$, then a gauge transformation by h at v conjugates the plaquette holonomy based at v :

$$H_{p,v} \mapsto h H_{p,v} h^{-1}. \quad (26.20)$$

The property $H_{p,v} = e$ is invariant under conjugation. Hence $A_v^h B_p^e = B_p^e A_v^h$ for every h , and averaging over h gives $[A_v, B_p] = 0$. $\square$

Remark 26.3 (Why the identity flux is special). The operator B_p^k with fixed noncentral k does not commute with a gauge transformation at the basepoint. Instead,

$$A_v^h B_p^k A_v^{h^{-1}} = B_p^{hkh^{-1}}. \quad (26.21)$$

The gauge-invariant flux observable is therefore the conjugacy-class projector

$$B_p^C = \sum_{k \in C} B_p^k, \quad (26.22)$$

where $C \subset G$ is a conjugacy class. The Hamiltonian chooses the trivial class $C = \{e\}$.

26.5 Recovery of the toric code

Set $G = \mathbb{Z}_2 = \{0, 1\}$ with addition modulo two. The group algebra $\mathbb{C}[\mathbb{Z}_2]$ is a qubit Hilbert space with computational basis $|0\rangle, |1\rangle$. Left multiplication by the nontrivial group element flips the basis state:

$$L^1 |g\rangle = |g + 1\rangle = \sigma^x |g\rangle. \quad (26.23)$$

The vertex projector becomes

$$A_v = \frac{1}{2} \left(\mathbf{1} + \prod_{\ell \ni v} \sigma_\ell^x \right). \quad (26.24)$$

The plaquette flatness condition says that the sum of edge variables around a plaquette is zero modulo two. In the basis where $\sigma^z |g\rangle = (-1)^g |g\rangle$, this is

$$B_p = \frac{1}{2} \left(\mathbf{1} + \prod_{\ell \in \partial p} \sigma_\ell^z \right). \quad (26.25)$$

Therefore [Eq. \(26.16\)](#) becomes the toric-code Hamiltonian up to an additive constant and an overall normalization of the couplings. In this precise sense,

$$\text{toric code} = D(\mathbb{Z}_2). \quad (26.26)$$

26.6 Ground states as flat gauge fields

Let the lattice be embedded on a closed orientable surface Σ_g . The constraint $B_p = 1$ imposes trivial holonomy around every contractible plaquette. Therefore a ground-state configuration is a flat lattice G -connection. The constraint $A_v = 1$ then identifies configurations related by local gauge transformations.

Equivalently, the ground-state data are representations of the fundamental group into G , modulo global conjugation:

$$\text{Flat}_G(\Sigma_g) \simeq \text{Hom}(\pi_1(\Sigma_g), G)/G. \quad (26.27)$$

Here G acts by simultaneous conjugation. The ground-state Hilbert space is the space of gauge-invariant complex functions on flat connections:

$$\mathcal{H}_{\text{GS}}(\Sigma_g) \simeq \text{Fun}(\text{Hom}(\pi_1(\Sigma_g), G))^G. \quad (26.28)$$

Thus

$$\dim \mathcal{H}_{\text{GS}}(\Sigma_g) = \#(\text{Hom}(\pi_1(\Sigma_g), G)/G), \quad (26.29)$$

where the right-hand side denotes the number of conjugation orbits of flat G -connections.

For a torus T^2 ,

$$\pi_1(T^2) = \langle a, b : ab = ba \rangle. \quad (26.30)$$

Therefore a flat connection is a commuting pair $(x, y) \in G \times G$, modulo simultaneous conjugation:

$$\dim \mathcal{H}_{\text{GS}}(T^2) = \#\{(x, y) \in G \times G : xy = yx\}/G. \quad (26.31)$$

For Abelian G , conjugation is trivial, and this reduces to

$$\dim \mathcal{H}_{\text{GS}}(\Sigma_g) = |G|^{2g}. \quad (26.32)$$

Taking $G = \mathbb{Z}_2$ gives the toric-code result 2^{2g} .

Remark 26.4 (Ground-state counting is not stabilizer counting). For $G = \mathbb{Z}_2$, the model is a Pauli stabilizer code and the degeneracy can be counted by independent stabilizers. For a general finite non-Abelian group, there is no Pauli stabilizer structure. The correct replacement is the counting of gauge-invariant flat connections, or equivalently the representation theory of the quantum double algebra.

26.7 Electric, magnetic, and dyonic excitations

Excitations are local violations of the commuting projectors.

- A violation of A_v is an electric charge. It transforms nontrivially under local gauge transformations.
- A violation of B_p is a magnetic flux. Its gauge-invariant label is a conjugacy class of plaquette holonomy.
- In a non-Abelian group, charge and flux cannot in general be separated into independent labels. The elementary superselection sectors are dyons.

The precise label of a dyon is a pair

$$(C, \rho), \quad (26.33)$$

where $C \subset G$ is a conjugacy class and ρ is an irreducible representation of the centralizer of a representative element of C .

To define this label, choose an element $c \in C$. Its centralizer is

$$Z_c = \{h \in G : hc = ch\}. \quad (26.34)$$

If $c' = xcx^{-1}$ is another representative of the same conjugacy class, then $Z_{c'} = xZ_cx^{-1}$, so the set of irreducible representations is naturally identified up to conjugation. Hence the anyon label is well-defined as

$$\mathcal{A}_{D(G)} = \{(C, \rho) : C \in \text{Conj}(G), \rho \in \text{Irr}(Z_c), c \in C\}. \quad (26.35)$$

Definition 26.1 (Quantum-double anyon). An anyon type of the quantum double model $D(G)$ is a pair (C, ρ) , where C is a magnetic flux conjugacy class and ρ is an electric charge representation of the centralizer of that flux.

This formula has several important special cases.

- (i) The vacuum is $(\{e\}, \mathbf{1})$, where $\mathbf{1}$ is the trivial representation of G .
- (ii) Pure electric charges have trivial flux:

$$(\{e\}, \rho), \quad \rho \in \text{Irr}(G). \quad (26.36)$$

- (iii) Pure magnetic fluxes have nontrivial conjugacy class but trivial centralizer representation:

$$(C, \mathbf{1}_{Z_c}). \quad (26.37)$$

- (iv) Genuine dyons have both nontrivial flux and nontrivial charge representation.

26.8 The local quantum double algebra

The origin of the label (C, ρ) is the finite-dimensional Hopf algebra known as the Drinfeld double $D(G)$. At a puncture or excitation site, there are two kinds of local operations: measuring flux and applying gauge transformations. They obey a crossed relation.

Let P^g denote the projector onto flux g , and let T^h denote a gauge transformation by h . Then

$$T^h P^g = P^{hgh^{-1}} T^h. \quad (26.38)$$

Thus flux labels are not invariant under gauge transformations; only conjugacy classes are invariant. The algebra generated by P^g and T^h , with

$$P^g P^{g'} = \delta_{g,g'} P^g, \quad T^h T^{h'} = T^{hh'}, \quad \sum_{g \in G} P^g = \mathbf{1}, \quad (26.39)$$

is the quantum double algebra $D(G)$ in its standard lattice form.

Proposition 26.2 (Irreducible sectors of $D(G)$). *The irreducible representations of the local quantum double algebra are labelled by pairs (C, ρ) , where C is a conjugacy class of G , and ρ is an irreducible representation of the centralizer Z_c of any representative $c \in C$.*

Sketch. First diagonalize the commuting flux projectors P^g . The gauge action T^h moves a flux g to the conjugate flux hgh^{-1} . Therefore an irreducible representation cannot be supported on a single noncentral group element; it is supported on an entire conjugacy class C . After choosing a representative $c \in C$, the transformations that preserve this representative are precisely the centralizer elements Z_c . The remaining internal charge quantum numbers must therefore form an irreducible representation ρ of Z_c . This gives the pair (C, ρ) . $\square$

The quantum dimension of the anyon (C, ρ) is

$$d_{(C, \rho)} = |C| \dim \rho. \quad (26.40)$$

The total quantum dimension is

$$\mathcal{D} = \sqrt{\sum_{(C, \rho)} d_{(C, \rho)}^2} = |G|. \quad (26.41)$$

Indeed,

$$\sum_{(C,\rho)} d_{(C,\rho)}^2 = \sum_C |C|^2 \sum_{\rho \in \text{Irr}(Z_C)} (\dim \rho)^2 = \sum_C |C|^2 |Z_C| = \sum_C |C| |G| = |G|^2. \quad (26.42)$$

The topological spin of (C, ρ) is

$$\theta_{(C,\rho)} = \frac{\chi_\rho(c)}{\dim \rho}, \quad c \in C, \quad (26.43)$$

where χ_ρ is the character of ρ . Since c lies in the center of its own centralizer Z_c , Schur's lemma ensures that this ratio is a phase for a unitary irreducible representation.

26.9 Abelian case and the toric-code anyons

If G is Abelian, every conjugacy class contains one element and every centralizer is the whole group. Therefore

$$\mathcal{A}_{D(G)} = G \times \widehat{G}, \quad (26.44)$$

where $\widehat{G} = \text{Irr}(G)$ is the character group. An anyon is labelled by a magnetic flux $g \in G$ and an electric character $\chi \in \widehat{G}$. All anyons have quantum dimension one:

$$d_{(g,\chi)} = 1, \quad \mathcal{D} = |G|. \quad (26.45)$$

The braiding phase between (g, χ) and (h, ψ) is determined by the Aharonov–Bohm pairing between charge and flux:

$$M_{(g,\chi),(h,\psi)} = \chi(h) \psi(g). \quad (26.46)$$

For $G = \mathbb{Z}_2$, there are two group elements $0, 1$ and two characters $\mathbf{1}, \chi$, with $\chi(1) = -1$. The four anyons are

$$1 = (0, \mathbf{1}), \quad e = (0, \chi), \quad m = (1, \mathbf{1}), \quad \epsilon = (1, \chi). \quad (26.47)$$

This reproduces the toric-code fusion rules

$$e^2 = m^2 = \epsilon^2 = 1, \quad e \times m = \epsilon. \quad (26.48)$$

The mutual phase between e and m is -1 , because

$$M_{e,m} = \chi(1) = -1. \quad (26.49)$$

26.10 Ribbon operators

In the toric code, open string operators create pairs of anyons at their endpoints. In the non-Abelian quantum double model, the correct generalization is a ribbon operator. A ribbon is a strip made from a direct-lattice path and a nearby dual-lattice path. It carries both electric and magnetic information.

A ribbon operator has two conceptual components.

- (i) A magnetic component changes group variables on crossed edges and creates plaquette fluxes at the ribbon endpoints.

- (ii) An electric component parallel-transport representation indices along the direct path and creates gauge-charge violations at the endpoints.

Closed ribbon operators commute with the Hamiltonian and measure topological charge. Open ribbon operators commute with all projectors away from their endpoints and create a pair of excitations whose total topological charge is trivial.

For pure electric probes, one obtains Wilson-loop operators. If γ is a closed oriented loop and R is a representation of G , define

$$W_R(\gamma) = \chi_R \left(\prod_{\ell \in \gamma} g_\ell^{\epsilon_\ell} \right), \quad (26.50)$$

where χ_R is the character of R . The trace makes $W_R(\gamma)$ invariant under conjugation of the loop holonomy. On a contractible loop in the ground state, flatness forces $W_R(\gamma) = \dim R$. On noncontractible loops, Wilson operators distinguish different topological sectors.

Remark 26.5 (Why ribbons replace strings). For Abelian G , electric and magnetic string operators can be separated cleanly, and their algebra is essentially a phase. For non-Abelian G , moving flux changes the charge frame by conjugation. A ribbon keeps track of both the flux transport and the charge indices. This is why the excitation label is not “flux plus representation of G ”, but rather “flux conjugacy class plus representation of the corresponding centralizer.”

26.11 Example: $D(S_3)$

The smallest non-Abelian example is $G = S_3$, the permutation group of three objects. Its conjugacy classes are

$$C_e = \{e\}, \quad C_t = \{(12), (23), (13)\}, \quad C_c = \{(123), (132)\}. \quad (26.51)$$

The corresponding centralizers have sizes

$$|Z_e| = 6, \quad |Z_{(12)}| = 2, \quad |Z_{(123)}| = 3. \quad (26.52)$$

The irreducible representations are therefore distributed as follows:

$$\#\mathcal{A}_{D(S_3)} = \#\text{Irr}(S_3) + \#\text{Irr}(\mathbb{Z}_2) + \#\text{Irr}(\mathbb{Z}_3) = 3 + 2 + 3 = 8. \quad (26.53)$$

Thus $D(S_3)$ has eight anyon types. Their quantum dimensions are

$$1, 1, 2, \quad 3, 3, \quad 2, 2, 2, \quad (26.54)$$

where the first group comes from pure charges over C_e , the second from transposition fluxes, and the third from three-cycle fluxes. The dimension check gives

$$1^2 + 1^2 + 2^2 + 3^2 + 3^2 + 2^2 + 2^2 + 2^2 = 36 = |S_3|^2. \quad (26.55)$$

This example illustrates the genuinely non-Abelian feature: some anyons have quantum dimension larger than one.

26.12 Relation to topological quantum field theory

The model $D(G)$ is the Hamiltonian lattice realization of finite-group Dijkgraaf–Witten gauge theory with trivial cocycle. Its input data are purely group-theoretic, but its low-energy physics is topological. The ground states only depend on the topology of the spatial surface, and quasiparticles are classified by the representation category of the Drinfeld double,

$$\text{anyon category of } D(G) = \text{Rep}(D(G)). \quad (26.56)$$

The toric code corresponds to $\text{Rep}(D(\mathbb{Z}_2))$.

The commuting-projector Hamiltonian should be viewed as a fixed-point Hamiltonian in the same way that the toric code is a fixed-point Hamiltonian of $\mathbb{Z}_2$ topological order. Perturbing the Hamiltonian generally destroys exact solvability, but small local perturbations cannot immediately remove the topological phase as long as the bulk gap remains open.

26.13 Summary and comparison with the toric code

The quantum double construction replaces every toric-code $\mathbb{Z}_2$ ingredient by its finite-group analogue.

Table 26.1: Toric code versus finite-group quantum double models.

Structure	Toric code	Quantum double $D(G)$
Local edge Hilbert space	Qubit $\mathbb{C}[\mathbb{Z}_2]$	Group algebra $\mathbb{C}[G]$
Vertex term	$\mathbb{Z}_2$ star constraint	Gauge-invariance projector $A_v = G ^{-1} \sum_h A_v^h$
Plaquette term	$\mathbb{Z}_2$ zero-flux constraint	Flatness projector B_p^e
Ground states	Flat $\mathbb{Z}_2$ gauge fields modulo gauge transformations	Flat G -connections modulo gauge transformations
Anyon labels	$1, e, m, \epsilon$	Pairs (C, ρ) with C a conjugacy class and $\rho \in \text{Irr}(Z_c)$
Abelian or non-Abelian	Abelian	Abelian iff G is Abelian; generally non-Abelian
Total quantum dimension	2	$ G $

The toric code is not an isolated construction. It is the $G = \mathbb{Z}_2$ member of the finite-group quantum double family. The general replacement is

$$\mathbb{Z}_2 \text{ gauge theory} \longrightarrow G \text{ lattice gauge theory,}$$

with electric charges, magnetic fluxes, and dyons unified by the representation theory of $D(G)$.

The next chapter studies a different kind of two-dimensional exact solvability: the Kitaev honeycomb model. There the Hamiltonian is not a commuting-projector Hamiltonian. Instead, spin interactions fractionalize into free Majorana fermions moving in a static $\mathbb{Z}_2$ gauge field.

Chapter 27

The Kitaev Honeycomb Model

The toric code and the quantum double models are exactly solvable because their Hamiltonians are sums of commuting projectors. The Kitaev honeycomb model is solvable by a different, but closely related, mechanism. It is not a commuting-projector Hamiltonian. Instead, its spin degrees of freedom can be represented by Majorana fermions coupled to a static $\mathbb{Z}_2$ gauge field. In each fixed gauge-flux sector, the Hamiltonian reduces to a free Majorana hopping problem.

This chapter therefore serves as the bridge between two ideas developed earlier in these notes:

- (i) free-fermion/BdG solvability, as in the transverse-field XY chain and the Kitaev chain;
- (ii) gauge-sector solvability, as in the toric code and finite-group quantum double models.

The special feature of the honeycomb model is that these two structures coexist in a two-dimensional spin system. The resulting phase diagram contains gapped Abelian topological phases, a gapless Majorana spin liquid, and, after a small time-reversal-breaking perturbation, a gapped non-Abelian Ising-anyon phase.

27.1 Honeycomb lattice and bond-dependent exchange

Consider a honeycomb lattice. Each vertex carries a spin-1/2 degree of freedom, and the nearest-neighbour bonds are colored by three labels $\alpha = x, y, z$. Every site touches exactly one bond of each type. The Kitaev honeycomb Hamiltonian is

$$H_{\text{KHC}} = -J_x \sum_{\langle ij \rangle_x} \sigma_i^x \sigma_j^x - J_y \sum_{\langle ij \rangle_y} \sigma_i^y \sigma_j^y - J_z \sum_{\langle ij \rangle_z} \sigma_i^z \sigma_j^z. \quad (27.1)$$

Here $\langle ij \rangle_\alpha$ denotes a nearest-neighbour bond of type α . Unless otherwise stated, we take

$$J_x, J_y, J_z \geq 0. \quad (27.2)$$

Changing signs of the couplings can often be absorbed by sublattice-dependent spin rotations, but keeping the positive-coupling convention makes the phase diagram transparent.

Remark 27.1 (Why the model is not trivially solvable). Each local term in [Eq. \(27.1\)](#) is an Ising exchange, but different bonds use different spin components. Therefore terms sharing one site usually do not commute. For example, if an x -bond and a y -bond meet at site i , then the local Pauli matrices σ_i^x and σ_i^y anticommute. Thus the model is not

a commuting-projector model. Its exact solvability is more subtle: noncommuting spin interactions become quadratic Majorana hopping terms in a background of static $\mathbb{Z}_2$ link variables.

The honeycomb lattice is bipartite. We denote the two sublattices by A and B . Choose every bond orientation from A to B . A unit cell contains one A -site and one B -site. Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be primitive lattice vectors chosen so that, in the vortex-free gauge used below, the three bonds from an A -site at cell $\mathbf{r}$ connect to

$$B_{\mathbf{r}}, \quad B_{\mathbf{r}+\mathbf{a}_1}, \quad B_{\mathbf{r}+\mathbf{a}_2}, \quad (27.3)$$

with couplings J_z , J_x , and J_y , respectively. This choice fixes the later Bloch function

$$f(\mathbf{k}) = J_z + J_x e^{i\mathbf{k}\cdot\mathbf{a}_1} + J_y e^{i\mathbf{k}\cdot\mathbf{a}_2}. \quad (27.4)$$

Other conventions only multiply $f(\mathbf{k})$ by phase factors or relabel the momenta; the spectrum is unchanged.

27.2 Plaquette fluxes in the spin language

Before introducing Majorana fermions, one can already identify conserved plaquette operators directly in the spin model. Let p be a hexagonal plaquette and label its six vertices cyclically by 1, 2, 3, 4, 5, 6. With the usual bond coloring around a hexagon, define

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z, \quad (27.5)$$

up to a cyclic relabeling determined by the local bond convention. The precise sequence of x, y, z, x, y, z may be shifted, but the operator is always the product of the spin component not associated with the outgoing external bond at each vertex of the plaquette.

These operators obey

$$W_p^2 = \mathbf{1}, \quad [W_p, W_{p'}] = 0, \quad [W_p, H_{\text{KHC}}] = 0. \quad (27.6)$$

Thus every plaquette carries a conserved $\mathbb{Z}_2$ flux eigenvalue

$$W_p = w_p, \quad w_p = \pm 1. \quad (27.7)$$

A plaquette with $w_p = -1$ is called a vortex or $\mathbb{Z}_2$ flux excitation.

Remark 27.2 (Fluxes versus commuting projectors). The fluxes W_p commute with the Hamiltonian, but the Hamiltonian itself is not a sum of the W_p . They are conserved labels of superselection sectors, not the complete set of local energy projectors. After fixing the flux sector, one still has to diagonalize itinerant Majorana fermions.

On a closed surface, the plaquette fluxes are not all independent. For example, on a torus,

$$\prod_p W_p = \mathbf{1}, \quad (27.8)$$

so vortices are created in pairs on a closed lattice. There are also nonlocal Wilson-loop sectors associated with the two noncontractible cycles of the torus. These global sectors will later become important for the finite-size ground state degeneracy.

27.3 Four-Majorana representation

The exact solution begins by enlarging the Hilbert space. At every site i , introduce four Hermitian Majorana operators

$$b_i^x, \quad b_i^y, \quad b_i^z, \quad c_i, \quad (27.9)$$

satisfying

$$\{b_i^\alpha, b_j^\beta\} = 2\delta_{ij}\delta_{\alpha\beta}, \quad \{c_i, c_j\} = 2\delta_{ij}, \quad \{b_i^\alpha, c_j\} = 0. \quad (27.10)$$

The spin operators are represented as

$$\sigma_i^\alpha = ib_i^\alpha c_i, \quad \alpha = x, y, z. \quad (27.11)$$

This representation is redundant. Four Majoranas span a four-dimensional local Hilbert space, whereas one spin-1/2 has dimension two. The physical subspace is selected by the local gauge constraint

$$D_i |\psi\rangle_{\text{phys}} = |\psi\rangle_{\text{phys}}, \quad D_i = b_i^x b_i^y b_i^z c_i. \quad (27.12)$$

The operator D_i satisfies $D_i^2 = \mathbf{1}$. It generates a local $\mathbb{Z}_2$ gauge transformation:

$$D_i b_i^\alpha D_i = -b_i^\alpha, \quad D_i c_i D_i = -c_i, \quad (27.13)$$

while leaving Majoranas on other sites invariant. Since each spin operator $\sigma_i^\alpha = ib_i^\alpha c_i$ contains two Majoranas at site i , it is gauge invariant.

The projector onto the physical spin Hilbert space is

$$P_{\text{phys}} = \prod_i \frac{1 + D_i}{2}. \quad (27.14)$$

On finite lattices, especially with periodic boundary conditions, this projection must be imposed carefully because it constrains the allowed fermion parity in each gauge sector. For deriving the bulk spectrum and the phase diagram, it is usually sufficient to diagonalize in the enlarged Majorana space and then project physical states at the end.

Proposition 27.1 (Pauli algebra on the physical subspace). *The operators $\sigma_i^\alpha = ib_i^\alpha c_i$ satisfy the spin-1/2 Pauli algebra on the subspace $D_i = +1$:*

$$\sigma_i^\alpha \sigma_i^\beta = \delta_{\alpha\beta} \mathbf{1} + i \sum_\gamma \epsilon_{\alpha\beta\gamma} \sigma_i^\gamma. \quad (27.15)$$

Proof. For $\alpha = \beta$, one has $(ib_i^\alpha c_i)^2 = \mathbf{1}$. For distinct α, β, γ ,

$$\sigma_i^\alpha \sigma_i^\beta = (ib_i^\alpha c_i)(ib_i^\beta c_i) = b_i^\alpha b_i^\beta. \quad (27.16)$$

Using $D_i = b_i^x b_i^y b_i^z c_i = +1$, one obtains

$$b_i^\alpha b_i^\beta = i\epsilon_{\alpha\beta\gamma} ib_i^\gamma c_i = i\epsilon_{\alpha\beta\gamma} \sigma_i^\gamma, \quad (27.17)$$

with the sign fixed by the ordering (x, y, z) . This gives [Eq. \(27.15\)](#). $\square$

27.4 Static $\mathbb{Z}_2$ gauge field

Substituting Eq. (27.11) into the spin Hamiltonian gives

$$\begin{aligned} -J_\alpha \sigma_i^\alpha \sigma_j^\alpha &= -J_\alpha (ib_i^\alpha c_i)(ib_j^\alpha c_j) \\ &= iJ_\alpha u_{ij} c_i c_j, \quad \langle ij \rangle_\alpha, \end{aligned} \quad (27.18)$$

where the link operator is

$$u_{ij} = ib_i^\alpha b_j^\alpha, \quad \langle ij \rangle_\alpha. \quad (27.19)$$

The orientation convention is important: throughout this chapter $i \in A$ and $j \in B$, so $u_{ji} = -u_{ij}$. Since $u_{ij}^2 = \mathbf{1}$, its eigenvalues are

$$u_{ij} = \pm 1. \quad (27.20)$$

The Hamiltonian becomes

$$H_{\text{KHC}} = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j. \quad (27.21)$$

Proposition 27.2 (Static link variables). *The link operators satisfy*

$$[u_{ij}, u_{kl}] = 0, \quad [u_{ij}, H_{\text{KHC}}] = 0. \quad (27.22)$$

Therefore each u_{ij} is a conserved $\mathbb{Z}_2$ variable in the enlarged Majorana Hilbert space.

Proof. Two link variables either act on disjoint Majoranas or share one site but use Majoranas of different bond types. In both cases the total number of Majorana interchanges needed to swap the two bilinears is even, so they commute.

To prove $[u_{ij}, H_{\text{KHC}}] = 0$, note first that u_{ij} trivially commutes with the corresponding bond term $iJ_\alpha u_{ij} c_i c_j$. For any other bond term, either the two bonds are disjoint, or they share one site but involve different b -Majoranas and a single c -Majorana at that site. Again the relevant operator products differ by an even number of Majorana interchanges. Hence u_{ij} commutes with every term in Eq. (27.21). $\square$

The conserved u_{ij} are not themselves gauge-invariant observables. Under the local gauge transformation generated by D_i , all link variables adjacent to site i change sign:

$$u_{ij} \longmapsto -u_{ij}, \quad u_{ki} \longmapsto -u_{ki}. \quad (27.23)$$

Gauge-invariant information is contained in products around closed loops. For a plaquette p , define

$$W_p = \prod_{\langle ij \rangle \in \partial p} u_{ij}, \quad (27.24)$$

where the product follows the oriented boundary, with the sign convention fixed by the chosen bond orientations. This operator is the Majorana representation of the spin flux in Eq. (27.5). The eigenvalue $W_p = +1$ is called the vortex-free or zero-flux sector, and $W_p = -1$ is a $\mathbb{Z}_2$ vortex.

27.5 Reduction to free Majorana fermions

Because all u_{ij} commute with the Hamiltonian, the Hilbert space splits into static gauge sectors. In a fixed configuration $u = \{u_{ij} = \pm 1\}$, the Hamiltonian is quadratic:

$$H[u] = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j. \quad (27.25)$$

Equivalently, it can be written in the canonical Majorana quadratic form

$$H[u] = \frac{i}{4} \sum_{i,j} A_{ij}[u] c_i c_j, \quad A_{ij}[u] = -A_{ji}[u] \in \mathbb{R}. \quad (27.26)$$

For an α -bond oriented from $i \in A$ to $j \in B$, the matrix entries are

$$A_{ij}[u] = 2J_\alpha u_{ij}, \quad A_{ji}[u] = -2J_\alpha u_{ij}. \quad (27.27)$$

Diagonalizing the real antisymmetric matrix $A[u]$ gives pairs of eigenvalues $\pm \epsilon_n$ of $iA[u]$, with $\epsilon_n \geq 0$. The Hamiltonian in that sector has the form

$$H[u] = \sum_n \epsilon_n \left(f_n^\dagger f_n - \frac{1}{2} \right), \quad (27.28)$$

up to the physical projection discussed in Eq. (27.14). Thus the original interacting spin model is solved by the following sequence:

$$\begin{aligned} \text{spin model} &\longrightarrow \text{Majoranas plus static } \mathbb{Z}_2 \text{ gauge field} \\ &\longrightarrow \text{free Majorana band problem in each flux sector} \\ &\longrightarrow \text{physical projection.} \end{aligned} \quad (27.29)$$

Principle 27.1 (Majorana–gauge solvability). A spin model is Majorana–gauge solvable if, after an enlarged Majorana representation and a local gauge constraint, its Hamiltonian becomes quadratic in itinerant Majoranas with coefficients given by conserved $\mathbb{Z}_2$ link variables. The Kitaev honeycomb model is the canonical two-dimensional example of this mechanism.

27.6 Vortex-free sector and Bloch Hamiltonian

For the honeycomb lattice with positive couplings, the ground state lies in the vortex-free sector. This is a consequence of a flux theorem for reflection-symmetric planar lattices. We will not prove the theorem here; instead, we use it to obtain the bulk phase diagram.

Choose the translationally invariant gauge

$$u_{ij} = +1 \quad \text{for every bond oriented from } A \text{ to } B. \quad (27.30)$$

Using the unit-cell convention in Eq. (27.3), the Hamiltonian becomes

$$H_0 = i \sum_{\mathbf{r}} (J_z c_{A,\mathbf{r}} c_{B,\mathbf{r}} + J_x c_{A,\mathbf{r}} c_{B,\mathbf{r}+\mathbf{a}_1} + J_y c_{A,\mathbf{r}} c_{B,\mathbf{r}+\mathbf{a}_2}). \quad (27.31)$$

After Fourier transformation, the only gauge-invariant spectral data are encoded in

$$f(\mathbf{k}) = J_z + J_x e^{i\mathbf{k} \cdot \mathbf{a}_1} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2}. \quad (27.32)$$

The Majorana quasiparticle dispersion is

$$\epsilon(\mathbf{k}) = 2 |f(\mathbf{k})|. \quad (27.33)$$

Remark 27.3 (Factor of two convention). The factor of two in Eq. (27.33) follows from the Hamiltonian normalization in Eq. (27.1). Some references absorb this factor into the definition of J_α or into the canonical Majorana matrix A . The physically invariant statement is the location of zeros of $f(\mathbf{k})$ and the relative gap scale.

The gap closes if and only if there exists a momentum $\mathbf{k}$ such that

$$J_z + J_x e^{i\mathbf{k} \cdot \mathbf{a}_1} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2} = 0. \quad (27.34)$$

Geometrically, this means that three complex numbers of lengths J_x, J_y, J_z can form a triangle. Therefore the gapless condition is

$$J_x \leq J_y + J_z, \quad J_y \leq J_z + J_x, \quad J_z \leq J_x + J_y. \quad (27.35)$$

Inside this region, the spectrum has two Dirac cones. Outside it, one coupling dominates and the spectrum is gapped.

Proposition 27.3 (Bulk phase criterion). *For positive couplings, the vortex-free Majorana spectrum is gapless precisely inside the triangle region Eq. (27.35). It is gapped in the three regions*

$$J_x > J_y + J_z, \quad J_y > J_z + J_x, \quad J_z > J_x + J_y. \quad (27.36)$$

Proof. The only possible gap closing is $f(\mathbf{k}) = 0$. Since the phases $e^{i\mathbf{k} \cdot \mathbf{a}_1}$ and $e^{i\mathbf{k} \cdot \mathbf{a}_2}$ can vary over the unit circle subject to the Brillouin-zone periodicity, the equation $f(\mathbf{k}) = 0$ is equivalent to closing a triangle with side lengths J_x, J_y, J_z . Such a triangle exists if and only if the three triangle inequalities hold. If one inequality is violated, $|f(\mathbf{k})|$ has a strictly positive lower bound. $\square$

At the isotropic point

$$J_x = J_y = J_z, \quad (27.37)$$

the system lies in the gapless phase. The low-energy theory consists of two cones of massless Majorana fermions coupled to a gapped $\mathbb{Z}_2$ gauge field. Since the gauge field is static in the exactly solvable model, the low-energy fermions are free within each flux sector, but flux excitations remain massive.

27.7 Gapped Abelian phases and the toric-code limit

The three gapped regions in Eq. (27.36) are usually called A_x , A_y , and A_z . In the A_z phase, for example,

$$J_z \gg J_x, J_y. \quad (27.38)$$

The unperturbed Hamiltonian consists of decoupled z -dimers. Treating the x - and y -bond interactions perturbatively, the first nontrivial plaquette-dependent term appears at fourth order and has the form

$$H_{\text{eff}}^{(4)} = -J_{\text{eff}} \sum_p W_p + \text{constant}, \quad J_{\text{eff}} \sim \frac{J_x^2 J_y^2}{J_z^3}. \quad (27.39)$$

More precisely, the numerical prefactor depends on the perturbative convention, but the structure is fixed: the low-energy Hamiltonian is a $\mathbb{Z}_2$ commuting-projector theory generated by plaquette fluxes.

This shows that the gapped A phases are Abelian topological phases in the same universality class as the toric code. The elementary excitations can be identified as

- (i) $\mathbb{Z}_2$ fluxes, corresponding to plaquettes with $W_p = -1$;
- (ii) fermionic matter excitations from the Majorana band;
- (iii) bound states of fluxes and Majorana fermions.

Their fusion and braiding reduce, in the deep Abelian limit, to the toric-code anyon structure.

Remark 27.4 (Why the toric code reappears). The toric code describes a deconfined $\mathbb{Z}_2$ gauge theory with gapped charges and fluxes. The honeycomb model in a strongly anisotropic gapped phase has the same long-distance topological order, but its microscopic degrees of freedom are spins on sites rather than qubits on edges. The emergence of the toric code is therefore an infrared statement, not an exact microscopic equality at generic couplings.

27.8 Time-reversal breaking and the non-Abelian phase

The gapless phase is protected by the combination of lattice symmetries and time-reversal symmetry. A small magnetic field

$$V_h = - \sum_i (h_x \sigma_i^x + h_y \sigma_i^y + h_z \sigma_i^z) \quad (27.40)$$

breaks time reversal. The magnetic field itself destroys exact solvability, but its leading nonvanishing effect within perturbation theory is a solvable three-spin interaction of the schematic form

$$H_\kappa = -\kappa \sum_{\langle ijk \rangle} \sigma_i^\alpha \sigma_j^\beta \sigma_k^\gamma, \quad \kappa \propto \frac{h_x h_y h_z}{J^2}, \quad (27.41)$$

where the ordered triples $\langle ijk \rangle$ are length-two paths on the honeycomb lattice and α, β, γ are fixed by the adjacent bond types. In the Majorana representation this term adds next-nearest-neighbour Majorana hopping. It opens a mass gap at the Dirac cones.

The resulting gapped phase has a nonzero Majorana Chern number,

$$\nu = \pm 1, \quad (27.42)$$

with the sign determined by the sign of κ . The vortices of the $\mathbb{Z}_2$ gauge field bind Majorana zero modes. Consequently, vortices become non-Abelian anyons of Ising type.

The Ising-anyon fusion rules are

$$\sigma \times \sigma = 1 + \psi, \quad \sigma \times \psi = \sigma, \quad \psi \times \psi = 1. \quad (27.43)$$

Here σ denotes a vortex carrying a Majorana zero mode, and ψ is the fermionic Majorana excitation. The quantum dimensions are

$$d_1 = 1, \quad d_\psi = 1, \quad d_\sigma = \sqrt{2}. \quad (27.44)$$

The noninteger quantum dimension of σ is the hallmark of non-Abelian statistics.

Remark 27.5 (Relation to chiral topological superconductors). After fixing a gauge sector, the itinerant Majoranas form a two-dimensional class-D superconductor. The time-reversal-breaking mass converts the gapless Majorana semimetal into a chiral Majorana topological superconductor. Coupling this band topology to the dynamical $\mathbb{Z}_2$ fluxes produces the non-Abelian Ising topological order.

27.9 Fluxes, vortices, and physical projection

The Majorana solution contains two levels of description. First, one fixes a configuration of link variables u_{ij} and diagonalizes the free Majorana Hamiltonian $H[u]$. Second, one projects the resulting states onto the physical spin Hilbert space using P_{phys} . This projection has two important consequences.

- (i) Gauge-equivalent link configurations describe the same physical flux sector. A local sign flip generated by D_i changes adjacent u_{ij} and c_i , but leaves spin observables invariant.
- (ii) On a finite torus, the physical projection fixes a relation between the gauge-field sector and the fermion parity of the matter Majoranas. Thus not every free-Majorana eigenstate in the enlarged Hilbert space represents a physical spin state.

For bulk thermodynamic properties, the projection usually does not change the local dispersion or the phase boundaries. For exact finite-size spectra and for topological degeneracy on a torus, however, it is essential.

A useful way to summarize the structure is

$$\text{physical sector} = \frac{\text{link variables } u_{ij} = \pm 1}{\text{local } \mathbb{Z}_2 \text{ gauge transformations}} + \text{projected Majorana matter states.} \quad (27.45)$$

Gauge-invariant fluxes W_p and noncontractible Wilson loops label the gauge part, while the free Majorana occupation numbers label the matter part.

27.10 Spin correlations and fractionalization

A single spin operator is not diagonal in the flux sector. From $\sigma_i^\alpha = ib_i^\alpha c_i$, acting with σ_i^α changes link variables adjacent to site i . Equivalently, in the spin language it creates or moves a pair of fluxes on the two plaquettes adjacent to the α -bond touching site i . Therefore ordinary spin correlations are strongly constrained.

In the exactly solvable zero-field model, the equal-time two-spin correlator vanishes beyond nearest neighbours:

$$\langle \sigma_i^\alpha \sigma_j^\beta \rangle = 0 \quad \text{unless } i, j \text{ are nearest neighbours on an } \alpha\text{-bond and } \alpha = \beta. \quad (27.46)$$

This short-ranged spin correlation is not a sign of a trivial phase. It reflects spin fractionalization: a physical spin flip fractionalizes into an itinerant Majorana excitation and a pair of $\mathbb{Z}_2$ fluxes.

By contrast, gauge-invariant Majorana bilinears and flux operators can encode long-distance information about the Majorana band structure. In the gapless phase, the itinerant Majoranas have algebraic correlations, but local spin operators couple to them only together with flux excitations.

27.11 Comparison with the toric code and quantum double models

The following table summarizes the relation between the solvability mechanisms of the previous two chapters and the honeycomb model.

Model class	Exact-solvability mechanism	Excitations
Toric code	Commuting stabilizers A_s, B_p	Static Abelian anyons $1, e, m, \epsilon$
Quantum double $D(G)$	Commuting gauge and flatness projectors A_v, B_p	Electric, magnetic, and dyonic anyons labeled by (C, ρ)
Kitaev honeycomb model	Static $\mathbb{Z}_2$ gauge sectors plus free Majorana matter	Fluxes, itinerant Majorana fermions, and in the chiral phase non-Abelian Ising anyons

The toric code and quantum double models are fixed-point commuting-projector Hamiltonians. The Kitaev honeycomb model is instead a microscopic spin model with dispersive fractionalized matter. Its gapped Abelian regions flow to toric-code topological order at long distances, while its time-reversal-broken gapless region flows to Ising topological order.

27.12 Summary

The Kitaev honeycomb model is exactly solvable because the spin Hamiltonian can be rewritten as free Majorana fermions moving in a static $\mathbb{Z}_2$ gauge field. The essential steps are:

- (i) enlarge each spin into four Majoranas,

$$\sigma_i^\alpha = i b_i^\alpha c_i, \quad D_i = b_i^x b_i^y b_i^z c_i = +1; \quad (27.47)$$

- (ii) define conserved link variables,

$$u_{ij} = i b_i^\alpha b_j^\alpha, \quad u_{ij} = \pm 1; \quad (27.48)$$

- (iii) reduce the Hamiltonian in every gauge sector to

$$H[u] = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j; \quad (27.49)$$

- (iv) identify gauge-invariant fluxes

$$W_p = \prod_{\langle ij \rangle \in \partial p} u_{ij}; \quad (27.50)$$

- (v) diagonalize the vortex-free Majorana band with dispersion

$$\epsilon(\mathbf{k}) = 2 \left| J_z + J_x e^{i\mathbf{k} \cdot \mathbf{a}_1} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2} \right|. \quad (27.51)$$

The bulk phase diagram follows from the triangle inequalities [Eq. \(27.35\)](#). One obtains gapped Abelian phases when one coupling dominates, a gapless Majorana spin liquid inside the triangle, and a non-Abelian Ising-anyon phase when the gapless phase is gapped by a small time-reversal-breaking perturbation.

Chapter 28

Rokhsar–Kivelson Points and Quantum Dimer Models

The toric code and the quantum double models are exactly solvable because their Hamiltonians are sums of mutually commuting projectors. The Kitaev honeycomb model is solvable because the spin Hamiltonian reduces to free Majorana fermions in every static gauge sector. Rokhsar–Kivelson (RK) quantum dimer models provide a third mechanism in two dimensions: the full Hamiltonian need not be diagonalizable, but a special point in parameter space has an exact frustration-free ground state whose amplitudes are controlled by a classical statistical model.

The central idea is

$$\begin{aligned} \text{classical constrained configurations } \mathcal{C} &\implies \text{quantum Hilbert space } \text{span}\{|C\rangle : C \in \mathcal{C}\} \\ &\implies \text{local resonance Hamiltonian} \implies \text{exact equal-amplitude or weighted ground} \\ &\hspace{10em} \text{state at the RK point.} \end{aligned}$$

This chapter explains this mechanism for quantum dimer models. The model is not usually exactly solvable in the strong spectral sense. Its exact content is more specific: the ground-state wavefunction, equal-time ground-state correlators, and topological-sector structure can be obtained from a classical dimer ensemble.

28.1 Classical dimer coverings and the quantum Hilbert space

Let Λ be a two-dimensional lattice with vertex set V and edge set E . A *dimer covering*, or perfect matching, is a subset $D \subset E$ such that every vertex is incident on exactly one occupied edge. Equivalently,

$$\sum_{\ell \ni i} n_\ell = 1, \quad n_\ell \in \{0, 1\}, \quad i \in V. \quad (28.1)$$

The classical configuration space is therefore

$$\mathcal{C}_{\text{dim}} = \{D \subset E : D \text{ satisfies Eq. (28.1)}\}. \quad (28.2)$$

The quantum dimer Hilbert space is the vector space spanned by orthonormal basis states labelled by dimer coverings:

$$\mathcal{H}_{\text{QDM}} = \text{span}\{|D\rangle : D \in \mathcal{C}_{\text{dim}}\}, \quad \langle D|D'\rangle = \delta_{D,D'}. \quad (28.3)$$

This Hilbert space is already constrained. It is not the full tensor product of independent qubits on edges. The local condition Eq. (28.1) plays a role analogous to a Gauss-law constraint: every site must be touched by exactly one dimer.

Remark 28.1 (Dimer constraint as an emergent gauge constraint). On a bipartite lattice one can orient every edge from the A sublattice to the B sublattice and define a background charge. The dimer constraint can then be rewritten as a lattice divergence equation. This is why height fields, Coulomb phases, and gauge-theory language naturally appear in quantum dimer models. The constraint is not optional; it defines the Hilbert space.

The elementary local move in the square-lattice quantum dimer model is a plaquette flip. A plaquette p is called *flippable* if it contains two parallel dimers on opposite edges. There are two such local states, denoted abstractly by

$$|h_p\rangle, \quad |v_p\rangle, \quad (28.4)$$

where h and v stand for the two parallel dimer orientations on the plaquette. The plaquette flip exchanges these two states and leaves all other edges unchanged.

28.2 The square-lattice quantum dimer Hamiltonian

The standard square-lattice Rokhsar–Kivelson quantum dimer Hamiltonian is

$$H_{\text{QDM}} = \sum_p H_p, \quad (28.5)$$

with local plaquette term

$$H_p = -t (|h_p\rangle \langle v_p| + |v_p\rangle \langle h_p|) + v (|h_p\rangle \langle h_p| + |v_p\rangle \langle v_p|). \quad (28.6)$$

The first term is kinetic: it resonates two dimer coverings that differ by one plaquette flip. The second term is potential: it counts flippable plaquettes. The parameters t and v are usually taken real, and we will assume $t > 0$.

In the two-dimensional subspace $\text{span}\{|h_p\rangle, |v_p\rangle\}$, the local operator is the matrix

$$H_p \hat{=} \begin{pmatrix} v & -t \\ -t & v \end{pmatrix}. \quad (28.7)$$

Thus the symmetric and antisymmetric local combinations have energies

$$E_+ = v - t, \quad E_- = v + t, \quad (28.8)$$

where

$$|+_p\rangle = \frac{1}{\sqrt{2}}(|h_p\rangle + |v_p\rangle), \quad |-_p\rangle = \frac{1}{\sqrt{2}}(|h_p\rangle - |v_p\rangle). \quad (28.9)$$

The RK point is the special point

$$v = t. \quad (28.10)$$

At this point, the local Hamiltonian becomes a positive semidefinite projector up to a scalar factor:

$$H_p^{\text{RK}} = t (|h_p\rangle - |v_p\rangle) (\langle h_p| - \langle v_p|). \quad (28.11)$$

Therefore

$$H_{\text{QDM}}^{\text{RK}} \geq 0. \quad (28.12)$$

This is the frustration-free structure hidden in the model.

Remark 28.2 (Not a commuting-projector model). Although each H_p^{RK} is a local positive-semidefinite projector, the different plaquette terms do not generally commute. Thus the RK point is not a toric-code-type commuting-projector point. Exact solvability here means exact ground-state solvability, not simultaneous diagonalization of the full spectrum.

28.3 The equal-amplitude RK ground state

The plaquette flip dynamics decomposes $\mathcal{C}_{\text{dim}}$ into connected components. On a torus these components are labelled, in part, by winding numbers. Let $\mathcal{S} \subset \mathcal{C}_{\text{dim}}$ be one connected sector under local plaquette flips. Define

$$|\Psi_{\mathcal{S}}\rangle = \frac{1}{\sqrt{N_{\mathcal{S}}}} \sum_{D \in \mathcal{S}} |D\rangle, \quad N_{\mathcal{S}} = |\mathcal{S}|. \quad (28.13)$$

This is the equal-amplitude superposition of all dimer coverings in the sector $\mathcal{S}$.

Proposition 28.1 (RK ground state). *At $v = t$, the state $|\Psi_{\mathcal{S}}\rangle$ is annihilated by every local RK plaquette term:*

$$H_p^{\text{RK}} |\Psi_{\mathcal{S}}\rangle = 0 \quad \text{for all } p. \quad (28.14)$$

Consequently,

$$H_{\text{QDM}}^{\text{RK}} |\Psi_{\mathcal{S}}\rangle = 0. \quad (28.15)$$

Proof. Fix a plaquette p . The only basis states on which H_p^{RK} acts nontrivially are pairs of dimer coverings (D_h, D_v) that differ by flipping p . In the equal-amplitude state, $|D_h\rangle$ and $|D_v\rangle$ occur with the same coefficient. The local operator Eq. (28.11) projects onto the antisymmetric combination $|D_h\rangle - |D_v\rangle$. Therefore the contribution from each flippable pair cancels. Nonflippable configurations are annihilated because neither $|h_p\rangle$ nor $|v_p\rangle$ is present locally. Hence every H_p^{RK} annihilates the state. $\square$

The proof also explains the logic of the RK construction. The Hamiltonian is built so that every allowed local move imposes an equality of amplitudes between configurations connected by that move. If the sector $\mathcal{S}$ is connected under the local moves, those local equality conditions force a unique equal-amplitude state within that sector.

Theorem 28.1 (Sector-wise RK ground space). *Assume a connected component $\mathcal{S}$ of the dimer configuration graph is connected by the plaquette flips appearing in the Hamiltonian. At $v = t > 0$, the zero-energy ground state in that component is the equal-amplitude state $|\Psi_{\mathcal{S}}\rangle$. The full RK ground space is spanned by one such state for each dynamically disconnected sector, together with possible isolated frozen configurations that contain no flippable plaquettes.*

Remark 28.3 (Frozen states). Some lattices or boundary conditions admit staggered dimer configurations with no flippable plaquettes. These states are also annihilated by $H_{\text{QDM}}^{\text{RK}}$, because every local term acts only on flippable plaquettes. They are exact zero-energy states, but they are not liquid-like RK superpositions. When discussing universal long-distance physics, one usually focuses on the liquid sectors rather than these frozen sectors.

28.4 Topological sectors and winding numbers

On a simply connected domain with fixed boundary conditions, the local plaquette flips typically connect all dimer coverings compatible with the boundary. On a torus, however, local flips cannot change global winding numbers.

For a bipartite square lattice, choose a reference dimer configuration D_0 . Given another dimer covering D , the superposition of D and D_0 forms oriented transition loops. The net number of transition-loop crossings through two noncontractible cuts defines two winding numbers

$$(W_x, W_y) \in \mathbb{Z} \times \mathbb{Z}. \quad (28.16)$$

A local plaquette flip changes the dimer configuration only inside a contractible region. Therefore it cannot change (W_x, W_y) .

Proposition 28.2 (Conservation of winding sectors). *For the square-lattice quantum dimer model on a torus,*

$$[H_{\text{QDM}}, W_x] = [H_{\text{QDM}}, W_y] = 0, \quad (28.17)$$

where W_x and W_y denote the winding-sector labels regarded as operators that are diagonal in the dimer basis.

Proof. Each Hamiltonian term flips dimers around a single plaquette. Such a move adds or removes an elementary contractible loop in the transition-graph representation. A contractible modification has zero intersection number with any noncontractible cut. Hence the winding numbers are unchanged by every local term. $\square$

The existence of winding sectors is similar in spirit to the topological sectors of the toric code, but the physical meaning is different. In the toric code the fourfold degeneracy on a torus is robust throughout a topologically ordered phase. In the square-lattice RK dimer model, the number of winding sectors can grow with system size, and the RK point is critical rather than a stable gapped $\mathbb{Z}_2$ topological phase.

28.5 Relation to classical dimer partition functions

The RK wavefunction converts quantum equal-time expectation values into classical statistical averages. Let $\hat{O}$ be an operator diagonal in the dimer basis:

$$\hat{O} |D\rangle = O(D) |D\rangle. \quad (28.18)$$

Then in the equal-amplitude RK state,

$$\langle \Psi_S | \hat{O} | \Psi_S \rangle = \frac{1}{N_S} \sum_{D \in \mathcal{S}} O(D). \quad (28.19)$$

Thus diagonal ground-state correlations are exactly the correlations of a classical dimer model with uniform Boltzmann weight in the same sector.

More generally, one may introduce positive classical weights $w(D) > 0$ and the wavefunction

$$|\Psi_w\rangle = \frac{1}{\sqrt{Z_w}} \sum_{D \in \mathcal{C}_{\text{dim}}} \sqrt{w(D)} |D\rangle, \quad Z_w = \sum_{D \in \mathcal{C}_{\text{dim}}} w(D). \quad (28.20)$$

Then diagonal observables satisfy

$$\langle \Psi_w | \hat{O} | \Psi_w \rangle = \frac{1}{Z_w} \sum_{D \in \mathcal{C}_{\text{dim}}} w(D) O(D). \quad (28.21)$$

This is the broader classical-to-quantum RK construction.

Principle 28.1 (RK classical–quantum correspondence). If a quantum ground state has amplitudes equal to the square roots of classical Boltzmann weights, then the norm of the quantum state is the classical partition function, and all equal-time diagonal quantum correlators are classical thermal correlators.

For the uniform dimer RK state,

$$\langle \Psi | \Psi \rangle = \sum_{D \in \mathcal{C}_{\text{dim}}} 1 = Z_{\text{dimer}}. \quad (28.22)$$

This relation is the reason why classical dimer techniques, especially Kasteleyn orientations and Pfaffians, enter the quantum problem.

28.6 Kasteleyn orientation and Pfaffian solvability

Classical close-packed dimers on a planar graph are exactly solvable by the Kasteleyn–Pfaffian method. Assign an antisymmetric matrix K to the graph, with entries

$$K_{ij} = \begin{cases} \kappa_{ij}, & \text{if the oriented edge is } i \rightarrow j, \\ -\kappa_{ij}, & \text{if the oriented edge is } j \rightarrow i, \\ 0, & \text{if } i \text{ and } j \text{ are not adjacent,} \end{cases} \quad (28.23)$$

where $\kappa_{ij} > 0$ is the dimer fugacity on edge (ij) . A Kasteleyn orientation is an orientation for which each face has an odd number of clockwise oriented edges, with a standard modification for boundary and topology.

For a planar graph with a Kasteleyn orientation, the classical dimer partition function is

$$Z_{\text{dimer}} = |\text{Pf } K|. \quad (28.24)$$

Since

$$(\text{Pf } K)^2 = \det K, \quad (28.25)$$

one can compute the free energy from a free-fermion determinant. Moreover, dimer occupation correlations are obtained from entries of K^{-1} . For example, for two edges $e = (ij)$ and $e' = (kl)$, connected dimer correlations can be expressed by finite determinants built from matrix elements of K^{-1} .

Remark 28.4 (Why Pfaffians matter for the quantum RK point). The RK Hamiltonian itself is not a free-fermion Hamiltonian. However, its exact ground-state norm and diagonal equal-time correlators reduce to a classical dimer problem. If that classical dimer problem is Pfaffian-solvable, then those quantum ground-state quantities are exactly computable by Pfaffian methods. This is a ground-state correspondence, not a full spectral fermionization of the quantum Hamiltonian.

On a torus, one needs four Pfaffians corresponding to different boundary signs around the two noncontractible cycles. Schematically,

$$Z_{\text{dimer}}^{\text{torus}} = \frac{1}{2} | -\text{Pf } K_{00} + \text{Pf } K_{10} + \text{Pf } K_{01} + \text{Pf } K_{11} |, \quad (28.26)$$

where the subscripts indicate twists along the two cycles. The precise sign convention depends on the chosen Kasteleyn orientation, but the need for four spin structures is universal.

28.7 Height representation and the square-lattice RK point

On bipartite planar lattices, close-packed dimer configurations can be mapped to integer-valued height configurations on the dual lattice. The construction is local: crossing an edge changes the height by an amount that depends on whether the edge is occupied by a dimer and on the sublattice convention. The exact microscopic increments are conventional, but the long-distance structure is not. The coarse-grained height field $h(\mathbf{r})$ has an effective classical action of Gaussian form

$$S_{\text{cl}}[h] = \frac{K}{2} \int d^2r (\nabla h)^2 + \text{irrelevant or locking terms}. \quad (28.27)$$

For the square-lattice uniform dimer model, this height theory is critical. Consequently, the square-lattice RK ground state has algebraic equal-time dimer correlations.

At the RK point, the quantum dynamics adds a kinetic term to the height field. The long-wavelength quantum theory is the quantum Lifshitz theory

$$S_{\text{QL}}[h] = \frac{1}{2} \int d\tau d^2r [(\partial_\tau h)^2 + \kappa^2 (\nabla^2 h)^2]. \quad (28.28)$$

The dispersion relation is therefore

$$\omega(\mathbf{q}) \sim \kappa |\mathbf{q}|^2, \quad (28.29)$$

so the dynamic critical exponent is

$$z = 2. \quad (28.30)$$

Remark 28.5 (Why this is not a relativistic critical point). In most quantum critical points discussed earlier in one-dimensional free fermion chains, the low-energy theory has $z = 1$ and spacetime scaling is relativistic. The square-lattice RK point is different: its effective action contains $(\nabla^2 h)^2$, not $(\nabla h)^2$, in the quantum potential energy. The resulting $\omega \sim q^2$ dispersion is a hallmark of RK-type criticality.

The square-lattice phase diagram near the RK point is often summarized as follows. For $v/t > 1$, flippable plaquettes cost energy and staggered frozen states dominate. For sufficiently smaller v/t , crystalline valence-bond solid phases appear. The RK point $v = t$ is a fine-tuned critical point separating the staggered regime from resonating dimer physics.

28.8 Triangular lattice and gapped $\mathbb{Z}_2$ dimer liquid

The physics of the RK point depends strongly on the lattice. On the triangular lattice, the elementary move again flips two parallel dimers around a rhombus, but the lattice is nonbipartite. The absence of a bipartite height field changes the long-distance behavior. At the triangular-lattice RK point, the equal-weight dimer liquid is gapped and realizes $\mathbb{Z}_2$ topological order rather than a critical height phase.

The distinction may be summarized as

Lattice type	Classical dimer ensemble	RK quantum phase
Bipartite square lattice	Height/Coulomb phase with algebraic correlations	Critical RK point with $z = 2$
Nonbipartite triangular lattice	Short-ranged dimer correlations	Gapped $\mathbb{Z}_2$ topological dimer liquid

On a torus, a gapped $\mathbb{Z}_2$ dimer liquid has four topological sectors, analogous to the four ground states of the toric code. A useful way to label them is by the parity of the number of dimers crossing two noncontractible cuts:

$$(P_x, P_y) \in \mathbb{Z}_2 \times \mathbb{Z}_2. \quad (28.31)$$

Local plaquette flips preserve these parities. In the thermodynamic limit, the four corresponding RK ground states become locally indistinguishable if the phase is gapped and topologically ordered.

Remark 28.6 (Relation to the toric code). The triangular-lattice RK dimer liquid and the toric code are not identical microscopic Hamiltonians. However, they lie in the same broad topological universality class when the dimer liquid is gapped and has $\mathbb{Z}_2$ topological order. The toric code is the fixed-point commuting-projector representative; the quantum dimer model is a constrained Hilbert-space realization with resonance dynamics.

28.9 General RK construction from stochastic dynamics

The dimer RK point is a special case of a general construction. Let $\mathcal{C}$ be a classical configuration space with positive weights $w(C) > 0$. Suppose there is a set of local reversible moves $C \leftrightarrow C'$. Define local operators of the form

$$H_{C,C'} = \gamma_{C,C'} \left(\sqrt{\frac{w(C')}{w(C)}} |C\rangle - \sqrt{\frac{w(C)}{w(C')}} |C'\rangle \right) \left(\sqrt{\frac{w(C')}{w(C)}} \langle C| - \sqrt{\frac{w(C)}{w(C')}} \langle C'| \right), \quad (28.32)$$

with $\gamma_{C,C'} > 0$. Then the state $|\Psi_w\rangle$ in [Eq. \(28.20\)](#) is annihilated by every such term. Hence

$$H_{\text{RK}}[w] = \sum_{(C,C')} H_{C,C'} \quad (28.33)$$

is frustration-free and has $|\Psi_w\rangle$ as an exact ground state in every connected component of the move graph.

This construction is closely related to classical Markov dynamics satisfying detailed balance. A Markov generator M with stationary distribution

$$\pi(C) = \frac{w(C)}{Z_w} \quad (28.34)$$

can often be transformed by a similarity transformation into a Hermitian RK Hamiltonian:

$$H_{\text{RK}} = -\Pi^{-1/2} M \Pi^{1/2}, \quad \Pi_{C,C'} = \pi(C) \delta_{C,C'}. \quad (28.35)$$

Consequently, relaxation modes of the classical stochastic process map to excitation modes of the quantum RK Hamiltonian.

Principle 28.2 (Stochastic-matrix form). RK Hamiltonians often have a stochastic-matrix form: after a diagonal similarity transformation, the quantum Hamiltonian becomes a classical Markov generator. The exact ground state is the square root of the stationary distribution, while low-energy quantum modes mirror slow classical relaxation modes.

28.10 Excitations: resonons, visons, and monomers

Although the RK point gives an exact ground state, its excitations are generally more difficult than those of a free-fermion or commuting-projector model. Nevertheless, several important excitation types can be identified.

Resonons. On the square-lattice RK point, low-energy collective modes of the height field are sometimes called resonons. Their energy vanishes as

$$E(\mathbf{q}) \sim |\mathbf{q}|^2, \quad (28.36)$$

consistent with the quantum Lifshitz exponent $z = 2$.

Visons. In a gapped $\mathbb{Z}_2$ dimer liquid, a vison is a $\mathbb{Z}_2$ vortex excitation. Operationally, a vison can be detected by a string operator that measures the parity of dimers crossing a path on the dual lattice. Its mutual statistics with a monomer excitation is analogous to the mutual statistics of m and e particles in the toric code.

Monomers. A monomer is a violation of the close-packing constraint: a site is left unmatched or has the wrong dimer occupancy. In a strict quantum dimer Hilbert space, monomers are excluded. They can be introduced by enlarging the Hilbert space. Their confinement or deconfinement diagnoses whether the dimer phase is crystalline, critical, or topologically ordered.

28.11 Comparison with earlier exact models

Quantum dimer models clarify an important distinction in these notes. Exact solvability can refer to the full spectrum, to a sector-wise reduction, or only to the ground-state wavefunction and its equal-time correlations. The RK point belongs to the last category.

Model class	Solvable structure	What is exactly controlled
TFIM/XY/Kitaev chain	Jordan–Wigner plus BdG diagonalization	Full spectrum and Gaussian correlators
XXZ/XYZ chains	Yang–Baxter algebra and Bethe ansatz	Spectrum through Bethe equations
Toric code	Commuting stabilizers/projectors	Full spectrum and anyon algebra at the fixed point
Kitaev honeycomb	Static $\mathbb{Z}_2$ gauge fields plus free Majoranas	Free-fermion spectrum in each flux sector
RK quantum dimer model	Frustration-free stochastic-matrix form	Exact ground state and classical equal-time correlators

Thus the RK model is exactly solvable in a weaker but conceptually important sense. It shows that a nontrivial many-body wavefunction can be known exactly even when the full many-body spectrum is not.

28.12 Summary

The essential points of this chapter are:

- (i) A quantum dimer model is defined on the constrained Hilbert space of perfect matchings.
- (ii) The square-lattice QDM has kinetic plaquette flips and a potential term that counts flippable plaquettes.
- (iii) At the RK point $v = t$, the Hamiltonian becomes a sum of positive semidefinite local terms.
- (iv) In every connected sector, the equal-amplitude superposition of dimer coverings is an exact zero-energy ground state.
- (v) The norm and diagonal equal-time correlators of the RK state are those of a classical dimer model.
- (vi) Classical dimer Pfaffian methods therefore compute quantum ground-state correlations at the RK point.
- (vii) Bipartite square-lattice RK physics is critical and described by a height/quantum-Lifshitz theory with $z = 2$.
- (viii) Nonbipartite triangular-lattice RK physics can realize a gapped $\mathbb{Z}_2$ topological dimer liquid.

The next chapter moves from RK ground-state solvability to more general exact ground-state constructions, including valence-bond-solid states, stabilizer cluster states, and parent Hamiltonians of tensor-network states.

Chapter 29

Exact Ground-State Models: AKLT, Cluster, and Parent Hamiltonians

The preceding chapters presented several different notions of exact solvability in two-dimensional quantum many-body systems. The toric code and quantum double models are commuting-projector models, so their full spectra reduce to local constraint eigenvalues. The Kitaev honeycomb model is not a commuting-projector model, but it reduces to free Majorana fermions in each static gauge sector. Rokhsar–Kivelson quantum dimer models provide exact ground states at special frustration-free points, while the full spectrum is generally not solved.

This chapter develops a broader exact-ground-state viewpoint. The central objects are Hamiltonians whose ground states are known exactly because the local terms are projectors annihilating a prescribed many-body state. The paradigms are

- (i) the AKLT chain, whose valence-bond-solid ground state is an exact matrix-product state;
- (ii) the cluster model, whose ground state is a stabilizer state and a simple example of symmetry-protected topological order;
- (iii) the general parent-Hamiltonian construction for MPS and PEPS.

The common mechanism is

$\begin{aligned} \text{exactly known wavefunction} &\implies \text{local reduced-density-matrix support} \implies \text{positive} \\ &\text{local projector} \implies \text{frustration-free Hamiltonian.} \end{aligned}$

This is a different sense of solvability from Bethe ansatz or free-fermion diagonalization. It is exact and nonperturbative, but it usually controls the ground-state subspace rather than every excited level.

29.1 Frustration-free exact ground-state solvability

Let a local Hamiltonian be written as

$$H = \sum_X h_X, \quad h_X \geq 0, \tag{29.1}$$

where each term acts on a finite region X of the lattice. A state $|\Psi\rangle$ is a frustration-free ground state if

$$h_X |\Psi\rangle = 0 \quad \text{for every local term } X. \tag{29.2}$$

Then

$$H|\Psi\rangle = 0, \quad H \geq 0, \quad (29.3)$$

so $|\Psi\rangle$ is a ground state with energy zero. If the kernel of H is known exactly, the ground-state problem is solved.

Definition 29.1 (Frustration-free Hamiltonian). A positive local Hamiltonian $H = \sum_X h_X$, with $h_X \geq 0$, is called frustration-free if

$$\ker H = \bigcap_X \ker h_X. \quad (29.4)$$

Equivalently, a global ground state minimizes every local term separately.

This definition should be compared with commuting-projector solvability. If all local projectors commute, then their common eigenspaces give the full spectrum. If the terms are frustration-free but noncommuting, the ground space can still be known exactly, but the excited spectrum is generally nontrivial.

Remark 29.1 (Three strengths of solvability). It is useful to distinguish three levels.

- (a) *Full-spectrum commuting-projector solvability*: all local terms commute, as in the toric code.
- (b) *Free-particle or Bethe-ansatz spectral solvability*: the full spectrum is generated by quasiparticles or rapidities, as in the XY and XXZ chains.
- (c) *Exact ground-state solvability*: the ground-state manifold is exactly known, but the full spectrum is not generally diagonalized. AKLT, cluster, RK, and generic MPS/PEPS parent Hamiltonians mostly belong to this class.

The exact-ground-state construction is especially powerful because it produces nontrivial phases with simple wavefunctions. In one dimension, it naturally leads to matrix-product states. In two dimensions, it leads to projected entangled-pair states, string-net states, dimer liquid states, and other tensor- network fixed points.

29.2 The AKLT chain

The Affleck–Kennedy–Lieb–Tasaki (AKLT) chain is the canonical exactly soluble spin-1 model. It realizes the Haldane phase by an explicit valence-bond-solid (VBS) wavefunction. Its solvability is not based on free fermions or Bethe rapidities. Instead, it follows from a projector decomposition of two adjacent spin-1 degrees of freedom.

29.2.1 Spin-1 degrees of freedom and two-site projectors

Let $\mathbf{S}_j = (S_j^x, S_j^y, S_j^z)$ be spin-1 operators satisfying

$$[S_j^\alpha, S_\ell^\beta] = i\delta_{j\ell}\epsilon^{\alpha\beta\gamma}S_j^\gamma, \quad \mathbf{S}_j^2 = 2\mathbf{1}. \quad (29.5)$$

For two neighboring spin-1 sites, the tensor product decomposes as

$$1 \otimes 1 = 0 \oplus 1 \oplus 2. \quad (29.6)$$

Let $P_{j,j+1}^{(S)}$ denote the projector onto total spin S in the pair $(j, j+1)$. Since

$$\mathbf{S}_j \cdot \mathbf{S}_{j+1} = \frac{1}{2} [(\mathbf{S}_j + \mathbf{S}_{j+1})^2 - \mathbf{S}_j^2 - \mathbf{S}_{j+1}^2], \quad (29.7)$$

its eigenvalues in the $S = 0, 1, 2$ sectors are

$$x_0 = -2, \quad x_1 = -1, \quad x_2 = 1. \quad (29.8)$$

Therefore the projector onto the total-spin-2 sector is the polynomial

$$P_{j,j+1}^{(2)} = \frac{1}{6} (\mathbf{S}_j \cdot \mathbf{S}_{j+1} + 2) (\mathbf{S}_j \cdot \mathbf{S}_{j+1} + 1). \quad (29.9)$$

Equivalently,

$$P_{j,j+1}^{(2)} = \frac{1}{6} (\mathbf{S}_j \cdot \mathbf{S}_{j+1})^2 + \frac{1}{2} \mathbf{S}_j \cdot \mathbf{S}_{j+1} + \frac{1}{3}. \quad (29.10)$$

The AKLT Hamiltonian on a periodic chain is

$$H_{\text{AKLT}} = \sum_{j=1}^L P_{j,j+1}^{(2)}, \quad L+1 \equiv 1. \quad (29.11)$$

Up to an overall factor and additive constant, this is the more familiar bilinear-biquadratic form

$$H'_{\text{AKLT}} = \sum_{j=1}^L \left[\mathbf{S}_j \cdot \mathbf{S}_{j+1} + \frac{1}{3} (\mathbf{S}_j \cdot \mathbf{S}_{j+1})^2 + \frac{2}{3} \right] = 2H_{\text{AKLT}}. \quad (29.12)$$

The projector form [Eq. \(29.11\)](#) is the most transparent for exact solvability.

29.2.2 Valence-bond-solid construction

The VBS construction replaces every physical spin-1 by two virtual spin- 1/2 degrees of freedom. At site j , the two virtual spins are projected onto their symmetric triplet subspace:

$$\mathcal{H}_j^{S=1} = \text{Sym} (\mathbb{C}_{j,L}^2 \otimes \mathbb{C}_{j,R}^2). \quad (29.13)$$

Neighboring virtual spins form singlets

$$|\omega\rangle_{j,R;j+1,L} = \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_{j,R} |\downarrow\rangle_{j+1,L} - |\downarrow\rangle_{j,R} |\uparrow\rangle_{j+1,L} \right). \quad (29.14)$$

The AKLT state is obtained by placing singlets on all virtual bonds and applying the triplet projection at every site:

$$|\Psi_{\text{AKLT}}\rangle = \prod_j P_j^{\text{sym}} \prod_j |\omega\rangle_{j,R;j+1,L}. \quad (29.15)$$

For an open chain, two unpaired virtual spin-1/2 degrees of freedom remain at the ends. They produce a four-dimensional edge multiplet before boundary terms are added. This is the simplest microscopic origin of the spin- 1/2 edge states in the Haldane phase.

Proposition 29.1 (AKLT ground state). *The VBS state $|\Psi_{\text{AKLT}}\rangle$ is annihilated by every local projector $P_{j,j+1}^{(2)}$. Hence it is a zero-energy ground state of H_{AKLT} .*

Proof. Consider two neighboring physical spin-1 sites. In the virtual construction, the right virtual spin of site j and the left virtual spin of site $j+1$ form a singlet. Thus the total spin carried by the four virtual spin- 1/2 variables on the two sites cannot reach total spin 2. After projection to the physical triplet subspace at each site, the two physical spin-1's therefore have no component in the total-spin-2 sector. Consequently $P_{j,j+1}^{(2)} |\Psi_{\text{AKLT}}\rangle = 0$ for every bond. Since the Hamiltonian is a sum of positive projectors, the state is a ground state. $\square$

29.2.3 AKLT state as a matrix-product state

The VBS state can be written as an MPS with bond dimension $D = 2$. In the basis $|+1\rangle, |0\rangle, |-1\rangle$ of S^z , one convenient convention is

$$A^{+1} = \sqrt{\frac{2}{3}} \sigma^+, \quad A^0 = -\frac{1}{\sqrt{3}} \sigma^z, \quad A^{-1} = -\sqrt{\frac{2}{3}} \sigma^-. \quad (29.16)$$

The periodic-chain wavefunction is

$$|\Psi_{\text{AKLT}}\rangle = \sum_{m_1, \dots, m_L} \text{Tr} (A^{m_1} A^{m_2} \dots A^{m_L}) |m_1 m_2 \dots m_L\rangle. \quad (29.17)$$

For an open chain, the trace is replaced by boundary spinors in the virtual space. The different boundary spinors describe the edge spin-1/2 degrees of freedom.

The MPS transfer channel is

$$\mathbb{E}(X) = \sum_{m=-1}^1 A^m X (A^m)^\dagger. \quad (29.18)$$

With the convention Eq. (29.16), the identity is the leading fixed point and the three Pauli matrices form subleading eigenoperators with eigenvalue $-1/3$. Thus connected spin correlations decay as

$$\langle S_i^\alpha S_j^\alpha \rangle_c \propto \left(-\frac{1}{3}\right)^{|i-j|}, \quad \xi_{\text{AKLT}} = \frac{1}{\log 3}. \quad (29.19)$$

This is an exact statement obtained directly from the finite-dimensional MPS transfer matrix.

Remark 29.2 (Hidden order). Although ordinary spin correlations are short-ranged, the AKLT state has nonzero string order. A standard string correlator is

$$\mathcal{O}_{\text{string}}^z = - \lim_{|i-j| \rightarrow \infty} \left\langle S_i^z \exp \left(i\pi \sum_{k=i+1}^{j-1} S_k^z \right) S_j^z \right\rangle. \quad (29.20)$$

For the AKLT state this quantity is finite. It detects the hidden antiferromagnetic order associated with the Haldane symmetry-protected topological phase.

29.3 The cluster model

The cluster model is the simplest stabilizer example of an exact ground-state model with symmetry-protected topological structure. Unlike the AKLT chain, its local terms commute, so it is also full-spectrum soluble as a stabilizer Hamiltonian. It is nevertheless conceptually useful in this chapter because its exact ground state is a graph state and therefore a tensor-network fixed point.

29.3.1 One-dimensional cluster Hamiltonian

Consider a chain of qubits with Pauli operators X_j, Y_j, Z_j . On a periodic chain, define the stabilizers

$$K_j = Z_{j-1} X_j Z_{j+1}, \quad j = 1, \dots, L, \quad (29.21)$$

with site labels understood modulo L . The one-dimensional cluster Hamiltonian is

$$H_{\text{cl}} = - \sum_{j=1}^L K_j. \quad (29.22)$$

The stabilizers commute:

$$[K_i, K_j] = 0, \quad K_j^2 = \mathbf{1}. \quad (29.23)$$

Indeed, if K_i and K_j overlap nontrivially, they either share only commuting Pauli matrices or have two anticommuting overlaps, whose signs cancel. Therefore the full spectrum is labelled by eigenvalues $k_j = \pm 1$.

The cluster ground state $|C\rangle$ is the unique common $+1$ eigenstate on a periodic chain:

$$K_j |C\rangle = |C\rangle, \quad j = 1, \dots, L. \quad (29.24)$$

Equivalently, it is obtained by applying controlled- Z gates to the product state $|+\rangle^{\otimes L}$:

$$|C\rangle = \prod_{\langle j, j+1 \rangle} CZ_{j, j+1} |+\rangle^{\otimes L}, \quad X_j |+\rangle_j = |+\rangle_j. \quad (29.25)$$

This formula follows because conjugation by $CZ_{j, j+1}$ maps

$$X_j \mapsto Z_{j-1} X_j Z_{j+1} \quad (29.26)$$

when all nearest-neighbor controlled- Z gates are applied.

Remark 29.3 (Open-chain edge degeneracy). For an open chain, one may define bulk stabilizers $K_j = Z_{j-1} X_j Z_{j+1}$ for $2 \leq j \leq L-1$. Then the two boundary qubits are not fully fixed by the bulk terms, and protected edge degrees of freedom appear. This parallels the AKLT edge spin- $1/2$'s, although the cluster model is built from qubits and commuting stabilizers rather than spin-1 projectors.

29.3.2 Symmetry-protected topological interpretation

The one-dimensional cluster state is a fixed point of a nontrivial $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry-protected topological phase. A convenient choice of symmetry generators is

$$G_{\text{even}} = \prod_{j \text{ even}} X_j, \quad G_{\text{odd}} = \prod_{j \text{ odd}} X_j. \quad (29.27)$$

On a closed chain, these commute with the Hamiltonian. On an open chain, the symmetry acts projectively on the edge degrees of freedom. This projective edge action is the one-dimensional SPT invariant.

The cluster state also has a string order parameter. For example, products of stabilizers imply that

$$Z_i \left(\prod_{k=i+1}^{j-1} X_k \right) Z_j \quad (29.28)$$

has long-range expectation value in the cluster ground state for suitable parity choices of i and j . The precise endpoint operators depend on whether one chooses the even or odd symmetry sector, but the qualitative structure is robust: nonlocal string order replaces ordinary local order.

29.3.3 Graph-state generalization

The cluster construction extends immediately to any graph $\Gamma = (V, E)$. Place a qubit on every vertex and define

$$K_v = X_v \prod_{u:(u,v) \in E} Z_u. \quad (29.29)$$

The graph-state Hamiltonian is

$$H_\Gamma = - \sum_{v \in V} K_v. \quad (29.30)$$

All K_v 's commute, and the graph state is the common $+1$ eigenstate. For a two-dimensional square-lattice graph, this gives the two-dimensional cluster state, an important stabilizer resource state for measurement-based quantum computation.

Remark 29.4 (Cluster model versus toric code). Both the cluster model and the toric code are stabilizer Hamiltonians, but their orders are different. The cluster state is short-range entangled and becomes a trivial product state after a finite-depth circuit if the protecting symmetry is ignored. The toric code is long-range entangled and has intrinsic topological order. Thus stabilizer solvability by itself does not determine the phase; one must also examine entanglement, symmetry, and nonlocal operators.

29.4 Parent Hamiltonians of matrix-product states

The AKLT chain is not an isolated miracle. It is an example of a general MPS parent-Hamiltonian construction. The input is an exact tensor-network state; the output is a local frustration-free Hamiltonian for which that tensor network is an exact ground state.

29.4.1 Matrix-product states

Let each site have physical dimension d , with basis $|s\rangle$, $s = 1, \dots, d$. A translation-invariant MPS with bond dimension D on a periodic chain is

$$|\Psi_A\rangle = \sum_{s_1, \dots, s_L=1}^d \text{Tr}(A^{s_1} A^{s_2} \dots A^{s_L}) |s_1 s_2 \dots s_L\rangle, \quad (29.31)$$

where each A^s is a $D \times D$ matrix. The transfer channel is

$$\mathbb{E}(X) = \sum_{s=1}^d A^s X (A^s)^\dagger. \quad (29.32)$$

The state is called injective, possibly after blocking a finite number of sites, if the blocked tensors generate the full matrix algebra on the virtual space. More explicitly, for some finite length ℓ , the map

$$\Gamma_\ell : \text{Mat}_{D \times D}(\mathbb{C}) \rightarrow (\mathbb{C}^d)^{\otimes \ell}, \quad X \mapsto \sum_{s_1, \dots, s_\ell} \text{Tr}(X A^{s_1} \dots A^{s_\ell}) |s_1 \dots s_\ell\rangle \quad (29.33)$$

is injective.

Injectivity is the tensor-network analogue of having a unique locally generated ground state. Noninjective MPS describe symmetry-breaking or topologically ordered fixed-point structures and require a slightly refined parent construction.

29.4.2 Local support and the parent projector

Choose a block length ℓ at least as large as the injectivity length. Let $\mathcal{S}_\ell$ be the support subspace of the ℓ -site reduced density matrix of $|\Psi_A\rangle$. Equivalently,

$$\mathcal{S}_\ell = \text{Im } \Gamma_\ell. \quad (29.34)$$

Let Π_ℓ be the orthogonal projector onto $\mathcal{S}_\ell$. The local parent-Hamiltonian term is

$$h_{j,j+\ell-1} = \mathbf{1} - \Pi_{j,j+\ell-1}, \quad (29.35)$$

where $\Pi_{j,j+\ell-1}$ is the translated projector onto the corresponding support on sites $j, \dots, j + \ell - 1$. The parent Hamiltonian is

$$H_{\text{par}} = \sum_j h_{j,j+\ell-1}. \quad (29.36)$$

Because every local reduced density matrix of $|\Psi_A\rangle$ is supported in $\mathcal{S}_\ell$, one has

$$h_{j,j+\ell-1} |\Psi_A\rangle = 0 \quad \text{for all } j. \quad (29.37)$$

Thus $|\Psi_A\rangle$ is an exact frustration-free ground state.

Theorem 29.1 (Parent Hamiltonian of an injective MPS). *For an injective translation-invariant MPS, the parent Hamiltonian Eq. (29.36) has the MPS as its unique ground state on a periodic chain, for sufficiently large system size. On an open chain, the ground-state space is controlled by the virtual boundary degrees of freedom.*

Remark 29.5 (AKLT as an MPS parent Hamiltonian). For the AKLT MPS, the two-site physical Hilbert space decomposes into total spin $S = 0, 1, 2$. The two-site support of the VBS state contains only $S = 0$ and $S = 1$, while the $S = 2$ subspace is absent. Therefore the parent local term is precisely the spin-2 projector $P_{j,j+1}^{(2)}$. The AKLT Hamiltonian is an MPS parent Hamiltonian.

29.4.3 Spectral gap and correlation length

For injective MPS in one dimension, the parent Hamiltonian is generically expected to be gapped, and the standard injective parent construction indeed produces a finite spectral gap in the thermodynamic limit. The finite bond dimension also implies exponential decay of connected correlations. Both features are controlled by the transfer channel $\mathbb{E}$: if the leading eigenvalue is normalized to one and the subleading eigenvalue has magnitude $|\lambda_2| < 1$, then typical connected correlators decay as

$$C(r) \sim \lambda_2^r, \quad \xi = -\frac{1}{\log |\lambda_2|}. \quad (29.38)$$

The parent Hamiltonian and the transfer channel are therefore two sides of the same exact-ground-state structure: the parent Hamiltonian fixes the state as a ground state, while the transfer channel computes its equal-time observables.

29.5 Parent Hamiltonians of PEPS

Projected entangled-pair states are the natural two-dimensional generalization of MPS. A PEPS assigns a tensor to each vertex of a lattice. Its physical index labels the local Hilbert space, while its virtual indices are contracted along edges. The AKLT construction itself has a two-dimensional analogue on suitable lattices: place virtual spin-1/2's on incident bonds, form virtual singlets on edges, and project the virtual degrees of freedom at each site into a physical spin representation.

The PEPS parent-Hamiltonian construction parallels the MPS case. For a finite region R , let $\mathcal{S}_R$ be the support of the reduced density matrix of the PEPS on R . Define a local term

$$h_R = \mathbf{1} - \Pi_R, \quad (29.39)$$

where Π_R projects onto $\mathcal{S}_R$. Then

$$H_{\text{PEPS}} = \sum_R h_R \quad (29.40)$$

is frustration-free and has the PEPS as a ground state.

In two dimensions, however, the physics is more subtle than in injective one-dimensional MPS. PEPS parent Hamiltonians can describe

- (i) short-range-entangled trivial phases;
- (ii) symmetry-protected topological phases;
- (iii) intrinsic topological order, when the tensors have virtual gauge symmetries;
- (iv) critical states, where correlations may decay algebraically;
- (v) gapless or gapped parent Hamiltonians, depending on the tensor.

Thus the PEPS parent construction gives an exact ground state, but it does not by itself guarantee a simple spectrum or even a finite gap.

Remark 29.6 (Virtual symmetries and topological order). In PEPS, intrinsic topological order is often encoded by symmetries acting on the virtual indices. The toric code, string-net fixed points, and many quantum double states admit PEPS descriptions with such virtual gauge structures. In this language, topological degeneracy arises from nontrivial virtual boundary conditions rather than from ordinary local order parameters.

29.6 Comparison of AKLT, cluster, RK, and toric-code solvability

The models in this part use related but distinct exact mechanisms. The following table summarizes the differences.

Model or construction	Exact mechanism	What is exactly controlled	Full spectrum?
Toric code	Commuting stabilizer projectors	Ground space, anyon sectors, all stabilizer energies	Yes, in stabilizer variables
Cluster model	Commuting graph-state stabilizers	Unique or edge-degenerate stabilizer ground state, SPT edge structure	Yes, in stabilizer variables
AKLT chain	Noncommuting projector parent Hamiltonian	VBS/MPS ground state, edge spin-1/2's, correlations	Not by the parent construction alone
RK dimer model	Frustration-free projector at RK point	Equal-amplitude or weighted dimer ground state, classical correlators	Generally no
Injective MPS parent Hamiltonian	Local support projector	Unique exact MPS ground state and exponential correlations	Generally no
PEPS parent Hamiltonian	Local support projector with possible virtual symmetry	Exact tensor-network ground space, possibly topological sectors	Generally no

The important lesson is that projector methods are broader than commuting-projector methods. Commutativity solves the spectrum. Frustration-free parent Hamiltonians solve the ground state. Tensor-network transfer matrices then compute equal-time observables, much as the row-to-row transfer matrix computed classical correlations in earlier chapters.

29.7 A unified parent-Hamiltonian principle

The parent-Hamiltonian construction can be stated abstractly.

Principle 29.1 (Parent-Hamiltonian principle). Given a many-body state $|\Psi\rangle$, choose local regions R and let Π_R project onto the support of the reduced density matrix $\rho_R = \text{Tr}_{\bar{R}} |\Psi\rangle\langle\Psi|$. Then

$$H_{\text{par}}[\Psi] = \sum_R (1 - \Pi_R) \quad (29.41)$$

is positive semidefinite and satisfies

$$H_{\text{par}}[\Psi] |\Psi\rangle = 0. \quad (29.42)$$

If the intersection of the local supports is exactly the desired ground-state space, then $H_{\text{par}}[\Psi]$ is an exact frustration-free Hamiltonian for that state or phase.

This principle explains why the AKLT chain, graph states, PEPS string-net fixed points, and many RK-like constructions fit naturally into the same lecture-note architecture. They are not solved by the same algebra, but they share the same logic:

local tensor or local constraint $\implies$ finite-dimensional support condition $\implies$
projector Hamiltonian $\implies$ exact ground state and controlled correlations.

29.8 Common pitfalls

Several points are worth keeping separate.

- (i) *Frustration-free does not mean commuting.* The AKLT local projectors do not commute on overlapping bonds, although the exact ground state is annihilated by all of them.
- (ii) *Exact wavefunction does not imply exact spectrum.* Parent Hamiltonians usually determine the ground-state subspace, not all excited eigenstates.
- (iii) *Short-range entangled does not mean trivial under symmetry.* AKLT and cluster states are short-range entangled, but they represent nontrivial SPT phases when the relevant symmetries are preserved.
- (iv) *Tensor-network exactness is algebraic, not variational.* In an exact parent-Hamiltonian construction, the tensor network is not merely a good approximation; it is annihilated exactly by local projectors.
- (v) *Two-dimensional parent Hamiltonians are more delicate than one-dimensional ones.* PEPS may have topological sectors, virtual gauge symmetries, critical correlations, or gapless parent Hamiltonians.

29.9 Connection to the rest of the notes

The AKLT, cluster, and parent-Hamiltonian constructions complete the set of solvability mechanisms emphasized in these notes. They should be read in parallel with the previous chapters as follows:

XY/Kitaev/SSH	: quadratic fermions and BdG diagonalization,
2D Ising	: transfer matrices and free-fermion structure,
XXZ/six-vertex	: Yang–Baxter algebra and Bethe ansatz,
toric code/quantum double	: commuting projectors and topological constraints,
Kitaev honeycomb	: Majorana fermions in static gauge sectors,
RK/AKLT/cluster/parent models	: exact frustration-free ground states.

(29.43)

Together these examples show that exact solvability is not a single technique. It is a family of algebraic reductions, each exposing a different kind of structure in many-body quantum systems.

Part VI

Concluding Synthesis

Chapter 30

Mechanisms of Exact Solvability in Quantum Many-Body Models

The previous chapters developed several families of exactly solvable models: one-dimensional free-fermion spin chains, transfer-matrix formulations of two-dimensional classical systems, Yang–Baxter integrable quantum chains, commuting-projector topological phases, Majorana–gauge solvable models, Rokhsar–Kivelson points, and parent-Hamiltonian constructions. The purpose of this final chapter is to put those examples into one conceptual map.

The central lesson is simple but important:

Exact solvability is not a single technique. It is the manifestation of hidden structure.

Different models are solvable for different reasons. Sometimes the hidden structure is a nonlocal fermionization. Sometimes it is a commuting family of transfer matrices. Sometimes it is factorized scattering. Sometimes it is a local stabilizer algebra, a static gauge field, a frustration-free projector structure, or a tensor-network parent construction. These mechanisms are not equivalent, but they are comparable. Together they form a small set of organizing principles for a large part of exactly solvable many-body physics.

30.1 Several Meanings of Exact Solvability

The phrase *exactly solvable* is often used broadly. In a many-body context it is useful to distinguish several levels of exactness.

Definition 30.1 (Exact solvability). A many-body model is exactly solvable, in a specified sense, if some nonperturbative part of its many-body structure can be determined in closed or systematically controllable form. The controlled object may be the full spectrum, a transfer-matrix spectrum, the partition function, ground states, correlation functions, quasiparticle scattering, topological sectors, or a complete set of commuting projectors.

This definition is deliberately plural. For example, the transverse-field Ising chain and the AKLT chain are both exactly solvable, but not in the same sense. The transverse-field Ising chain can be mapped to a quadratic fermion Hamiltonian and spectrally diagonalized. The AKLT chain has an exactly known valence-bond-solid ground state and a frustration-free parent Hamiltonian, but its full excitation spectrum is not solved by the same elementary diagonalization.

Table 30.1: Several useful meanings of exact solvability.

Type of solvability	Controlled structure	Representative examples
Full quadratic spectral solution	The Hamiltonian is reduced to independent fermionic or bosonic modes by an exact transformation.	TFIM, XY chain, Kitaev chain
Transfer-matrix solution	The partition function or quantum spectrum is encoded in the eigenvalues of a transfer matrix.	2D Ising model, six-vertex model, eight-vertex model
Yang–Baxter integrability	There is a commuting family of transfer matrices and factorized many-body scattering.	XXX, XXZ, XYZ, Hubbard chain
Commuting-projector solution	The Hamiltonian is a sum of mutually commuting local terms, so the spectrum is organized by simultaneous eigenvalues.	Toric code, quantum double models, cluster stabilizers
Majorana–gauge solution	The spin problem decomposes into static gauge sectors, and each sector contains a quadratic Majorana Hamiltonian.	Kitaev honeycomb model
Frustration-free ground-state solution	The ground state is annihilated by every local positive term.	RK point, AKLT chain, MPS/PEPS parent Hamiltonians

The distinction matters because different exact solutions answer different physical questions. A free-fermion solution gives precise quasiparticle dispersions and correlation functions. A Bethe-ansatz solution gives interacting quasiparticles and thermodynamic Bethe equations. A commuting-projector solution gives topological degeneracy and anyonic superselection sectors with almost no conventional quasiparticle dispersion. A parent-Hamiltonian construction gives exact ground states and entanglement structure, but usually not a closed expression for all excited states.

Principle 30.1 (Solvability must specify what is solved). A statement that a model is exactly solvable should be accompanied by the object that is exactly controlled: the spectrum, the ground state, the partition function, the transfer-matrix eigenvalues, the scattering matrix, the correlation functions, or the topological sector decomposition.

30.2 The Free-Fermion Mechanism

The first major solvability mechanism in these notes is nonlocal fermionization followed by quadratic diagonalization. In one dimension, the Jordan–Wigner transformation converts spin flips into fermionic operators by attaching a parity string:

$$c_j = \left(\prod_{\ell=1}^{j-1} \sigma_{\ell}^z \right) \sigma_j^+, \quad c_j^{\dagger} = \left(\prod_{\ell=1}^{j-1} \sigma_{\ell}^z \right) \sigma_j^-. \quad (30.1)$$

The string is not an optional decoration. It is the operator that converts spin commutation relations on different sites into fermionic anticommutation relations.

After fermionization, many spin chains in these notes take the general quadratic form

$$H = \sum_{i,j} \left(c_i^\dagger A_{ij} c_j + \frac{1}{2} c_i^\dagger B_{ij} c_j^\dagger + \frac{1}{2} c_i B_{ji}^* c_j \right) + E_{\text{const}}, \quad (30.2)$$

where $A = A^\dagger$ and $B = -B^\top$. Equivalently, in Nambu notation,

$$H = \frac{1}{2} \begin{pmatrix} c^\dagger & c \end{pmatrix} \begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} c \\ c^\dagger \end{pmatrix} + E_0. \quad (30.3)$$

The remaining steps are linear algebra: Fourier transform, block diagonalization in momentum space, and Bogoliubov rotation.

30.2.1 Jordan–Wigner as nonlocal linearization

The Jordan–Wigner transformation solves an apparent paradox. The transverse-field Ising and XY chains are interacting spin systems, but their interactions are special: nearest-neighbor spin terms become quadratic fermion hopping and pairing terms. For example, spin operators on adjacent sites combine with the Jordan–Wigner strings so that the nonlocal strings cancel locally. This is why the method is powerful but also dimension-specific. In two spatial dimensions there is no simple one-dimensional ordering for which all local spin interactions remain local quadratic fermion terms.

The essential pattern is

$$\text{local spin interaction} \xrightarrow{\text{JW}} \text{quadratic fermion hopping/pairing}, \quad (30.4)$$

provided the spin interaction belongs to the appropriate Ising/XY family. Once the Hamiltonian is quadratic, its eigenstates are fermionic Gaussian states.

30.2.2 Bogoliubov diagonalization and criticality

For translation-invariant chains, the quadratic Hamiltonian decomposes into independent momentum blocks. A typical BdG block has the form

$$\mathcal{H}(k) = d_x(k)\tau^x + d_y(k)\tau^y + d_z(k)\tau^z, \quad (30.5)$$

with quasiparticle energy

$$\epsilon(k) = \sqrt{d_x(k)^2 + d_y(k)^2 + d_z(k)^2}. \quad (30.6)$$

A quantum critical point occurs when the gap closes:

$$\min_k \epsilon(k) = 0. \quad (30.7)$$

This is the common mechanism behind the critical transverse-field Ising chain, the gapless XX chain, the anisotropic XY chain, and the topological transition of the Kitaev chain. The universal low-energy theory is obtained by expanding $\epsilon(k)$ near the gapless momentum. Linear gap closing gives relativistic Majorana or Dirac modes with dynamical exponent $z = 1$; quadratic band touching gives $z = 2$ behavior.

30.2.3 Boundary sectors and parity

The free-fermion solution is exact only after the boundary and parity sectors are handled correctly. On a ring, the Jordan–Wigner string crossing the boundary produces a fermion-parity dependence:

$$c_{L+1} = \pm c_1, \quad (30.8)$$

where the sign is fixed by the total fermion parity sector. Thus the finite-size spectrum is not obtained by a single Fourier transform with an arbitrary momentum grid. One must solve the even and odd parity sectors separately and then impose the physical spin boundary condition.

This caveat is not a minor technicality. It is the finite-size remnant of the fact that fermionization is nonlocal. In the thermodynamic limit, the difference between the sectors often disappears in bulk observables, but it remains essential for exact finite-size spectra, degeneracies, and edge-mode physics.

30.2.4 Classical–quantum correspondence

The same free-fermion mechanism appears in the two-dimensional classical Ising model. The row-to-row transfer matrix T can be interpreted as an imaginary-time evolution operator,

$$T \sim e^{-\delta\tau H_{\text{eff}}}. \quad (30.9)$$

In the strongly anisotropic limit, the two-dimensional classical Ising model reduces to the one-dimensional transverse-field Ising chain with coupling dictionary of the form

$$K_x = \delta\tau J, \quad K_y = -\frac{1}{2} \log \tanh(\delta\tau h). \quad (30.10)$$

The Onsager critical curve

$$\sinh(2K_x) \sinh(2K_y) = 1 \quad (30.11)$$

then becomes the quantum critical point $h = J$ in the Hamiltonian limit.

This illustrates a recurring principle:

A d -dimensional quantum model at zero temperature is often related to a $(d + 1)$ -dimensional classical statistical model through imaginary time.

The relation is not merely formal. It identifies quantum criticality with anisotropic classical criticality and explains why transfer matrices are natural bridges between statistical mechanics and quantum many-body theory.

30.3 Transfer Matrices, Yang–Baxter Structure, and Bethe Ansatz

The second major solvability mechanism is integrability by commuting transfer matrices. This mechanism is more general than free fermions. The XXZ chain is interacting after Jordan–Wigner transformation, yet it remains exactly solvable because its many-body scattering factorizes into two-body scattering processes.

30.3.1 Transfer matrices as generating functions

The basic construction starts from an R -matrix acting on the tensor product of two two-dimensional spaces. The monodromy matrix is

$$M_a(u) = R_{aL}(u)R_{a,L-1}(u) \cdots R_{a1}(u), \quad (30.12)$$

where a is an auxiliary space and $1, \dots, L$ are quantum spaces. The transfer matrix is the auxiliary trace

$$\tau(u) = \text{Tr}_a M_a(u). \quad (30.13)$$

The Yang–Baxter equation implies

$$[\tau(u), \tau(v)] = 0 \quad \text{for all } u, v. \quad (30.14)$$

Therefore $\tau(u)$ is a generating function for commuting conserved quantities:

$$Q_n \propto \left. \frac{d^n}{du^n} \log \tau(u) \right|_{u=u_0}. \quad (30.15)$$

For a regular R -matrix satisfying $R(0) \propto P$, the first logarithmic derivative gives a local quantum Hamiltonian.

30.3.2 Coordinate Bethe ansatz

In coordinate Bethe ansatz, one works directly in the basis of magnon positions. In a sector with M down spins, the wavefunction is written as a superposition of plane waves in each ordering region:

$$\psi(x_1, \dots, x_M) = \sum_{P \in S_M} A(P) \exp \left(i \sum_{a=1}^M k_{P_a} x_a \right), \quad x_1 < \cdots < x_M. \quad (30.16)$$

The amplitudes $A(P)$ are related by two-body scattering. For the XXZ normalization used in these notes,

$$H_{\text{XXZ}}^{(+)} = J \sum_{j=1}^L \left(\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \Delta (\sigma_j^z \sigma_{j+1}^z - 1) \right), \quad (30.17)$$

with one-magnon dispersion

$$\varepsilon(k) = 4J(\cos k - \Delta). \quad (30.18)$$

Periodic boundary conditions give Bethe equations

$$e^{ik_j L} = \prod_{\ell \neq j} S(k_j, k_\ell), \quad j = 1, \dots, M. \quad (30.19)$$

The conceptual content of [Eq. \(30.19\)](#) is that taking one magnon around the ring is equivalent to scattering it through all other magnons.

30.3.3 Algebraic Bethe ansatz

Algebraic Bethe ansatz repackages the same physics in operator language. Write

$$M_a(u) = \begin{pmatrix} A(u) & B(u) \\ C(u) & D(u) \end{pmatrix}_a. \quad (30.20)$$

A reference state $|0\rangle$ is chosen so that

$$C(u) |0\rangle = 0, \quad A(u) |0\rangle = \alpha(u) |0\rangle, \quad D(u) |0\rangle = \delta(u) |0\rangle. \quad (30.21)$$

Bethe vectors are then created by the operator $B(u)$:

$$|\lambda_1, \dots, \lambda_M\rangle = B(\lambda_1) \cdots B(\lambda_M) |0\rangle. \quad (30.22)$$

The RTT algebra determines the action of $\tau(u) = A(u) + D(u)$ on this state. The unwanted terms vanish precisely when the rapidities satisfy the Bethe equations. In the hyperbolic XXZ convention used earlier, these equations take the representative form

$$\left[\frac{\sinh(\lambda_j + \eta)}{\sinh \lambda_j} \right]^L = \prod_{\ell \neq j} \frac{\sinh(\lambda_j - \lambda_\ell + \eta)}{\sinh(\lambda_j - \lambda_\ell - \eta)}. \quad (30.23)$$

30.3.4 One structure, several models

The transfer-matrix viewpoint explains why classical vertex models and quantum spin chains appear in pairs.

Table 30.2: Transfer-matrix and Bethe-ansatz families.

Classical/statistical object	Quantum Hamiltonian limit	Integrability structure
Six-vertex model	XXZ chain	Trigonometric R -matrix; $U(1)$ conservation
Rational six-vertex limit	XXX chain	Rational $R(u) = u\mathbf{1} + P$; enhanced $SU(2)$ symmetry
Eight-vertex model	XYZ chain	Elliptic R -matrix; Baxter TQ equation
Hubbard vertex structure	Hubbard chain	Nested Bethe ansatz; spin and charge rapidities
Long-range inverse-square structure	Haldane–Shastry / Calogero–Sutherland	Yangian symmetry; fractional statistics

The most compact statement is:

Bethe-ansatz integrability is the exact solvability of interacting many-body motion by factorized scattering.

Coordinate Bethe ansatz makes the factorized scattering visible in wavefunctions. Algebraic Bethe ansatz makes the same structure visible in the R -matrix algebra.

30.4 Topological Exact Solvability in Two Dimensions

Two-dimensional quantum models introduce a different class of exact solvability. Instead of diagonalizing a one-dimensional quadratic Hamiltonian or solving one-dimensional factorized scattering, one often exploits local constraints, gauge structure, and topological sectors.

30.4.1 Commuting projectors and the toric code

The toric code is the cleanest example. Spins live on links of a two-dimensional lattice, and the Hamiltonian is

$$H_{\text{TC}} = -J_e \sum_s A_s - J_m \sum_p B_p, \quad A_s = \prod_{\ell \ni s} \sigma_\ell^x, \quad B_p = \prod_{\ell \in \partial p} \sigma_\ell^z. \quad (30.24)$$

The exact solution follows from

$$[A_s, A_{s'}] = [B_p, B_{p'}] = [A_s, B_p] = 0. \quad (30.25)$$

The ground-state conditions are

$$A_s = +1, \quad B_p = +1. \quad (30.26)$$

On a torus, two global constraints reduce the number of independent stabilizers, leaving a four-dimensional ground-state space:

$$\dim \mathcal{H}_{\text{GS}}(\mathbb{T}^2) = 4. \quad (30.27)$$

This degeneracy is not caused by spontaneous symmetry breaking. It is topological: local operators cannot distinguish the four ground states in the thermodynamic limit.

The elementary excitations are e charges, m fluxes, and their composite $\epsilon = e \times m$. Their fusion rules are

$$e \times e = m \times m = \epsilon \times \epsilon = 1, \quad e \times m = \epsilon. \quad (30.28)$$

The mutual statistics of e and m is semionic: braiding one around the other gives a phase -1 .

30.4.2 Quantum doubles

Kitaev's quantum double models generalize the toric code from $G = \mathbb{Z}_2$ to an arbitrary finite group G . Each oriented link carries a group algebra Hilbert space

$$\mathcal{H}_\ell = \mathbb{C}[G]. \quad (30.29)$$

Vertex terms impose gauge invariance, while plaquette terms impose flatness. The Hamiltonian has the same schematic form:

$$H_{D(G)} = -J_e \sum_v A_v - J_m \sum_p B_p. \quad (30.30)$$

The exact solution again follows from commuting projectors. On a closed surface, ground states are gauge-invariant flat G -connections. On a torus they are labeled, roughly, by commuting pairs of group elements modulo simultaneous conjugation.

The anyons of the untwisted quantum double $D(G)$ are labeled by

$$(C, \rho), \quad (30.31)$$

where C is a conjugacy class of G , and ρ is an irreducible representation of the centralizer of a representative element of C . Their quantum dimensions are

$$d_{(C, \rho)} = |C| \dim \rho, \quad \mathcal{D} = |G|. \quad (30.32)$$

The toric code is the Abelian special case where all conjugacy classes and irreducible representations are one-dimensional.

30.4.3 Kitaev honeycomb model

The Kitaev honeycomb model is not a commuting-projector Hamiltonian. Its solvability is subtler. It uses spin fractionalization into four Majorana fermions,

$$\sigma_i^\alpha = i b_i^\alpha c_i, \quad D_i = b_i^x b_i^y b_i^z c_i = +1, \quad (30.33)$$

with link variables

$$u_{ij} = i b_i^\alpha b_j^\alpha \quad \text{on an } \alpha\text{-type bond.} \quad (30.34)$$

These link variables commute with the Hamiltonian, so they define static $\mathbb{Z}_2$ gauge sectors. In each fixed sector, the matter Majorana Hamiltonian is quadratic:

$$H[u] = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j. \quad (30.35)$$

Thus the honeycomb model combines two earlier mechanisms:

$$\text{static gauge-field decomposition} + \text{free Majorana diagonalization.} \quad (30.36)$$

Its gapped Abelian phases reduce, in the strong anisotropy limit, to toric-code topological order. With time-reversal-breaking perturbations, the gapless phase can be gapped into a chiral non-Abelian Ising topological phase.

Principle 30.2 (Two-dimensional exact solvability). In two dimensions, exact solvability often comes from decomposing the Hilbert space into gauge or topological sectors before diagonalizing the remaining degrees of freedom.

30.5 Frustration-Free and Parent-Hamiltonian Solvability

Another form of exact solvability controls the ground state rather than the entire spectrum. The general structure is frustration-free:

$$H = \sum_{\alpha} h_{\alpha}, \quad h_{\alpha} \geq 0, \quad h_{\alpha} |\Psi_0\rangle = 0 \quad \text{for every } \alpha. \quad (30.37)$$

It follows immediately that $|\Psi_0\rangle$ is an exact ground state with energy zero.

30.5.1 Rokhsar–Kivelson points

For a quantum dimer model, the Hilbert space consists of close-packed dimer coverings. On a flippable plaquette p , the RK Hamiltonian at $v = t$ is

$$H_p^{\text{RK}} = t(|h_p\rangle - |v_p\rangle)(\langle h_p| - \langle v_p|). \quad (30.38)$$

The equal-amplitude state in a fixed topological or winding sector $\mathcal{S}$,

$$|\Psi_{\mathcal{S}}\rangle = \frac{1}{\sqrt{N_{\mathcal{S}}}} \sum_{D \in \mathcal{S}} |D\rangle, \quad (30.39)$$

is annihilated by every local RK projector. Equal-time quantum correlations in this state reduce to correlations of the corresponding classical dimer ensemble. On the square lattice this connects the RK point to height models and quantum Lifshitz criticality. On the triangular lattice the RK point realizes a gapped $\mathbb{Z}_2$ dimer liquid.

30.5.2 AKLT chain

The spin-1 AKLT chain is another frustration-free model. Its Hamiltonian is a sum of projectors onto the total spin-2 subspace of neighboring spin-1's:

$$H_{\text{AKLT}} = \sum_j P_{j,j+1}^{(2)}. \quad (30.40)$$

Using the spin-1 algebra, this projector can be written as

$$P_{j,j+1}^{(2)} = \frac{1}{6}(\mathbf{S}_j \cdot \mathbf{S}_{j+1})^2 + \frac{1}{2}\mathbf{S}_j \cdot \mathbf{S}_{j+1} + \frac{1}{3}. \quad (30.41)$$

The exact ground state is the valence-bond-solid state: each spin-1 is decomposed into two virtual spin-1/2's, neighboring virtual spins form singlets, and the two virtual spins on each site are projected back to the symmetric spin-1 subspace. This gives an exact matrix-product state and explains the spin-1/2 edge modes of an open chain.

30.5.3 Cluster states and stabilizer parent Hamiltonians

The one-dimensional cluster Hamiltonian is

$$H_{\text{cl}} = - \sum_j K_j, \quad K_j = Z_{j-1} X_j Z_{j+1}. \quad (30.42)$$

The stabilizers commute and satisfy $K_j^2 = 1$. Hence the ground state is exactly characterized by

$$K_j |\Psi_{\text{cl}}\rangle = |\Psi_{\text{cl}}\rangle \quad \text{for all } j. \quad (30.43)$$

This is both a commuting-projector model and a parent Hamiltonian for a graph state. In one dimension, with suitable symmetry, it is also a canonical example of symmetry-protected topological order.

30.5.4 MPS and PEPS parent Hamiltonians

The parent-Hamiltonian idea generalizes these examples. Suppose $|\Psi\rangle$ is an MPS or PEPS. For a region R , let Π_R be the projector onto the support of the reduced density matrix ρ_R . A local parent Hamiltonian is

$$H_{\text{parent}} = \sum_R (\mathbf{1} - \Pi_R). \quad (30.44)$$

By construction,

$$(\mathbf{1} - \Pi_R) |\Psi\rangle = 0 \quad \text{for every } R. \quad (30.45)$$

The state $|\Psi\rangle$ is therefore an exact ground state.

The important distinction is:

A parent Hamiltonian gives exact ground-state solvability. It does not automatically give full spectral solvability.

This separates AKLT, cluster, RK, and PEPS parent models from free-fermion and Bethe-ansatz systems.

30.6 A Map of Mother Models and Their Descendants

The organizing philosophy of these notes is that many named models are best understood as descendants of a smaller number of mother structures. The following table summarizes the resulting map.

Table 30.3: Mother structures and representative descendants.

Mother structure	Representative model	Descendants or limits	Main method
Quadratic spin-chain fermions	Transverse-field XY chain	TFIM, XX chain, Kitaev chain	Jordan–Wigner + Bogoliubov
BdG two-band structure	Kitaev / SSH chains	Majorana chain, chiral two-band insulator	Nambu spinor and winding number
Classical transfer matrix	2D classical Ising model	1D TFIM Hamiltonian limit	Row-to-row transfer matrix
Free Majorana/Pfaffian structure	2D Ising and dimer models	Ising fermions, Kasteleyn dimers	Pfaffians and covariance matrices
Trigonometric R -matrix	Six-vertex / XXZ	XXX rational limit, six-vertex phases	Bethe ansatz
Elliptic R -matrix	Eight-vertex / XYZ	XXZ degeneration	Baxter transfer matrix
Nested integrability	Hubbard chain	Spin-charge rapidities	Nested Bethe ansatz

Mother structure	Representative model	Descendants or limits	Main method
Commuting projectors	Toric code	Stabilizer codes, $D(\mathbb{Z}_2)$	Projector algebra
Finite-group gauge theory	Quantum double $D(G)$	Abelian and non-Abelian anyons	Flat connections and representation theory
Static gauge Majoranas	Kitaev honeycomb model	Toric-code limit, Ising topological order	Majorana + $\mathbb{Z}_2$ gauge field
RK projector construction	Quantum dimer models	Quantum Lifshitz point, $\mathbb{Z}_2$ dimer liquid	Classical-to-quantum mapping
Tensor-network parent structure	AKLT, cluster, MPS/PEPS	SPT states, exact tensor-network states	Frustration-free parent Hamiltonian

The mother-model viewpoint also clarifies what should be learned from a model. The TFIM teaches fermionization, parity sectors, and quantum criticality. The XXZ chain teaches interacting integrability and factorized scattering. The toric code teaches topological degeneracy and anyon statistics. The AKLT chain teaches exact tensor-network ground states and symmetry-protected edge degrees of freedom. The Kitaev honeycomb model teaches fractionalization into matter Majoranas and static gauge fluxes.

30.7 Criticality, Topology, and Fractionalization

Exact solvability is valuable because it gives sharply defined mechanisms for phases and excitations. Three themes recur throughout the notes: criticality, topology, and fractionalization.

30.7.1 Criticality

At a quantum critical point, the excitation gap closes:

$$\Delta_{\text{gap}} \rightarrow 0. \quad (30.46)$$

Consequently, the correlation length and correlation time diverge:

$$\xi \rightarrow \infty, \quad \tau \sim \xi^z \rightarrow \infty. \quad (30.47)$$

Exactly solvable models show this mechanism without approximation. The TFIM has an Ising critical point. The XY chain contains several gapless regimes and anisotropic transitions. The 2D classical Ising model gives the exact Onsager critical point. The XXZ chain is gapless for $-1 < \Delta \leq 1$ in the antiferromagnetic convention and is described at low energy by a Luttinger liquid. The square-lattice RK point realizes a $z = 2$ quantum Lifshitz critical point.

Thus solvable models are not merely examples with closed-form spectra; they are laboratories for the exact relation between gap closing, long-distance field theory, and universality.

30.7.2 Topology

Topological phases do not fit into the Landau paradigm of local order parameters. The toric code ground states on a torus cannot be distinguished by any local operator. Quantum double models organize topological charge by conjugacy classes and centralizer representations. The Kitaev honeycomb model realizes Abelian topological order in its strong-coupling phases and non-Abelian Ising topological order after time-reversal breaking. The Kitaev chain and SSH chain show one-dimensional topological phases through boundary zero modes and winding numbers. AKLT and cluster models illustrate symmetry-protected topology rather than intrinsic topological order.

A useful distinction is:

$$\text{intrinsic topological order} \neq \text{symmetry-protected topological order.} \quad (30.48)$$

The former has long-range entanglement and anyonic excitations in two dimensions. The latter is short-range entangled but nontrivial in the presence of protecting symmetries.

30.7.3 Fractionalization

Fractionalization occurs when the elementary degrees of freedom of the Hamiltonian are not the elementary excitations of the phase. Several versions appeared in these notes:

- (i) In Jordan–Wigner solvable chains, spin flips are represented by nonlocal fermions.
- (ii) In the Kitaev chain, the fermion can be decomposed into Majorana components, and boundary Majorana zero modes appear in the topological phase.
- (iii) In the toric code, local spin operators create pairs of anyons rather than isolated local particles.
- (iv) In the Kitaev honeycomb model, a spin operator fractionalizes into a matter Majorana excitation together with a change of $\mathbb{Z}_2$ flux sector.
- (v) In Bethe-ansatz chains, stable quasiparticles scatter elastically with nontrivial two-body phase shifts.
- (vi) In the Haldane–Shastry chain, spinon excitations behave as fractionalized spin-1/2 objects with fractional statistics.

The recurring lesson is that exact solvability often reveals the correct quasiparticles. These quasiparticles may be nonlocal relative to the microscopic variables.

30.8 What Exact Solvability Does Not Mean

It is equally important to state what exact solvability does *not* imply.

Remark 30.1 (Exact ground state versus exact spectrum). An exact ground state does not imply an exact solution of the full spectrum. AKLT, RK, MPS, and PEPS parent Hamiltonians are exact ground-state models, but generic excitations need not be solvable by free-particle or Bethe-ansatz methods.

Remark 30.2 (Free fermions can be physically nontrivial). Free-fermion solvability does not mean trivial physics. The TFIM contains a nontrivial quantum critical point. The Kitaev chain contains topological boundary modes. The Kitaev honeycomb model is quadratic only after decomposition into gauge sectors and supports topological order and non-Abelian anyons.

Remark 30.3 (Integrability is fragile). Bethe-ansatz integrability depends on an infinite set of commuting charges. Generic perturbations, such as arbitrary next-nearest-neighbor interactions, usually destroy this structure. The resulting model may still be physically close to an integrable point at intermediate scales, but it is no longer exactly solvable in the same sense.

Remark 30.4 (Commuting projectors are not conventional order). Commuting-projector models often have zero correlation length in local operators, but they can have nontrivial topological order. The absence of a local order parameter does not imply a trivial phase.

Remark 30.5 (Thermodynamic and finite-size solutions require different care). Finite-size exact solutions require attention to parity sectors, boundary conditions, Bethe quantum numbers, string deviations, gauge constraints, and topological sectors. These details may disappear from bulk thermodynamics, but they are essential for exact finite-size spectra and degeneracies.

These cautions are part of the discipline of exact solvability. A correct solution is not just a formal diagonalization; it must also specify the Hilbert space, boundary conditions, constraints, and sectors.

30.9 Outlook: From Solvable Models to Modern Many-Body Physics

Exactly solvable models remain central because they provide calibration points for theory, numerics, and physical intuition. They are used to test field-theory predictions, benchmark tensor-network algorithms, verify quantum Monte Carlo simulations, construct topological quantum codes, study integrability breaking, and train intuition about phases that survive away from the exactly solvable point.

Several modern directions grow naturally from the models studied here.

Conformal field theory. Critical solvable models such as the TFIM, XXZ chain, and 2D classical Ising model flow to conformal field theories in the infrared. Exact lattice solutions identify the microscopic origin of scaling dimensions, central charges, operator content, and universal finite-size spectra.

Tensor networks. AKLT, cluster, MPS, and PEPS parent Hamiltonians show how exact wavefunctions can be constructed from local tensors. This is the conceptual basis for modern tensor-network variational methods. Exact parent models provide fixed points and benchmarks for numerical algorithms.

Topological quantum matter. Toric code, quantum double models, and Kitaev honeycomb models provide microscopic realizations of anyons, topological degeneracy, and fault-tolerant logical operators. They are foundational models for topological quantum computation and quantum error correction.

Integrability breaking and thermalization. The XXZ and related Bethe-ansatz chains are starting points for studying what happens when integrability is weakly broken. The contrast between integrable dynamics and generic thermalizing dynamics is sharp only because the integrable limit is exactly controlled.

Numerical and machine-learning benchmarks. Exactly solvable models provide ground-truth data for algorithms. Free-fermion correlation matrices, Bethe-ansatz energies, toric-code sectors, and RK wavefunctions give controlled tests for tensor networks, Monte Carlo methods, neural quantum states, and generative models.

The final lesson can be summarized as follows:

Solvability is hidden structure made explicit.

The structures encountered in these notes—nonlocal fermionization, BdG linear algebra, transfer matrices, Yang–Baxter equations, factorized scattering, stabilizer algebras, gauge constraints, topological sectors, and parent Hamiltonians—are not merely solution tricks. They are organizing principles for quantum many-body physics.

Final Summary

The models studied in these notes can be arranged into one compact hierarchy:

$$\begin{aligned}
 &\text{spin algebra} \longrightarrow \text{Jordan–Wigner fermions} \longrightarrow \text{BdG/free-fermion mother models}, \\
 &\text{transfer matrices} \longrightarrow \text{classical–quantum correspondence} \longrightarrow \text{Ising and vertex models}, \\
 &\text{Yang–Baxter equation} \longrightarrow \text{commuting transfer matrices} \longrightarrow \text{Bethe-ansatz integrable chains}, \\
 &\quad \text{local projectors and gauge constraints} \longrightarrow \text{topological exact solvability}, \\
 &\quad \text{frustration-free tensor structures} \longrightarrow \text{exact ground-state parent Hamiltonians}.
 \end{aligned}
 \tag{30.49}$$

Each line represents a different answer to the question: *why can this many-body problem be solved?*

The answer is never simply that the model is special. It is special because it possesses a structure strong enough to reorganize the Hilbert space. Recognizing that structure is the main skill developed by studying exactly solvable quantum many-body models.

Essential References

These notes are intended to be largely self-contained. The references below are not exhaustive; they are selected as minimal original sources and standard texts for the main exactly solvable structures discussed in the notes. The emphasis is therefore on representative sources for the solvability mechanisms rather than on a complete historical bibliography.

Bibliography

- [1] P. Jordan and E. Wigner, “Über das Paulische Äquivalenzverbot,” *Zeitschrift für Physik* **47**, 631–651 (1928).
- [2] E. H. Lieb, T. Schultz, and D. Mattis, “Two soluble models of an antiferromagnetic chain,” *Annals of Physics* **16**, 407–466 (1961).
- [3] L. Onsager, “Crystal statistics. I. A two-dimensional model with an order-disorder transition,” *Physical Review* **65**, 117–149 (1944).
- [4] W. P. Su, J. R. Schrieffer, and A. J. Heeger, “Solitons in polyacetylene,” *Physical Review Letters* **42**, 1698–1701 (1979).
- [5] A. Y. Kitaev, “Unpaired Majorana fermions in quantum wires,” *Physics-Uspekhi* **44**, 131–136 (2001).
- [6] H. Bethe, “Zur Theorie der Metalle. I. Eigenwerte und Eigenfunktionen der linearen Atomkette,” *Zeitschrift für Physik* **71**, 205–226 (1931).
- [7] R. J. Baxter, *Exactly Solved Models in Statistical Mechanics*, Academic Press, London (1982).
- [8] L. D. Faddeev, “How algebraic Bethe ansatz works for integrable models,” in *Quantum Symmetries / Symetries Quantiques*, edited by A. Connes, K. Gawedzki, and J. Zinn-Justin, North-Holland, Amsterdam (1998), pp. 149–219; also available as arXiv:hep-th/9605187.
- [9] E. H. Lieb and F. Y. Wu, “Absence of Mott transition in an exact solution of the short-range, one-band model in one dimension,” *Physical Review Letters* **20**, 1445–1448 (1968).
- [10] F. D. M. Haldane, “Exact Jastrow–Gutzwiller resonating-valence-bond ground state of the spin- $\frac{1}{2}$ antiferromagnetic Heisenberg chain with $1/r^2$ exchange,” *Physical Review Letters* **60**, 635–638 (1988).
- [11] B. S. Shastry, “Exact solution of an $S = \frac{1}{2}$ Heisenberg antiferromagnetic chain with long-ranged interactions,” *Physical Review Letters* **60**, 639–642 (1988).
- [12] B. Sutherland, *Beautiful Models: 70 Years of Exactly Solved Quantum Many-Body Problems*, World Scientific, Singapore (2004).
- [13] A. Y. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003).

- [14] A. Y. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006).
- [15] P. W. Kasteleyn, “The statistics of dimers on a lattice: I. The number of dimer arrangements on a quadratic lattice,” *Physica* **27**, 1209–1225 (1961).
- [16] D. S. Rokhsar and S. A. Kivelson, “Superconductivity and the quantum hard-core dimer gas,” *Physical Review Letters* **61**, 2376–2379 (1988).
- [17] I. Affleck, T. Kennedy, E. H. Lieb, and H. Tasaki, “Rigorous results on valence-bond ground states in antiferromagnets,” *Physical Review Letters* **59**, 799–802 (1987).
- [18] R. Raussendorf and H. J. Briegel, “A one-way quantum computer,” *Physical Review Letters* **86**, 5188–5191 (2001).
- [19] M. Fannes, B. Nachtergaele, and R. F. Werner, “Finitely correlated states on quantum spin chains,” *Communications in Mathematical Physics* **144**, 443–490 (1992).
- [20] S. Sachdev, *Quantum Phase Transitions*, 2nd ed., Cambridge University Press, Cambridge (2011).
- [21] E. Fradkin, *Field Theories of Condensed Matter Physics*, 2nd ed., Cambridge University Press, Cambridge (2013).
- [22] P. Di Francesco, P. Mathieu, and D. Senechal, *Conformal Field Theory*, Springer, New York (1997).
- [23] A. Altland and B. D. Simons, *Condensed Matter Field Theory*, 2nd ed., Cambridge University Press, Cambridge (2010).